



**GREEN
CLIMATE
FUND**

Meeting of the Board

15 – 18 July 2024

Songdo, Incheon, Republic of Korea

Provisional agenda item 10

GCF/B.39/02/Add.10

24 June 2024

Consideration of funding proposals - Addendum X

Funding proposal package for FP235

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled “Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador”;
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat’s assessment;
- e) Independent Technical Advisory Panel’s assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel’s assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador
Country(ies):	Ecuador
Accredited Entity:	Conservation International Foundation
Date of first submission:	<u>2023/05/01</u>
Date of current submission	<u>2024/06/05</u>
Version number	<u>V.12</u>



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill **out** information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]"

Acronyms and Abbreviations

AAE	Asesoramiento Ambiental Estratégico
AGB	Aboveground biomass
ASC	Aquaculture Stewardship Council
ASOTEP	Asociación de Terminales Portuarios Privados del Ecuador (Association of Private Port Terminals of Ecuador)
AUSCEM	Acuerdos para el Uso Sostenible y Custodia del Manglar (Agreements for Sustainable Use and Custody of the Mangrove Ecosystem)
BGB	Belowground biomass
BUR	Biennial Update Report
CAF	Corporación Andina de Fomento (Andean Development Corporation)
CBD	Convention on Biological Diversity
CI	Conservation International Foundation
CI-E	Conservation International's Ecuador country program
CIIFEN	Centro Internacional para la Investigación del Fenómeno de El Niño (International Research Centre for El Niño Phenomenon)
CITES	Convention on International Trade in Endangered Species of Wild Fauna and Flora
CLIRSEN	Centro de Levantamientos Integrados de Recursos Naturales por Sensores Remotos (Center for Integrated Survey of Natural Resources by Remote Sensing)
COA	Código Orgánico del Ambiente (Organic Environmental Code)
CSS	Climate Smart Shrimp
dbh	Diameter at Breast Height
DW	Deadwood
EFU	Estuary Field Unit
ENF	Evaluación Nacional Forestal (National Forest Inventory)
ENSO	El Niño-Southern Oscillation
ERU	Emissions Reductions Unit
ESL	Extreme Sea Level
FIAS	Fondo de Inversión Ambiental Sostenible (Sustainable Environmental Investment Fund)
FPIC	Free, Prior and Informed Consent
FREL	Forest Reference Emissions Level

GAD	Gobierno Autónomo Descentralizado (Autonomous Decentralized Government)
GAP	Gender Action Plan
GCF	Green Climate Fund
GEF	Global Environment Facility
GHG	Greenhouse gas
GMA	Global Mangrove Alliance
INEC	Instituto Nacional de Estadística y Censos (National Institute for Statistics and Census)
INOCAR	Instituto Oceanográfico y Antártico de la Armada (Oceanographic Institute of the Navy)
ITCZ	Inter-Tropical Convergence Zone
IUCN	International Union for Conservation of Nature
KPI	Key Performance Indicator
MAATE	Ministerio del Ambiente, Agua y Transición del Ecuador (Ministry of Environment, Water and Transition of Ecuador)
MTOP	Ministerio de Transporte y Obras Públicas (Ministry of Transport and Public Works)
MRV	Monitoring, Reporting and Verification
NAP	National Adaptation Plan
NCA	National Chamber of Aquaculture
NDA	Nationally Designated Authority
NDC	Nationally Determined Contributions
NRT	Near Real-Time
PA	Protected Area(s)
Project Management Unit	Project Management Unit
PAN-Manglares Ecuador	National Action Plan for the Conservation of the Mangroves of Continental Ecuador
PAR-Manglares	Regional Action Plan for the Conservation of Mangroves in the Southeast Pacific
PDOT	Plan de Desarrollo y Ordenamiento Territorial (Development and Land Use Plan)
PECC	Programa Ecuador Carbono Cero (Ecuador Zero Carbon Program)
PUCESE	Pontificia Universidad Católica del Ecuador Sede Esmeraldas (Pontifical Catholic University of Ecuador Esmeraldas Extension)
RBP	Results Based Program

RCP	Representative Concentration Pathway
RE	Reserva Ecológica (Ecological Reserve)
REM	REDD Early Movers
RPF	Reserva de Producción de Flora y Fauna (Flora and Fauna Production Reserve)
RVS	Reserva de Vida Silvestre (Wildlife Refuge)
REDD+	Reducing emissions from deforestation and forest degradation, and fostering conservation, sustainable management of forests, and enhancement of forest carbon stocks
SATA	Sistema de Alerta Temprana Ambiental
SBP	Programa Socio Bosque (Forest Partner Incentive Program)
SC	Steering Committee
SDG	Sustainable Development Goals
SGMC	Subsecretaría de Gestión Marina y Costera (Undersecretariat of Marine and Coastal Management)
SLR	Sea Level Rise
SM	Socio Manglar (Mangrove Partner)
SNGR	Servicio Nacional de Gestión de Riesgos y Emergencias (National Service for Risk Management and Emergencies)
SOC	Soil Organic Carbon
SOF	Althelia Sustainable Ocean Fund
SPA	South Pacific Anticyclone
SSP	Sustainable Shrimp Partnership
SUIA	Sistema Único de Información Ambiental (Unique Environmental Information System)
TULSMA	Texto Unificado de Legislación Secundaria Ambiental (Unified Text of Secondary Environmental Legislation)
UNCCD	United Nations Convention to Combat Desertification
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UOCV	Unidades Operativas de Control y Vigilancia (Operational Units for Control and Surveillance)
VMAP	Viceministerio de Acuicultura y Pesca (Viceministry for Aquaculture and Fisheries)
WB	World Bank
WRI	World Resources Institute

PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Project	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	<u>Not applicable</u>			
A.4. Result area(s)				
		GCF contribution	Co-financers' contribution¹	
	Mitigation total			
	☐ Energy generation and access			
	☐ Low-emission transport			
	☐ Buildings, cities, industries and appliances			
	☒ Forestry and land use	41 %	100 %	
	Adaptation total			
	☒ Most vulnerable people and communities	29.5 %		
	☐ Health and well-being, and food and water security			
☐ Infrastructure and built environment				
☒ Ecosystems and ecosystem services	29.5 %			
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	4,609,470 tCO ₂ e over 20-year project lifespan ² ; 732,111 tCO ₂ e by the end of the implementation period.	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	Indicate total number of direct and indirect beneficiaries	
			89,600 direct beneficiaries	3.4 million indirect beneficiaries
			1.8% of population of coastal provinces; 0.6 % of Ecuador's population	44.7% of coastal provinces; 20.1% Ecuador's population
A.7. Total financing (GCF + co-finance³)	\$45.9 million USD	A.9. Project size	Small (Upto USD 50 million)	
A.8. Total GCF funding requested	\$36.4 million USD			

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e., GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² The total lifespan of the project/programme is defined as the maximum number of years over which the outcomes of the investment are expected to be effective. This is different from the project/programme implementation period.

³ Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant <u>36.4 million USD</u> ☐ Equity ☐ Loan ☐ Results-based payment ☐ Guarantee		
A.11. Implementation period	6.5 years	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	<i>12/31/2023</i>	A.14. ESS category	B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐ <i>The CN was included within the application for PPF-009</i>	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ⁴ No
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	Conservation International Foundation (CI)		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

⁴ Ecuador has not yet prepared a GCF Country Programme, but does have a pipeline of expected projects, in which this project is included.

Impacts of climate change⁵

Climate change is projected to create unprecedented changes in conditions along Ecuador's coast, including increases in sea level, extreme El Niño-Southern Oscillation events, higher intensity and variability of precipitation, and increased atmospheric temperatures. The combination of these changes is expected to cause significant and more frequent coastal flooding, which will damage local communities and economic activity. Recent flood events have caused death, homelessness, increased disease risk and over US\$1 billion of damages and economic losses (EM-DAT, 2021)⁶. Mangroves serve as natural structural barriers that reduce coastal erosion and provide important flood-control functions, by dissipating wave energy and storm damage. Current mangrove cover directly provides flood protection for coastal populations and is estimated to prevent approximately US\$250 million in losses annually from flooding in Ecuador's coastal region. Conversely, ongoing deforestation of mangroves in proximity to coastal populations magnifies exposure to climate-related flood risks and other climate impacts. Nearly 25% of Ecuador's mangroves have been deforested since 1969 with conversion to shrimp farms as the largest driver. As of 2018, there were 156,633 hectares (ha) of mangrove remaining in mainland Ecuador, with an average of 990 ha of primary mangrove forests lost each year.

There are opportunities to consolidate recent mangrove conservation successes and for mangrove restoration that would strengthen natural flood protection while also providing other ecosystem service co-benefits (especially supporting artisanal fisheries). Ecuador's mangroves have been identified as part of the world's irrecoverable carbon stock – stores of carbon in nature that are vulnerable to release from human activity but, if lost, could not be restored by 2050 when the world must reach net-zero emissions to avoid the worst impacts of climate change (Noon *et al.*, 2021). Protecting these carbon stocks and harnessing the long-term ability of mangroves to sequester carbon will provide important global climate change mitigation benefits.

Despite recent stabilization of net mangrove coverage, in large part due to one-off restoration efforts during 2008-2012 and community-based mangrove management, mangrove areas are expected to be a net source of emissions from lost biomass and soil carbon for decades to come (see **Annex 2**). This project aims to transform the situation by coordinating public sector, private sector and community efforts to protect and restore mangroves, thereby reducing mangrove loss, resulting in a reduction of 4.6 M tCO₂e of greenhouse gas (GHG) emissions⁷, and accelerating Ecuador's mangrove recovery to maintain and enhance flood protection for 89,600 people⁸ from annual flooding risk and enhance the livelihoods of 41,500 people to increase their resiliency.

Proposed interventions

The current paradigm underestimates climate risks and undervalues the ecosystem services provided by mangroves, augmenting both exposure to climate risk and greenhouse gas emissions. By fully valuing the adaptation and mitigation benefits of mangrove ecosystems and integrating these into governance and economic frameworks across Ecuador's coastal region, the project will shift this paradigm with an innovative approach that will create public, private and community partnerships to reduce GHG emissions and improve the livelihoods of some of the country's most vulnerable people.

If local communities are provided with knowledge and resources for mangrove management and livelihoods development, and **if** the private sector and government actively collaborate on mangrove protection and restoration, and the enabling environment for mangrove protection is strengthened, **then** coverage and quality of mangrove ecosystems will be increased, resulting in reduced climate change impacts on vulnerable coastal populations, increased economic resilience, and reduced GHG emissions **because** healthy and more extensive mangroves reduce flood impacts and sequester carbon.

By integrating the participation of local communities, private sector businesses, Ministry of Environment, Water and Ecological Transition (MAATE) and subnational governments, the project will deliver mutually reinforcing, multi-scale actions across the three project components to address key barriers.

Component 1. Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and protected areas (PAs) implementing climate adaptation plans. This component focuses on engaging local communities to strengthen community management of mangroves, restore mangroves, increase the area of mangroves under protection by local community stakeholders, and improve community economic sustainability.

Component 2. The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations. This component focuses on engaging the private sector, particularly shrimp aquaculture, to become transformational agents for change, reversing their previous role as an agent of mangrove

loss, by integrating climate-smart production practices to increase climate resilience, reduce pressures for mangrove deforestation, and restore mangrove ecosystems, as well as catalyzing new sources of financing for long-term sustainability of mangrove conservation. An endowment of US\$4 million is requested from GCF to build up a dedicated financial mechanism (the recently created Socio Manglar subaccount of the Socio Bosque Fund⁹) to provide long-term support to communities managing mangroves through the Agreements for Sustainable Use and Custody of the Mangrove Ecosystem (AUSCEM) model. The US\$4 million will be held in an interest-bearing account and the interest used to fund community incentive payments. Alongside this, CI and MAATE will seek additional financial contributions to build up the capital of the investment account of the Socio Manglar subaccount. These additional contributions for community management of mangroves will be sought from the private sector during the project as part of innovative Government of Ecuador programs to encourage private investment in climate and nature conservation initiatives¹⁰. The project will promote more sustainable shrimp production methods but will not invest directly in shrimp farms. Investment on shrimp farms to implement more sustainable production methods will be financed by the private sector and will not use GCF funds.

Component 3. Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement. This component focuses on creating the enabling conditions to increase adaptation and mitigation, through improved governance and the generation of timely, targeted information, which are the scaffolding supporting mangrove conservation, planning, and regulation.

The project will contribute to achieving the GCF's Strategic Plan (2024-2027) through its impact on the following four GCF 'targeted results' for climate mitigation and adaptation efforts.

1. **T1 Readiness** (GCF Strategic Priority on Country Programming): National Climate Plan implementation. Ecuador will be directly supported to advance implementation of its Nationally Determined Contribution (NDC) as set out in the national NDC implementation plan (which also currently serves as the implementation plan for the new (2023) National Adaptation Plan regarding adaptation interventions). The project will support Ecuador's conditional GHG reduction goals and adaptation goals to increase the cover of mangrove and expand the use of community-based management of mangroves through AUSCEM. The project is expected to reduce national GHG emissions by 730,000 tCO₂e over the project implementation period and 4.6 MtCO₂e over the project lifespan. The project will strengthen management of 70,000 ha of mangroves under community management through establishing new areas under management and strengthening existing ones as part of the Socio Bosque program (SBP)¹¹. Ecuador's current NDC goal is to increase the forest area in the SBP by 127,500 ha, including mangroves. As part of strengthening mangrove management, the project will establish an additional 10 community-held AUSCEMs covering mangroves, in addition to strengthening the existing 59 community associations with AUSCEM agreements. (Ecuador's current NDC goal is to establish 30 new AUSCEM agreements covering mangrove areas.)
2. **T4 Food** (GCF Strategic Priority on Food). Mangroves provide a range of ecosystem services that directly and indirectly contribute to the livelihoods and food security of local people. In addition to the flood protection services that provide important climate change adaptation benefits, mangroves serve as important nurseries and breeding grounds for species of fish, shrimp, crabs and clams that support artisanal and commercial/industrial fisheries. 41,500 people are expected to benefit from low-emission, climate resilient fisheries management practices that underpin their livelihoods.
3. **T5 Ecosystems** (GCF Strategic Priority on Ecosystems). The project will contribute to the protection and sustainable management of 156,633 ha of mangrove ecosystems. This will be achieved through a combination of increasing the area under community management, strengthening PA planning to integrate climate change adaptation, improved coastal land-use planning and strengthening the capacity of government agencies and local government to enforce existing regulations designed to protect mangroves.

⁵ This section summarizes key information presented in more detail, along with references, in Chapter 2 of Annex 2, the Feasibility Study.

⁶ Full references for citations in the Funding Proposal Document can be found in Annex 2, the Feasibility Study

⁷ The 4.6 MtCO₂e emissions reductions are projected over the 20-year lifespan of the project. A total of 732,000 tCO₂e are projected to be achieved during the 7-year implementation period.

⁸ This is the number of people expected to benefit annually from reduced flooding risk due to project interventions (see Annex 25)

⁹ The Socio Manglar subaccount of the existing Socio Bosque Fund will be used to provide incentive payments to AUSCEM associations who are part of the Socio Manglar program, which is an extension of the existing Socio Bosque program to promote long-term management of mangrove areas.

¹⁰ Specifically, we are referring to the government's "Program Zero"/PECC (<https://www.ambiente.gob.ec/programa-carbono-cero-se-pone-en-marcha-en-el-ecuador/>) and the "Doble Deducibilidad" Program (<https://www.ambiente.gob.ec/beneficio-de-la-doble-deducibilidad/>)

¹¹ the project will support the Socio Manglar program, which is the part of the Socio Bosque program focused on mangroves.

4. **T6 Infrastructure** (GCF Strategic Priority infrastructure target). The project adopts a nature-based solutions approach to infrastructure because mangroves are expected to provide flood protection benefits that reduce risks of coastal flooding and related damage. The maintenance and improvement of 156,633 ha of mangrove ecosystems will be through conservation, sustainable use and restoration activities.
5. **T9 Adaptation and locally-led action** (GCF Strategic Priority on Adaptation proposals). The project will promote locally led adaptation action through community management of mangroves and action by the private sector to reduce deforestation and support mangrove restoration. The project will also support locally-led adaptation through the support to land-use planning processes included in Component 3 of the project.

In addition to the project's contributions towards advancing national climate change adaption and mitigation plans, the project will also contribute to the aims of the Global Mangrove Alliance (GMA)¹², which brings together a global community of practitioners to halt mangrove loss, restore lost mangroves and secure mangrove protection. CI is a founding member along with other international organizations and the project will contribute in two ways to the GMA's work:

1. The project will contribute to knowledge sharing through the GMA, especially related to mangrove restoration and "blue carbon" information. One of the functions of GMA is as a knowledge hub to share information between practitioners engaged in mangrove conservation and restoration. The mangrove restoration approach is informed by GMA toolkits and best practice guides. Data collected under activity 3.1.1 will also be shared with the GMA as a matter of course (through the Mangrove Restoration Tracker Tool and Global Mangrove Watch).
2. The project itself makes an important contribution to the 'Mangrove Breakthrough', an initiative launched by the GMA and the UN Climate Change high level Champions during the UNFCCC COP27. This high-level initiative aims to secure the future of 15 million ha of mangroves globally (through conservation, restoration and securing long-term financing of mangrove protection). This project alone represents about 1% of this goal.

¹² <https://www.mangrovealliance.org/>

PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

B.1.1 Overview of Ecuador's Coast (further detail and full references for citations are provided in **Annex 2, the Feasibility Study**)

1. Ecuador's complex coastal geography comprises 3,631 km of coastline, which is home to seven million people and Ecuador's largest city, Guayaquil. Mangroves extend along 36% of Ecuador's continental coastlines and play a central role in sustaining both export-oriented industries and local livelihoods by providing habitat to commercial fish species and maintaining healthy coastal zones. The coast is where a significant portion of the nation's economic output is concentrated, much of which is directly dependent on coastal and marine resources, especially shrimp and small-scale fisheries.

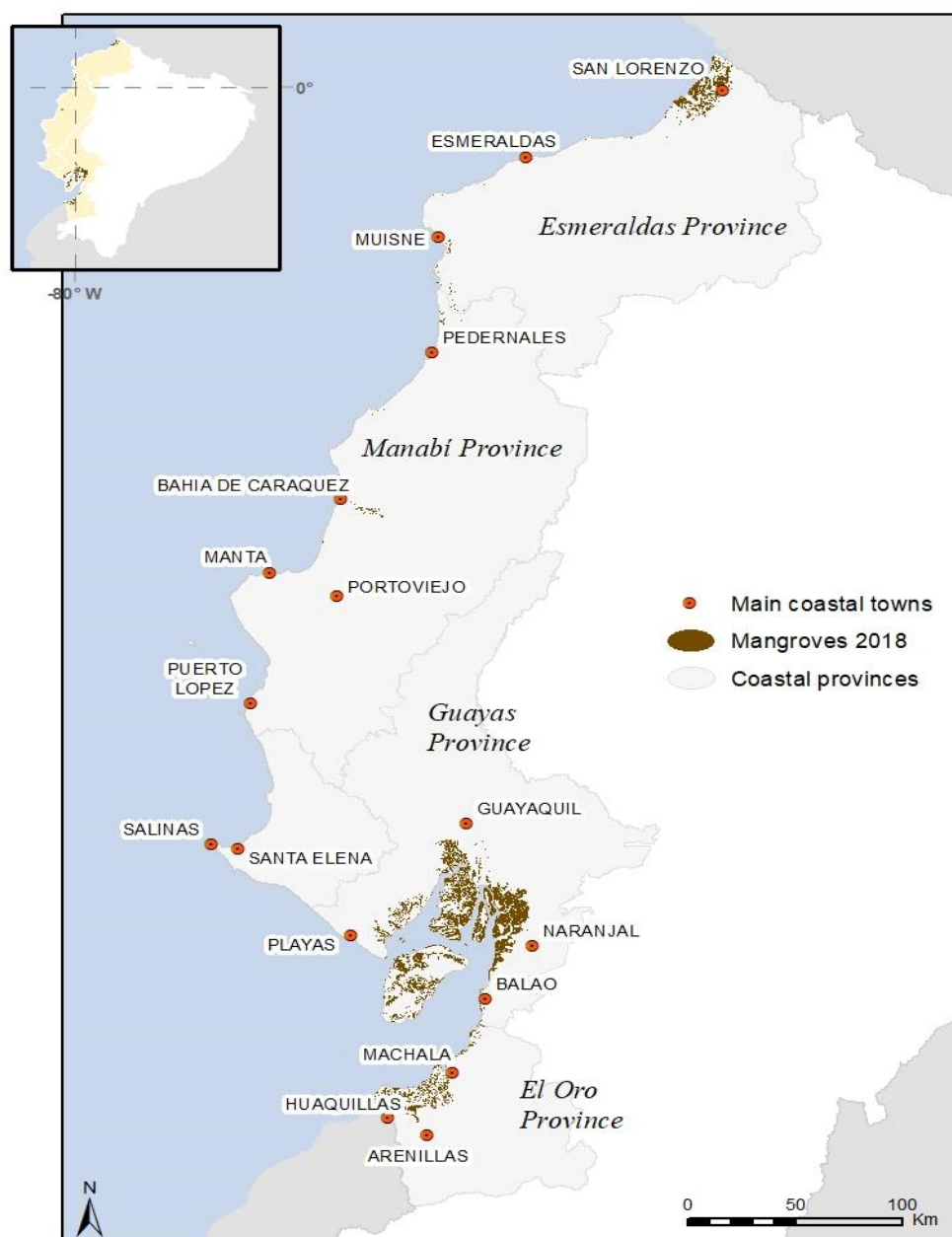


Figure 1. Ecuador's coastal region: 5 provinces (with 35 municipalities) and main urban centers and mangrove areas (in 2018). Data sources: Towns from IGM (2018), provinces boundaries from CONALI (2016), mangroves from MAAE (2018).

2. Shrimp: Ecuador is the world's second largest exporter of farmed shrimp, with approximately 174,000 ha of shrimp ponds and annual exports of 1,060,000 MT worth US\$6.653 billion (Cámara Nacional de Acuacultura, 2022)¹³. The volume of exports has more than doubled in the past 10 years. Production is dominated by small and medium producers who constitute 97% of farms and 76% of the total area (GPS, 2020). Ecuador is recognized in the market for slightly higher quality shrimp, due in part to lower production intensities, but there is still substantial room for intensification and modernization. Demand for certified shrimp is strong in the North American and European markets – while price premiums are not always present, environmental certification is often seen as a requirement to access these markets.
3. Small scale fisheries: Based on the most detailed national data, there are 28,399 artisanal fishers in Ecuador, organized into 420 associations (Vice Ministry of Aquaculture and Fisheries (VMAP), 2013), with an estimated 113,596 people directly or indirectly dependent on artisanal fishing¹⁴. These fishers have low incomes; in the Gulf of Guayaquil, 74% of fishers and gatherers earn less than US\$400 per month, and 37% have monthly incomes less than US\$200 (Herrera et al., 2017).

B.1.2 Current and projected climate change impacts in the coast of Ecuador (see Annex 2, Feasibility Study for more information)

4. In the continental coastal region of Ecuador, the climate is strongly influenced by the ocean. During the rainy season from December to May, the air temperatures are slightly higher due to the warming of the sea (intensification of the Panama current). The dry season coincides with the intensification of the South Pacific Anticyclone, which in turn strengthens the Humboldt Current and the cold waters in the region. In the north-central region, rainfall exceeds 2,000 mm annually, and air temperatures are between 25°C and 39°C. Further south, the coastal zone is under the influence of the Humboldt Current, so its precipitation values are between 300 and 1,000 mm per year, and the average temperatures are between 24°C and 26°C.
5. Climate change is projected to create significant changes in local environmental conditions along Ecuador's coast including **increases in sea level, increased intensity of El Niño-Southern Oscillation events, increased intensity and variability of precipitation, and atmospheric temperatures**, all of which will affect Ecuador's vulnerable populations and economy.
6. By 2100, mean **sea level** is projected to rise an additional 49-64 cm on Ecuador's coast¹⁵. An increase in mean sea level will also be compounded by increased frequency of extreme sea level (ESL) events (storm surge and tidal flooding), which are expected to occur annually by mid-century in low-lying coastal areas (IPCC, 2019). Increases in sea level rise (SLR) and ESL are expected to increase the risks of coastal flooding and consequently impact rural and urban populations, and aquaculture in coastal areas, especially when compounded by periodic El Niño events.
7. **El Niño-Southern Oscillation (ENSO)** is a phase of a large-scale, recurring climate pattern associated with above average sea surface temperatures in the eastern Pacific, which contributes to increased sea levels and extreme rainfall events in Ecuador. El Niño events have already led to observed ocean warming and dramatic SLR due to seawater expansion (Cai et al., 2014; Wang et al., 2019). During the 1997-98 season, sea surface temperatures off the coast of Ecuador increased by 5°C above normal and sea level increased up to 42 cm (CAF, 2000). In the equatorial Pacific region, sea levels can deviate from the global mean sea level by as much as 40 cm because of El Niño (Walsh et al., 2012). ENSO may be considered a natural climatic driver, and the effect of climate change on future ENSO dynamics is unclear although rainfall variability is very likely to increase over the east-central tropical Pacific (IPCC, 2014; 2021). There are indications that extreme El Niño events may increase in frequency due to climate change (McPhaden et al., 2020).
8. Between 1960 and 2016, the average annual **precipitation** for the coast of Ecuador increased 33% (INAMHI, from Ministerio del Ambiente del Ecuador, 2017). Some models project significant increases in annual precipitation for coastal Ecuador (9-10% under RCP 4.5 by 2071-2100). Analysis focused on the Santa Elena peninsula by García-Garizábal et al. (2017) documented increasing temperatures (0.038°C/year) and precipitation (0.196 mm/year) for the period 1982-2011 with modeling (ECHOG A2) predicting increases in temperature (+2.7°C) and precipitation (+8,2%) by 2100. Other analyses indicate that while total annual rainfall is not expected to increase significantly, greater variability is projected, with heavier localized rains (up to 15 more days per year of extreme rainfall events) and periodic dry spells expected to increase (Ministerio del

¹³ See Annex 2 (Feasibility Study) for references

¹⁴ Assuming family size of four people.

¹⁵ <https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool> The SLR range cited is based on projections for scenario SSP2-4.5 (lower estimate) and SSP3-7.0 (higher estimate).

Ambiente de Ecuador, 2019; UNDP, 2013), increasing the risk of flooding and drought (Ministerio del Ambiente del Ecuador, 2012).

9. From 1960-2016, average **temperature** for Ecuador's coast increased 0.6°C, and an increase in atmospheric temperature of 2.2°C-2.9°C is expected by 2100 under RCP 4.5 and RCP 8.5, respectively. Though outside the scope of this project, these temperature increases may affect livelihoods from agricultural activities and increase health risks in project areas adjacent to or outside of mangroves. This variable is included for reference purposes but is not directly expected to impact flood risks, which are the climate change adaptation focus of this project.
10. Analyses conducted for this project by the International Research Center on El Niño (CIIFEN) indicate that dissolved oxygen, primary production and chlorophyll would have reductions in their values under projected climate change scenarios (RCP 4.5 and 8.5). The hydrogen potential off the coast of Ecuador would be reduced, showing signs of possible ocean acidification, which could impact fisheries and associated livelihoods.

Climate-related flooding events in Ecuador

11. Flood risks are a significant concern for Ecuador's coast and in the last 45 years, the coastal provinces have suffered the highest degree of flooding in the country. Guayas has been the most affected, with more than 100 floods occurring where water levels exceeded the drainage canals designed to convey stormwater runoff and thus inundated adjacent lands (Secretaría Nacional de Gestión de Riesgos (SNGR), 2018). Augmented sea level, coupled with the exceptionally high precipitation and associated with El Niño in coastal Ecuador, limits the discharge of river systems into the ocean, leading to inland flooding. This results in impacts to drainage throughout the coastal zone, which are not fully captured in existing coastal and river flooding models. Under current assessments, 39% of areas within 5 km of Ecuador's coastline are already at high risk of annual flooding (SNGR, 2018).
12. Flooding events registered since 1965 on the coast of Ecuador have affected over 1.8 million people and caused over US\$1.5 billion in damages and economic losses, of which 96% occurred during El Niño events (EM-DAT, 2021). During those flooding events, 97,031 people became homeless and 836 people were killed (EM-DAT, 2021; see Figure 6). Those numbers, even if considered the best available, are underestimated as only events that have impacted more than 100 people, and have killed more than 10 people, are recorded.
13. Historical El Niño events provide a reference point for the scale of future climate-related damages for Ecuador's coast. The El Niño event in 2015 generated damage to infrastructure and economic activities associated with agricultural production. Ecuador's agricultural losses were estimated at US\$3.5 million, with more than 2,000 small- and medium-sized enterprises and 27,000 people affected by landslides and flooding (CIIFEN, 2017). Limited government preparation and response to previous El Niño events, in areas such as land-use planning, management, and decision making, are indicative of inadequate adaptive capacity in the face of climate change.
14. Extreme El Niño events, related precipitation variability on regional scales and increases in sea level will likely intensify (Pachauri et al., 2015). Combined with the expected increase in SLR due to climate change but unrelated to ENSO events, flooding of coastal areas is expected to increase. Future El Niño events may become more frequent and more intense (Magrin et al., 2007; Wang et al., 2019; McPhaden et al, 2020), but even an El Niño of similar magnitude to historical events will affect larger areas and more people as its flooding impacts are compounded by SLR. These conclusions are based on several models that project flood risks under diverse future scenarios and assumptions (see Section B 1.6 below).

B.1.3 Mangroves in Ecuador

Mangrove Ecosystem

15. Ninety-eight percent (98%) of Ecuador's mangroves are concentrated along the shores of four main estuaries - Cayapas Mataje, Muisne Cojimies, Guayas and Jambelí (Figure 2), which spread across eight municipalities within the project area, with over 1,000 ha of mangrove each, in three provinces¹⁶. These municipalities are

¹⁹ Mangrove management associations (AUSCEM), with 4,556 registered members, are almost exclusively dedicated to fisheries activities in mangrove areas (Ministerio del Ambiente del Ecuador, 2020). These management associations cover approximately 43% of Ecuador's mangroves and do not include unaffiliated 'independent' fishers who are not part of these associations, implying that the actual number of people fishing and gathering shellfish from mangroves is likely more than double the number of registered members in associations (Zambrano, N. *pers. comm.*). Total estimated beneficiary numbers are based on 4.04 family members dependent on each of these fishers (INEC 2010).

home to approximately 3.4 million people, with very high rates of poverty¹⁷ (see Table 1). Of this total population, over 2.18 million people live within 5 km of mangroves (Instituto Nacional de Estadística y Censos - INEC, 2022¹⁸), with significant vulnerable populations living in poverty (49% of urban population, 95% of rural).

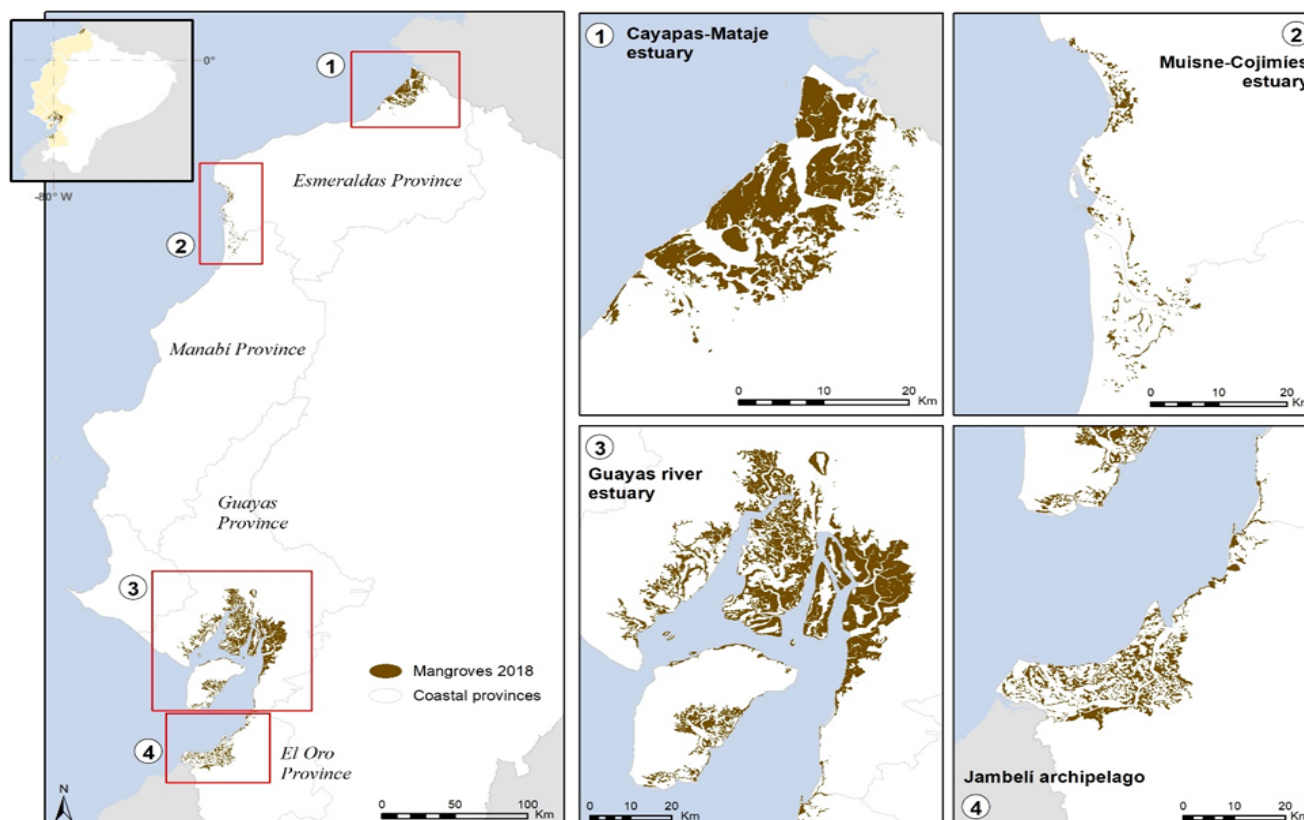


Figure 2. Principal Mangrove Areas of Ecuador. Data sources: Provincial boundaries from CONALI (2016), mangroves from MAAE (2018).

Table 1. Municipalities of Coastal Ecuador with >1,000 hectares of Mangrove

Province	Municipality	Estuary	Population ¹	% Poverty ²	Mangrove Area 2018 (ha) ³
El Oro	El Guabo	Jambelí	59,536	74%	1,377
El Oro	Machala	Jambelí	306,309	56%	3,434
El Oro	Santa Rosa	Jambelí	80,299	56%	10,164
Esmeraldas	Eloy Alfaro	Cayapas Mataje	46,305	94%	10,454
Esmeraldas	Muisne	Muisne Cojimies	36,426	98%	1,507

¹⁹ Mangrove management associations (AUSCEM), with 4,556 registered members, are almost exclusively dedicated to fisheries activities in mangrove areas (Ministerio del Ambiente del Ecuador, 2020). These management associations cover approximately 43% of Ecuador's mangroves and do not include unaffiliated 'independent' fishers who are not part of these associations, implying that the actual number of people fishing and gathering shellfish from mangroves is likely more than double the number of registered members in associations (Zambrano, N. *pers. comm.*). Total estimated beneficiary numbers are based on 4.04 family members dependent on each of these fishers (INEC 2010).

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Esmeraldas	San Lorenzo	Cayapas Mataje	48,391	84%	10,296
Guayas	Guayaquil	Guayas	2,746,403	47%	90,059
Guayas	Naranjal	Guayas	83,691	74%	22,774
Total			3,407,360		150,065

Sources: ¹Instituto Nacional de Estadística y Censos - INEC. 2022; ² Instituto Nacional de Estadística y Censos - INEC. 2010 ³Ministerio del Ambiente del Ecuador 2020

Importance of mangroves for livelihoods of Ecuador's coastal populations

16. Mangroves provide many valuable ecosystem services that contribute to human well-being, including provision of timber and fuelwood, breeding and nursery grounds for fisheries and protection from coastal flooding and erosion. People in these mangrove areas, including both urban and rural dwellers, are highly reliant on fishing and gathering shellfish. The total population dependent on mangrove fisheries is approximately 42,000 people, most of whom live in the project area¹⁹. This number does not include the larger coastal open-water fisheries and associated economic activity, which are indirectly dependent on mangroves' ecological function as critical nurseries for a variety of economically important fish species. It is estimated that the national market for fish and seafood products generated by small-scale fisheries is approximately US\$200 million per year (Monnier *et al.*, 2020).

Historical changes to the area of mangrove land cover

17. Mangroves have experienced significant losses in Ecuador, which has resulted in GHG emissions from the trees' lost biomass and through the release of soil carbon. The most extensive losses have been linked to the conversion of the intertidal land to shrimp aquaculture, an industry that was first introduced in 1968 and quickly led to large scale deforestation. This trend has been consistent over time, and shrimp farms continue to be the prominent driver of mangrove loss, with shrimp ponds during 2008-2018 accounting for 51% of mangrove loss²⁰ (CIIFEN-MAE, 2020). As of 2018 (the date of the last national forest cover assessment) there were 156,633 ha of mangrove remaining in the four key estuaries of mainland Ecuador²¹.
18. Recent years have seen growth in new mangrove areas replacing other land uses, with a total net gain in mangrove area of 15,258 ha from 2008-2018. Despite gross gains of 15,161 ha in mangrove coverage, there is also significant ongoing gross loss of mangroves, totaling an area of 9,903 ha lost during 2008-2018, resulting in the overall 5,258 ha net gain. The ongoing loss of mangroves in these areas is the result of illegal logging of mangroves (mostly for shrimp ponds but also for agriculture and building), which continues to occur due to weak application of environmental regulations and poor territorial planning for mangrove conservation by local governments. Mangroves are one of the nine forest strata included in the Forest Reference Emission Level (FREL) and referenced in the REDD+ Action Plan.

Community-based management of mangroves and effectiveness for mangrove conservation and restoration

19. Strengthening community management of mangroves has been an important pillar of Ecuador's approach to mangrove conservation for twenty years. Strengthening the capacity of local mangrove users as a strategy for climate change adaptation has been highlighted by local studies and is an important priority in Ecuador's climate change strategies (see D.5).
20. While coastal mangroves are legally public lands, AUSCEM²², first established in 2000, grant traditional community users formal rights linked to conservation commitments, allowing them to benefit from improved livelihoods activities and living conditions. These agreements grant use rights and stewardship responsibilities

¹⁹ Mangrove management associations (AUSCEM), with 4,556 registered members, are almost exclusively dedicated to fisheries activities in mangrove areas (Ministerio del Ambiente del Ecuador, 2020). These management associations cover approximately 43% of Ecuador's mangroves and do not include unaffiliated 'independent' fishers who are not part of these associations, implying that the actual number of people fishing and gathering shellfish from mangroves is likely more than double the number of registered members in associations (Zambrano, N. *pers. comm.*). Total estimated beneficiary numbers are based on 4.04 family members dependent on each of these fishers (INEC 2010).

²⁰ Data from the more recent 2014-2018 period also allows for classification of post-deforestation land use: an additional 13% was converted to other agricultural use, 22% converted to open water (due to natural dynamics of estuarine areas), with the remainder transformed for infrastructure, scrub and other vegetation, and land without vegetation (e.g. beaches and mud flats).

²¹ And 1,730 ha in fragments outside these estuaries and on coastal islands.

²² *Acuerdos de Uso Sostenible y Costodia del Ecosistema Manglar*

to organized groups of traditional mangrove users to restrict open access and overuse, reduce deforestation and illegal extraction, and increase the ability of local users to realize economic benefits from conservation actions. Community control and participation in conservation of mangrove areas, as well as national PAs, have proven effective tools for mangrove protection. Gross deforestation rates under AUSCEMs and PAs are 86% lower than areas without these protections; while mangrove areas without community stewardship or PA status (totaling 36,186 ha or 23% of total mangrove area) account for 69% of all recent mangrove deforestation. The coverage of mangrove areas under community-based management is, however, still insufficient. AUSCEMs currently only cover 60,217 ha of mangroves (Figure 3 and Table 2), this means that significant mangrove areas and their traditional users are not covered by active AUSCEMs, leaving them at risk of encroachment and uncontrolled resource exploitation.

21. In addition, approximately one quarter of the areas under AUSCEM have lapsed due to lack of support and technical assistance for legal proceedings and management activities, leaving these mangrove areas without effective protection. Furthermore, despite most mangrove residents, being highly dependent on natural resources and ecosystems, they have limited awareness of climate change. Training, participatory planning and strengthening of livelihoods alternatives in mangrove areas can therefore contribute to enhancing climate change resilience.

Table 2. Area and deforestation rates of mangroves by land-tenure and use category

Category	Total Mangrove Area (2008)	Total Mangrove Area 2018 (ha)	% National total mangrove area (2018)	Gross Loss (ha)	Gross Gain (ha)	Net Change (ha)	Net Change (%)
Protected Areas (PAs) ¹	62,339	64,916	41%	1,733	4,309	2,577	4.13%
AUSCEM, active June 2020 ²	56,522	60,217	38%	1,389	5,084	3,695	6.54%
No Protection ³	36,956	36,186	23%	6,868	6,099	-770	-2.08%
Overlap AUSCEM and PAs	-4,441	-4,686	-3%				
Total	151,376	156,633	100%				

¹ Includes 4,441 ha (2008) and 4,686 ha (2018) of mangrove in AUSCEM overlapping with PAs. 2018 data includes 10,589 ha of mangrove in AUSCEM, inactive/pending renewal as of June 2020.

² Includes 4,441 ha (2008) and 4,686 ha (2018) of mangrove in AUSCEM overlapping with PAs.

³ Includes 3,307 ha (2008) and 3,640 ha (2018) of mangrove in AUSCEM, inactive/pending renewal as of June 2020.

22. The Socio Manglar program (meaning “Mangrove Partner”) builds on AUSCEMs and was established by Ministerial Decree No. 198 on July 9, 2014, by the Ministry of the Environment, as part of the Socio Bosque program (SBP) of conservation incentives, which has been transformative within Ecuador as a mechanism for engaging local communities in the stewardship of native forests²³. The Socio Manglar program is designed to provide direct economic payments to beneficiary AUSCEM groups to support the conservation of mangroves and supplement communities’ income. The objective of the Socio Manglar program is to complement and consolidate the results achieved through the AUSCEMs granted to communities and ancestral groups of users, to guarantee the conservation of mangroves while improving the living conditions of the population dependent on fisheries resources. (Note: Socio Manglar payments are only used in the protection of mangroves, it is not financial support for shrimp cultivation.)

23. In 2020, Socio Manglar payments by the MAATE to AUSCEMs totaled US\$379,563. Current Socio Manglar payments come from MAATE’s core budget and contributions from the REDD Early Movers program supported by the German government (see **Annex 2**, Appendix 12). Currently only 33,467 ha of mangroves under 25 AUSCEMs receive Socio Manglar incentive payment agreements, leaving significant scope for expansion within remaining active AUSCEMs (26,750 ha) as well as in potential new AUSCEM areas.

24. Expanding the coverage of AUSCEMs, Socio Manglar incentives and enhancing the economic viability of sustainable community enterprises have been identified by stakeholders as interventions that would contribute to further reducing deforestation and strengthening of mangrove stewardship. The Socio Manglar subaccount,

²³ <https://sociobosque.ambiente.gob.ec/>

a subaccount of the existing Socio Bosque Fund, has been established in preparation for this project and as a mechanism for developing the long-term financial sustainability of the Socio Manglar Program. The Socio Manglar subaccount is managed as part of the Socio Bosque Fund with the intent that it operates as an endowment, with the interest and investment income from its investments used to pay for incentives to AUSCEMs. Component 2 of the project includes a request for \$4 million from GCF to help capitalize the Socio Manglar subaccount. The GCF funds will be kept separately from the other Socio Manglar subaccount funds in an interest-bearing account, while other funds will be kept in an investment account and invested according to the investment strategy of the fund manager, *Fondo de Inversión Ambiental Sostenible* (FIAS), which is Ecuador's Environmental and Sustainable Investing Fund. Additional contributions will be sought for the Socio Manglar subaccount from private sector companies operating in Ecuador as part of project activities in Component 2.

25. While there have not yet been evaluations of the Socio Manglar program, due to its recent establishment, several evaluations of the SBP highlight that deforestation rates are lower on land included within the SBP and that communities gain important monetary and non-monetary benefits (Arriaga et al. 2018, Perafán and Pabón, 2019). The most important impacts identified by the evaluations were increased land and rights-use, security, economic development, and organizational support. Communities also reported non-monetary benefits due to collective decision making and social cohesion from working together to achieve a common goal. However, the evaluations noted the need to strengthen financial sustainability of the program and the need to address conflicts that can arise if some communities receive incentive benefits while others do not (Etchart et al., 2020). In the case of mangrove areas, there is a stronger possibility to monetize the traditional economies because the economic value of non-timber products (i.e., from fisheries) is much higher than for terrestrial forests.

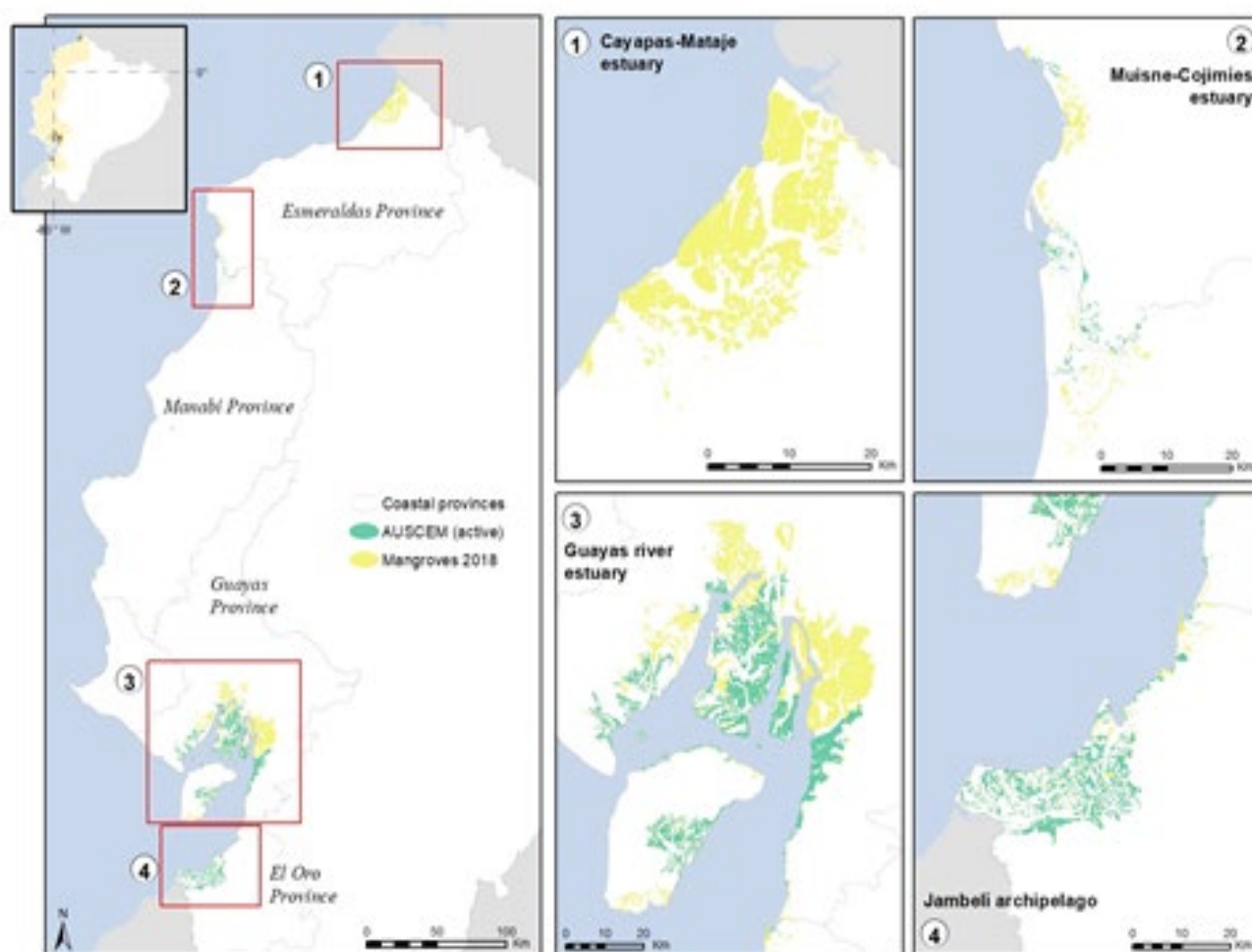


Figure 3. Coverage of current AUSCEMs and Mangroves in Ecuador, indicating significant scope for expansion to mangrove areas currently outside active AUSCEMs Source of data: Mangrove cover from MAAE (2018), AUSCEMs areas from MAAE (2020).

26. This project will scale and expand the results and lessons learned from previous initiatives, including complementarity with GEF projects. As included in table 25 of **Annex 2**, five GEF projects have contributed to the creation of capacities and field experience on marine protected areas (MPA), enforcing local associations, local planning and financial sustainability. Conservation International – Ecuador (CI-E) is also currently implementing a project with the GMA that seeks to create networks among organizations working with mangrove ecosystems and support AUSCEMs in the management of fisheries.

National Protected Areas

27. Forty-one percent of all mangroves of mainland Ecuador lie within the boundaries of nine PAs, making these an important strategy for mangrove management and conservation. These nine PAs cover three different categories of PAs: i) Ecological Reserves (Reserva Ecológica – RE); ii) Flora and Fauna Production Reserves (Reserva de Producción de Flora y Fauna – RPF); and iii) Wildlife Refuges (Reserva de Vida Silvestre – RVS).
28. Four PAs, prioritized for inclusion in the project contain 94.6% of mangroves in PAs:
- Reserva Ecológica Manglares Cayapas Mataje (includes 20,012 ha of mangrove)
 - Reserva Ecológica Manglares Churute (includes 28,467 ha of mangrove)
 - Reserva de Producción de Flora y Fauna Manglares El Salado (includes 11,659 ha of mangrove)
 - Reserva de Vida Silvestre Manglares Estuario del Río Muisne (includes 1,295 ha of mangrove)

Mangrove reforestation and restoration

29. Ecuador has established and executed successful mangrove reforestation efforts. Techniques for mangrove reforestation are well-proven in Ecuador, and from other sites around the world (see **Annex 2, Feasibility Study**). An effective policy intervention that led to an increase in mangrove cover was Executive Decree 1391 in 2008, which required many shrimp farms that had been illegally established in public coastal areas, to restore significant areas of mangrove on or adjacent to their farms, and allowed a three-year grace period for registration and starting reforestation activities, before legalization of the shrimp farms. This resulted in a surge in mangrove restoration (totaling 3,110 ha) immediately following its issue. Unfortunately, since the policy ended, no new areas are being registered and reforested under this mechanism. Sustaining or increasing the rate of mangrove recovery now depends on other regulatory instruments and on voluntary actions, which are the principal focus of this project.
30. Conservation International (CI) has identified over 10,000 ha of land in the four estuaries appropriate for potential reforestation based on criteria including size, form, connection with remaining mangroves, and vulnerability of populations. Of these areas, 4,850 ha will be restored as part of the GCF project (see **Annex 2** for further details of the prioritization and restoration methods) to provide long-term flood protection benefits, carbon sequestration and ecosystem service co-benefits in the form of improved fisheries for vulnerable people.

B.1.4 Governance

Planning and zoning

31. Local governments are known as Decentralized Autonomous Governments (GADs²⁴) in Ecuador and include subnational entities at the provincial, municipal (or cantonal) and parish levels with a degree of administrative and regulatory autonomy. GADs need to play an important role in climate governance and, in coordination with the national government, collaborate on integrated coastal management. A fundamental instrument for planning and investment of the GADs are Development and Land Use Plans (*Planes de Desarrollo y Ordenamiento Territorial* - PDOT). PDOTs are a key tool to ensure that decision-making processes are based on information collected at the territorial level, and the implementation of measures that reduce the impact of climate threats on planned programs or projects is considered. PDOTs also provide the opportunity to ensure a strong marine and coastal spatial planning approach in the mangrove zone of the coastal region of Ecuador. These PDOTs can be complemented, under Ecuadorian law (Organic Code for the Environment – CODA), by Management Plans for Beaches and the Coastal Strip addressing the specific needs and characteristics within and across multiple GADs. Three municipalities have so far established such plans. Mangrove areas and climate change vulnerabilities are generally not currently adequately addressed in PDOTs.

Enforcement

32. Legal enforcement has faced a series of setbacks when trying to enforce mangrove conservation measures with shrimp farm owners who expanded shrimp farms without prior authorization from the National

²⁴ *Gobiernos Autónomos Descentralizados*

Environmental Authority. Although regulatory frameworks are in place, enforcement is inadequate for protection and restoration of mangroves by public, private and community-based actors.

33. Operational Control and Surveillance Units (UOCVs – *Unidades Operativas de Control y Vigilancia*) were piloted for detection and law enforcement of illegal activities affecting mangroves as part of the Program for the Management of Coastal Resources project. As a joint effort between the Navy, MAATE, Vice-ministry of Aquaculture and Fisheries (VMAP), NGOs and the National Chamber of Aquaculture (NCA), the UOCVs' main goal was to avoid and control the clear-cutting of mangrove forest areas through the cooperation of public, private and NGO entities involved in this environmental issue.
34. The application of sanctions for environmental crimes under the Organic Code for Criminal Processes (*Código Orgánico Procesal Penal*) is also limited by gaps in knowledge about regulations, including administrative procedures and fines. When environmental crimes and sanctions are taken up by the courts, there is frequently a lack of knowledge on the part of prosecutors and judges regarding marine and coastal laws and regulations, and a lack of coordination and collaboration between prosecutors and the MAATE.

B.1.5 Private sector initiatives: Shrimp aquaculture

35. Shrimp farming has historically been the single biggest driver of mangrove deforestation in Ecuador and continues to account for around half of anthropogenic mangrove loss. In 2020, there were 3,933 registered shrimp farms in Ecuador operating on 216,000 ha (Seafood Watch, 2021). Effectively engaging with the shrimp aquaculture sector, through better enforcement and through voluntary action, is an important element of effective mangrove conservation and restoration.
36. Shrimp aquaculture in Ecuador is recognized for using low stocking density production (extensive production) to decrease disease risks, although the sector remains GHG intensive. The low stocking density strategy, widely adopted after disease outbreaks in the 1990s severely impacted national production but does produce a high-quality product with lower environmental impacts. Overall growth of the sector has increased from about 50,000 Mt in 2000 to 700,000 Mt in 2020 (Chase, 2021). Shrimp production intensity averages 1.92 t/ha/y in Ecuador, which is significantly lower than intensive production seen in Thailand (18.2 t/ha/y) and below production intensity averages of China, India, and Indonesia (4.5 t/ha/y, 3.6 t/ha/y, and 3.3 t/ha/y, respectively) (CEA Consulting, 2018).

Opportunities for engagement with the private sector

37. The shrimp aquaculture industry has had limited engagement with mangrove conservation, despite the role of mangroves in protecting their vulnerable coastal infrastructure, regulatory requirements for reforestation, and increasing market demand for sustainable production. Increasing private-sector participation is constrained mainly by lack of information about the benefits of mangroves, limited technical capacity to identify and adopt practices that restore or conserve mangroves, and in the case of small shrimp farmers, limited access to affordable finance for all aspects of their operations. However, previous efforts by aquaculture companies to restore mangroves following Decree 1391 were generally successful and survival rates of mangrove seedlings were high despite localized problems (Pesante & Rosero, 2020). These previous experiences of mangrove restoration in Ecuador mean there is capacity to expand restoration efforts.
38. Beyond the need for legal compliance, market opportunities and barriers are also driving a shift within Ecuador's shrimp aquaculture sector. Voluntary, market-facing certification and labeling schemes are drawing increasing attention as a way to differentiate Ecuadorian shrimp as sustainable, healthy and good for the environment. Depending on the retailer, certification is seen as: i) a requirement for sourcing (without additional pricing), or ii) a value-added product that retailers and consumers concerned with environmental sustainability and social responsibility are willing to pay more for on a per kilogram basis. While there has been growing interest in sustainable practices in the shrimp aquaculture sector, less than 1% of all shrimp farms in Ecuador currently operate under certification schemes that align with climate resilience and emissions reductions. Several complementary initiatives are underway that present an opportunity for scaling and improving practices, including Aquaculture Stewardship Council (ASC) certification and the Sustainable Shrimp Partnership (SSP).
39. ASC provides a certification standard to minimize unintended social and environmental impacts of aquaculture production around the world. ASC certified shrimp farms must minimize impacts on their neighboring ecosystem in several ways, such as the partial restoration of lost mangrove forest, the development and implementation of a biodiversity-focused environmental impact assessment (B-EIA) and ensuring farms are not sited in critical habitats. Furthermore, since shrimp farming often occurs along coastal areas, a permanent coastal barrier must be in place between the farm and the coastline. Ecuador has one of the largest areas of shrimp farms certified under ASC; approximately 15% of production is certified, with a total volume (including shrimp and tilapia) of

75,000 Mt sold in 2020. Currently, 15 companies are participating in ASC with 28 certified farms (49 sites) in Ecuador. Products from ASC-certified farms and chains of custody can display the ASC logo to improve market access and the perceived value to consumers of their products.

40. The NCA has also been working on the development and implementation of the national SPP initiative to create a brand to differentiate the Ecuadorian product in the global seafood market. The SSP is an industry led initiative that builds on the ASC criteria adding additional criteria, including a neutral impact on water²⁵. With approximately 3,000 shrimp farms in Ecuador, there is significant scope for growth for both the ASC and SSP, leveraging market access and differentiation to achieve environmental outcomes, including carbon neutrality and mangrove restoration and protection; with their requirements to eliminate mangrove deforestation, the ASC and SSP can contribute directly to reducing emissions from mangrove conversion if widely adopted (see **Annex 2 Feasibility Study** for further details).
41. CI's Climate-Smart Shrimp (CSS) initiative provides a model for climate change adaptation within the sector. CSS is complementary to the ASC and SSP standards and is a novel set of practices that prevent deforestation and incentivize mangrove restoration while also increasing the amount of shrimp a farm can produce. CSS involves working with shrimp pond farmers to sustainably intensify shrimp production, construct treatment wetlands, and restore coastal ecosystems, while securing buyers for the 'CSS' produced (Figure 4). For small and medium farmers, the primary incentives to participate are access to capital and technical assistance, while for larger farmers, who often already have access to capital and expertise, the incentive is connections to sustainability-oriented buyers in North American and European markets who preferentially source responsibly produced shrimp. The CSS approach can realize climate change resilience and adaptation benefits while also increasing production and profitability to meet development goals.
42. Establishing the shrimp industry as a partner in mangrove restoration will increase funding for restoration and expand the number and diversity of stakeholders conserving, protecting, and rebuilding natural infrastructure at scale, and eliminate pressures for deforestation. The project includes activities with Ecuadorian shrimp producers to expand the use of ASC/SPP certification, directly restore mangroves and generate additional financial contributions for the Socio Manglar Program that will support community-based mangrove conservation and management beyond the implementation period of the project.

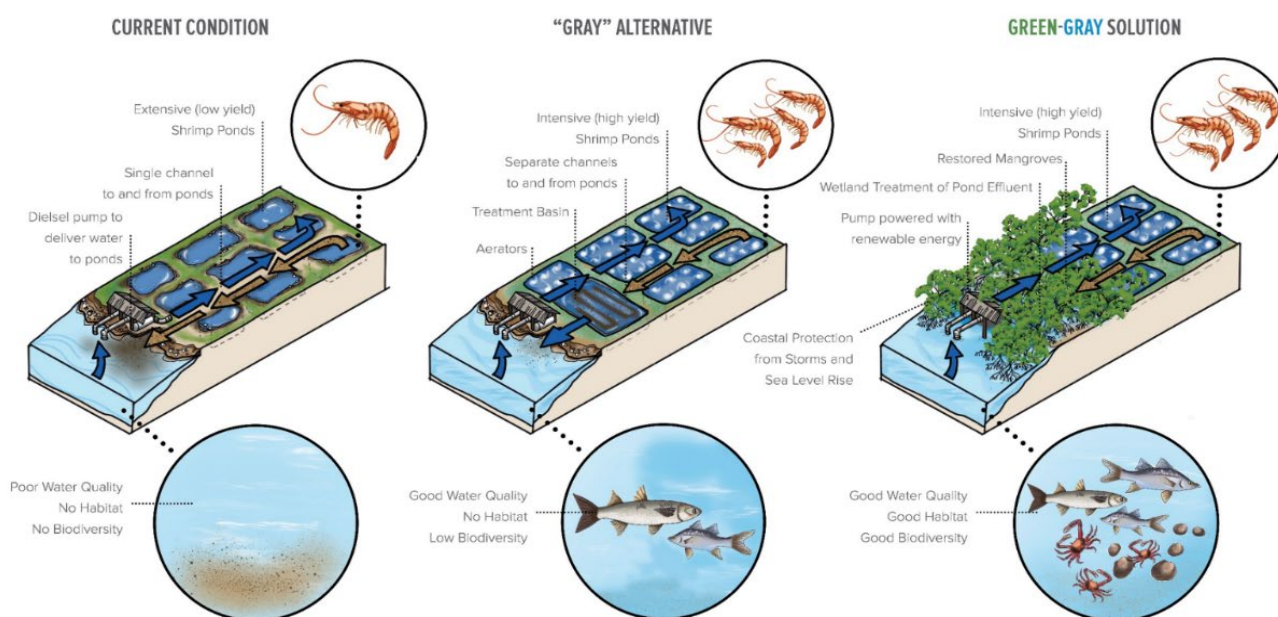


Figure 4. A visual comparison of the "gray" alternative to intensification (middle) versus applying a green-gray climate adaptation approach to sustainably intensify shrimp production using the CSS model (right).

Conditions for private sector engagement

43. To promote CSS or any type of certification, a key element is ensuring that adequate and accessible financing exists to address the needs of shrimp farmers. An important barrier to access finance is the lack of technical

²⁵ <https://www.sustainableshrimppartnership.org/>

knowledge within financial institutions about the industry and its sustainability pathways. Establishing partnerships that allow access to information, training and implementation of demonstration cases facilitates the learning curve so that the shrimp sector becomes more attractive and less risky for financial institutions. This is the approach of the eco.business Fund²⁶, a US\$390 million fund that provides financial services to financial intermediaries in the Latin America region, in support of green economy transformation. The fund has ambitious targets for the aquaculture sector in Ecuador; and helps financial institutions develop tools, skills and demonstration cases for key investments. Alliances with key stakeholders, including CI and its model of CSS, ASC and SPP, are critical to strengthening the content and scientific approach and influencing the target audience for those trainings.

44. Besides lack of knowledge, there are other important barriers to access finance, such as early-stage companies with unstable cash flows or lack of collateral to access traditional funding sources like the banking system. Investment funds fill some of these gaps with a diversity of instruments such as debt and equity that can adapt to specific needs and business stages. To further fill this gap, CI is developing a fund with a specific focus on supporting adoption of the CSS approach. The Climate Smart Shrimp Fund (CSSF)²⁷ will provide loan packages, supported by a technical assistance facility, that will enable shrimp farmers to transition to more sustainable and efficient production systems while simultaneously restoring mangrove ecosystems. The initial objective is to create a US\$100 million fund²⁸ with a focus on supporting CSS in Indonesia, Ecuador and the Philippines. Pilot testing of the CSSF is currently underway with a company in Ecuador.

Other opportunities to work with the private sector

45. In addition to Ecuador's shrimp sector, there may be opportunities to work with international companies whose interest in investing in mangrove conservation and restoration is driven by their own Environmental, Social, Governance strategies. CI will seek out such opportunities to build long term support for community-based conservation and forest restoration (see project activity 2.2.1).

B 1.6 The impacts of coastal flooding events in the project area

46. Based on review of local data sets and a series of models that were applied to Ecuador's coast (see Appendix 6 of **Annex 2, Feasibility Study**), the project will reduce flood risk for at least 89,600 people:
- In coastal mangrove areas, approximately 198,000 people are exposed annually to high flood risks from either coastal or riparian flooding (SNGR, 2018). Of those, approximately 22,400 are estimated to have an annual risk of flooding from coastal storm surges even with current mangrove cover (based on the methodology of Menéndez *et al.*, 2020; see below and **Annex 25** for more details).
 - Without the protective functions of the current mangrove cover, an additional 86,200 people would have an annual risk of flooding²⁹ due to coastal storm surge events (based on the methodology of Menéndez 2020; see **Annex 25** for more details of calculations). These 86,200 people will benefit from the project since it will strengthen the protection of existing mangroves and therefore ensure that the flood protection continues.
 - By ensuring that current mangrove areas are conserved, deforestation due to shrimp farming is reduced by 50% and mangrove coverage is expanded by 4,850 ha through restoration, the project will benefit an estimated 3,400³⁰ of the 22,400 people currently exposed to annual flood risk (see **Annex 25**). The figure of 3,400 is based on a calculation of the average number of people that derive flood protection benefit from one hectare of mangroves in Ecuador. The project will therefore provide improved flood protection to 89,600 people (assumed to be 50% female, 50% male).

Demography of the population in the project area (See **Annex 2, Feasibility Study** for further detail on the demographics in the project area and the **Gender Assessment and Action Plan (Annex 8)** for specific gendered uses and management of mangroves.)

47. Three million, four hundred thousand people live in the municipalities of the estuaries targeted by the project. The sex ratio is 50.9% males; 49.1% females. The population is 83% urban, due to the inclusion of some urban

²⁶ <https://www.ecobusiness.fund/en/>

²⁷ <https://www.climatefinancelab.org/ideas/climate-smart-shrimp-fund/>

²⁸ The CSS Fund is being developed as a blended finance vehicle, with approximately 30% concessional finance and 70% commercial finance. The fund's long-term goal is to make the case to local financial institutions that they can/should make CSS loans. The purpose of the CSS Fund is to create a track record of successful CSS loans (as defined by loan repayment at commercial rates, farm profitability, and mangrove restoration) that show the attractiveness of making CSS loans to commercial investors (including local financial institutions).

²⁹ For the Menéndez model, the term "people impacted by flooding" refers to people who live in areas where the water height is estimated to reach higher than 0.5m above the land height.

³⁰ The exact figure from the calculation is 3,465 but has been rounded down to 3,400.

areas of Guayaquil and smaller towns in the project area. Poverty levels average 72%. Unstructured, rapid urbanization in Ecuador's coastal towns and cities have pushed poor urban dwellers into low-lying areas around waterways that are the most prone to flooding, leaving them highly vulnerable to climate change. Across all project sites, women and men depend on the natural resources they obtain from the mangroves, including fish and shellfish such as crabs and mollusks (especially black clam). In exceptional circumstances, such as during the closed fishing season, men find they need to earn an income outside the home. Further information on specific gendered uses and management of mangroves is detailed in the **Gender Assessment and Action Plan (Annex 8)**.

The impacts of flooding events on local communities and key economic sectors

48. The hazard level of coastal flooding in all areas of the project is classified as the highest hazard level, by the Global Facility for Disaster Reduction and Recovery (GFDRR), meaning that potentially damaging waves are expected to flood the coast at least once in the next ten years³¹. The southern coastal region of Ecuador (the region of the two southern estuaries included in the project) has been identified as the location with the highest coastal risk in the Latin America and Caribbean region due to a combination of coastal hazards, geographic exposure, and socio-economic vulnerability (Calil *et al.*, 2017).
49. Flooding events have become more frequent over the last 30 years; and are particularly acute during El Niño events in which rainfall can rise as high as 3,500 mm per year. The Borbor-Cordova *et al.* (2020) study recommends that coastal cities in Ecuador must recognize that they are part of larger scale ecosystems and that they need to integrate coastal ecosystem services from mangroves and other wetlands into urban and landscape design.
50. In a global assessment, Ecuador's shrimp production has been highlighted as having the highest vulnerability to climate change of the world's brackish aquaculture producers, based on exposure (temperature, water balance, population density, precipitation change, flood risk, drought risk and cyclone risk), adaptive capacity and sensitivity (Handisyde *et al.*, 2017). Eighty-two percent of shrimp farms along Ecuador's coast are sited in areas classified as high flood risk under current conditions (MAP, 2018; SNGR, 2018), risks which will increase dramatically with increased sea level and intensified precipitation projected due to climate change. During the 1997-98 El Niño event, flooding, disruption of supply chains and damage to transport infrastructure affected the shrimp aquaculture sector; around 4,500 ha of ponds were severely damaged (CAF, 2000). Heavy rains and flooding can also affect the harvest cycle, causing shrimp to molt early, resulting in up to a 20% drop in yields per pond (Sackton, 2016), while severe weather events can cause partial or total loss of harvest. These damages have direct consequences in the form of loss of income. The shrimp aquaculture sector should therefore have an incentive to support measures and adopt practices that help reduce current and future flood risk.

Table 3. Project populations at risk of flood and other climate-change related impacts in the project area (see detailed explanation and calculations in Annexes 22 and 25).

	Description	Number of people	Flood and other climate change-related impacts
Fishers	Fishers and families directly dependent on mangrove resources	~41,500	Loss of income and food insecurity due to disrupted logistics, changes in fish populations and distribution Property loss due to flood damage Displacement and disruptions to transport Increased risk of infectious disease

³¹ <https://thinkhazard.org/en/report/73-ecuador/CF> , accessed February 2022.

Populations directly at risk of annual flooding during storm surge events in mangrove coastal zone	Populations in flood-prone areas currently at high risk of annual flooding events (people with annual expected risk of flooding as defined by Menendez <i>et al.</i> , 2020; see below and Annex 25)	108,600 in areas where mangroves reduce flood risk (22,400 already at annual risk and 86,200 at risk without current mangrove cover)	Loss of food security and income from fisheries, agriculture and tourism Property loss due to flood damage Displacement and disruptions to transport Increased risk of infectious disease
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B 1.7 Importance of mangroves in providing coastal protection benefits against flooding and climate change

51. While mangroves provide a diversity of ecosystem services, their value for coastal protection and the reduction of flood risks is particularly well documented and relevant for Ecuador's coast. The use of "green infrastructure" in the form of natural ecosystems to reduce the effects of climate change hazards, for example to dissipate wave energy and reduce the intensity of storm surges, has been well documented (see additional detail in **Annex 2**). For example, mangrove species have dense roots to reduce wave energy and height (Figure 5) such that wave height is decreased as the waves travel through the mangroves. Several studies highlight the importance of re-establishing wide and intact tracts of mangrove to maximize their coastal flood protection benefit. Storm surges can raise water levels on the coast for long periods (hours or days), compared to waves that come from normal winds or tides and that last for shorter times (seconds or minutes). This further highlights the importance of having wide tracts of mangroves to reduce storm surges.

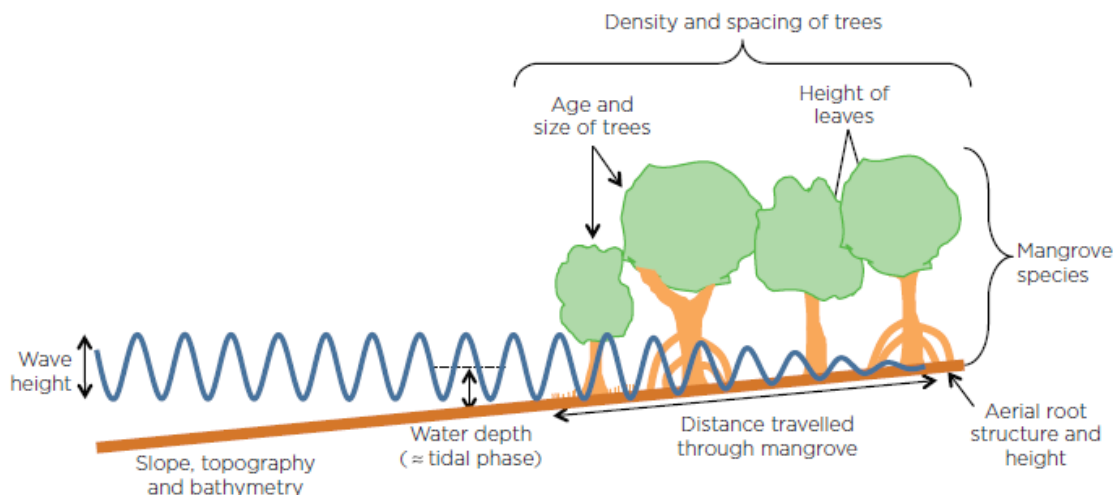


Figure 5. Key factors contributing to wave attenuation. Source: McIvor *et al.*, 2012b

Modelling the flood protection benefits of mangroves

52. To model the flood protection benefits of mangroves we applied the 'avoided damage' valuation approach described by Menéndez *et al.* (2020), which uses the cost of damages prevented by mangroves to estimate their value. (see Appendix 6 of **Annex 2**). The 'avoided damage' analysis using the Menéndez model does not evaluate future climate scenarios, nor does it directly estimate benefits of adding mangroves, only the damage caused if existing mangroves are removed. However, it does provide per hectare estimates of the value of mangroves in terms of annual expected risk of flooding and reduced damage, which has been used in the preparation of this project to estimate the benefits of increasing mangrove cover.

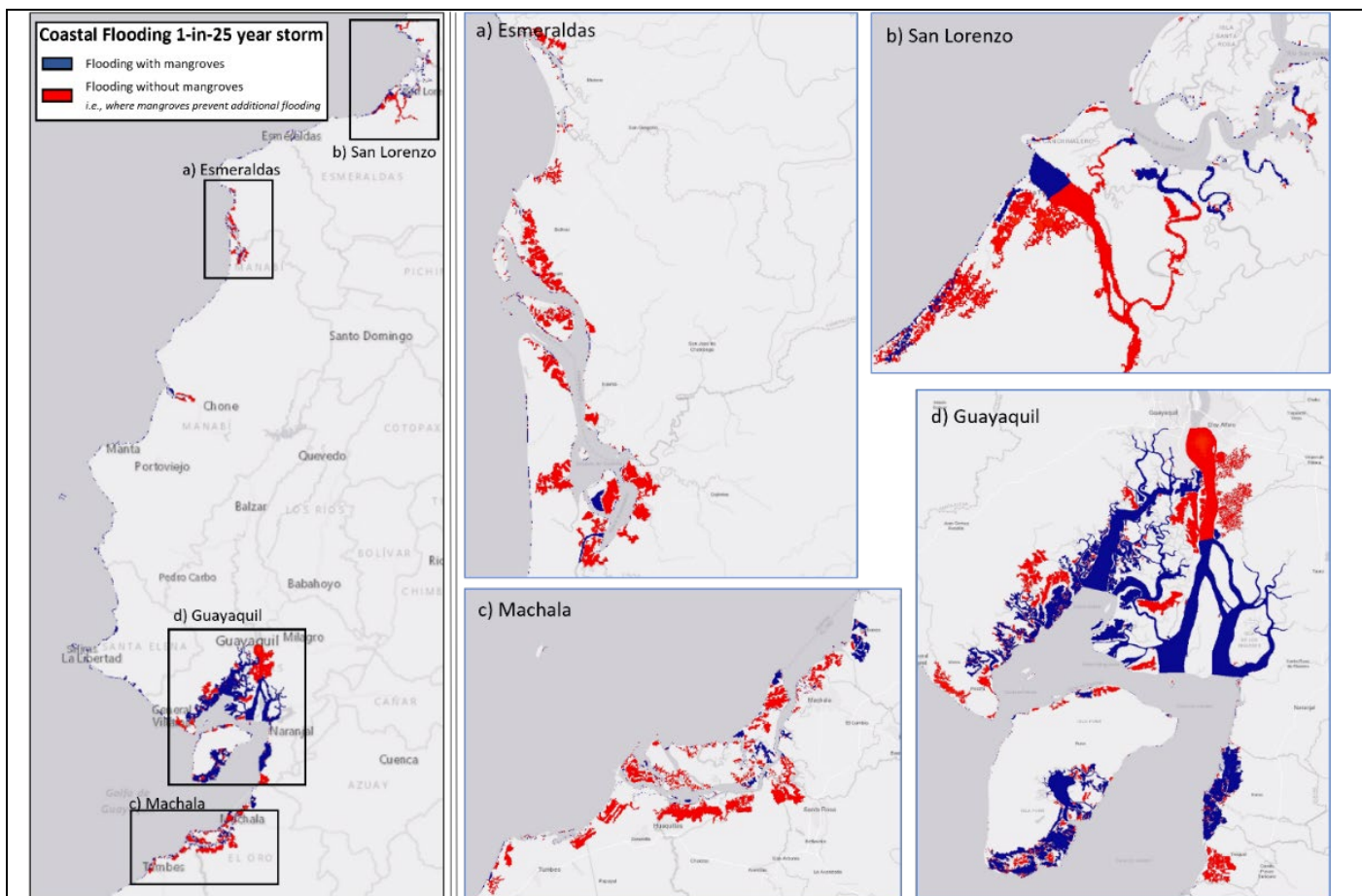


Figure 6. Projected flooding in Ecuador with (in blue) and without (in red) mangroves for a 1 in 25-year storm event based on the methodology of Menéndez *et al.* (2020).

53. The Menéndez model analysis concludes that based on current conditions, 22,400 people have an annual expected risk of coastal flooding caused by storm surge (16,900 in urban areas and 5,500 in rural areas). Under these current climate conditions, but without the protections that mangroves provide, approximately 86,200 more people would be at annual risk of coastal flooding in the project area. The analysis also shows that the mangroves prevent approximately US\$250 million of property loss from flooding annually. Based upon mangrove coverage in Ecuador from 2018 (approximately 156,633 ha), every hectare of mangrove prevents approximately US\$1,600 of property loss from coastal flooding every year in Ecuador. Property loss in this case includes both industrial and residential stock where damage and property loss values increase with increasing flood depth. Based on these figures, approximately one person has reduced annual expected flood risk for each two hectares of mangrove in Ecuador³². As noted above, currently urban areas account most of the people at risk and therefore concentrating mangrove protection and restoration efforts between the sea and urban areas will have the most impact. This is particularly the case in the Gulf of Guayaquil, where most urban inhabitants reside.

³² Or, more precisely 1 hectare of mangrove provides protection for 0.58 people on average.

54. In preparation of the project, an alternative modelling approach has also been used (using the Aqueduct Flood model, which does include future climate change related flooding risk) and is presented in **Annex 2, Feasibility Study**. This alternative model concurs with the Menéndez *et al.* (2020) in projecting that most people and property impacted by annual coastal floods in Ecuador live in Guayas. The important conclusion from this model is that with climate change, the relative value of mangroves to reduce coastal flooding increases. The two modelling approaches used also converge on a similar number of people being currently affected by coastal flooding during storm surges. Neither of the models includes potential changes in population due to migration patterns, which would include an additional large degree of uncertainty in the estimates for the future population at risk of flooding. We have therefore assumed for the purposes of this project, that the figure of 89,600 people with reduced annual expected risk from coastal surge flooding is a conservative estimate of the number of people benefitting from the additional coastal flood protection that the project will achieve by reducing forest loss and increasing mangrove cover through restoration.

Summary of Climate Change Hazards and Risk/Impacts

Hazard:	1. Increase in flooding Coastal Ecuador is already exposed to frequent flooding from both riparian floods due to high rainfall and coastal surges associated with storms. The frequency of coastal surges will increase as sea level rise increases (see Hazard 2 below) and extreme El Niño events (see Hazard 3 below) exacerbate the impacts of storms.
Risk/ Impact³³:	In coastal mangrove areas, approximately 198,000 people are currently exposed annually to high flood risks from either coastal or riparian flooding (SNGR, 2018). By ensuring that current mangrove areas are conserved, deforestation due to human pressures (mostly shrimp farming) is reduced by 50% and restoration expanding mangrove coverage by 4,850 ha, the project will benefit at least 89,600 by protecting them from flood risk from coastal surges (see Annex 25). Hallegatte et al. (2013) estimated the relative risk of vulnerability to flooding (economic average annual losses/city's GDP) in 136 coastal port cities, by assessing city-level flood risk and a database of urban coastal protection in 2005 and showed that the city of Guayaquil (which is protected by the mangroves included in the project) is ranked number four globally for flood risk. When future socio-economic changes are included, as well as a modest sea level rise scenario (20 cm increase), the assumption of adaptation measures by 2050, the increase in the economic average losses increases by 13%, putting the city of Guayaquil within the top 14 of flooding-related economic losses expected between 2005 and 2050 (Hallegatte et al., 2013) Ecuador's shrimp production has been highlighted as having the highest vulnerability to climate change of the world's brackish aquaculture producers in a global assessment, based on exposure (temperature, water balance, population density, precipitation change, flood risk, drought risk and cyclone risk), adaptive capacity and sensitivity (Handisyde et al., 2017). Eighty-two percent of shrimp farms along Ecuador's coast are sited in areas classified as high flood risk under current conditions (Ministerio de Acuacultura y Pesca, 2018; Secretaría Nacional de Gestión de Riesgos, 2018), risks which will increase with increased sea level and intensified precipitation projected due to climate change. During the 1997-98 El Niño, flooding, disruption of supply chains and damage to transport infrastructure affected the shrimp aquaculture sector and around 4,500 ha of ponds were severely damaged³⁴ (CAF, 2000). Heavy rains and flooding can also affect the harvest cycle, causing shrimp to molt early, resulting in up to a 20% drop in yields per pond (Sackton, 2016). These damages have direct consequences for the more than 60,000 people employed (approximately 0.45 jobs/ha of shrimp farm) in shrimp aquaculture farming and processing (Yahira Piedrahita, 2018)³⁵.

³³ Assess the *impact*, that is the effects on natural and human systems of extreme weather and climate change events. Impacts generally refer to effects on lives, livelihoods, health, ecosystems, economies, societies, cultures, services, and infrastructure due to the interaction of climate changes or hazardous climate change events occurring within a specific time period and the **vulnerability** of an **exposed** society or system.

³⁴ Clark Labs data estimates the presence of 1,571,298 km² of aquaculture ponds in Ecuador in 1999 (source: <http://www.aquaculture.earth/coastal/index.html>)

³⁵ Other estimates for the shrimp industry reference 100,000 - 220,000 directly or indirectly employed.

	Direct consequences include loss of income from lower production or when severe weather events cause partial or total loss of harvest. See section 2.2.3 “Communities at Highest Risk from Flooding” in Annex 2: Feasibility Study for more details.
Hazard:	2. Sea Level rise Sea Level Rise (SLR) will have significant direct impacts on Ecuador’s coast and its increase is a climate-change driven phenomenon. By 2100, mean sea level is projected to rise an additional 49-64 cm on Ecuador’s coast³⁶ depending on the Shared Socioeconomic Pathways (SSP) scenario (see Table 6; Annex 2: Feasibility Study). Measured mean sea level increased at an average annual rate of approximately 0.8 cm (1990-2009) based on the tide gauge of La Libertad (Alavera & Nath, 2013), implying that these projections are conservative (Figure 7; Annex 2: Feasibility study). An increase in mean sea level will also be compounded by increased frequency of extreme sea level (ESL) events (storm surge and tidal flooding), which will become common by the end of the century and occur annually mid-century in low-lying coastal areas (IPCC, 2019).
Risk/ Impact:	Increases in SLR and ESL are expected to increase the risks of coastal flooding and consequently impact rural and urban populations, and aquaculture in coastal areas, especially when compounded by periodic El Niño events. Celemin (2018), in an analysis of Ecuador’s northern coastal province of Esmeraldas, posited extensive damage related to a 5 m increase in mean sea level, though this extreme scenario is outside the range of most other projections.
Hazard:	3. Extreme El Niño Events El Niño events have already led to observed ocean warming and dramatic rises in sea level due to seawater expansion (Cai et al., 2014; Wang et al., 2019). During the 1997-98 season, sea surface temperatures off the coast of Ecuador increased by 5°C above normal, and sea level increased up to 42 cm (CAF, 2000). In the equatorial Pacific region, sea levels can deviate from the global mean sea level by as much as 40 cm because of El Nino (Walsh et al., 2012). ENSO is a natural climatic driver, and it is very likely that rainfall variability related to ENSO will be amplified by the second half of the 21st century under the SSP 2-4.5, SSP 3-7.0 and SSP 5-8.5 scenarios (IPCC, 2021). There are indications that extreme El Niño events may increase in frequency due to climate change (McPhaden et al., 2020). Future El Niño events are likely to become more frequent and more intense (Magrin et al., 2007; Wang et al., 2019), but even an El Niño of similar magnitude to historical events, will affect larger areas and more people as its flooding impacts are compounded by sea level rise.
Risk/ Impact:	Historical El Niño events provide a reference point for the scale of future climate-related damages for Ecuador’s coast. The Dartmouth Flood Observatory registered 22 major flooding events since 1987, with over 506,000 people displaced (DFO, 2020). Flooding events were particularly severe during El Niño events of 1972/73, 1982/83, 1997/98, as well as in 2008 and 2012, with 184,000-222,000 ha flooded (Ocles Padilla, 2018). Flooding events registered since 1965 on the coast of Ecuador have affected over 1,822,704 people, who required immediate assistance during those events, with 68% of those occurring during El Nino years. During those flooding events, 97,031 people became homeless. Flooding events recorded since 1965 have killed 836 people, with 622 of those occurring during El Nino years (Em-DAT, 2021). Those flooding events led to over US\$ 1.5 billion in damages and economic losses, of which 96% occurred during years when El Niño events occurred (EM-DAT, 2021). Furthermore, one quarter of the total coastal population was exposed to increased health risks in the form of infectious diseases such as malaria, diarrhea and cholera, which are associated with floods and damages to sanitary infrastructure caused by the 1997/98 El Niño event (Vos et al., 1999).

³⁶ <https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool> The SLR range cited is based on projections for scenario SSP2-4.5 (lower estimate) and SSP3-7.0 (higher estimate).

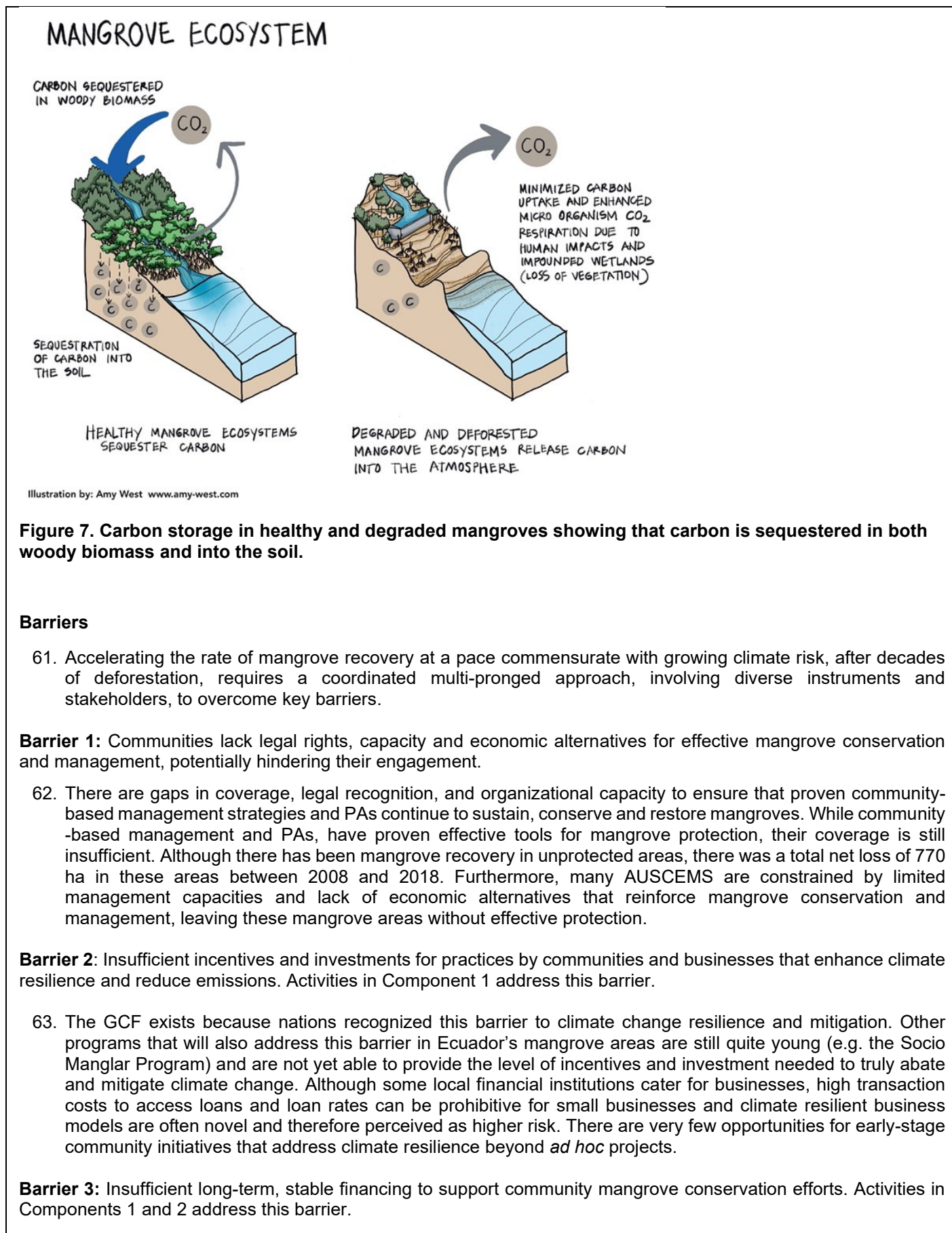
B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Current Conditions and Trends

55. Current trends and drivers of emissions and vulnerability are described in greater detail in section B1 and **Annex 2**, but key points for the Theory of Change are summarized here for clarity.
56. Sea level rise (~12 cm by 2040 and 49-64 cm by 2100)³⁷, increased impacts of El Niño events, and augmented intensity and variability of precipitation due to climate change are projected to increase flood frequency, intensity and damages, thereby threatening coastal infrastructure, settlements and economic activities.
57. Flood impacts are concentrated in recurring El Niño events and other extreme sea level events. SLR will magnify El Niño damages.
58. In addition to maintaining biodiversity and many ecosystem services, mangroves play a dual role, both in reducing climate change vulnerability of coastal communities to flooding events, and in increasing carbon storage. Although mangrove areas are slowly recovering and increasing (+0.35% per year)³⁸, an average of 990 ha of primary mangrove forests are still lost each year. Areas under community management and PAs are gaining mangrove coverage while unprotected areas continue to have net loss of mangrove.
59. Nearly 96% of all mangrove areas are concentrated in a subset of eight municipalities providing important flood protection for people and assets. With measures to protect and restore mangroves, the current flood protection benefits will be secured for the future and 89,600 people are expected to benefit annually by the end of the project (see **Annex 25**). The project is also expected to provide more resilient livelihoods for 41,500 climate-vulnerable direct beneficiaries. CI conservatively assumes that all these 41,500 people will be included within the 89,600 direct beneficiaries and therefore, to avoid double counting, estimates the overall number of direct beneficiaries of the project to be 89,600.
60. Shrimp aquaculture, which accounted for half of mangrove loss this past decade, also accounted for US\$6.2 billion in exports in 2022 (MercoPress, 2023). Increasing recognition of the threats that climate change poses to the sector and growing demand for sustainably produced shrimp have created an opportunity for shrimp aquaculture operations to play a pivotal role in protecting and restoring mangroves. Other private sector actors, including investors and companies seeking to become 'carbon-neutral', create additional opportunities for private sector finance for mangroves.

³⁷ <https://sealevel.nasa.gov/ipcc-ar6-sea-level-projection-tool> with 2100 range calculated based on the SSP2-4.5 and SSP3-7.0 scenarios.

³⁸ The recovery is in part due to short-lived regulatory measures adopted in 2008 that no longer apply.



64. The current conservation payments incentive program, currently covers only 44% of active AUSCEMs and longer-term financing mechanisms are needed to sustain the Socio Manglar program. There is also insufficient access to financial incentives, both public and market-based, to incentivize communities' continued commitment and private-sector action, particularly in shrimp aquaculture, to decisively drive a model of economic development that reinforces mangrove conservation and recovery.

Barrier 4: Businesses, communities and local governments lack information or access to technical support needed to adopt practices to reduce emissions and/or exposure to climate risks. Activities in Component 2 address this barrier.

65. Weaknesses in governance, including gaps in knowledge of legal and regulatory instruments, limited institutional capacity, poor inter-institutional coordination and lack of articulated stakeholder engagement, allow for the continued illegal destruction of mangroves and hinder proactive planning, enforcement and policy formation.

Barrier 5: Insufficient finance targeted at sustainable aquaculture production methods due to lack of knowledge among financial institutions and demand from producers. Activities in Component 2 address this barrier.

66. Prior to the development of this project, significant resources from the private sector, most notably the shrimp aquaculture industry had not yet been mobilized to support mangrove conservation and restoration. Although there has been growing interest in sustainable practices in the shrimp aquaculture sector, less than 1% of all shrimp farms in Ecuador currently operate under certification schemes that include actions leading to climate resilience of coastal communities and emissions reductions objectives. As part of the project, the NCA will work with CI to roll out the SSP approach, which includes a no-deforestation requirement, and to promote CSS production methods that includes mangrove restoration.

Barrier 6: Regulatory frameworks and enforcement are inadequate for protection and restoration of mangroves. Activities in Component 3 address this barrier.

67. There is a lack of enforcement that allows the drivers of mangrove deforestation (Shrimp Aquaculture sector, small scale land clearance for crops or pasture) to continue. Also, information for climate resilience is deficient and poorly integrated into decision-making and planning across all structures and scales of governance.
68. Governance, knowledge of regulatory frameworks and enforcement are inadequate for protection and restoration of mangroves by public, private and community-based actors. Detection of mangrove deforestation often lags far behind the occurrence of events and usually depends primarily on ground-based observation from government agents and complaints filed by citizens. MAATE field staff are limited and there is poor coordination among overlapping government agencies, which means that relatively few deforestation events are identified, and action taken. Despite the high rates of ongoing clearing of mangroves, only 13 judicial and administrative proceedings were processed for deforestation in 2011-2016, and of these only three had been resolved by 2017. The lack of effective enforcement and sanction has been emphasized by community stakeholders during the FPIC process of this project.
69. Overall, businesses, communities and local governments do not have information or access to technical support needed to adopt practices to reduce emissions and/or exposure to climate risks. Growing awareness of the importance of mangroves for the economies, populations and infrastructure of the coast has led to changes in public policies (including the Socio Manglar incentive program and legal requirements to restore mangroves), which has contributed to more recent net gains in mangrove coverage. Many key stakeholders, including the general public, still lack information about laws and regulations, the economic value that mangrove ecosystem services provide, and technical aspects of conservation, restoration and resource management.

70. Proposed project activities will address these barriers to achieve the overarching goal of the project.

Goal Statement: If local communities are provided with knowledge and resources for mangrove management and livelihoods development, and if the private sector and government actively collaborate on mangrove protection and restoration, and the enabling environment for mangrove protection is strengthened, **then** coverage and quality of mangrove ecosystems will be increased, resulting in reduced climate change impacts on vulnerable coastal populations, increased economic resilience, and reduced GHG emissions **because** healthy and more extensive mangroves reduce flood impacts and sequester carbon.

Outcomes and the project contribution:

Outcome 1. The area of mangroves under effective climate-adapted management is increased. The project will expand the mangrove areas under effective community management, support adaptation planning in existing PAs and strengthen management of other areas through support for improved land-use planning by local government and improved application of existing regulations on mangroves. The project targets 156,633 ha of mangroves under effective climate-adapted management by the end of the implementation period.

Outcome 2. Flood risks associated with climate change are reduced by expanding mangrove areas under effective climate-adapted management. The project will increase mangrove cover through reduced deforestation and restoration activities, which will result in reduced flood risk for an estimated 89,600 people.

Outcome 3. GHG emissions from deforestation are reduced and carbon sequestered by expanding mangrove areas under effective climate-adapted management, including mangrove restoration. Reduced deforestation and restoration activities will result in an estimated net reduction of 732,000 tCO₂e over the project implementation period and expected 4.6 MtCO₂e over a 20-year project lifespan period.

Outcome 4. Institutional framework for mangrove protection and coastal planning is strengthened. Project activities will result in improved regulatory systems or incentives for climate resilience and their effective implementation at the level of the national government, nine subnational governments and 60 AUSCEM-holding groups.

Co-benefit 1. Biodiversity and ecosystem service benefits are increased. The increased forest cover from reduced deforestation and restoration will provide additional habitat for biodiversity (5,725 ha) and increase the ecosystem services from the area gained. While the project does not propose to try and directly measure the value of ecosystem co-benefits (due to the complexity and methodological challenges involved), global estimates of the biodiversity ecosystem service value of one hectare of mangroves average US\$21,100 per hectare per year (in 2018 prices) according to a recent review of 250 attempts at valuing the ecosystem services of mangroves (Getzner et al., 2020), highlighting the importance of mangrove habitat. See also D.3 and **Annex 3** for further discussion of ecosystem service value.

Co-benefit 2. Economic resilience of coastal communities is increased. Project activities to secure mangrove management rights for local communities, including for fishing rights, to improve livelihood activities and to support small-scale community businesses will result in increased economic resilience for 41,500 people from coastal communities.

Co-benefit 3. Uptake of sustainable shrimp production practices is increased. The project will contribute to international and national initiatives to increase sustainability of shrimp production practices and to increase adoption of nationally and internationally recognized aquaculture standards and/or improved practices that include commitments to no deforestation covering at least 20,000 ha of shrimp farms.

Co-benefit 4. Economic value of fisheries for artisanal fishers is increased. The increased forest cover from reduced deforestation and restoration (5,725 ha) will provide additional habitat for fisheries areas and nurseries. Based on average estimates of catch, sale value and production, the additional habitat will increase the economic value of artisanal fisheries by US\$12.6 million per year (see **Annex 3** for fisheries value calculations).

Project Components and Project Results

Project Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and PAs implementing climate adaptation plans.

Output 1.1. Reduced exposure to flood risk for vulnerable people and reduced GHG emissions from mangrove restoration are achieved by strengthening community-based management through AUSCEMs and PAs.

Output 1.2. Improved livelihood activities and more economically productive community businesses enable local people to become more resilient to climate change and incentivized to participate in, and maintain, mangrove conservation and restoration.

Project Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.

Output 2.1. Shrimp aquaculture farms adopt practices and production standards that require elimination of deforestation and active reforestation in coastal and mangrove areas.

Output 2.2. Sustainable management of mangroves is improved through agreements with the private sector, including direct financial support for mangrove conservation and restoration.

Project Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.

Output 3.1. Decision making for mangrove management by national government agencies and local governments is based on generation and provision of accurate and up-to-date data on mangrove condition and socio-economic information on mangrove dependent communities.

Output 3.2. Legal and regulatory frameworks at local and sectoral level are harmonized and include climate resilience and mitigation strategies and enforcement.

71. The interventions also need to be designed in a way that minimize potential technical risks that could arise from the proposed interventions. The main risks to address within the design of technical interventions are:
 - Some community members might not benefit equally from the project, exacerbating inequalities; and,
 - Poorly implemented project activities could create tensions between different stakeholders who may have conflicting interests regarding mangrove protection.
72. In addition, socio-political disruption, health risks and natural disasters could cause interruptions to activities and changes in local, regional or national priorities. Other more general project risks and associated mitigation measures are explained in section F.1.
73. The project's theory of change also relies on the following assumptions (all are important but the most important have been highlighted in bold):
 - **Community associations feel sufficiently incentivized by project activities to engage in mangrove conservation and restoration.**
 - When community associations have resource rights over mangroves, they are more likely to implement sustainable practices.
 - **Equipping community associations with the necessary knowledge and skills for mangrove management and restoration will enhance their ability to implement effective conservation practices.**
 - There are sufficient business opportunities for mangrove community associations that can be developed as part of incentives for better mangrove management.
 - **Exposure of private sector actors to market drivers and climate risks motivates increased investments in mangrove protection and climate smart practices.**
 - Commercial bank lending rates remain at levels that would not discourage investment in sustainable aquaculture practices.
 - Local governments will make use of improved data availability in their decision making and land-use planning.
 - **Government agencies will have at least their current resources and capacities to support the project.**
 - **There will be political will to strengthen the application of legal and regulatory frameworks during and after project implementation.**

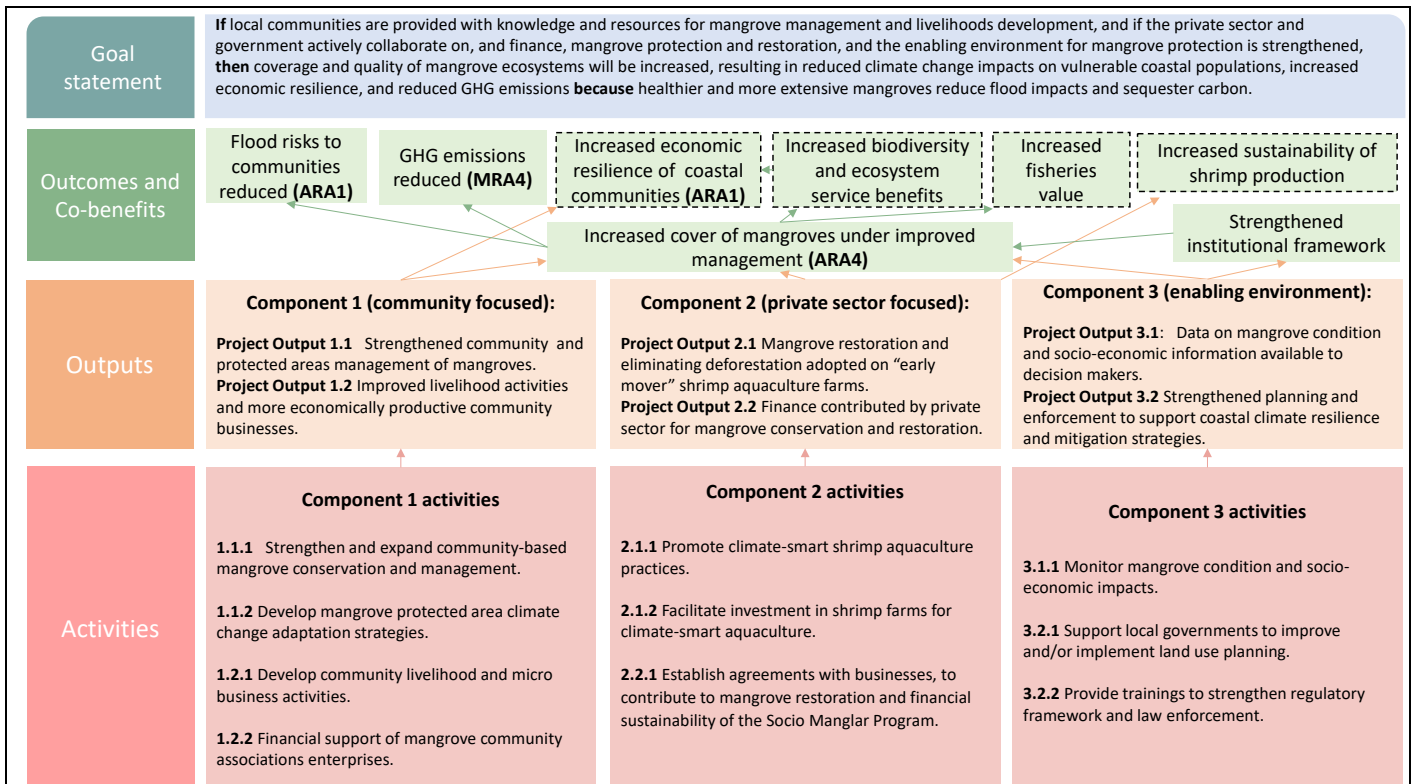
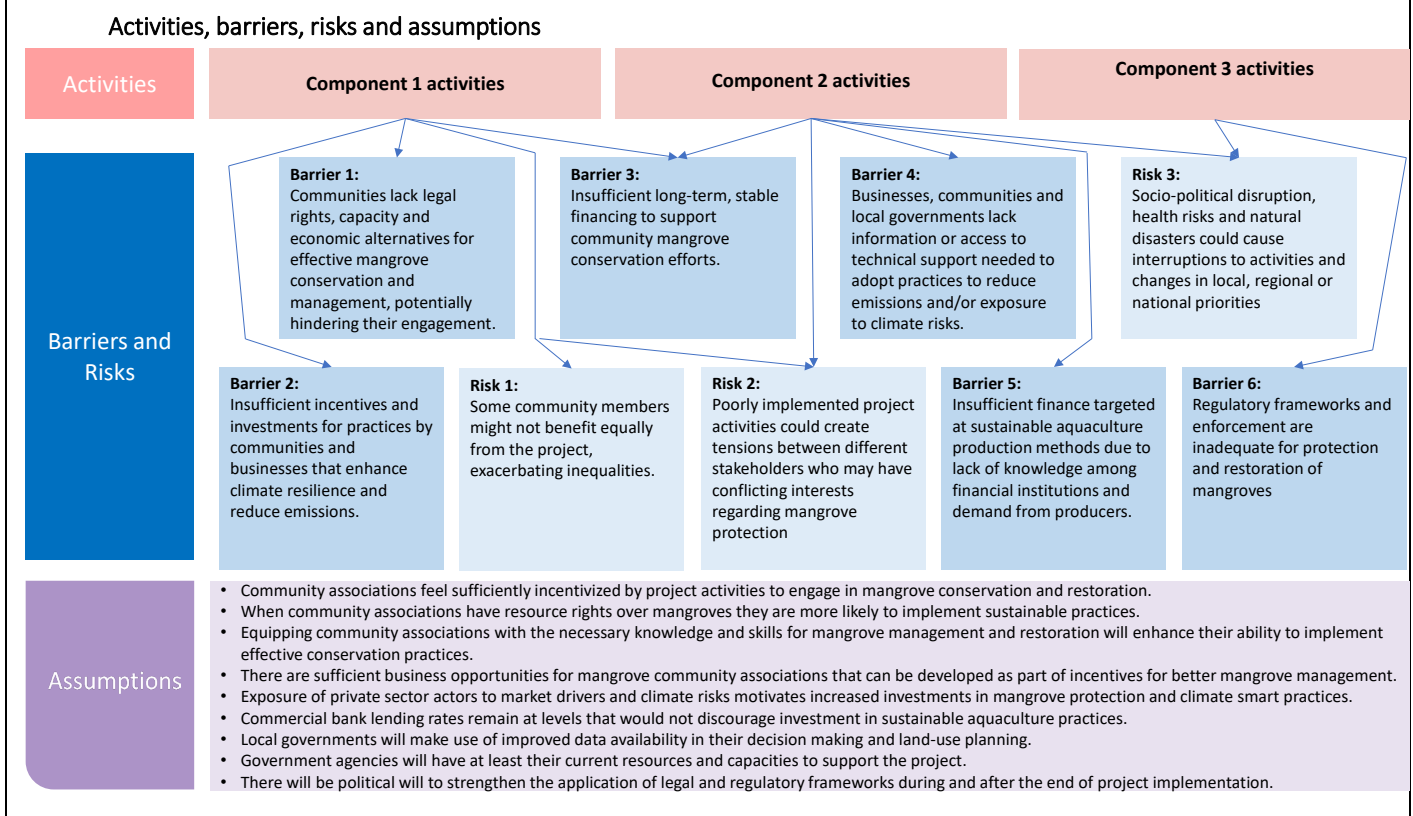


Figure 8. Simplified Theory of Change diagram for the Project



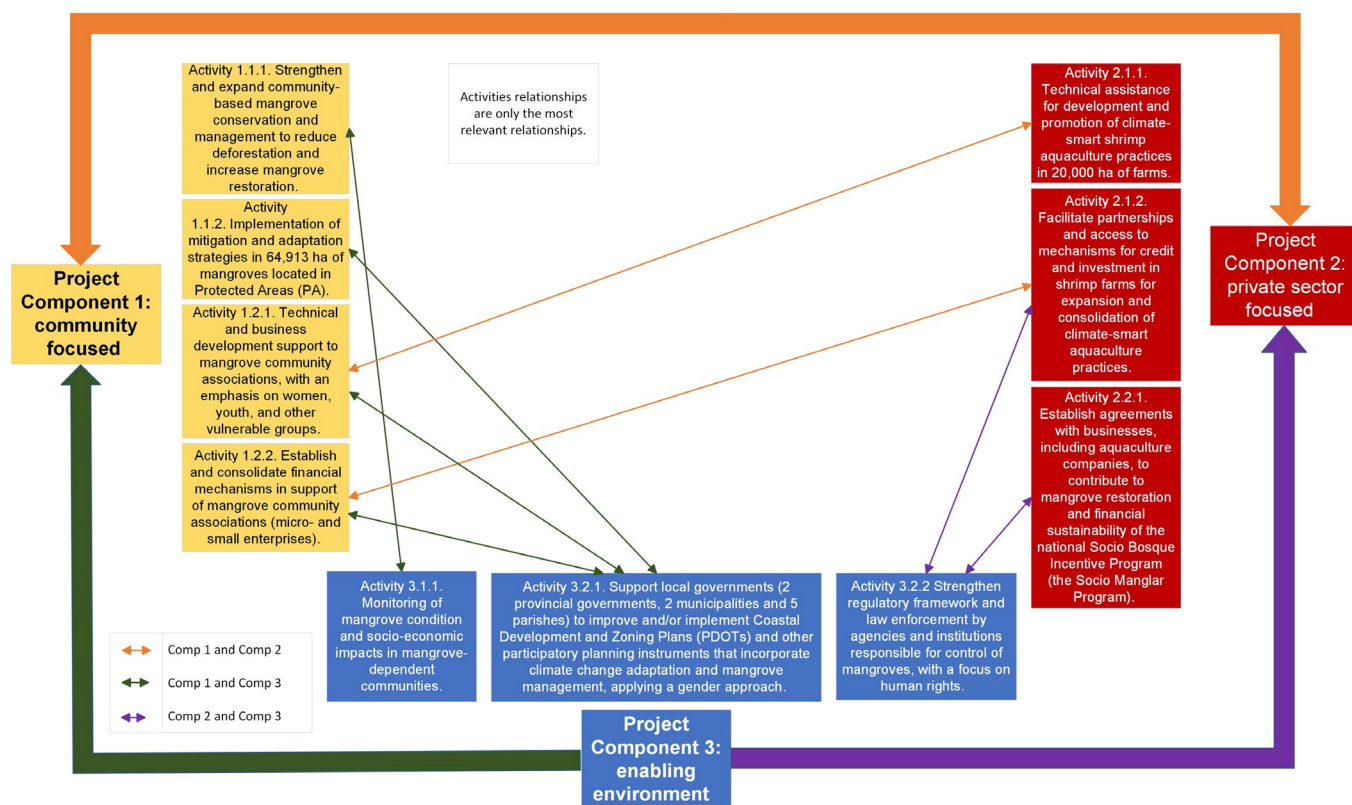


Figure 9: Linkages between project components and activities

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1. The area of mangroves under effective climate-adapted management is increased.	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 2. Flood risks associated with climate change are reduced by expanding	☐	☐	☐	☐	☒	☐	☐	☐

mangrove areas under effective climate-adapted management.								
Outcome 3. GHG emissions from deforestation are reduced and carbon sequestered by expanding mangrove areas under effective climate-adapted management, including mangrove restoration.	☐	☐	☐	☒	☐	☐	☐	☒
Outcome 4. Institutional framework for mangrove protection and coastal planning is strengthened.	This outcome maps to core indicator 5 (enabling environment) of GCF's IRMF: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low-emission climate-resilient development pathways in a country-driven manner.							

Co-benefit number	Co-benefit					
	Environment al	Social	Economic	Gender	Adaptation	Mitigation
1. Biodiversity and ecosystem service benefits are increased	☒	☐	☐	☐	☐	
2. Economic resilience of coastal communities is increased	☐		☒			☐
3. Uptake of sustainable shrimp production practices is increased	☒	☐		☐		
4. Economic value of fisheries for artisanal fishers is increased	☐		☒	☐		☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

B.3.1 Overview

74. Mangroves are among the planet's most important habitats for carbon sequestration and Ecuador's mangroves have been identified as important stocks of irrecoverable carbon – carbon that once lost cannot now be recovered before 2050 when the world must reach net-zero emissions to avoid the worst impacts of climate change (Noon et al., 2021). At the same time, because mangroves sequester carbon into the soil over centuries, mangrove restoration is an important nature-based solution over both the short- and long-term. Mangroves also provide essential functions for coastal protection and reduction of flood risks for vulnerable populations with a high incidence of poverty as well as for vital economic sectors for Ecuador's economy, principally shrimp aquaculture, of which 97% are small and medium producers. Mangroves also provide a suite of ecosystem services that support livelihoods, particularly due to their importance in natural fisheries, making their conservation and restoration multi-benefit, no-regret approaches to addressing climate change.
75. As described in section B1, loss of Ecuador's mangroves has slowed in the last decade, but unprotected areas continue to lose mangroves annually. This project will transform the situation by consolidating the recent gains in community managed areas, address remaining drivers of deforestation and scale-up mangrove restoration by catalyzing partnerships between community, private and public sector actors, and long-term incentives to protect mangroves will be provided to AUSCEM through the Socio Manglar program.
76. Reducing gross deforestation rates of mangroves by 250 ha per year (50% of man-made deforestation) and restoring 4,850 ha of mangroves are the principle direct strategies for generating flood-protection benefits for 89,600 people (coastal flood protection from mangroves), livelihoods benefits and economic resilience for 41,500 people from mangrove-dependent communities, achieving net emissions reductions and sequestration of 4.6 MtCO₂e over 20 years³⁹, and integrating climate-change adaptation strategies into regional development plans and policy instruments for 3.4 million people.

The project will focus on a combination of measures relating to reducing mangrove deforestation and increasing reforestation, centered on community-based conservation approaches, increased private-sector engagement and improved government agency and public-sector capacity centered on three main Components and six project Outputs as described in Section B.2., the Theory of Change. Specific Project Results and Activities are summarized below, with additional detail provided in Section 5.4.3 of **Annex 2** and in Section E.6 of this proposal (Log Frame, Activities).

Component 1 focuses on actions to increase both the area of mangrove under protection by local community stakeholders, their economic sustainability and the quality and effectiveness of management for these areas, as well as national PAs, to reduce flood risks and provide multiple benefits.

Component 2 focuses on engaging the private sector, particularly shrimp aquaculture, to become a transformational agent for change, reversing its previous role as an agent of mangrove loss. This component will promote the integration of climate-smart production practices to increase climate resilience and reduce pressures for mangrove deforestation as well as catalyze new sources of financing for the long-term sustainability of community-led mangrove conservation.

Component 3 focuses on creating the enabling conditions, through improved governance and the generation of timely, targeted information, which are the scaffolding supporting mangrove conservation, planning, regulation and benefits to the broader coastal and national communities.

77. Each component addresses critical barriers; none is sufficient alone to produce the desired climate outcomes of maintaining mangroves to reduce vulnerability, especially to floods, and to reduce emissions. Local action by communities requires responsive public institutions, reliable information, accessible finance and constructive relationships with the private sector. For the private sector, realizing sustainable growth opportunities and contributing to recovery of mangroves cannot be achieved in atomized fashion, but requires clear and consistent regulation, and its application, as well as supportive government action to position the country's shrimp sector as a pioneer in climate-smart production.

B.3.2 Outcomes' contribution to GCF Integrated Results Management Framework (IRMF)

³⁹ 732,000 tCO₂e over the 7-year project implementation period

78. A detailed description of project impacts and outcomes with regards to the GCF IRMF is included in sections E.3 and E.4 of the Logical Framework. The project outcomes will be measured according to the core indicators of the GCF IRMF and the project's contribution to the IRMF indicators is briefly summarized here:

ARA 1 Most vulnerable people and communities

79. All three components, by working synergistically to conserve and expand mangrove cover to reduce flood risk, contribute to reducing the impact of coastal flooding for 89,600 vulnerable people, 68% of whom live in poverty. Component 1 specifically targets work with 41,500 people whose livelihoods directly depend on mangroves, primarily through artisanal fisheries.

ARA4 Ecosystems and ecosystem services

80. All three components of the project, with convergent activities reinforcing sustainable management, conservation and restoration, contribute to improved resilience of 156,633 ha mangrove ecosystems, by ensuring maximum contiguous coverage and sustainable practices that do not undermine the ecological and structural integrity of the ecosystem. Community-based approaches will directly focus on a total of 70,000 ha⁴⁰, PAs on 64,916 ha (Component 1), while private sector and governance-oriented approaches (Components 2 and 3) ensure the overall enabling conditions reducing pressures on mangroves in coastal Ecuador.

MRA 4 Forestry and land use

81. All three components converge on a primary goal of reducing mangrove deforestation by 250 ha per year. Components 1 and 2 will result directly in 4,850 ha of mangrove reforestation in priority areas, which, over time, will also reduce risks for populations in their areas of influence. These combined mangrove-based mitigation activities will result in 4.6 MtCO₂e in emissions reductions over 20 years (732,000 tCO₂e by end of project implementation).

82. Additionally to the IRMF, the project contributes to the GCF USP2 Targets, as follows:

Strategic Priority	Description	Contribution
Country Programming	More than 100 developing countries directly supported to advance NDCs, NAPs or LTS	NDC and NAP: Ecuador will be directly supported to advance implementation of its NDC as set out in the national NDC implementation plan (which also currently serves as the implementation plan for the new (2023) National Adaptation Plan). The project will support Ecuador's conditional GHG reductions goals and adaptation goals to increase the cover of mangrove and expand the use of community-based management of mangroves through AUSCEMs. The project is expected to reduce national GHG emissions by 732,000 tCO ₂ e over the project implementation period and 4.6 MtCO ₂ e over the project lifespan. The project will strengthen management of 70,000 hectares of mangroves under community management as part of the SBP ⁴¹ . Ecuador's current NDC goal is to increase the forest area in the SBP by 127,500 hectares, including mangroves. The project will establish an additional 10 community-held AUSCEMs covering mangroves in addition to strengthening the existing 59 community associations with AUSCEMs (Ecuador's current NDC goal is to establish new 30 AUSCEMs covering mangrove areas).

⁴⁰ Including 60,217 ha in active AUSCEMs and an additional ~10,000 ha to be added.

⁴¹ the project will support the Socio Manglar program, which is the part of the SBP focused on mangroves

Food	190 to 280 million beneficiaries adopting low-emission climate-resilient agricultural and fisheries	89,600 people. Mangroves provide a range of ecosystem services that directly and indirectly contribute to the livelihoods and food security of local people. In addition to the flood protection services that will provide important climate change adaptation benefits to 89,600 people, mangroves serve as important nurseries and breeding grounds for species of fish, shrimp, crabs and clams that support artisanal and commercial/industrial fisheries. 41,500 people are expected to benefit from low-emission, climate resilient fisheries management practices that underpin their livelihoods. Note that CI is conservatively assuming that the 41,500 people are included within the 89,600 beneficiaries of flood protection.
Ecosystems	120 to 190 million hectares of terrestrial and marine areas conserved, restored or brought under sustainable management	156,633 hectares The project will contribute to the protection and sustainable management of a total of 156,633ha of mangrove ecosystems by increasing the area under community management, strengthening PA planning to integrate climate change adaptation, improved coastal land-use planning and strengthening the capacity of government agencies and local governments to enforce existing regulations designed to protect mangroves.
	Total (no double-counting)	156,633 ha (excludes restored areas)
	1.Sustainable management (sub-total)	91,733 ha
	2.Conserved/protected (subtotal)	64,900 ha
	3.Restored/natural regen/agroforestry etc (subtotal)	4,850 ha
Infrastructure	45 to 60 developing countries develop climate resilient infrastructure; including ecosystem-based approaches	Mangroves will provide flood protection benefits that reduce risk of coastal flooding and related damage.
Adaptation and locally-led action	40-70 approved proposals for adaptation projects, including for locally-led adaptation action.	The project will promote locally led adaptation action through community management of mangroves and action by the private sector to reduce deforestation and support mangrove restoration.

Degree to which GCF investments contribute to strengthened institutional and regulatory frameworks for low-emission climate-resilient development pathways in a country-driven manner.

83. Strengthening institutional and regulatory frameworks is the primary goal of Component 3, working particularly with local (municipal and parish-level governments) as well as national-level institutions responsible for planning, regulation, enforcement and information. Nine subnational governments and a variety of national government institutions (e.g., MAATE and the Judiciary) will be strengthened through project activities, including training, regulatory reform, technical support for planning, and improved inter-institutional coordination. At the grassroots level, 60 mangrove-based associations will benefit from training and technical support to strengthen their capacity for climate-responsive planning and development. The project also addresses current barriers and deficiencies by supporting the capacity of government entities to monitor and report mangrove forest cover and carbon stock information to enhance national mitigation efforts.

84. All three components contribute to maintaining and expanding mangrove coverage as a strategy to reduce flood risks through mangrove protection for 89,600 people.
85. Three million and four hundred thousand people living in the eight priority municipalities will be reached as indirect beneficiaries by risk reduction measures established as part of PDOTs, public investments and other policies and programs adopted by these subnational governments.

B.3.3 Project Description (see Annex 2; Section 5.4.3 for additional detail; please see E.5. Project/programme for a chart on specific outputs and their results)

Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and PAs implementing climate adaptation plans.

86. Legally recognized, community-based stewardship, despite having demonstrated its effectiveness in protecting mangroves (see Table 3), still has significant gaps in coverage that can be resolved through outreach and assistance in organizing, planning and a strengthened relationship with the government programs that recognize AUSCEM. For existing AUSCEM associations, ensuring that they have the organizational capacity to sustain their stewardship and renew their management plans and legal agreements is critical to ensuring that effectiveness of management is maintained.
87. These community-based mechanisms are also frequently economically precarious. Generating stronger local economic benefits based on mangrove conservation, in the form of both public incentives (from the Socio Manglar Program), finance for restoration activities, and local sustainable enterprises and livelihoods activities is indispensable to ensuring that sustainable management and conservation of mangroves is economically viable and contributes to dignified, equitable livelihoods.
88. Restoration of 4,600 ha under community stewardship will contribute to enhancing mangrove coverage and associated benefits⁴².
89. By ensuring that AUSCEMs, as well as national PAs, grow both quantitatively (in area covered) and qualitatively, in terms of robust management that integrates climate-change resiliency, mangrove areas will provide flood-protection benefits for 89,600 vulnerable people, as well as directly benefiting communities who are dependent on mangrove resources.

Output 1.1. Reduced exposure to flood risk for vulnerable people and reduced GHG emissions from mangrove restoration are achieved by strengthening community-based management through AUSCEMs and PAs.

Activity 1.1.1. Strengthen and expand community-based mangrove conservation and management to reduce deforestation and increase mangrove restoration.

90. CI and PUCSE (Pontifical Catholic University of Ecuador Esmeraldas Extension) will provide training, exchanges, and targeted material support as key tools to strengthen the inclusive governance of existing AUSCEMs, focusing on a variety of topics and tools that will empower these associations to better protect mangroves at risk of deforestation, enhance their livelihood benefits and adapt to climate risk. As a result, at least 60 associations, representing 4,600 families (24% with women-headed households), will have stronger human, organizational and operational capacity reflected in Management Capacity Assessments and the effective maintenance or renewal of AUSCEM agreements.
91. Existing AUSCEM groups will receive ongoing technical assistance from CI Ecuador or PUCSE for the management, renewal and reporting of their agreements with the MAATE (Undersecretary of Natural Heritage), including support for preparing semiannual reporting to MAATE and development and/or updating of management plans for the mangrove areas under their responsibility. In each of the southern estuaries, CI will have two project staff: one Mangrove Specialist and one Social Specialist responsible for daily activities in the field. For the two northern estuaries, PUCSE will employ a Project Coordinator, a Mangrove specialist, a Social Specialist and 10 community agents to support work in the field.

⁴² Note that an additional 250 ha of mangrove restoration is also planned under Activity 2.2.1 to be supported directly by shrimp farms/aquaculture companies.

92. CI and PUCSE will provide training sessions in the field with all association members and the fishers' spouses interested in participating in the trainings⁴³. For the leaders of the associations, follow-up sessions will be implemented to reinforce knowledge.
93. To ensure that the AUSCEMs can carry out their mangrove management and protection role, CI will also provide equipment via in-kind grants to support community patrols, threats and environmental monitoring and to assist with reporting infractions. To this end, the associations will be provided with basic monitoring equipment (for measuring fish sizes as part of mangrove fisheries monitoring), GPS devices and demarcation signage.
94. As shown in section B.1., mangrove areas without AUSCEMs are at high risk of deforestation and this activity is designed to increase community protection of mangroves. The areas prioritized for expansion of AUSCEMs are those that are most likely to contribute quickly to the project's deforestation reduction targets because they have high threat, larger areas with good existing or potential governance structures; and areas most likely to shield human settlements from coastal flooding. CI and PUCSE will support groups that have a lapsed AUSCEM agreement with MAATE to renew their agreements (13,000 ha). In addition, for 10,000 ha of mangroves not yet under AUSCEMs, CI and PUCSE will support community associations to establish new AUSCEM in close collaboration with MAATE (Undersecretary of Natural Heritage) for the selection of the new areas and the enforcement of the authority in the field. Technical support will be provided to the community associations to develop the management plans and legal documents needed for the AUSCEM application process and to undertake a community consultation process that ensures FPIC. These activities will start in the second year and continue until the end of Year 5.
95. CI and PUCSE will also provide technical assistance to current and new AUSCEM groups to prepare and apply for Socio Manglar incentive payments, with the aim of expanding coverage from the current baseline of 33,467 ha of mangroves under 25 AUSCEM groups receiving incentive payments to 62,243 ha during the project implementation. In preparation for the application process, support will be provided to develop an investment plan, and financial accounting training will be provided to ensure adequate capacity for managing funds received from the Socio Manglar program (including verification that necessary bank accounts and legal documents are in place). For successful applicants, CI and PUCSE will provide follow-up training on budgeting practices and on requirements for monitoring and reporting to MAATE. The trainings and technical support will also include diversity, equity and inclusion trainings with a particular focus on gender, governance and decision making, accountability, financial planning, tax reporting and legal procedures for the associations and models for investing the incentive in economic alternatives. This technical assistance will be provided in Years 2-5 of the project.
96. GCF funding (targeting 4,600 ha restored) will be invested to cover costs of mangrove restoration with AUSCEMs' involvement, for which CI will enter into a combination of grant agreements with the associations and service agreements with service providers for restoration activities. In addition, CI and PUCSE will support training, monitoring and technical follow-up. This will be done from Years 2 to 5 and with the support of the restoration specialist and the social technician based in each estuary.
97. Details of the process for identifying priority areas for restoration and further details on restoration methods are provided in Appendices 3 and 4 of **Annex 2, the Feasibility Study**.

Activity 1.1.2. Implementation of mitigation and adaptation strategies in 64,913 ha of mangroves located in PAs.

98. Currently, PAs do not have enough human and financial resources to effectively implement conservation and management actions nor to include adaptation or mitigation practices. As described in Section B1 and the Feasibility Study, well-managed mangrove PAs have lower deforestation and therefore GHG emissions than unprotected areas. Improving PA effectiveness can further reduce GHG emissions from loss and degradation of the areas and increase new carbon sequestration through active restoration activities and natural mangrove recovery. Mangroves themselves are potentially vulnerable to climate change and therefore PA managers need to ensure that they are managing the PAs consistent with promoting ecosystem resilience. For example, one issue with SLR is the need for mangroves to be able to expand landward. Ensuring that there is space to do so could require management action in surrounding habitats. PAs usually don't include adequate adaptation practices in the management plans, so working to improve the planning and implementation of climate adaptation actions is important.

⁴³ There is one association in Jambeli Estuary that only has women as active members, the others are mainly formed by men.

99. CI and PUCESE will work with MAATE staff (paid with MAATE in-kind co-financing) in four PAs – Manglares Cayapas Mataje, Manglares Churute, Manglares El Salado and Manglares Estuario des Rio Muisine – which are the PAs with the largest area of mangrove and the most deforestation (in terms of area), and therefore have the highest potential for further reductions in deforestation. CI, PUCESE and MAATE staff will work to integrate climate-change into planning instruments (management plans and annual operating plans) for each of the PAs. In Year 1, workshops will be organized by CI to identify the conservation objectives and the specific climate change threats that each PA faces. During Year 2, the partners will develop assessments of each PA management plan, that will include recommendations for climate resilience and adaptation and identify strategies and actions with MAATE staff and local stakeholders. The assessments and recommendations are intended to be integrated into updated PA management plans at their next update (typically management plans are updated every five years); and will be developed under service agreements through a competitive bid process. During Years 2 and 3, twice-yearly training events will be developed in each of the PAs to make PA managers aware of climate change risks and vulnerabilities and how to explicitly incorporate climate change adaptation into MPA management plans and objectives.
100. Management of PAs to reduce deforestation will also be strengthened by improving staff training and providing equipment for threat monitoring, enforcement, and community engagement, including monitoring equipment such as data loggers, gear for underwater surveys, software and computers to ensure monitoring in line with the climate adapted management plans and analysis of information collected. This equipment will be managed by PA staff, with resources from the national environmental fund (FIAS) covering costs of operation and maintenance.
101. Project impacts and learning at the four PAs will be amplified through the national network of PAs (*Red de Areas Marinas y Costeras Protegidas del Ecuador*). PAs and PA networks need to be adaptively designed and managed to address altered coastal and oceanic conditions and habitat shifts due to climate change, which may affect future boundaries, locations, and sizes. Networks help reduce risk and promote resiliency. A network analysis and action plan will be developed in Year 3. For this activity, CI will work closely with the Undersecretary of Natural Heritage and the provincial offices of MAATE.
102. In subsequent years, knowledge on the design, implementation and monitoring practices will be presented to the PA Network to amplify action. For this, CI will hold a workshop in Year 4 with all the marine PAs in the PA network, produce and distribute guidelines on how to integrate climate change into management plans and organize activities with local associations to highlight the importance of the topic. The PA network is critical to maintaining climate change resilience and rebuilding ecological and social resilience.

Output 1.2. Improved livelihood activities and more economically productive community businesses to enable local people to become more resilient to climate change and incentivized to participate in, and maintain, mangrove conservation and restoration.

Activity 1.2.1. Technical and business development support to mangrove community associations, with an emphasis on women, youth, and other vulnerable groups.

103. By supporting mangrove-dependent communities, through groups formally linked to mangrove conservation, ongoing conservation and stewardship can be made economically viable, with improved incomes reinforcing conservation, climate change mitigation and resilience objectives. By improving the economic productivity of community enterprises, the value of mangroves to the communities is also increased and this creates incentives for mangrove conservation. Increasing incomes to communities increases their resilience to climate change shocks and provides more opportunities to adopt adaptation measures. CI and PUCESE will provide support to at least 60 community associations linked to protection of mangroves to design and implement business plans and strategies, including strategies for improving enterprise governance and administration, access to finance and to markets for more resilient livelihood strategies from Year 1 of the project, with additional technical capacity provided through specialist consultants. The estuary-based CI sustainable production specialist and the social specialist will work directly with the associations. The main strategy of this Activity is focused on adding value to mangrove-dependent fisheries products. Product quality and food safety will be enhanced by monitoring cold chains, classification and selection of product sizes and classes according to market requirements.
104. Other community economic activities not directly related to seafood products will also be supported to enforce resilience of local communities and diversify income sources. These initiatives will receive technical support from CI, market analysis and study visits to other initiatives for one-to-one training.

Activity 1.2.2. Establish and consolidate financial mechanisms in support of mangrove community associations (micro- and small enterprises).

105. Activity 1.2.2 is designed as a follow-on to support early-stage businesses (either new ones created in Activity 1.2.1 or existing ones).
106. CI will award small cash grants on a competitive basis for productive projects originating with the associations legally established in the areas of the project and supported under Activity 1.2.1. Various organizations with legal status in the provinces of Esmeraldas, Manabí, Guayas and El Oro will participate, including both AUSCEM groups and other organizations with links to mangroves. The objective is to support promising micro- and small-scale enterprises by helping them to scale their operations with the aim of improving the income of these organizations and the standard of living of their members, while strengthening the incentives for sustainable use of mangrove environmental goods and services. To be eligible for the grants, the enterprises will require a simple business plan that includes a financial analysis demonstrating how the grant will transform the business. These enterprises are artisanal fishing production association or cooperatives that report to the Superintendence of Popular and Solidarity Economy. Also, in this category are included microenterprises that have an annual income less than US\$100,000 and between 1 to 9 workers.
107. This grant mechanism, totaling US\$500,000, will be administered by CI and will provide an estimated 20-25 cash grants to community associations (US\$10,000-50,000 per grant). Grant support can be invested in processing equipment, operating costs and other needs to scale up enterprises. Selection criteria for the grants have been designed (See Appendix 5 of **Annex 2, Feasibility Study**), and a selection committee will include representatives from MAATE, associations and partner organizations of the project. CI will enter into grant agreements with grantees and monitor sub-grants using CI's policies and procedures.
108. To access credit and investment from existing private and public financial institutions, CI will support associations to develop cost and market analysis to increase the degree of formality of the mangrove community business, define baseline indicators (e.g., operational efficiency and profitability of the operation, profit/ha/day, feed conversion index). This information and technical support will be used to provide assistance in the identification and application to national sources of finance, such as cooperatives, public financial institutions (BanEcuador) and other private financial entities that provide credit and investment to micro- and small enterprises.
109. CI will hire a Bioentrepreneur Specialist with expertise in small enterprise development to provide the technical support described in activity 1.2.2.

Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.

110. Engagement with the private sector in the project consists of promoting and enabling the adoption of improved shrimp production methods to reduce mangrove deforestation (and achieve other GHG emissions reductions that are not the focus of the project) and advancing mangrove conservation and restoration through philanthropic support. CI will work with Ecuador's NCA to provide training on the SSP approach, which includes a no-deforestation requirement and to promote CSS production methods that include mangrove restoration (250 ha planned on farms in the project). CI will engage consultants to support small- and medium-sized shrimp farms to access private finance.
111. In preparation for the project, the ASC and CI provided seed funding (US\$50,000 and \$100,000, respectively) in 2021 to establish the Socio Manglar subaccount of Ecuador's Socio Bosque Fund. This funding will be available to support the Socio Manglar program during project implementation. ASC will request that their members contribute to growing this subaccount through voluntary contributions.⁴⁴ Additional fundraising by CI and the Government of Ecuador will grow the subaccount to provide long-term community incentives for mangrove conservation. Four million dollars (US\$4 million) is requested from GCF to contribute to the Socio Manglar subaccount and will be managed in an interest-bearing account separate from other contributors' funds, which will be managed in an investment account in accordance with the investment policy of the fund manager, FIAS.

⁴⁴ These funds will be reported as parallel co-financing for the project during implementation.

112. Adoption of improved management and production practices in the shrimp sector would lead to reduced GHG emissions, improved climate resilience, and reduced environmental impacts, all while increasing production and profitability of the sector (see Section B1 for discussion of shrimp aquaculture and the reasons these changes have not yet occurred). By working on sector-wide policy and market changes, and directly working with a core group of 'early adopters' comprising at least 20,000 ha (approximately 10% of all shrimp farms in Ecuador) to adopt an integrated package of CSS practices, the project will:
- Reduce deforestation on current shrimp farms and build climate resiliency by reducing the overall environmental footprint in terms of energy use, water quality, and biodiversity loss, and increased profitability (through increased yields). Since the project will be working to protect all mangroves in the project area (and almost all mangroves nationally) through either community AUSCEM agreements, PAs or improved land-use planning with subnational governments, there is very little opportunity for leakage of deforestation caused by improved management on farms.
 - Restore 250 ha of mangrove forests on current shrimp farm areas (resulting in removal of 0.1 MtCO_{2e}).
 - Catalyze an estimated US\$5 million in investment in the transition to CSS from private-sector sources (see section B1 for background on other investment institutions interested in the aquaculture sector). Note that GCF grant funding will not be used to directly invest in shrimp production using CSS methods on shrimp farms. Shrimp farms needing to invest will do so from their own resources or from private sector finance sources.
113. The rollout and adoption of the SSP approach and/or ASC certification nationally alongside the use of CSS will push the shrimp aquaculture sector to adhere to zero deforestation and promote mangrove restoration. The increased profitability of the CSS model will act as an incentive for farms to invest in it without the need for future public grant funding. As farmers modernize their production ponds by adding aerators, autofeeders, and water quality sensors and by utilizing biosecurity protocols, water quality management systems, and improved data management and analytics, they are able to reduce the cost of producing shrimp on a per kilogram basis and produce more shrimp per hectare of ponds. This enables farms to increase their profitability and output. Sustainable intensification requires increasing technical capacity to enable 'best management practices' (see Activity 2.1.1), financing of capital improvements (see Activity 2.1.2), and policy alignment (see Activity 2.1.3). The CSS strategy will be applied to farms in different geographies, in areas of high climate vulnerability, and near mangrove AUSCEMs and areas associated with Socio Manglar incentives. The project will specifically target areas where risks, vulnerability and ecosystem services overlap. More broadly, this project will strengthen the enabling conditions in policy, capacity, and market conditions to drive a sector-wide shift in production practices to favor zero-deforestation and restoration of mangroves in and around shrimp aquaculture areas.

Project output 2.1. Shrimp aquaculture farms adopt practices and production standards that require elimination of deforestation and active reforestation in coastal and mangrove areas.

Activity 2.1.1. Technical assistance for development and promotion of climate-smart shrimp aquaculture practices in 20,000 ha of farms.

114. The activity will involve the design, implementation and monitoring of mangrove restoration and conservation plans in and around shrimp farms to achieve biodiversity, water quality and coastal adaptation objectives (led by CI, with partners from NCA, farm holders, members of local farming associations, and local communities).
115. CI and the NCA will provide direct technical support to aquaculture farms for adoption of the SSP standard and CSS. The NCA will advise on training materials and training sessions. CSS involves working with shrimp farmers to sustainably intensify shrimp production, constructing treatment wetlands and restoring coastal ecosystems. NCA and ASC will act in an advisory role and provide input to training materials and trainings. Activities include dissemination of sustainable management practices, dissemination of best restoration practices, and ongoing socialization of CSS.
116. Intensification of production using a CSS approach creates the necessary conditions and incentives to integrate mangrove restoration into shrimp farming operations while also enhancing climate change resilience of farms by protecting against SLR and storm surge and mitigating and sequestering carbon emissions. Successful uptake and implementation of CSS program activities in select regions of Ecuador will require substantial coordination and capacity development at multiple levels of government, the private sector (farmers, supply chain, and other companies), and civil society. CI will develop guidelines for sustainable intensification best practices to engage with and train farmers, government agencies, and other private and public sector actors to strengthen and internalize technical capacity building. Rather than working only with individual farms to

implement the guidelines and foster adoption of best practices, CI and NCA will utilize a 'jurisdictional approach' to coordinate alignment among relevant stakeholders throughout the region. Jurisdictional Approach projects have been shown to foster wider, faster adoption of best practices, with a particular efficacy at enabling coordinated disease management, enhanced climate resilience, and environmental stewardship (Kittinger et al., 2021). Implementing a jurisdictional approach to best intensification practices requires coordinated activities, including:

- Partner selection (led by CI, in partnership with NCA), based on an assessment tool⁴⁵ (see criteria in Appendix 8 of **Annex 2, Feasibility Study**), applicable regionally and across geographies, to evaluate candidate farms.
- Engagement to identify CSS project participants, and a final partner screening to select shrimp farms (small, medium, or large) willing to participate in the project.
- Develop a collaborative, time-bound workplan with all CSS project partners based on the baseline analyses, final partner screening, and engagement strategy with explicit commitments from each participating farm to produce shrimp meeting the CSS criteria and each participating entity to support the advancement of CSS with defined roles and outcomes (e.g. offtake agreements, policy support, conservation use agreements). Led by CI, in partnership with NCA.
- Workplan execution and stewardship (led by CI, in partnership with NCA). Conduct trainings and technical sessions for 200-250 farmers from small- and medium-sized farms and other stakeholders (e.g. universities and NGOs working on the topic).

117. Technical assistance from CI and NCA will enable farmers to design and implement restoration initiatives on portions of their farms, which will confer water quality benefits within the farm and provide broader climate change mitigation and adaption benefits, including effective and efficient flood protection. The costs of the mangrove restoration on farms will be financed by the farms.

118. A target of 250 ha of mangroves will be restored in the immediate vicinity of shrimp ponds and in areas where ponds are no longer viable. Restoration designs as part of climate-smart business models will be developed by consultants hired by CI to establish hydrology and landscape conditions within the restored area while allowing the shrimp farm operations to be maintained or improved. This innovative approach improves upon the conventional approaches to coastal protection, by integrating ecosystems to improve the overall outcome for biodiversity, production, and coastal adaptation. The resulting designs, details and case studies generated through this project will provide tools to replicate at scale elsewhere in the region and the world, with potential to transform business-as-usual for global shrimp-farming practices and as a model for other agricultural production sectors.

119. There are many examples of exploitation and mismanagement of mangroves and shrimp farms, and hence the need to address these challenges through effective public policy that embraces partnerships between government agencies, local communities and the private sector, all of whom operate in the same space. Stakeholder consensus around the benefits and importance of CSS is a critical enabling condition of implementation of CSS at project sites and project initiatives and is critical to achieving program sustainability. Consensus requires alignment among a diverse set of stakeholders, ranging from community members, local and national government, public institutions, universities, local organizations, national and international financial institutions, supply chain actors, and most importantly, shrimp farmers and technicians. Additional efforts to communicate project activities and outcomes will promote climate-smart shrimp amongst other producers in Ecuador while signaling for supplemental interest and investments.

120. To disseminate knowledge gathered by the project, collections of key documents and resources will be housed within existing knowledge portals including: i) the website and training initiatives led by NCA as part of the SSP initiative; ii) the eco.business Sustainability Academy platform; iii) Escuela Politécnica del Litoral (ESPOL) University, with its course on Marine engineering and Aquaculture engineering (a key stakeholder of the CSS initiative); and iv) the GMA knowledge hub.

121. A CI Communications & Knowledge Management Manager, the Manager of Component 2 (Incentives, Financial Mechanisms and Climate Finance) and the Climate Smart Aquaculture specialist will lead activities that will target specific stakeholder groups. A total of 35 workshops are planned to socialize CSS production models (with an estimated average of 20 people per training). A further 19 workshops are planned for CSS technical training aimed at small- and medium-sized companies (with an estimated average of 20 people per training).

⁴⁵ See <https://ci-aquafarm-mapping.web.app/> for the CSS assessment tool developed for the Philippines and Indonesia that will be adapted for Ecuador.

Activity 2.1.2. Facilitate partnerships and access to mechanisms for credit and investment in shrimp farms for expansion and consolidation of climate-smart aquaculture practices.

122. CI will work with different actors of the shrimp supply chain to address finance access barriers and facilitate the flow of credit and investment for farm operations that seek to transition their production models to more sustainable ones.
123. The limited offer of financial products, both from the financial system and other sources, is a significant barrier for shrimp farms wanting to adopt the SSP and CSS models. The eco.business Fund is financing green credit lines to key banks such as Produbanco, Banco de Guayaquil, Banco del Pacifico and Banco Pichincha, and is an important ally in promoting a sustainable model of shrimp production, training bank staff on impact and risk management in relation to the shrimp business and testing models so that the institutions feel confident in investing in a shrimp-related business.
124. As new financial instruments are being developed, business demand for technical support to ensure successful investments is also increasing. CI will coordinate with the eco.business Fund to support and expand initiatives such as the Sustainability Academy, to small- and medium-sized farms, and other service providers in sustainable production models and investments that derive into more sustainable operations (the platform is open to the general public). The Sustainability Academy portal combines education materials, workshops, self-assessment tools, and case documentation on several sectors, including shrimp aquaculture. The Sustainability Academy content will be expanded to include CSS, good aquaculture practices, mangrove conservation and restoration, safeguards implementation, and other climate and conservation-oriented programs, such as the Socio Manglar program. The project will bring together stakeholders to make the Sustainability Academy and the existing financial products more available to small- and medium-sized farmers that are not necessarily clients of the banking system to date, expanding the base of beneficiaries of the program.
125. Implementation of demonstration cases, from diagnosis to work plan development, implementation and measurement, is critical to develop the skills of investors, suppliers and farmers. To strengthen these successful investment cases, CI will provide technical support to businesses in the project feasibility and implementation phases. Criteria for selecting beneficiaries will be developed to include conservation priorities (mangrove conservation or restoration component), market opportunities in connection with SSP and/or ASC, and particularly links to investors or banks that can provide the required funding for making an operation more sustainable. Support from the project will be conditional on the business's commitment to invest in mangrove restoration and/or conservation.
126. CI will also promote the implementation of a roundtable created by the National Mangrove Plan with industry actors to include the CSS concept into the agenda and priority discussion topics and the solutions to address finance access barriers described in the project approach. Links to the commercial sector will be critical to reduce risks of investments in the shrimp industry, so working in connection with the NCA as well as international buyers to secure commitments to CSS is important.
127. To address investment needs, the project will seek to identify and support shrimp farmers on the design of sustainability efforts as a tool to mobilize capital towards CSS, ASC and SSP models. CI will assist these needs through technical support from consultants. Shrimp farmers/companies will be eligible for this technical support if they comply with initial requirements: i) a concrete investment opportunity that can improve climate resilient practices and ii) a link to a financial institution interested in financing such investment. CI will assess all requests and prioritize opportunities based on a series of eligibility criteria (further details in Section 5.4.3 of **Annex 2, Feasibility Study**).
128. The technical support from the project will be used with the goal of unlocking private finance to specific geographies with a role in mangrove conservation or restoration but not necessarily those farmers being the most investable in the sector. Furthermore, there is an understanding that many small- and medium-sized farmers operate in the area, and they are the ones that face more challenges in their capacity to attract investment into their operations. This technical support will: a) strengthen weaknesses in management areas that limit the capacity of attracting capital to promote sustainable intensification at farm level, or b) refine an investment opportunity through a feasibility assessment, business plan development, impact targets such as efficiencies and improved conditions, and definition of Key Performance Indicators (KPIs) to monitor advancement towards those goals.

129. The CI project staff (Climate Smart Aquaculture Specialist) will facilitate agreements between retailers, importers/exporters, processors, and farmers to guarantee purchase of climate smart shrimp.
130. CI will work with NCA and ASC to continually strengthen training materials for the CSS concept and to refine the concept itself based on industry feedback. CI will engage a consultant in Year 3 to develop a marketing strategy for CSS in Ecuador. This will include assessing market engagement for CSS with retailers, including market traceability, market recognition and establishing buyer assurance mechanisms. CI will also collaborate with NCA and ASC and third-party independent auditors that confirm farm adherence to the CSS standard.

Output 2.2. Sustainable management of mangroves is improved through agreements with the private sector, including direct financial support for mangrove conservation and restoration.

Activity 2.2.1. Establish agreements with businesses, including aquaculture companies, to contribute to mangrove restoration and financial sustainability of the national Socio Bosque Incentive Program (the Socio Manglar Program).

131. Ecuador's Socio Manglar incentive mechanism provides a tangible, government-supported pathway for shrimp farms and other enterprises operating on Ecuador's coast to make effective their commitments to mangrove conservation. CI, in preparation for this project, has worked with the ASC to create and contribute to the capital of the Socio Manglar subaccount of Ecuador's Socio Bosque Fund, with an initial donation of US\$150,000 (US\$50,000 from ASC and US\$100,000 from CI). The Socio Bosque Fund is managed by Ecuador's Environmental and Sustainable Investment Fund (FIAS)⁴⁶, and the new Socio Manglar subaccount is managed using the same operating procedures. Each fund managed by FIAS has a governance mechanism for decision making and FIAS has a board made up of representatives from the public and private sectors. All funds are audited, and information is presented to the board and to each governance mechanism. FIAS also has an investment committee that oversees the policies and strategies implemented with the funds based in Ecuador and with the funds invested internationally. Appendix 12 of **Annex 2** provides further background on the Socio Manglar subaccount.
132. FIAS is a private, non-profit entity that has a track record of the management, mobilization, investment and implementation of public and private funds to finance nature conservation and mitigation and adaptation to climate change, complementing the efforts of the Government of Ecuador. FIAS currently has more than US\$100 million in five funds under management, with contributions from the Government of Ecuador and international donors, including CI. CI has a permanent voting seat on the "Technical Committee" responsible for the management of the Socio Manglar subaccount. Through its role in the Socio Manglar subaccount governance and ongoing financial and technical monitoring, CI will ensure compliance with GCF requirements both during the project implementation period and beyond. After the project implementation ends, but during the 20-year project lifespan, CI will provide an annual 'testament' to GCF confirming the continued existence of the Socio Manglar subaccount and that its purpose remains consistent with the objectives of supporting community management of mangrove areas. CI will also obtain from MAATE and FIAS written confirmation that the remaining GCF funds in the Socio Manglar subaccount after project lifespan will continue to be used for the objectives of supporting community management of mangrove areas. CI will monitor financial and technical reports of the Socio Manglar subaccount, including the interest-bearing account in which GCF funds will be held (financial reports) and broader Socio Manglar program (technical reports). Further details on CI's role in supervision, monitoring and compliance with GCF requirements after the project implementation period are included in the Term Sheet, Annex 14.
133. The Government of Ecuador (through MAATE) currently finances the Socio Manglar program directly from its annual budget and the REDD Early Movers project. The objective for the project is to build up the Socio Manglar subaccount to provide a more stable and long-term sustainable source of financing for the Socio Manglar program. Three main sources of funding for the Socio Manglar subaccount will be targeted: i) aquaculture enterprises looking to expand their environmental commitments⁴⁷; ii) Ecuadorian businesses seeking to compensate for their carbon footprint under Ecuador's Carbon Zero program (PECC); and iii) a US\$4 million contribution from GCF (see below). CI will work with ASC, the corporate partners of their value chain and their membership (aquaculture enterprises) to secure voluntary commitments to contribute to the Socio Manglar program. CI will also work closely with MAATE to promote support for mangrove conservation through the PECC program.

⁴⁶ FIAS is a private, non-profit entity to support finance for environmental and conservation purposes in line with MAATE priorities.

⁴⁷ Such commitments would be voluntary and not be a condition for support from the project.

134. To strengthen private-sector action, CI will develop a communications strategy, with the support of consultants, highlighting the importance of mangroves and the role the private sector can take in protecting and restoring them. The strategy will be developed for several audiences, including international companies in the ASC value chain, national shrimp farms and other Ecuadorian companies (i.e., others not linked to the aquaculture sector) looking to compensate their carbon footprint (through the PECC program). Meetings to present the Socio Manglar program (including the Socio Manglar subaccount) and the opportunities for private sector engagement with it will be organized in Years 2 and 3 in both Quito and Guayaquil.
135. GCF funds of US\$4 million are requested to capitalize the Socio Manglar subaccount managed by FIAS. The GCF funds will be granted to FIAS as an endowment in perpetuity for the subaccount, and will be held separately from other funds in an interest-bearing account. The interest income will be used to pay annual incentives to AUSCEMs in the Socio Manglar program both during the project and after the project implementation period ends.⁴⁸ The Socio Manglar subaccount is the principal mechanism for providing long-term sustainable finance to support AUSCEMs. The US\$4 million will capitalize the Socio Manglar subaccount at a level that, along with other contributions (see next section), will generate enough income to provide incentive payments for the additional AUSCEMs eligible for payments resulting from Component 1 of the Project. FIAS will manage the GCF resources in a separate interest-bearing bank account alongside other contributions to the Socio Manglar subaccount and in accordance with its existing management procedures, which have been assessed to be compliant with GCF requirements. (See Appendix 12 of **Annex 2** for details of the Socio Manglar subaccount and projections for growing its capital). CI-Ecuador, with support from CI's Conservation Finance Division, will monitor the use of GCF funds within the subaccount managed by FIAS.
136. In addition to capitalizing the Socio Manglar subaccount, the US\$4 million in GCF funds are expected to catalyze mobilization of private investment into the subaccount. Increasing the subaccount's capital early in the project (Year 2) is expected to provide a level of confidence from potential private sector donors because the subaccount will already be functioning with sufficient capital to provide incentives to some AUSCEM groups. Private-sector donors are more likely to contribute alongside public funding, such as from GCF, since it will demonstrate there are realistic long-term prospects for the subaccount to act as a mechanism to provide incentive payments and to attract sufficient additional capital to allow the Socio Manglar program to expand to conserve more mangroves. The US\$4 million is requested from GCF as a grant, rather than a reimbursable grant or loan, since i) the Socio Manglar subaccount is intended to operate in perpetuity, ii) interest and investment income earned on it will be used to provide incentive payments in the form of grants to AUSCEM groups and iii) there is no revenue to the Socio Manglar subaccount (other than the interest or investment income on the capital) with which to repay a reimbursable grant or loan.

Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.

137. Actions by local communities and the private sector to protect mangroves and maintain flood-protection benefits as a nature-based climate change adaptation and mitigation strategy depend significantly on the government creating the necessary governance, regulatory and enabling conditions. These both dissuade and control illegal activities and ensure that a full range of climate considerations are incorporated into planning and investment decisions.
138. By generating and disseminating better climate-change and mangrove-forest information to decision makers and stakeholders, project activities, particularly those focused on public entities, the project will contribute directly to increasing climate knowledge for use in decision making and will provide the foundation for achieving project Outcomes 1 and 2 and their associated impacts.
139. Improved legal and regulatory capacity, focused on planning, preparedness and enforcement, will contribute to sustaining mangroves as well as enhancing awareness and resilience of direct and indirect beneficiaries, primarily concentrated in the eight priority municipalities of the project area and will contribute directly to strengthened institutional and regulatory systems for climate responsive planning and development.

⁴⁸ This endowment grant mechanism is a Grant financing instrument within the scope of CI's AMA accreditation. The \$4 million endowment will be disbursed from CI to FIAS via a subgrant. FIAS will place the funds in an interest-bearing bank account, and any interest generated from these funds will be used by FIAS to pay for the community incentive payments under the Socio Manglar Program. Further details are included in Appendix 12 of Annex 2.

Output 3.1. Decision making for mangrove management by national government agencies and local governments is based on generation and provision of accurate and up-to-date data on mangrove condition and socio-economic information on mangrove dependent communities.

Activity 3.1.1. Monitoring of mangrove condition and socio-economic impacts in mangrove-dependent communities.

140. The reduction of emissions and the environmental and social benefits associated with the implementation of project activities will be monitored integrating remote sensing and community participation. This activity will provide the data needed for monitoring of the GHG emission reductions achieved by the other components.

Monitoring, Reporting and Verification (MRV) of Mangrove Cover and Carbon Stocks

141. CI will enter into subgrants with two local universities or NGOs (competitively chosen) to collaboratively generate data and capacity at multiple levels to improve MRV systems for mangroves, to be integrated both into project M&E and the national REDD+ MRV system. Ecuador's NDC is strongly linked with the terrestrial ecosystems. Mangroves along with other fragile ecosystems are included as a key priority in the NDC (2019). The information that this project will generate will be included in the national MRV system and the various initiatives that Ecuador is implementing to reach the targets included in the NDCs. The information generated in this activity is also fundamental to the implementation of the PECC with the private sector (linked to activity 2.2.1).

142. As described in **Annex 2**, natural changes in mangrove cover due to the complex hydrological dynamics of mangrove ecosystems are an important contributor to mangrove losses and gains. Improving the understanding of these natural changes is essential for future mangrove cover monitoring. MAATE, local universities and community stakeholders such as mangrove associations will be trained by CI to monitor the dynamics and structure of the mangroves and the biodiversity they contain. The areas that will be monitored will include: i) areas affected by natural disturbances; ii) restored areas of mangroves and other natural forests; and iii) areas of infrastructure development and deforestation not controlled by project activities. A subgrant will be awarded to a local university or NGO to collect the mangrove 'blue carbon' monitoring information. CI will also provide training on the collection of mangrove 'blue carbon' data to the sub-grantee and other stakeholders, including MAATE staff.

143. Updated geospatial data on mangrove coverage and change will be gathered and integrated into national forest inventory and FRELS. CI will enter into a subgrant with a local university or NGO to analyze mangrove cover and provide results to the national MRV system.

144. CI will disseminate information to decision makers at the local and national level, through policy reports for local governments, creating maps for local communities and extended reports to the MAATE and other national decision makers.

Socio economic impact evaluation (SIE)

145. The SIE's main objective is to assess the effects from project proposed activities on environmental and socioeconomic outcomes (i.e., climate mitigation and emissions reductions, resilience, enhanced livelihoods) and to detect an effect between treatment and control groups within the project implementation period for the project activities.

146. CI will conduct the SIE in Year 2 following project start-up for the baseline and in Year 6 for the final assessment. CI's Moore Center for Science will support the establishment of the monitoring system and the processing and analysis of the information. CI will also enter into grant agreements with local universities in Ecuador for the data gathering and training. Eligibility criteria for selection of the local universities is provided as Appendix 7 of **Annex 2**.

Output 3.2. Legal and regulatory frameworks at local and sectoral level are harmonized and include climate resilience and mitigation strategies and enforcement.

Activity 3.2.1. Support local governments (2 provincial governments, 2 municipalities and 5 parishes) to improve and/or implement PDOTs and other participatory planning instruments that incorporate climate change adaptation and mangrove management, applying a gender approach.

147. CI will work with local governments to enforce their capacity to manage their territories with a clear perspective of integrated coastal zone management and marine spatial planning that are critically important to realizing flood protection benefits from mangroves as well as other ecosystem services from coastal ecosystems. CI will work with nine local governments/administrations to integrate natural climate change adaptation measures into their planning, including mangrove conservation and restoration. For this, CI will work following PDOT's guidelines from the National Planning Secretariat that includes adaptation and mitigation actions⁴⁹. The local governments have been prioritized based on the extent of their mangrove cover. To support this work, CI will provide a subgrant in Year 2 to a local organization (university or NGO to be competitively selected) to generate climate risk information to inform local planning.
148. CI will work with five parish administrations (*Parroquias*) to develop climate change risk-management plans that value impacts and involvement of men and women. CI will provide technical support to these local governments for planning that will help them increase the economic, social, health and cultural resilience of individuals and communities. The risk-management plans will include activities for monitoring and evaluation of the adaptive capacity of men and women considering their access and control of resources, participation and roles within organizational processes.
149. CI will work with two municipal governments (Guayaquil and Machala), responsible for land-use planning and zoning through their PDOTs. CI will provide technical input and facilitation of participatory planning processes to integrate strategies for climate-change adaptation, including strategies for conservation and restoration of mangroves.
150. CI will support two provincial governments (Guayas and El Oro) with technical support to integrate climate change considerations into the provincial planning and to support budget planning to promote additional climate change investment. These activities will support management of priority conservation areas, local fishermen, strengthen capacities for the preparation of management plans for the coastal areas, and develop land use and management plans as well as integrated coastal management plans. The project will also support the integration into the provincial plans of actions to create the enabling conditions necessary for the long-term conservation and restoration of mangroves.
151. CI will enter into service agreements with consultants beginning in Year 2 to provide territorial planning support to local governments for these activities. The CI Local Governance & Integrated Coastal Management Specialist will provide support to these government entities across the four estuaries of the project.
152. CI will provide ongoing support, monitoring and public communication for the implementation of plans, corresponding public investments and municipal ordinances and other legal instruments determined by PDOTs.

Activity 3.2.2 Strengthen regulatory framework and law enforcement by agencies and institutions responsible for control of mangroves, with a focus on human rights.

153. CI staff and consultants engaged by CI will conduct an analysis of multi-sectoral legal framework for mangroves and climate change and the creation of multi-sectoral working groups ("*mesas técnicas*") to generate proposals for regulatory changes in Environment, Vice-ministry of Aquaculture and Fisheries (VMAP), Navy, Ports, and GADs. Examples of such changes may include updates to the regulations regarding illegal shrimp farms not covered by Executive Decree 1391, which have not been expropriated or approved, procedures for the restoration of mangroves, and regulation of areas that were formerly mangroves.
154. CI will also support the discussion of legal reforms and new regulations in the context of the Roundtable for Sustainable Shrimp, to be convened by the MAATE and VMAP. CI will provide staff time and cover workshop costs to implement this roundtable.
155. In Years 2 to 5, CI's Legal Specialist and Local Governance & Integrated Coastal Management Specialist will work with partners to re-establish the strategy of the UOCVs for law enforcement and sanctions for illegal activities affecting mangroves. A joint effort between the Oceanographic Institute of the Navy (INOCAR), MAATE, VMAP, NGOs and the NCA, the UOCVs' main goal is to avoid and control the clear-cutting of mangrove forest areas through the cooperation of public, private and NGO entities. The implementation of this initiative

⁴⁹ (<https://www.planificacion.gob.ec/guias-para-la-formulacion-actualizacion-de-los-pdot/>)

includes the surveillance technical system, the legal control technical system and the report and communication system.

156. Also in Years 2 to 5, CI's Legal Specialist and Component 3 Coordinator will support MAATE staff members with responsibilities for mangrove administration, planning, PAs, and provincial districts by providing them training on the laws and regulations relating to mangroves and the administrative procedures for enforcement and sanctions. A system to follow up deforestation complaints will be improved by CI project staff and implemented in collaboration with government officials so complaints from local associations, local governments and citizens will have a transparent follow-up process.
157. In Year 3, CI will engage a consultant to develop a training curriculum relating to mangroves specifically aimed at judges and other staff of the judiciary system, including from the Public Prosecutor, Navy, and Ministry of Fisheries and Aquaculture. Training workshops will be conducted by CI's Legal Specialist and the Local Governance & Integrated Coastal Management Specialist for judges and judiciary staff. This training will be integrated into the regular training curriculum of the named institutions.
158. In Years 1 to 3, CI will engage a consultant to design an online and mobile app to facilitate reporting of deforestation and other impacts on mangroves. The consultant will also provide training to users of the app (MAATE staff and AUSCEMs).

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

NDA and Government Partner

159. Ecuador's MAATE, as the GCF National Designated Authority (NDA), will ensure that activities implemented by the project align with strategic national objectives, priorities, and standards, including the National Climate Change Strategy, and help advance ambitious action on adaptation and mitigation in line with national goals and needs. The Executing Entity (EE) will engage with the NDA throughout project implementation. The NDA will contribute to the development of the multiyear workplan and will be provided with detailed reporting on the status of project activities and impacts. MAATE, through its Undersecretariat of Natural Heritage and Undersecretariat of Climate Change, will also contribute to project activities and provide grant and in-kind co-financing as described elsewhere in this proposal.

Accredited Entity

160. CI, through its CI-GCF Agency, will serve as the Accredited Entity (AE) for the project. The CI-GCF Agency will be responsible for the overall oversight of this project as defined in the Accredited Master Agreement (AMA) between the GCF and CI, including technical, financial, and administrative monitoring and supervision (through reporting, audits, and annual site visits) and review and approval of the EE's annual workplans and budgets. CI-GCF will also be responsible for providing support, guidance and backstopping to the EE; monitoring the achievement of project results and Outputs; reporting to the GCF; and project closure and evaluation. CI-GCF will conduct these responsibilities and disburse GCF funds to the EE. The CI-GCF Agency currently serves as AE for FP026, *Sustainable Landscapes in Madagascar*, a GCF project addressing mitigation, adaptation, and sustainable livelihoods, and for FP158, *Ecosystem-based Adaptation and Mitigation in Botswana's Communal Rangelands*.

Project Governance

161. The project's governance structure includes the Project Steering Committee and the Project Management Committee. The Project Steering Committee will be comprised of the following individuals: i) the Undersecretary of Climate Change of MAATE or his/her delegate; ii) the Undersecretary of Natural Heritage of MAATE or his/her delegate; and iii) the Vice President of CI-Ecuador or his/her delegate; and will be chaired by the Undersecretary of Climate Change.
162. The principal functions of the Project Steering Committee will be to provide strategic guidance and support adaptive management of project implementation, review progress and evaluation reports, discuss problems or strategic issues that might arise during implementation, and provide support for the necessary inter-institutional coordination and contributions to project activities. The Steering Committee will also participate in the selection of the Project Director, through a competitive process.

163. The Project Steering Committee will meet at least twice per year, to review the progress of the ongoing semester or year and to advise the Project Director on strategic and policy-related decisions. The Project Steering Committee will be convened by the Project Director in advance to give the members sufficient time to schedule the meeting and agree on the agenda. The Project Director will prepare minutes of each meeting. Extraordinary meetings of the Project Steering Committee will be convened when deemed necessary and at the request of one of its members. The Project Steering Committee may also invite key stakeholders to support specific themes.
164. The Project Management Advisory Committee will comprise the following individuals: i) the Director of the Marine, Coastal and Oceanic⁵⁰ (MAATE); ii) the Director of Climate Change Adaptation⁵¹ (MAATE); iii) a representative from VMAP; d) the Director of the Coastal and Marine Program of CI-Ecuador; and e) the Project Director employed by CI; and will be chaired by the Director of Marine, Coastal and Oceanic from MAATE.
165. The Project Management Advisory Committee will meet at least four times per year to advise the Project Director and the National Project Director (NPD) on technical matters, and to discuss challenges and collaboration opportunities during implementation. The Project Management Committee may invite key partners to provide advice on specific themes.
166. For both governance committees, members who are government employees will not be remunerated by GCF funds.
167. The NPD will be appointed by MAATE's Undersecretary of Natural Heritage, who will be employed and funded by MAATE. The NPD will advise the Project Director on government policies and priorities; review coherence of the project activities, including results, risks, planning and procurement processes; advise on the project's annual Procurement Plan for project services and goods; and review the technical and financial quarterly project reports to the AE.
168. Technical Advisory Committees will be created for each of the Components and each of the estuaries of the project. In the case of Component 1, the Technical Advisory Committee will include MAATE, the Risk Secretariat, Universities working in climate change adaptation and mitigation, and representatives from AUSCEM. The Component 2 committee will include MAATE, the VMAP, the NCA, the Fisheries and Aquaculture Public Research Institute, shrimp farmers, and other value chain members. The Component 3 committee will include MAATE, the VMAP, the Planning National Secretariat, Provincial, municipal, and Parish governments, and representatives of the INOCAR, the Prosecutor's Office, and Judiciary Council. At the level of each estuary, a committee will be created to integrate the Zonal Directors of MAATE with the planning activities of local governments and AUSCEMS representatives.

Executing Entity

169. CI will self-execute this project. CI, acting through its country office in Ecuador (referred to in this proposal as "CI" or "CI-Ecuador"),⁵² will be the EE for all activities of this project. CI-Ecuador will be responsible for project execution, management of sub-grantees and their activities, reporting to the AE, and ensuring optimal alignment with the Government of Ecuador policies and priorities in coordination with MAATE to achieve project outcomes and Fund-level goals. As EE, CI, acting through its country office in Ecuador, will enter into legal agreements (including sub-grant agreements, services agreements, and MOUs) for this project. As a branch of CI, CI-Ecuador will follow CI policies and procedures in the execution of this project, including terms included in CI's Operations Manual (see **Annex 21**). The CI-GCF Agency has assessed the capacity of CI-Ecuador and has determined it to be capable of applying CI and CI-GCF standards and policies in the execution of this Project. Throughout Project implementation, CI-Ecuador will be supported by various CI divisions which will lend specific expertise, including CI's Americas Division, Conservation Finance Division (supporting private-sector funding),

⁵⁰ This Direction manages marine, coastal and oceanic heritage.

⁵¹ This Direction seeks to increase adaptive capacity and reduce environmental, social and economic vulnerability to climate change, through the management and implementation of policies, plans, programs, projects, measures and actions to achieve resilient communities and ecosystems to climate change.

⁵² All references to "CI Ecuador" denote the actions undertaken by the project's CI staff employed through CI Ecuador, a branch office of Conservation International Foundation registered in Ecuador as an external company. The legal entity for this reference is Conservation International Foundation. CI Foundation will be the legal entity entering into any agreements, sub-agreements, or MOUs used in this project.

and CI's Center for Oceans (supporting blue carbon and mangrove restoration), and Project Delivery and Monitoring (supporting safeguards and gender).

170. For more than 30 years, CI has been protecting nature for the benefit of all. CI employs more than 1,000 people and works with more than 2,000 partners in 30+ countries. Since 1987, CI has supported more than 1,200 PAs and undertaken interventions across 77 countries, protecting more than 601 million hectares of land, marine and coastal areas. For over two decades, CI has been operating in Ecuador in the Galápagos, marine and coastal areas of continental Ecuador, Andes Choco and the Amazon to implement conservation solutions within priority landscapes. CI has supported the creation and management of marine and coastal PAs, working in close collaboration with artisanal fisher associations to promote capacity building and sustainable management measures of resources (in Galápagos with the spiny lobster, tuna, and prawn fisheries and in mainland Ecuador, with black shell and red crab fisheries).
171. CI-Ecuador will establish the main Project Management Unit (PMU) at CI's office in Guayaquil. The PMU will be headed by a full-time Project Director employed by CI, who will be responsible for coordination with all stakeholders and successful implementation of the project and attainment of results specified in the project's Funding Proposal, to the required standards of quality and within the specified constraints of time and cost. The PMU will be responsible for overall project management and planning, providing support to the execution of day-to-day activities, coordinating with the national government and project partners, coordinating with the AE, managing and overseeing grants, and coordinating project execution across two project offices and four estuary sites. The PMU will also include the Operations and Finance Director and the Project Finance Manager and will receive support from the CI-Ecuador Senior Operations Director. Upon project inception, the Project Director in coordination with the Operations and Finance Director will prepare a Project Operations Manual, including responsibilities, procedures and details for a smooth and effective implementation, which will be approved by the Project Steering Committee. The project will also have dedicated full-time staff, including Monitoring & Evaluation Manager, Safeguards Manager, Gender Manager, Communications & Knowledge Manager, Procurement Manager and 3 Component Leads. Additional Project staff as detailed in **Annex 4** will be based in Guayaquil and Esmeraldas.
172. In addition, CI, in coordination with MAATE, will establish an Estuary Field Unit (EFU) in each of the Project's four target estuaries (Cayapas-Mataje, Muisne-Cojimíes, Gulf of Guayaquil, and Jambelí) as regional execution offices. These EFUs will be housed in the MAATE Provincial Directions in Esmeraldas, Guayaquil, and Machala (as in-kind support from MAATE), and will each be led by two project staff: an Estuary Coordinator and a Social Technician (CI staff for the southern estuaries and PUSECE staff for the northern estuaries). Establishing EFUs in the estuaries directly involved in the day-to-day activities of the local associations is a cost-effective strategy for achieving the goals of the project. Evaluations on mangrove areas have shown that frequent technical support leads to sustainable change in behavior and practices in local communities to change behavior and practices. The Estuary Coordinator will ensure effective liaison and coordination with local stakeholders and local government, Component Leads, the PMU, and the other EFUs in implementation of the project activities, and the supporting staff will support implementation of Conservation and Stewardship Agreements with fishing associations and in-kind grants to local NGOs and universities.

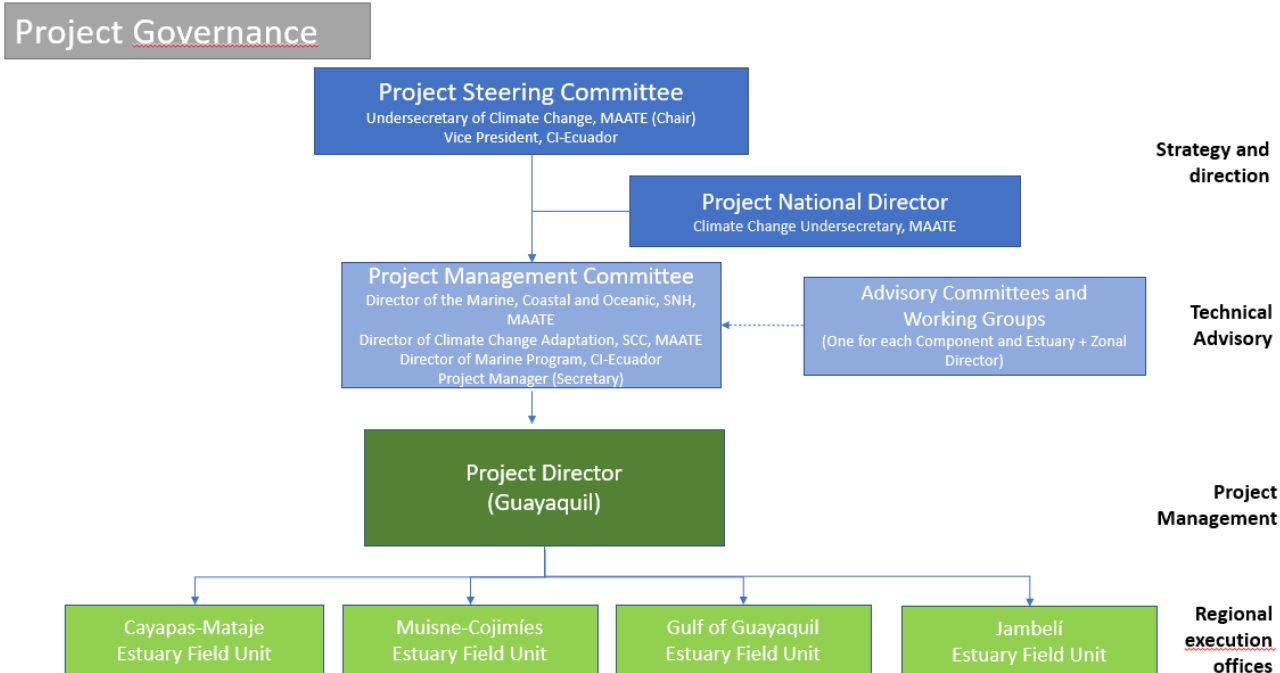


Figure 10. Project Governance

Subgrantee Organizations

173. CI will provide GCF funding to several sub-grantee organizations to implement certain project activities or sub-activities (see **Annex 23**). CI will enter into a grant agreement compliant with GCF requirements with each of the sub-grantees. CI Ecuador will manage and monitor these subgrants, will approve grantees' annual workplans and budgets, and will have ultimate approval authority over their activities. The resources will be granted based on selection criteria included Appendix 5 in **Annex 2**.
174. PUCSE will execute activities under Component 1 in the two northern estuaries in the project: Cayapas Mataje and Muisne Cojimies. PUCSE has prior experience working with local communities and local associations that depend on mangrove resources as well as developing research on mangroves and climate change. PUCSE is based close to the project area and has opened a new campus near the target Cayapas Mataje estuary. CI will enter into a cash grant agreement with PUCSE for the implementation of its designated activities in Component 1. CI Ecuador will apply risk mitigation measures and manage and monitor in-kind sub-grants to ensure that goods are used in a manner consistent with CI and GCF policies, including safeguards and prohibited practices.
175. FIAS manages the Socio Manglar subaccount of the existing Socio Bosque Fund. Under Output 2.2., CI will provide a \$4 million grant to FIAS that FIAS will hold in an interest-bearing account and use the interest generated to pay for community incentive payments under the Socio Manglar program. FIAS is a private, non-profit entity that has a track record of the management, mobilization, investment and implementation of public and private funds to finance conservation of natural resources and biodiversity, mitigation and adaptation to climate change and environmental quality, complementing the efforts of the Government of Ecuador. FIAS currently has more than US\$100 million in five funds under management, with contributions from the Government of Ecuador, Germany, Italy and other international sources, including GCF. CI has contributed to three FIAS funds to date, including a US\$4 million in grant funds for the Ecuador Azul Fund managed by FIAS.
176. CI will also enter into sub-grant agreements with community AUSCEM associations as described under Component 1 (selection criteria in **Annex 2**, Appendix 5). CI will provide goods via in-kind grants to the associations that follow the terms of related conservation agreements between CI and the associations for conservation, management and restoration activities as described under Activity 1.1.1. CI will also enter into small cash grant agreements with the community associations for the development of enterprises and livelihood activities as described under Activity 1.2.2.

177. CI will enter into sub-grant agreements with local universities or NGOs in Ecuador to support data gathering related to monitoring on blue carbon, mangrove forest cover and socio-economic impacts of the project. Selection criteria for these sub-awardees are included in Appendices 7, 10 and 11 of **Annex 2**.

Other Implementation Partners

178. As described in section B.3, important partners will contribute to the implementation of the project but will not receive GCF resources. CI will coordinate implementation of the project with these Implementation Partners (detailed in **Annex 23**) and will enter into agreements with them as appropriate to ensure clarity of roles and responsibilities of each party.
179. CI will work with key government partners, in addition to the NDA. The VMAP will support the CSS initiative in Output 2.1 and will support the harmonization of the legal framework under Output 3.2. Provincial, municipal and Parish governments/administrations are key partners who will receive technical assistance to integrate climate-change adaptation and mangrove management measures into land-use planning as part of Activity 3.2.1. The INOCAR, Prosecutor's Office, and Judiciary Council will participate in training and capacity building under Activity 3.2.2.
180. Partners from the private sector will contribute to and support activities in Component 2, including the NCA, ASC, financial institutions, and the eco.business Fund.

Financial and Contractual Arrangements

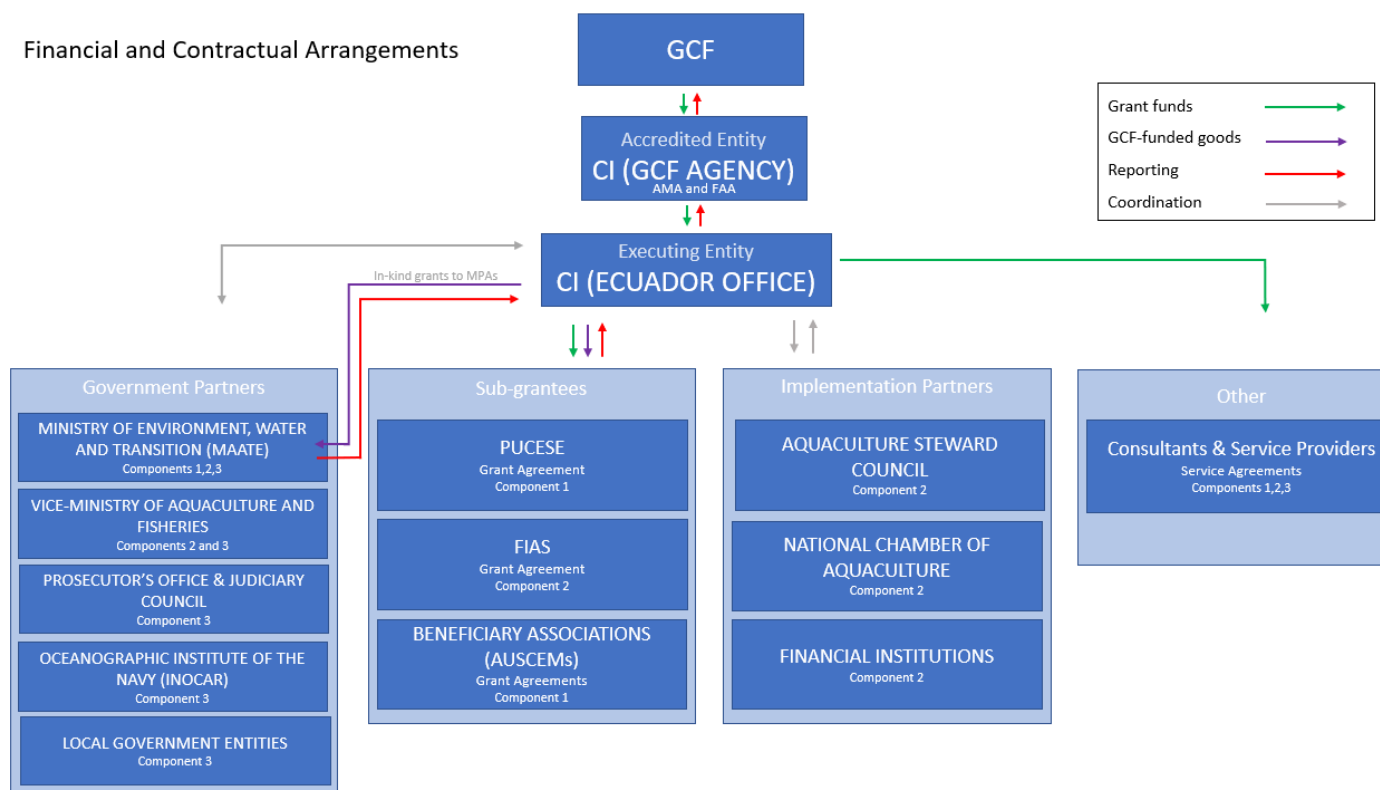


Figure 11: Project implementation arrangements and financial flows by Component

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

181. Public and private funding for the activities proposed in this project are limited and insufficient to tackle the ongoing challenge of mangrove deforestation and restoration at the scale required. GCF funding is necessary to implement the project and catalyze change. Project beneficiaries are among the most vulnerable populations in Ecuador, with direct exposure to heightened flood risks and livelihood disruptions associated with climate change compounded by low adaptive capacity due to poverty.
182. Grant funding is warranted to create the enabling conditions for sustained capacity and investment. The important ecosystem services that mangroves provide are not currently adequately valued by public and private sector actors and receive limited investment. In particular, mangroves have proven effectiveness in reducing

escalating coastal flood risks due to climate change and they are among the world's most valuable ecosystems for mitigating GHG emissions, especially through the accumulation of carbon in soil as organic matter.

183. Gaps in the generation and dissemination of data regarding climate change scenarios and mangrove ecosystem services for adaptation in coastal Ecuador have resulted in public entities, particularly subnational governments (provincial, municipal and parishes), not adequately incorporating climate change risks and the protection/restoration of mangroves into investment decisions. Similarly, while the carbon sequestration benefits of mangroves, particularly in soils, are well documented globally, better Ecuador-specific data and monitoring protocols to integrate these climate benefits fully into national public investment, mitigation strategies and GHG accounting are needed to better mobilize both national and international finance. These gaps have historically led to under-investment in mangrove conservation and restoration in Ecuador, despite the strong public benefits of mangroves.
184. Private-sector investment in mangroves has also been constrained. Private sector investment would in principle be motivated by: i) the potential for private returns; ii) regulatory requirements; and iii) market drivers (e.g. demand for sustainable shrimp). To date, these factors have had limited effects in driving private investment, and the project seeks to address these constraints.
185. Private sector returns from mangrove conservation and restoration have been undervalued, leading in the aquaculture sector to extensive production models that have driven approximately half of mangrove deforestation in Ecuador historically. There is, however, strong potential for key actors to internalize benefits through reduced costs and/or increased revenues. Shrimp aquaculture, using CSS models (see section B.3 and **Annex 2**), can increase yields, profitability and resilience through sustainable intensification, including increases in on-farm mangrove areas, requiring a shift away from Ecuador's traditional, extensive methods of shrimp aquaculture. Shifting a critical mass of early movers to CSS will leverage additional private finance and drive a broader sector-wide change in practice.
186. For community associations linked to protection of mangroves, reinforcing and expanding enterprises and revenue streams such as crab and shellfish fisheries or ecotourism also provide a way to internalize the returns of their mangrove conservation commitments under stewardship agreements (AUSCEM, see section B.3 and **Annex 2**). Currently, these livelihoods and income generating activities are not optimized, limiting the scale and effectiveness of these conservation strategies.
187. Regulatory requirements have contributed to reversing Ecuador's mangrove deforestation trends, but gross deforestation continues due to insufficient investment in enforcement and oversight, while market signals, especially from international demand for more sustainable products in shrimp markets, are rapidly strengthening.
188. Motivating private sector investment, which has to date been marginal for mangrove conservation/restoration requires better alignment of the potential for private benefits and regulatory frameworks and incentives that recognize the public benefits and externalities of mangroves. By generating and providing data, facilitating technical support and building the capacity of both the public and private sectors, the project will support long-term integration into policy and institutional frameworks governing mangroves as well as strengthening the financial mechanisms for stable investment flows.
189. The project objectives align well with GCF's function as demonstrated by its contribution to three of GCF's priority results areas (forestry and land-use, most vulnerable people and communities and ecosystems and ecosystem services) and the significant GHG emissions reductions and adaptation benefits it will provide. Given the existing challenges for Ecuador's government budget, which have been exacerbated by the economic impacts of Covid, significant national public funding for the project is not realistic. Nevertheless, the project will benefit from some government co-financing for support to community mangrove conservation. The project provides an important opportunity to catalyze current interest in improving sustainability within the shrimp farming industry such that it adopts a pathway towards zero mangrove deforestation and mangrove restoration, which will provide important climate change mitigation and coastal flood protection benefits. Further, there is a substantial unrealized but demonstrated opportunity to protect mangroves through community conservation, which would strengthen protection of existing areas and provide opportunities for restoration while also creating new economic opportunities for some of Ecuador's poorest and most vulnerable coastal communities. However, to capitalize on these opportunities, grant funding is critical to overcome the barriers to mangrove conservation and restoration. CI has raised funding for successful pilot projects on community conservation and promoting

climate-smart shrimp, but these are insufficient to tackle the issues for mangroves at the scale required. GCF funding for this project would not be duplicating work supported by other donors.

B.6. Exit strategy (max. 500 words, approximately 1 page)

[See **Annex 2, Feasibility Study** for more details.]

190. Project investments will drive a paradigm shift in the institutional, organizational and financial frameworks governing mangroves as a climate-change adaptation and mitigation tool, by the private and public sector. The project includes various activities that build sustainability, outlined below by component.

Project Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and PAs implementing climate adaptation plans.

191. Project investments in strengthening the management of AUSCEMs and PAs, expanding their coverage, and explicitly incorporating climate risks and benefits into management will consolidate a foundation that will be institutionalized and extend beyond the project lifespan.
192. AUSCEMs and Socio Manglar agreements create legally binding terms and agreements for community-based mangrove conservation and incentives for up to two decades. Project investments will support training, planning and administrative support to overcome initial establishment costs for agreements and thus scale up long-lasting coverage of these programs.
193. AUSCEMs establish (renewable) ten-year commitments from communities (with contracts established voluntarily and with mutual consent, between the government and community beneficiaries) to conserve and sustainably manage mangroves, recognized and enforced by the environmental authority. Areas restored to mangroves under this component will be within AUSCEMs or within formal PAs to ensure continued protection of these areas beyond project implementation. The project design assumes that the Government of Ecuador will continue to use AUSCEMs as a mechanism for engaging communities in mangrove conservation and sustainable management.
194. The project will expand the coverage of Socio Manglar agreements to cover an additional 20,000 ha. These Socio Manglar agreements embody a 10-year renewable commitment on the part of the Government of Ecuador to sustain these areas and benefits, while including additional legal provisions in their contracts beyond the terms of AUSCEMs, including financial planning, accounting, tax obligations and reporting to the environmental authority and to the members. As noted in the project Theory of Change, an important assumption of the project is that the government continues to be committed to the use of AUSCEMs and Socio Manglar agreements.
195. The project also includes incorporating climate adaptation into the strategies of government-run mangrove-dominated Marine Protected Areas (MPAs). By integrating these strategies into the management plans of these MPAs, any long-term costs associated with them will be integrated into the regular MPA budgets. Ecuador's MPAs already have consistent funding annually from the government but there are also additional ongoing actions to increase the sustainability of funding for the MPAs through two dedicated trust funds. This work on MPA sustainable funding is not directly part of this GCF project but will contribute to the long-term sustainability of the GCF investments.
196. Legal commitments and financial commitments will contribute to reinforcing a solid foundation of grassroots capacity built by the project. Social capital and local livelihoods require legal and financial instruments, and vice versa. Mangrove-based associations already have a strong predisposition and interest in mangrove conservation, driving the signing of AUSCEM agreements covering 60,000 ha to date. By investing in organizational and planning capacity across 60 associations with AUSCEMs, the project will strengthen human capacity of these local organizations and PA staff to value benefits of mangroves, including climate resilience, based on a participatory, gender-sensitive approach. Further reinforcing local commitments to mangrove management, local enterprises will perceive increased economic benefits through the implementation of business plans and strategies that are economically viable and have continuity after the project ends.

Project Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.

197. Long-term sustainability is the underlying rationale for this project outcome. By recognizing and internalizing the value of mangrove conservation and restoration in private-sector practices and investment decisions, the project will lay the foundation for ongoing action.
198. In the shrimp aquaculture sector, shifting a significant proportion of production area to climate-smart practices will lead to on-farm improvements in mangrove conservation and other climate-related practices over 20,000 ha of aquaculture. By linking these actions by early adopters to public-private dialogue and market engagement, the project aims to shift policies, commercial strategies and common practice broadly across the sector, positioning Ecuador's shrimp sector internationally and driving demand and therefore further adoption of ASC/SSP certification and CSS practices. Once established as common, profitable practices and demand grows, any ongoing investment to maintain ASC/SSP certification status and CSS practices is expected to come from the companies themselves, either through self-financing made possible by enhanced profitability or from private financial institutions (see activity 2.1.2). In addition, in parallel with this project, CI is working with partners to establish a Climate Smart Shrimp Fund⁵³ that will create a track record of successful CSS loans (as defined by loan repayment at commercial rates, farm profitability, and mangrove restoration) to demonstrate attractiveness to local financial institutions.
199. The CSS Fund is being developed as a blended finance vehicle, with approximately 30% concessional finance and 70% commercial finance. The fund's long-term goal is to make the case to local financial institutions that they can and should make CSS loans. Although the CSS Fund is not being developed as part of this GCF-funded project, it will contribute to the sustainability of the work being supported by GCF. Should the CSS Fund be fully capitalized during project implementation, this funding will be considered parallel financing.
200. Certification schemes are designed to reward responsible farming practices. Globally, buyers in major markets are making strong commitments to purchase farmed seafood products from certified sources and ASC is a recognized leader in seafood certification. This growing demand provides the incentive needed for shrimp farms to invest in more sustainable practices and to undergo the assessments necessary to obtain certification. The project will remove the knowledge barriers that are currently blocking more widespread participation in certification and by the end of the project there will be more examples of small- and medium-sized farms that have obtained private financing that will serve as examples for the industry and for the financial institutions that invest. The market-based incentives created by certification, and the influence of best practice among early adopters, combined with improved regulation (through the activities of Component 3) should mean that more attention is paid to environmental concerns including ensuring no further clearing of mangrove beyond the project. Price premiums for certified shrimp can be elusive. Instead, market access via preferential sourcing is a common incentive for farms to attain certification⁵⁴. The project will support farmers to access these markets.
201. The project will expand a dedicated long-term finance mechanism to assure continuity of resources for the Socio Manglar incentive payments to communities managing mangroves, and leverage commitments from the Government of Ecuador, which will provide US\$449,000 per year for these incentives as cofinance. The project will contribute to capitalizing the Socio Manglar subaccount of the existing and highly successful Socio Bosque Fund. Four million dollars (US\$4 million) of GCF funding is requested to increase the capital of the Socio Manglar subaccount, which will leverage additional private investment including from companies involved in shrimp aquaculture and Ecuador's PECC. The GCF contribution of \$4 million will be held in an interest-bearing bank account while funds from other contributors will be held in an investment account and invested in accordance with the investment policy of FIAS. Projections for the growth of the capitalization and use of the subaccount are provided in Appendix 12 of **Annex 2**. By clearly demonstrating and documenting emissions mitigation benefits, the project will also contribute to creating the necessary conditions for results-based payments (RBPs) for mangrove conservation in the long-term.

⁵³ <https://www.climatefinancelab.org/ideas/climate-smart-shrimp-fund/>

⁵⁴ For more information on markets of certified shrimp, see FS section 4.6.1.3 and [From Company Commitments to Collective Global Action: What's Next for our Sustainable Seafood Strategy • The David and Lucile Packard Foundation](#)

Project Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.

202. Sustainability and long-term permanence of improvements to policies and public institutions will be ensured by a combination of legal/regulatory instruments and institutional capacity.
203. A key set of outputs of this Component relate to the inclusive development of policy, planning and legal instruments that will govern management, enforcement and government budget allocations. These instruments will extend beyond the end of GCF funding and will include instruments at various levels:
- GADs will have codified mangrove conservation and climate considerations in local development plans, regulations and policy frameworks (e.g. municipal ordinances, budgets, PDOTs, COA, integrated coastal development plans), ensuring the basis for long-term application of these regulatory instruments guiding development and public investment.
 - Harmonized public policy instruments for the management of the mangrove ecosystem (e.g., procedures for assuring transparency in regulation, decision making and enforcement action, inter-agency cooperation for establishing technical standards).
 - Improvements in procedures at MAATE for reception and processing of complaints, enforcement and sanctions, reflected in regulatory reforms formally adopted by the Ministry and/or other public entities.
204. In addition, capacity building for national and regional institutions will help ensure that institutions can continue to monitor the effectiveness and sustainability of their actions after project completion and improve their effectiveness. Key partners in project implementation have been, and will be, involved in all aspects of project design and execution to ensure that they can sustain these activities after the GCF funding concludes. For example, forest monitoring data, specific to mangroves, including robust accounting of soil organic carbon, will be generated by the project with the aim of including a robust set of data and methodologies for inclusion by Ecuador's MAATE in the national Forest Reference Emissions Level (FREL). By incorporating this very significant carbon pool and sequestration into national monitoring, Ecuador will be able to provide a more comprehensive accounting of emissions and reductions as well as potentially mobilizing additional carbon finance. Also, institutionalization of training programs, curricula and materials developed by the project will provide the basis for ongoing replication by MAATE and the Judicial Branch after the project concludes.

FINANCING INFORMATION

C.1. Total financing

(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	36.4			million USD (\$)		
GCF financial instrument	Amount	Tenor		Grace period	Pricing	
(i) Senior loans	36.4					
(ii) Subordinated loans						
(iii) Equity						
(iv) Guarantees						
(v) Reimbursable grants						
(vi) Grants						
(vii) Results-based payments						
(b) Co-financing information ⁵⁵	Total amount			Currency		
	9.5			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
MAATE	<u>Grant</u>	<u>2.7</u>	<u>million USD (\$)</u>			
MAATE	<u>In kind</u>	<u>1.4</u>	<u>million USD (\$)</u>			
CI	<u>In kind</u>	<u>5.5</u>	<u>million USD (\$)</u>			
Total financing (c) = (a)+(b)	Amount			Currency		
	<u>45.9</u>			<u>million USD (\$)</u>		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	In addition to the co-financing committed for the project, other contributions can be considered parallel finance complementing the project. Prior to implementation of the Project, the ASC provided US\$50,000 and CI Ecuador provided US\$ 100,000 in 2021 to establish the Socio Manglar subaccount of the Socio Bosque Fund in preparation for the project. During project implementation, the ASC will request that its members contribute to the Socio Manglar subaccount (support for Activity 1.2.2). CI also expects that shrimp farms will seek an estimated US\$ 5 million in financing from various financial institutions for investments in CSS (see Activity 2.1.2). During implementation, CI will make efforts to track such parallel financing for final project reporting as detailed in the GCF Policy on Co-financing.					

C.2. Financing by component

Component	Output	Indicative cost ⁵⁶ million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions

⁵⁵ Please see Annex 4 for details on co-finance.

⁵⁶ Columns may not add up to the total amounts due to rounding. Please see Annex 4 Detailed Budget Plan for exact figures.

	Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and protected areas implementing climate adaptation plans.	Output 1.1: Reduced exposure to flood risk for vulnerable people and reduced GHG emissions from mangrove restoration are achieved by strengthening community-based management through AUSCEMs and protected areas.	22.4	16.3	Grants	2.7	Grants	Government of Ecuador (MAATE)
					0.9	In-Kind	Government of Ecuador (MAATE)	
					2.5	In-Kind	CI	
		Output 1.2: Improved livelihood activities and more economically productive community businesses enable local people to become more resilient to climate change and incentivized to participate in, and maintain, mangrove conservation and restoration.	3.3	2.9	Grants	0.4	In-Kind	CI
	Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.	Output 2.1: Shrimp aquaculture farms adopt practices and production standards that require elimination of deforestation and active reforestation in coastal and mangrove areas.	4.0	3.5	Grants	0.5	In-Kind	CI
		Output 2.2: Sustainable management of mangroves improved through agreements with the private sector, including direct financial support for mangrove conservation and restoration.	6.2	5.4	Grants	0.8	In-Kind	CI
	Component 3: Create the enabling conditions for sustaining	Output 3.1: Decision making for mangrove management by national	2.5	2.2	Grants	0.3	In-kind	CI

	reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.	government agencies and local governments is based on generation and provision of accurate and up-to-date data on mangrove condition and socio-economic information on mangrove dependent communities.						
		Output 3.2: Legal and regulatory frameworks at local and sectoral level are harmonized and include climate resilience and mitigation strategies and enforcement.	3.8	3.3	Grants	0.5	In-kind	CI
	<u>M&E</u>	M&E	1.2	1.1	Grants	0.2	In-kind	CI
	<u>PMC</u>	PMC	2.5	1.7	Grants	0.5 0.2	In-kind In-kind	Government of Ecuador (MAATE) CI
Indicative total cost (USD)			45.9	36.4	9.5			

GCF Funding is requested for all Project Activities. In addition, co-financing will be provided by the Government of Ecuador, through MAATE, and CI. MAATE will provide in-kind staff time and associated operational cost to support work under Output 1.1 (see paragraphs 113 and 184). Staff time will include MAATE specialists in climate change, sustainable production and development, forestry direction, and PAs as well as regional personnel. Operational costs associated with the staff costs (e.g., rent, communications, services, supplies and maintenance) will also be provided in-kind. MAATE will also provide in-kind staff time and associated operational costs to support its PMC-related activities. MAATE will also provide grant funding in the form of direct incentive payments to beneficiaries for the Socio Manglar program contributing to A1.1.1 impacts.

CI will provide in-kind financing of indirect costs that are not eligible for GCF financing to support all components of this project (see **Annex 14** for details).

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

Capacity building is a cross-cutting element of this project across outcomes and with multiple stakeholder groups.

Communities: AUSCEM communities will receive training, technical assistance and financial support to better conserve mangroves and their vital ecosystem services for both climate change mitigation and adaptation. This capacity building also includes the strengthening of mangrove community associations' micro- and small-sized enterprises, with an

emphasis on women, youth and other vulnerable groups (Activities 1.1.1 and Output 1.2; GCF funding requested US\$561,675)

Shrimp farmers: Increasing the technical capacity of small- and medium-sized shrimp farmers to drive adoption of climate-smart practices (Project Output 2.1; GCF funding requested US\$289,390) is key to the success of the project.

Government:

- MAATE's capacities for planning and implementation of activities that improve climate resilience and mangrove conservation in PAs will also be improved (Activity 1.1.2; GCF funding requested US\$82,266);
- National entities (MAATE, judiciary) will attend training and technical assistance for complaints, enforcement and sanctions for infractions affecting mangroves (Activity 3.2.2; GCF funding requested US\$230,774).
- Government entities will have the capacity to generate and incorporate improved mangrove forest cover and carbon stock data for planning, policies, investments and enforcement through training and basic equipment (Activity 3.1.1; GCF funding requested US\$12,058);
- Local and provincial governments (GADs) will have the capacity to develop, update and/or adopt zoning and development plans including criteria and indicators relating to mangrove protection, climate change mitigation or adaptation (Activity 3.2.1; GCF funding requested US\$39,884);

Local NGOs and universities: The Project will also support the capacity building of partners that need to enforce their operation practices through Enforcement Plans to subgrantees. There will be a technology transfer in restoration practices with green-gray practices in Ecuador through the trainings provided to shrimp farmers and promotion of the CSS concept (Project Output 2.1).

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

205. The project will provide both adaptation and mitigation benefits. The direct adaptation benefits come from reduced flooding risk of vulnerable people and project activities that are expected to increase people's resilience through livelihood benefits associated with better mangrove management. The mitigation benefits come from a combination of reduced emissions by reducing mangrove deforestation and from carbon sequestration in areas of restored mangroves.
206. Eight priority municipalities, with a population of 3.4 million, were selected based on mangrove cover and population at risk of flooding related to climate change (see B.1.1.4). Flood risk modelling including the effect of mangrove cover to reduce risk was used to estimate flood risk beneficiaries and expected reduced losses of economic assets (see **Annex 25**). Based on the modelling, the flood risk for 89,600 people can be mitigated by reducing mangrove loss and enhancing mangrove restoration efforts to increase the rate of mangrove recovery as compared to baseline rates. The population at risk is likely to increase as population grows, however we conservatively assume benefits for the 89,600 people as modelling population growth and migration introduces great uncertainty into the estimations (See **Annex 25** for details of Adaptation beneficiaries).

Adaptation Impacts

Impact Indicator	Value	Data and assumptions
Change in expected losses of economic assets due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention (US\$) – ARA1 Most vulnerable people and communities	US\$ 280 million per year in avoided losses of economic assets due to the flood protection benefits of mangroves	During project implementation period. Based on US\$1,600 in avoided property damage/ha/y of additional mangrove by comparison to the baseline.
Direct and Indirect beneficiaries reached (ARA 1 Most vulnerable people and communities)	89,600 direct beneficiaries of flood protection benefits.	People at high annual risk of flooding benefit from enhanced mangrove coverage and management. See Annex 25
	3.4 million indirect beneficiaries of improved planning processes in mangrove-rich municipalities	People living in municipalities with high mangrove coverage to benefit from improved land-use planning.
Beneficiaries (female/male) adopting improved and/or new climate resilient livelihood options – ARA 1 Most vulnerable people and communities	41,500 people (50% female, 50% male) with diversified climate-resilient livelihoods, and/or reduced flood risk	Includes total population expected to benefit from AUSCEMs, including support to livelihood activities in Component 1. These beneficiaries are also expected to benefit from flood protection benefits and are therefore not double counted in the estimation of total project direct beneficiaries.
Hectares of terrestrial non-forest, freshwater and coastal-marine areas brought under restoration and/or improved ecosystems (ha)	156,633 ha of mangrove protected and strengthened	Includes all of Ecuador's mangroves. Although project activities are directly focused on four estuaries (including 96% of Ecuador's mangroves), the enabling conditions activities

– ARA 4 Ecosystems and ecosystem services

(component 3) are expected to strengthen mangrove management nationally.

Includes areas in PAs, AUSCEMs (existing and additional), and shrimp farms as well as currently unprotected areas that will be protected through improved planning and enforcement at diverse levels of government.

Mitigation Impacts

- GHG emissions from gross deforestation under the baseline scenario are projected to be 2,428,986 tCO₂e over the project implementation period and 11,062,246 tCO₂e over 20 years.
- Reductions in GHG emissions from reducing deforestation due to project interventions relative to this baseline scenario are projected to be 567,404 tCO₂e over the project period and 3,566,685 tCO₂e over 20 years.
- In addition, removals of GHG emissions from project reforestation activities are projected to be 164,707 tCO₂e over the project implementation and 1,042,785 tCO₂e over 20 years.
- Total net GHG reductions, from avoided deforestation and mangrove restoration are projected to be 732,111 tCO₂e over the project implementation period and 4,609,470 tCO₂e over 20 years.
- A detailed description of methods and data is included in **Annex 22** along with an Excel workbook with the calculations of GHG emission reductions, including all assumptions and equations used, to enable a technical assessment to retrace the calculations.
- Estimates will be refined through on-the-ground measurements during project implementation.

Impact Indicator	Value	Data and assumptions
Tonnes of GHG emissions (tCO ₂ e) reduced, avoided or removed/sequestered – MRA4 Forests and land use	732,111 tCO ₂ e reduced during project implementation period 4,609,470 tCO ₂ e reduced over 20 years	See above and spreadsheet in Appendix 22. Includes sequestration from mangrove restoration.

207. In addition to the project's contributions towards advancing national climate change adaption and mitigation plans, the project will also contribute to the aims of the GMA⁵⁷, which brings together a global community of practitioners to halt mangrove loss, restore lost mangroves and secure mangrove protection. CI is a founding member along with other international organizations and the project will contribute in two ways to the GMA's work.

208. The first relates to ongoing knowledge sharing of GMA, especially related to mangrove restoration and "blue carbon" information. One of the functions of GMA is as a knowledge hub to share information between practitioners engaged in mangrove conservation and restoration. The mangrove restoration approach is informed by GMA toolkits and best practice guides. Data collected under activity 3.1.1 will also be shared with the GMA as a matter of course (through the Mangrove Restoration Tracker Tool and Global Mangrove Watch).

209. Second, the project itself makes an important contribution to the 'Mangrove Breakthrough', an initiative launched by the GMA and the UN Climate Change high level Champions during the UNFCCC COP27. This high-level initiative aims to secure the future of 15 million hectares of mangroves globally (through conservation, restoration and securing long-term financing of mangrove protection). This project alone represents about 1% of this goal.

⁵⁷ <https://www.mangrovealliance.org/>

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

210. This project will drive a paradigm shift in the way that Ecuador values and benefits from critical natural infrastructure that protects coastal people and assets from flooding. By fully valuing the adaptation and mitigation benefits of mangrove ecosystems and integrating these into governance and economic frameworks across Ecuador's coastal region, the project presents an innovative approach to the goals of reducing GHG emissions and improving the livelihoods of some of the country's most vulnerable people. The current paradigm undervalues both climate risks and the ecosystem services provided by mangroves, particularly with regards to climate change adaptation and mitigation. Under the current paradigm, exposure to climate risk and GHG emissions are both augmented by lack of sufficient government, private sector and civil society action. This situation is due to lack of adequate information, institutional coordination, empowerment of local stakeholders, and mobilization of private sector investment.
211. The combination of bolstering public sector frameworks and capacity, catalyzing public-private partnerships and private investment and strengthening local community participation and economic benefits, with a focus on the 'green infrastructure' that mangroves provide is an innovative and cost-effective approach with strong potential for scaling up and replication. Successful approaches will be sustained by local communities and government entities over time and the project will create lessons, guidance and tools for application in other countries with communities living in coastal areas threatened by climate change.
212. Across all activities, the project will engage in comprehensive monitoring and reporting with the aim of strengthening learning and knowledge sharing. The project aims to carry out two comprehensive evaluations: at the project mid-term and at the final evaluation of the project. Each exercise will include the identification of lessons learned and their dissemination.
213. Complementing the strong focus on local action, community participation, gender equity and private sector engagement, the project will consistently work to integrate these into a strong enabling environment and policy framework. This enabling environment includes the strengthening of regulatory and policy frameworks at the national and subnational level, as well as creating mechanisms and successful ongoing processes for dialogue and collaboration across sectors.
214. The project includes a series of capacity building packages for local communities to strengthen their ability to participate in innovative business models, as well as to engage more closely in the land use planning process with local government. Building consensus among key stakeholders, including government, community groups, and private sector actors, is a difficult task, largely neglected because of its high transaction costs. However, many experiences confirm that sharing a common vision is a very efficient and effective mechanism through which to achieve social objectives. The project approach is to develop a pragmatic mechanism through which consensus can be reached on the operational management of mangroves. The project has selected several planning tools to be formulated through a transparent and open participatory process, guided by the entity responsible for their promulgation and facilitated through the project. Technical assistance, training, logistical support, and coordination are the responsibilities of the project, along with the facilitation process to allow all stakeholders to express their views, expectations, and contributions, and to reach agreements for the benefit of all. It is expected that after several similar participatory processes, community leaders, public officials, and private sector leaders will learn that such an approach is valid and worthwhile, despite the additional initial transaction costs, and that reaching agreements is an effective means of managing relationships with other stakeholders.
215. The project will strengthen the participation of the private sector through its support to the definition of clear land use plans and the adoption of climate-smart practices, especially conservation and restoration of mangroves.
216. The project involves the scaling up and consolidation of activities that have successfully been used to conserve and restore mangroves in Ecuador. The area where the project is located lacks the most basic land-use planning instruments that could guide the sustainable use of natural resources. These instruments will be built jointly between the local communities and the local government as a first step to ensure the long-term sustainable development of the provinces involved. The project will facilitate participatory, open and transparent planning processes. In addition, the participatory process for land use planning will be closely documented and the lessons from this experience will be shared with the national government. Legal and regulatory adjustments, as needed for effective planning, will be recommended through technical documents.

D.3. Sustainable development (max. 500 words, approximately 1 page)

Environmental co-benefits

217. The project will reduce deforestation, increase restoration and enhance management across Ecuador's entire mangrove area of 156,633 ha.
218. In addition to the mitigation and flood-risk abatement benefits highlighted under GCF's reporting framework (D.1), mangroves provide significant additional ecosystem services including biodiversity conservation and habitats for fishery nursing grounds. Ecuadorian mangroves are home to 179 plant species (Cornejo, 2014) and constitute important nesting grounds for numerous bird species. Mangroves are also an important habitat for spawning, breeding and nurseries of commercial and non-commercial fish species. The structural complexity and ecological conditions of mangroves provide both protection from predators and abundant food resources due to the high primary productivity of this ecosystem (Huxham *et al.*, 2004, Mirera *et al.*, 2010, Shinnaka *et al.*, 2007, Wang *et al.*, 2009). Globally, secondary production of fish is 1-11.8 t/ha/y (Mirera, Kairo, and Waweru, 2010). Fisheries production dependent on mangroves alone has been estimated to be US\$100-21,000/ha/y (Spalding *et al.*, 2010). According to UNEP (2014), the full range of ecosystem services are worth an average US\$33,000 to US\$57,000/ha/yr to the national economies of developing countries with mangroves (UNEP, 2014). A more recent review of 250 studies around the world that have estimated the biodiversity and ecosystem service value of one hectare of mangrove reported an average value of US\$21,100 ha/yr in 2018 prices (Getzner *et al.*, 2020). Irrespective of valuation methods used, the importance of mangrove fisheries as a food source for local communities is widely recognized.

Social co-benefits including health impacts

219. In addition to the reduction of flood risks, the project will result in additional co-benefits, especially for mangrove-dependent communities and associations:
- Improved social cohesion and organizational capacity for mangrove management by 60 associations will lead to enhanced ability to engage with government stakeholders, resolve internal conflicts and develop additional community activities.
 - Improved mangrove management and plans are likely to integrate other social and environmental considerations with benefits for human health and wellbeing including water management and waste disposal.
 - Improvements in control of illegal activities in mangrove areas, in partnership between community-based associations and government authorities will also generate collateral benefits in enhancing local security.

Economic co-benefits

220. Mangrove conservation and restoration is a relatively low-cost solution for climate change adaptation that will generate economic benefits for the government, the private sector, and the communities through the avoided cost of the impacts of extreme climate events. The shrimp sector will be able to access new markets by producing shrimp more sustainably and being able to brand and market the product accordingly, resulting in increased export revenue and employment.
221. Communities will see lower flooding and relocation costs, reduced impacts on infrastructure, housing, crops and livelihoods. Livelihood diversification and adaptive fishing techniques will increase the resilience, economic stability, and food security of marginalized coastal fishing communities. Mangrove ecosystems are the foundation for diversified local economies, with 80% of the human population of the mangrove ecosystem dedicated to artisanal fishing, obtaining corvina (*Cynoscion albus*), mullet (*Mugil curema*), sea catfish (*Arius kesslei*) and grunts (*Anisotremus caesius*), as well as red crab (*Ucides occidentalis*) and black clam (*Anadara spp.*). Several of these species migrate to the sea and are an important contribution to the composition of rich marine fisheries, frequently exploited by the marine fishing industry.

Gender-sensitive development impact

222. The project will contribute directly to increasing the empowerment of women and other vulnerable groups in economic activities and advancing their participation and leadership in organizations and enterprises. See G.2 and **Annex 8**.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

223. Economic growth in Ecuador has directly reduced poverty and inequality levels and increased the size of the middle class. Between 2007 and 2018, poverty decreased from 36.7% to 23.2%, while extreme poverty decreased from 16.5% to 8.4%. Similarly, inequality as measured by the GINI Coefficient has fallen from 0.55 in 2007 to 0.47 in 2018.
224. However, the economy is still heavily dependent on the oil sector, a vulnerability demonstrated in recent years during periods of low oil prices and the appreciation of the US dollar. The reduction of this dependency has been part of the national strategies implemented by the central government.
225. With more than 3,630 km of coastline, approximately 54% of Ecuador's population of 17 million people lives in Ecuador's five coastal provinces. These areas are particularly vulnerable to SLR and the associated risks of saltwater intrusion and storm surges, flooding and storms.
226. These trends put coastal areas in Ecuador at particular risk. With 45% of the population below the extreme poverty line in the project area, the poverty rate in coastal areas is well above the national average. An increasing number of predominantly poor and vulnerable people in the coastal areas live in unsafe housing, in part due to rapid urbanization and persistent poverty.
227. Beyond the basic challenges of coastal development, the risks of climate change are placing additional stresses on the Government of Ecuador's efforts to achieve sustainable development. Despite the high level of exposure and vulnerability of the country's coasts, there is no integrated plan for the protection of mangroves and wetlands to address the combined future impact of sea level rise and intensified storms.
228. The project addresses the needs of the government of Ecuador, coastal communities, and important economic sectors such as the shrimp and fisheries sector to address the risks and impacts of climate change as described in the National Climate Change Strategy and the National Climate Change Plan.

D.5. Country ownership (max. 500 words, approximately 1 page)

229. The project is fully aligned with national plans to combat climate change and with a set of national policies and strategies. In addition to climate policies, the project is well aligned with a major new initiative announced by the Government of Ecuador in March 2023 to protect its entire coastline and marine resources within eight nautical miles of the coast. This initiative is in the early stages of design and consultation, but the objective can't be achieved without protection and sustainable management of the country's rich mangrove resources, which this GCF project will be helping to achieve.
230. The MAATE is the national environmental authority in charge of designing environmental policies and coordinating strategies, programs, and projects for the conservation of ecosystems and the sustainable use of natural resources. It proposes and defines the rules for environmental quality, with development based on the conservation and adequate use of biodiversity and natural resources. The GCF Focal Point (NDA) is part of the MAATE.
231. MAATE has been involved in the whole process of the development of the proposal, including development of the PPF application, public consultations and review of drafts of the funding proposal. Several different Under-secretariats of MAATE have been involved, including Climate Change, Natural Heritage and the former Coastal and Marine Resources (now included within Natural Heritage).
232. The protection of mangroves in the context of climate risk management in coastal areas is included as a priority in all current Ecuadorian climate policies, strategies and plans:
- National Climate Change Strategy 2012-2025: The proposed project directly addresses seven of the eight national climate change priorities.
 - NDC prioritizes addressing the multi-million-dollar cost of extreme climate events and recognizes ecosystem conservation measures as important adaptation and mitigation measures.
 - Mangroves are declared as forest and protective vegetation under Ecuadorian law and are an integral part of the country's REDD+ efforts.

- In April 2017, the Organic Environmental Code was published in which there are 10 articles (Art. 99, Art. 100, Art. 101, Art. 102, Art. 103, Art. 104, Art. 105, Art. 106, Art. 107, Art. 108) related to the conservation of mangroves, among which it is established that mangroves (along with several other categories of non-tree vegetation) are part of the National Forest Estate. Conservation of the mangrove ecosystem is described according to Art. 99 of Chapter IV, which prohibits its affectation, felling, and change of land use, in accordance with the law, and at the same time strengthens the participation of the communities in the care of mangrove ecosystems.
- The National Adaptation Plan approved in February 2023 includes mangrove conservation and restoration as an adaptation strategy.

233. Ecuador's NDC does not provide detailed targets but highlights the following relevant mitigation actions that the project will contribute to: conservation of natural heritage, strengthening of sustainable forest management, strengthening of the restoration of natural heritage, strengthening forest control, and strengthening the national system of PAs. The adaptation component of the NDC highlights the importance of mangroves and the ecosystem functions that they provide. It identifies increasing the area of forests and coastal ecosystems as a priority action to maintain their ecosystem function in climate change scenarios. Further, the implementation of sustainable natural resource use is also highlighted as a priority. Finally, the NDC notes as a priority the 'creation and strengthening of capacities on climate change and natural heritage management in social, academic, research and government sectors.'

234. The implementation plan for the first NDC (2020-2025) does not cover the proposed project period and an implementation plan beyond 2025 has not yet been developed. However, the current 2020-2025 implementation plan provides a sense of level of ambition and how the project will contribute to achieving NDC goals. The current plan includes a conditional goal of reducing GHG emissions from the Land Use, Land-use Change and Forestry (LULUCF) sector of 62,074,310 MtCO₂e. The estimated contribution of this project to that goal would be 732,111 MtCO₂e, or 1.2%, over the project implementation period, rising to 4,609,470 MtCO₂e, or 7.4%, over the project lifespan. Other goals included in the NDC implementation plan to which the project would contribute are: i) Update at least three tools for the management of PAs with climate change adaptation criteria (four updated management plans, including adaptation measures, are planned under the project); i) Increase the area under conservation by at least 127,500 ha through Socio Bosque incentives to contribute to the maintenance of ecosystem functionality in climate change scenarios (10,000 new hectares are planned under the project in addition to strengthening management at 60,000 hectares of existing AUSCEMs with incentive payments); and iii) Deliver at least 30 AUSCEM agreements in mangrove areas.

235. Additional frameworks, such as the updated National Climate Change Plan, are still under development. However, it is expected that this document will highlight the importance of coastal ecosystems for mitigation and adaptation.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

236. The financial structure of the project will ensure efficient use of GCF funds by directing that funding at the key barriers that are currently hindering progress to reduce mangrove deforestation and by encouraging mangrove restoration efforts. The majority of the project funding is directed towards community-based and public sector actions to strengthen mangrove conservation and restoration. The approximately 11% of GCF funding used for activities with the private sector (i.e., Output 2.1) is aimed at small- and medium-sized shrimp producers to provide training, promote the use of certification standards and to facilitate the access to credit that these farmers need to invest in the CSS model and/or achieve certification SSP/ASC standards. Investment on shrimp farms to implement more sustainable production methods will be financed by the private sector and not using GCF funds. The GCF funds directed towards supporting the private sector (~US\$3.1 million) are expected to result in approximately US\$5 million investment by the private sector in climate-smart shrimp production methods and certification. Further uptake of these models beyond the project implementation period is expected once these models are broadly recognized by the industry to increase profitability and resilience as noted in section B5.

237. The overall efficiency of the Project with regards to expected mitigation benefits is extremely high with a projected 4.6 MtCO₂e of GHG emissions reduced and sequestered over the project lifespan, equivalent to

a cost of US\$10 per tCO_{2e}, which is in the typical range for avoided deforestation type mitigation projects. In addition, the project will be bringing multiple adaptation benefits to a wide range of people, including direct expected livelihood benefits for 41,500 people and reduced risk of flooding for 89,600 people. We conservatively assume that the 41,500 people will also be beneficiaries of flood protection and therefore use the figure of 89,600 people as the total direct beneficiaries for the project (i.e. we do not sum these two figures together to get the overall number of beneficiaries), which equates to a cost of US\$512 per person over the six-year implementation period.

238. Grant funding is appropriate for the project because its primary goals are to secure public goods, in the form of the ecosystem services that mangroves provide to achieve mitigation and adaptation benefits. Most of the population that will benefit is defined as poor based on Ecuador's unmet needs index and the direct activities to improve livelihoods are directed at poor communities that can play an essential role in improving mangrove management. The part of the project targeted at the private sector is designed to enable increased private sector investment in climate-smart practices by addressing barriers related to lack of knowledge and targeted support for the development of investment plans needed for unlocking further private sector investment.
239. An Economic Analysis for the Project is provided as **Annex 3a (model in Excel) and 3b (narrative)** and demonstrates the important economic value of the project. A Financial Analysis has not been included because it is less meaningful for this project since its objective is to secure public goods and services rather than generate a financial return. Based on the economic analysis assumptions, which were chosen to be conservative, the economic Net Present Value (NPV) of the project over the 20 years and applying a 6% discount rate is US\$158 million more than the NPV for the 'without project' scenario. The estimated economic benefits of the project outweigh the costs by a factor of 4.7. Further, the 'project' scenario as presented in this Funding Proposal provided the most economic value of four scenarios that were considered (based on the highest NPV, benefits to cost ratio and economic internal rate of return). The three other alternative scenarios included in the economic cost benefit analysis were the 'without project' scenario, a scenario that only included the project's proposed mangrove conservation activities to reduce deforestation and a scenario that only included the project's proposed mangrove restoration targets and activities. The high economic value of the proposed project reflects the very high economic value of mangroves to the broader economy of Ecuador. Much of the value is driven by the importance of fisheries production, which underpins the livelihoods of the coastal fisher families that make up much of the vulnerable population of coastal Ecuador. The sensitivity analysis included in the model shows that the economic benefits of the project (and in fact all the scenarios considered) are robust to increases in costs and reductions in benefits of +/- 30%. The model (**Annex 3a**) can be adjusted to consider the impacts of larger changes.

LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Integrated Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Project/Programme Focus

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Ecuador's mangroves have been decimated over the last 40 years, mainly due to aquaculture expansion, and remain in a precarious state despite a recent stabilization in mangrove loss. Recent initiatives to halt mangrove loss, notably measures to better regulate shrimp farm expansion and community-based mangrove management pilots, have been locally effective and provide an opportunity that needs to be consolidated and expanded upon.	<u>Medium</u>	The paradigm shift achievable by the project is to align government, private sector and community initiatives to increase mangrove coverage nationally by conserving remaining mangroves and restoring additional areas to accelerate net gain of mangrove area. The target scenario involves well protected mangroves with minimal further loss under a combination of state-controlled PA and community-managed areas and an expansion of restored areas.	The project will deliver 4.6 MtCO _{2e} over 20 years, excluding replication effects. This would represent a significant paradigm shift in Ecuador's GHG emissions. Further, by strengthening the protection of existing mangroves and restoring new areas, the project will maintain coastal flood protection functions for an estimated 89,600 people. The mangroves will also maintain other essential ecosystem services, including fisheries production that supports livelihoods of vulnerable people.

Replicability	Few aquaculture companies have adopted sustainable production techniques with high environmental standards, so there are insufficient models and there are knowledge and financial barriers to replication. Community management of mangroves has been piloted successfully but could be replicated throughout Ecuador and in other countries.	<u>Low</u>	Once the climate-smart shrimp model is more widely demonstrated to provide added economic value while delivering environmental benefits, and knowledge and financial barriers are removed, it can be replicated within Ecuador, the region and internationally. The AUSCEM community management model is already demonstrated but once it is made more effective and linked to sustainable community businesses, it could be replicated throughout the country and could be adopted in other countries.	The project will address knowledge and financial barriers to promote adoption of the CSS model and more widespread adherence to standards such as the ASC's and those advocated by Ecuador's SSP. The project will provide the necessary knowledge, remove financial barriers and provide demonstrations of the model that will allow other companies to replicate it both nationally and internationally. Further strengthening of the AUSCEM community management model and further demonstration of its benefits to communities will lead to increased demand for new AUSCEMs.
Sustainability	Government commitment to improving mangrove conservation is evidenced by recent decrees to ensure that aquaculture operations are operating legally, as well as the operation and the funding of four mangrove PAs covering 64,916 ha. However, legal requirements for shrimp farms regarding mangrove restoration and conservation are low and significant areas of mangroves remain outside the PA network and community-managed areas.	<u>Low</u>	The paradigm shift would see improved decision making and enforcement of existing legislation to protect mangroves at all levels of government. Improved enforcement of legislation would be accompanied by the widespread adoption of profitable, climate-smart shrimp production models linked to highly respected certification standards that command a premium price. Financial institutions would encourage investment in shrimp farming operations that integrate climate-smart practices.	The project will strengthen the enabling conditions necessary to sustain reductions in the loss of mangroves by strengthening governance, coastal management policies and enforcement. The project will be promoting the use of improved climate-smart production methods for shrimp farming. Once the methodology of this profitable approach is better known, existing private financing will provide the means necessary for it to be replicated. The project will be expanding the Socio Manglar subaccount (part of the existing Socio Bosque Fund) and the government will be seeking additional funding for it from various sources to grow it over time

	Private sector sources of finance for aquaculture operations are available but don't currently incentivize climate-smart practices, including mangrove conservation and restoration. Financial resources for supporting community-based mangrove management are limited. There are more requests from communities to manage mangrove areas than current resources can support. There are no long-term dedicated funds to support community management other than the US\$150,000 Socio Manglar subaccount of the Socio Bosque Fund established in preparation for this project.		The Socio Manglar subaccount of the Socio Bosque Fund, which has been created to support community-based mangrove management, would be large enough, and generate sufficient resources, to provide long-term sustainable finance to all participating communities.	so that it is able to support the number and extent of AUSCEMs for which there is demand.
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E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)

Select appropriate IRMF core and supplementary indicators to monitor project/programme progress. More than one IRMF (core and or supplementary) indicators may be selected as applicable for each GCF results area and project/programme outcome (as defined in the table in section B.2(b)). If IRMF indicators are unable to measure any given project/programme outcomes, project/programme-specific indicators should be developed under section E.5 (project/programme specific indicators).

GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁵⁸	

⁵⁸ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Direct: Project reports on updated maps for flood protection provided by new mangrove cover. Indirect: Census data for the coastal provinces where the project works.	86,200 0	87,900 0	89,600 3.4 M	Direct: Calculated based on the number of people with reduced exposure to flooding due to mangroves conserved or restored by the project Indirect: based on the population of the municipalities where mangroves occur and the project works (census data) Target figures include beneficiaries under Supplementary 2.1. Hence, the targets under Core 2 indicator is not to be added with Supplementary 2.1.
	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	<i>Report of the social impact evaluation of the Project that includes a gender approach. (MRV)</i>	0 males 0 females	20,750 direct beneficiaries adopting or continuing to have a diversified climate resilient option (sustainable fisheries and	41,500 direct beneficiaries adopting or continuing to have a diversified climate resilient option (sustainable fisheries and	Assumption: there is a good response from communities, businesses and local governments to adopt and replicate adaptation strategies including new livelihood options. The adaptation benefits for direct beneficiaries are increased

				sustainable shrimp farming)	sustainable shrimp farming)	access and control over mangrove resources associated with AUSCEM agreements and more resilient livelihoods (supported through activities 1.2.1 and 1.2.2).
	<u>Supplementary 3.1: Change in expected losses of economic assets due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention</u>	Calculated based on economic value of avoided damages (in US\$) per ha of mangrove restored or conserved by project, over projected baseline scenario. Area verified through satellite monitoring of mangrove cover.	US\$ 250.6 M in avoided property loss from mangrove area increase due to project activities.	US\$ 256.7 M per year in avoided property loss	US\$ 281.0 M per year in avoided property loss	Project interventions are effective in increasing mangrove areas over baseline scenario. Valuation per hectare per year based on Menéndez model described in Annex 2, which gives a flood prevention value for mangroves of \$1,600 per year Baseline: 156,633 ha Hectares additional to baseline at: Midterm: 2575 ha Final: 5975 ha
	<u>Core 4: Hectares of natural resources brought under improved</u>	Official records (contracts, agreements, management plans,	0 ha of 156,633 ⁵⁹ ha of mangrove	60,000 ha	156,633 ha	Political will of MAATE, local governments and local

⁵⁹ Total area of mangroves within priority subnational government areas targeted by project. This area encompasses areas of AUSCEM and protected areas targeted for management improvements (120,000 ha, Project Result 1.1) as well as the portion of total shrimp farm area targeted by project (22,000 ha, Project Result 2.1) comprised of mangrove cover, as well as additional areas outside of these categories or formal protection with improved coverage and strengthened by project interventions at scale of national and subnational government policies and activities.

<u>ARA4 Ecosystems and ecosystem services</u>	<u>low-emission and/or climate-resilient management practice</u>	local development and zoning plans) for mangrove areas under improved management. With areas confirmed by assessment of mangrove forest cover by remote sensing.	ecosystems with sufficient protection or CC strategies			associations to adopt and implement strategies.
	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under restoration and/or improved ecosystems</u>	Project records on mangrove restoration activities. Remote sensing, project GIS and database.	0 ha	2200 ha	4850 ha	Assumes average survival rates of mangrove seedlings and planting of restoration keeps to the original schedule.
<u>MRA4 Forestry and land use</u>	<u>Core 1: GHG emissions reduced, avoided or removed/sequestered</u>	Project MRV linked to national forest monitoring system. Verified through satellite monitoring of mangrove cover and forest inventory plots	0 tCO ₂ e	195,186 tCO ₂ e <i>reduced</i>	732,111 tCO ₂ e <i>reduced</i> during project term 4.6 MtCO ₂ reduced over 20 years	Financial support from private and public sources is sufficient to incentivize conservation and restoration. Local, regional and national governments maintain and strengthen planning and enforcement to reduce deforestation

Select at least two relevant IRMF core (enabling environment) indicators to monitor and elaborate the baseline context and project/programme's targeted outcome against the respective indicators. Rate the current state (baseline) vis-à-vis the target scenario and select the geographical scope of the outcome to be assessed. Describe how the project/programme will contribute towards the target scenario. Refer to a case example in the accompanying guidance to complete this section.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
<u>Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner</u>	National government, 20 subnational governments and 60 artisanal fisheries and mangrove management groups with limited or no regulatory systems or incentives for climate resilience and their effective implementation	<u>low</u>	National government, 9 subnational governments and 60 artisanal fisheries and mangrove management groups include improved regulatory systems or incentives for climate resilience and their effective implementation	The project will create the enabling conditions for sustaining the reductions in mangrove deforestation by strengthening governance, climate change adaptation strategies, coastal management policies and legal enforcement (see description of Component 3 in B3 for details)	<u>Multiple sub-national areas within a country</u>
<u>Core indicator 7: Degree to which GCF Investments contribute to market development/transformation at the sectoral, local, or national level</u>	50 farms ⁶⁰ have ASC certification, 10 companies are members or associates of the SSP initiative ⁶¹ and 1 company is experimenting with CSS practices ⁶² .	<u>low</u>	Increased adoption of nationally and internationally recognized aquaculture standards and/or improved practices by shrimp farms (e.g., ASC/SSP or CSS practices) that include commitments to no mangrove deforestation to cover at least	The project will be working with the shrimp aquaculture sector to encourage the adoption of standards that include commitments to no mangrove deforestation and practices that include restoring mangroves (see Outcome 2 in section B3.	<u>National level (one country)</u>

⁶⁰ According to ASC online searchable database, <https://www.asc-aqua.org/find-a-farm/>; accessed 28 Feb 2023

⁶¹ According to the SSP website, <https://www.sustainableshrimppartnership.org/ssp-members/>; accessed 28 Feb 2023

⁶² In collaboration with CI Ecuador

			20,000ha of shrimp farms.		
<u>Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards</u>	Low adoption of improved standards and practices in shrimp farms but there is an emerging interest.	<u>low</u>	See indicator above for shrimp farms. Improved management in 60,000 ha of existing mangrove AUSCEMs and good management in 10,000 ha of new AUSCEMs to put in place. Integration of climate change adaptation measures into the management plans and practices of the 4 PAs targeted by the project (which cover the majority of protected mangroves in the country).	The project includes three main interventions that encourage adoption of better practices, methods and standards. First is the work with the shrimp aquaculture sector covered also in the indicator above. The second is the improved management of mangroves by community associations through the AUSCEM mechanism. The third is through training to integrate adaptation to climate change into the protected areas that include mangroves. See details of activities in the Funding Proposal and/or Feasibility Study.	<u>National level (one country)</u>

E.5. Project/programme specific indicators (project outcomes and outputs)						
<i>This section should list out project/programme-specific performance indicators (outcomes and outputs) that are not covered in sections above (E.1-E.4). List down tailored indicators to monitor /track progress against relevant project/programme results (outcomes/outputs). AEs have the freedom to decide against which outcomes they would like to set project/programme specific indicators. If any co-benefits are identified in sections B.2(a)(b), and D.3, AEs are encouraged to add and monitor co-benefit indicators under the “Project/programme co-benefit indicators” section in table below. Add rows as needed.</i> <i>Please number each outcome and output as shown below to indicate association of outputs to the contributing outcome. The numbering for outputs under this section should correspond to the output numbering in annex 4 (detailed budget plan).</i>						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Component 1. Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and PAs implementing climate adaptation plans.						
Output 1.1. Reduced exposure to flood risk for vulnerable people and reduced GHG emissions from mangrove restoration are achieved by strengthening community-based management through AUSCEMs and protected areas.	Number of hectares under community stewardship and protected areas with management capacity assessment scores increasing by 50% or more	Registry of active AUSCEMs Project monitoring system capacity assessment of communities and protected areas	0	60,000 ha	120,000 ha	Communities and protected areas management motivated to participate.
	tCO ₂ e emissions reduced through restoration in areas under community stewardship or PAs	Forest monitoring data with ground truthing	0	52,673 tCO ₂ e	252,495 tCO ₂ e	Communities motivated to participate in restoration.
	tCO ₂ e emissions reduced from avoided deforestation over project implementation period	Forest monitoring data with ground truthing	0	140,005 tCO ₂ e	467,075 tCO ₂ e	Market trends favoring sustainable investment and climate-smart shrimp, as well as political climate, reinforce conservation objectives.

	People with reduced exposure to climate change related flooding events due to the project	Forest monitoring data. Number of beneficiaries is derived from mangrove coverage using the model used to make the initial estimates (see Annex 25)	86,200	87,500 people (50%M/50%F)	89,600 people (50%M/50%F)	Land use planning, weather events and socio-economic factors do not lead to unpredicted, large-scale shifts in human settlement patterns in the coastal region. Updated census data may also affect the estimate. Improved management of mangroves ensures that baseline flood protection benefits are maintained.
Output 1.2. Improved livelihood activities and more economically productive community businesses enable local people to become more resilient to climate change and incentivized to participate in, and maintain, mangrove conservation and restoration.	Number of males and females in mangrove areas benefiting from the adoption of diversified, climate resilient livelihood options and business practices linked to mangrove promoted by project	Project monitoring system assessment of livelihoods	0 males and 0 females	20,000 people (50%M/50%F)	41,500 people (50%M/50%F)	Business and livelihood models will be successful at achieving livelihood impacts and conservation Project Outcomes
Component 2. The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.						
Output 2.1. Shrimp aquaculture farms adopt practices and production standards that require elimination of deforestation and active reforestation in	Hectares of shrimp farms adopting practices and production standards that eliminate deforestation (e.g. ASC or CSS)	Mapping of shrimp aquaculture areas that have adopted climate resilient approaches	0	8,000 ha	20,000 ha	Aquaculture producers motivated by market signals, technical assistance and economic viability of climate-smart practices

coastal and mangrove areas.	Hectares of mangrove restoration achieved on shrimp farms	Surveys and ground truthing	0	100 ha	250 ha	Aquaculture producers motivated by market signals, technical assistance and economic viability of climate-smart practices
	tCO ₂ e emissions reduced through restoration in areas by shrimp farms	Forest monitoring data with ground truthing	0	22,574 tCO ₂ e	112,870 tCO ₂ e	Technical support and market linkages of CSS sufficient to motivate shrimp farm investment
Output 2.2. Sustainable management of mangroves is improved through agreements with the private sector, including direct financial support for mangrove conservation and restoration.	Amount of funding provided by the private sector to support mangrove conservation.	Area details from signed agreements	0	USD 150,000/year	USD 300,000/year	Aquaculture producers and other private sector companies motivated by market signals, CSR and reputation. Includes contributions through PECC.
Component 3. Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.						
Output 3.1. Decision making for mangrove management by national government agencies and local governments is based on generation and provision of accurate and up-to-date data on mangrove condition and socio-economic information on mangrove dependent communities.	Number of annual reports on mangrove cover and socio-economic data shared with national government agencies and local government	Remote sensing/GIS for mangrove coverage Surveys for socioeconomic information	0	4	8	Government entities have technical capacity, staff and will to integrate data generated by project.

Output 3.2. Legal and regulatory frameworks at local and sectoral level are harmonized and include climate resilience and mitigation strategies and enforcement.	Number of local and provincial governments with zoning and development plans including criteria and indicators relating to mangrove protection, climate change mitigation or adaptation	Planning and zoning instruments, officially adopted	Only 2 local governments have planning instruments including climate-change measures	2	9	Willingness of political leadership and technical staff in governments to participate in project across election cycles
	Number of people living in jurisdictions of local and provincial governments with zoning and development plans including criteria and indicators relating to mangrove protection, climate change mitigation or adaptation	Census data Project assessment of whether jurisdictions have zoning/development plans including the criteria and indicators	0	1.0M people (50%M/50%F) (indirect beneficiaries)	3.4 M people (50%M/50%F) (indirect beneficiaries)	Willingness of political leadership and technical staff in governments to participate in project across election cycles
Project/programme co-benefit indicators						
Co-benefit 1. Biodiversity and ecosystem service benefits are increased.	National mangrove cover (hectares)	Remote sensing analysis (see activity 3.1.1); reports of mangrove restoration.	156,633 ha	159,208 ha	166,110 ha	Assumes the baseline cover of 156,633 ha (2018). Baseline to be revisited once mangrove cover analysis is redone for the project start date. Targets include habitat restoration areas which will not yet be detectable in remote sensing

						analysis by end of project.
Co-benefit 2. Economic resilience of coastal communities is increased.	See Supplementary indicator 2.1 in the Outcomes indicator section above					
Co-benefit 3. Uptake of sustainable shrimp production practices is increased.	See core indicator 7 in the enabling conditions section above.					
Co-benefit 4. Economic value of fisheries for artisanal fishers is increased.	Economic value of fisheries in additional mangroves protected or restored by the project	Value is derived from additional hectares of mangrove due to the project and the data on fisheries production collected during project design (see Annex 3 , Economic and Financial Analysis)	0	US\$ 5.1 million	US\$ 12.7 million	The model assumes that fisheries benefits are accrued once the restoration is done but there will likely be a delay as the mangrove grows so these are longer term benefits of the increased mangrove coverage

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 1.1.1 Strengthen and expand community-based mangrove conservation and management to reduce deforestation and increase mangrove restoration.	Support community stewards to better conserve mangroves and their vital ecosystem services for both climate change mitigation and adaptation.	1.1.1.1 Strengthening governance capacity and planning of existing AUCEMs 1.1.1.2 Expand areas under active AUSCEMs 1.1.1.3 Expand areas covered by SA incentives 1.1.1.4 Restoration of mangrove areas within AUSCEMs and PAs	Improved governance and management of 60,000 has of existing AUSCEMs Integration of 10,000 has under new AUSCEMs New Socio Manglar agreements covering 20,000 has supported through planning and preparation

			4,600 ha of additional mangrove restoration on AUSCEMs and PAs (note that an additional 250 ha of restoration will be on farms under Activity 2.1.1)
Activity 1.1.2 PAs have implemented mitigation and adaptation strategies in 64,913 ha of mangroves	Support management, enforcement and planning capacities for 4 MPAs (Cayapas Mataje, Churute, El Salado and Estuario del río Muisne)	1.1.2.1 Integrate climate-change scenarios into planning of protected areas and local management strategies.	Management plans and climate change adaptation plans designed and in implementation in 4 MPAs. Improvement in management effectiveness in 4 MPAs, as reflected in annual effectiveness assessments.
Activity 1.2.1 Technical and business development support to mangrove community associations, with an emphasis on women, youth and other vulnerable groups.	Provide technical and business development support to at least 60 community associations linked to protection of mangroves to design and implement business plans and strategies, including strategies for improving governance and administration, access to finance and to markets for more resilient livelihood strategies. Activities to strengthen and diversify local livelihoods to create economic alternatives aligned with mangrove protection and to make livelihoods more resilient to impacts of climate change.	1.2.1.1 Technical and business development assistance to mangrove community associations for development of enterprises and livelihood activities, with an emphasis on women, youth and other vulnerable groups.	Value chain analysis and strategies for 60 community associations designed, and 20 in implementation with support of financial mechanisms (link to 1.2.2) or independently.
Activity 1.2.2 Establish and consolidate financial mechanisms in support of mangrove community associations (micro- and small enterprises)	Structure grant mechanisms and facilitate access to investments for community associations supporting mangrove conservation having received technical and business development support (link to 1.2.1)	1.2.2.1 Create and implement grant mechanisms for technical support to micro- and small enterprises of mangrove community associations. 1.2.2.2 Support access to mechanisms and institutions providing credit and investment to micro- and small enterprises of mangrove community associations.	20 community association enterprises supported by grant mechanism with business plans financed and in implementation. Credit/investment from at least 2 financial institutions (public and private banks, cooperatives and other financial international funds) accessed
Activity 2.1.1 Technical assistance for development and promotion of climate-smart shrimp aquaculture practices in 20,000 hectares of farms	Provide technical assistance to early adopters and innovators, especially small and medium producers, for the	2.1.1.1 Sustainable Intensification Practices	Improvements in production systems leading to improved mangrove coverage and management to reduce emissions

	development and validation of the production model.	2.1.1.2 Mangrove Restoration – demonstration farms (250 ha of restoration) 2.1.1.3 Education, Outreach and Enabling Conditions Implementation of Sustainable Shrimp Aquaculture	and/or vulnerability to climate change in 20,000 ha of farms Mangrove restoration undertaken on at least 250 ha within Shrimp farms Public private roundtable for promotion of climate-smart shrimp practices and market engagement.
Activity 2.1.2 Facilitate partnerships and access to mechanisms for credit and investment in shrimp farms for expansion and consolidation of climate-smart aquaculture practices	Credit to finance working and investment capital needs will be provided/leveraged with CI's support.	2.1.2.1 Education as a tool to facilitate access to credit and other investment to shrimp farms for expansion and consolidation of CSS practices 2.1.2.2 Project feasibility as a tool to mobilize capital towards CSS production	At least US\$ 5M from financial institutions is accessed by farms in support of climate-smart shrimp practices contributing to improved mangrove coverage and management to reduce emissions and/or vulnerability to climate change
Activity 2.2.1 Establish agreements with businesses, including aquaculture companies, to contribute to mangrove restoration and financial sustainability of the national Socio Bosque Incentive Program (Mangrove Chapter).	Voluntary partnerships established with companies ⁶³ to finance restoration and conservation of mangroves through contributions to the Socio Manglar program, including providing capital to the Socio Manglar subaccount. Also, US\$ 4 million of GCF funds added to the capital of the Socio Manglar subaccount to be held in a dedicated interest-bearing bank account. See Appendix 12 of Annex 2 for details of Socio Manglar subaccount management by FIAS and Annex 14 for CI's supervision and monitoring of use of GCF funds.	2.2.1.1 Grow the capital of the Socio Manglar subaccount of the Socio Bosque Fund to support long-term community management of mangroves.	Agreements with the private sector result in at least an additional US\$ 300,000/year by the end of the project being available for incentive payments to AUSCEMs involved in the Socio Manglar program. US\$4 million of GCF funds added to Socio Manglar subaccount (held in an interest-bearing account) and the interest used in accordance with the conditions set out in the term sheet, Annex 14 to provide incentive payments to AUSCEMs during the project implementation period and in the long term after the project implementation ends.
Activity 3.1.1 Monitoring of mangrove condition and socio-economic impacts in mangrove-dependent communities.	Improved monitoring of mangrove cover, deforestation rates and GHG emissions related to mangroves.	3.1.1.1 Demonstrate the impact of mangrove conservation and restoration on national mangrove cover, stocks, and socio-economic indicators through monitoring linked to the national MRV, and build long-term monitoring capacity.	Inventory of blue carbon used by the project M&E and integrated into national REDD+ monitoring and reporting system, including national forest and emissions inventories and FREL.

⁶³ Incentives for companies to participate in mangrove conservation and restoration include achieving their own Corporate Social Responsibility aims, achieving carbon neutrality aims through the PECC program and in some cases, tax incentives. Contributions to mangrove conservation and restoration could be directly with AUSCEMs or through contributions to the Socio Manglar subaccount. Financial projections for the funds available for the Socio Manglar program incentive payments and growth of the Socio Manglar subaccount capital are provided in Appendix 12 of Annex 2.

	Monitoring of social and economic benefits and impacts of mangrove conservation providing information to government agencies to improve policies and programs.		Framework for socio-economic monitoring of livelihood activities adopted and implemented, including its use to evaluate the project impacts on livelihoods.
Activity 3.2.1 Support local governments (2 provincial governments, 2 municipalities and 5 parishes) to improve and/or implement Coastal Development and Zoning Plans (PDOTs) and other participatory planning instruments that incorporate climate change adaptation and mangrove management applying a gender approach.	Integration of better climate change data and mangrove conservation measures in current and proposed Development and Land Management Plans.	3.2.1.1 Provide technical assistance to subnational governments for improvement of PDOTs to integrate climate-change adaptation and mangrove management measures.	At least 9 subnational governments with mangrove areas have PDOTs and/or budgets that incorporate mangrove climate change strategies aligned with equity agendas.
Activity 3.2.2 Strengthen regulatory framework and law enforcement by agencies and institutions responsible for control of mangroves, with a focus on human rights	Coordinated planning and implementation of mangrove protection and coastal adaptation measures through a network of national and provincial institutions responsible for coastal management and planning This will improve cross-sectoral planning	3.2.2.1 Provide technical and legal support for harmonization and adoption of improved sectoral policies and regulations and technical assistance for implementation of CODA (Código Orgánico del Ambiente)	3 proposals for harmonized public policy instruments for the management of the mangrove ecosystem (e.g., procedures for assuring transparency in regulation, decision making and enforcement action, inter-agency cooperation for establishing technical standards)
		3.2.2.2 Provide technical and legal support leading to reforms to Ministry of Environment and Water processes of complaints, enforcement and sanctions for infractions affecting mangroves	140 MAATE staff members trained
		3.2.2.3 Provide training for judges and other institutions regarding regulations and sanctions for crimes involving mangroves	

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

240. The Monitoring and Evaluation (M&E) Plan is provided as **Annex 11**. The plan includes GCF-level Impact and Outcome indicators based on the IRMF and Project-level Outcome and Activity indicators, metrics for each indicator, methodology for data collection, baseline information, frequency of data collection, and indicative resources needed to complete the plan. The Project monitoring and evaluation will be coordinated by a dedicated M&E specialist at CI-Ecuador in close coordination with subgrantees (notably PUCESE), partners and other CI units contributing to the project. Project partners will be responsible for providing timely and comprehensive reporting including technical results and financial data as necessary and appropriate. These responsibilities will form part of contractual agreements between CI and the project partners and detailed monitoring requirements will be established based on their roles. All monitoring data being collected by these partners will be aggregated for project level reporting.
241. The Key Project Indicators, including measures of GHG emissions reductions and vulnerability reduction, as outlined in sections E2 to E6 will be monitored by the Project's M&E specialist (in coordination with staff at partner organizations and with the aid of consultants where necessary) applying the methodologies and frequencies as outlined in the Monitoring and Evaluation Plan (**Annex 11**). The M&E specialist will be assisted by the project experts to be hired within CI and other CI units on gender and safeguards issues, blue carbon and project impact evaluation.
242. The M&E Plan is a living document, follows the templates specified by GCF, and focuses on measuring the project's delivery of outputs, outcomes and broader paradigm shift. More detail will be added to the M&E Plan during the inception stage of the project (the first six months of implementation) by the M&E specialist and in collaboration with government agencies and local partners. Development of the full M&E Plan will ensure that the M&E requirements are fully assimilated and that the plan is owned by the project staff. The full M&E plan will build on the information provided in this document but elaborate in more detail on the roles and responsibilities for data collection and management, information flows and reporting systems, finalized indicators and means of verification, monitoring protocols and tools, implementation plans and schedules, alignments and collaborations with existing national M&E systems. The detailed M&E Plan will also include participatory methods for data collection and learning and an impact evaluation plan that builds on the summary evaluation plan included in **Annex 11**.
243. The Project Steering Committee will meet annually to review and approve annual budget and workplans, discuss implementation issues and identify solutions, and increase coordination and communication between key project partners. The NDA will participate in these meetings as a member of the project steering committee.
244. CI-GCF Agency staff, as AE, will continuously monitor the work of the EE through quarterly EE reports and meetings to monitor progress towards deliverables. CI-Ecuador, as EE, will prepare an Annual Performance Report to monitor project progress and report to the CI-GCF Agency. The report will summarize the annual project results and progress. CI-Ecuador will also provide a final report at the end of the Project.
245. The Project will undergo an independent Interim Evaluation in Year 4, consistent with GCF's Evaluation Policy. The evaluation will determine progress being made towards the achievement of outcomes and outputs and will identify course correction if needed. The evaluation will incorporate an assessment of the Project's contributions to a paradigm shift in the enabling conditions using a three-point scale rating as specified in the IRMF and with reference to the expected changes in enabling conditions noted in section E4 above. Recommendations from the Interim Evaluation will be incorporated into the design of the Project to secure maximum results and sustainability.
246. An independent Final Evaluation will be conducted at the end of the Project and will be undertaken in accordance with the AE's guidance and consistent with GCF's Evaluation Policy. The Final Evaluation will focus on the delivery of the Project's results as initially planned (and as corrected after the Interim Evaluation, if any such correction took place). Once again, it will include an assessment of the project's contributions to a paradigm shift in the enabling conditions using a three-point scale rating.

247. In addition to the project specific monitoring of the indicators noted in section E for reporting to GCF, the Project will also feed relevant information into Ecuadorian government monitoring systems and to CI's institutional results monitoring. Some of the indicators used for the Project monitoring have direct relevance for the Government of Ecuador's reporting to the UNFCCC, the Sustainable Development Goals, targets related to the Convention on Biological Diversity and national targets.

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1 A changing climate may impact the feasibility, sustainability or success of the project.

Category	Probability	Impact
<u>Technical and operational</u>	<u>High</u>	<u>Low</u>

Description

The base line study confirms that Ecuador's coastal ecosystems are susceptible to climate change impacts. However, the impact on the project activities themselves is considered limited. Moreover, it is one of the main aims of the project to increase the resilience of the mangrove ecosystem to the impacts of climate change. The main climate change impacts that may have a direct bearing on the project activities are those that could change the ability of mangroves to survive such as SLR and physical changes to mangrove estuaries that could make some areas unsuitable for mangrove. As noted in section B1, mangroves are dynamic systems and natural changes in mangrove cover can occur as mud/substrate moves due to erosion or deposition. However, as also noted in section B1, projected SLR rates are below the rates at which mangroves can adapt to the change by expanding into newly created suitable habitat.

Mitigation Measure(s)

Several aspects of the design of the project will address the potential threat that some areas will become less suitable for mangroves over time due to climate change. First, the restoration approach that the project will follow will include consideration of the hydrology of restoration sites to consider the current and future suitability of areas for mangrove survival and growth. Second, the activities to develop adaptation plans for the PAs are intended to build resilience for these areas, and mangroves in particular, by ensuring that potential impacts for each of the PAs are identified and adaptation measures are integrated into management plans and current management. Finally, the work on PDOTs with subnational governments includes land-use planning in which potential climate impacts will be identified and used to identify appropriate land-use planning adaptations. For example, this would include identifying areas unsuitable for development as they may be needed for mangrove 'retreat' under future SLR scenarios.

Selected Risk Factor 2 A changing climate may impact the adaptation opportunities provided by the project.

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>

Description

One climate change impact that could influence the adaptation opportunities provided by the project is SLR, as this could influence the area available for reforestation. The base line study included an assessment of the vulnerability of the entire project area to rising sea levels. SLR is expected in all parts of the project area, but the River Guayas Estuary was identified as having the highest risk.

The vulnerability assessment included in the baseline study included the factor of SLR, particularly in the development of maps prioritizing areas for mangrove reforestation. In fact, areas with high vulnerability to SLR (in combination with other factors) were prioritized for reforestation. This is because experience with mangrove reforestation projects has shown that successful reforestation can lead to increased sediments, allowing the young mangrove trees to form a protective barrier against further climate change impacts, such as coastal erosion, and even protecting themselves against future SLR (Spalding *et al.* 2014; Van Lavieren *et al.* 2012).

Mitigation Measure(s)

The project addresses this risk through Activity 1.1.1 under which stakeholder meetings will be conducted as part of the detailed planning of restoration activities to record local knowledge on climate change impacts and potential future impacts on mangrove reforestation, considering the views of men and women, for use in the final confirmation of sites for reforestation, especially in the Guayas River Estuary.

Selected Risk Factor 3 There are risks to the project area from potential third parties' action.

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>
Description		
The project area suffers from pollution from cities and their productive activities, agrochemicals, the proximity of large ports and water contamination by ballast water. These forms of pollution could impact mangrove survival in parts of the project area although they are likely to be very localized and involve small areas (with respect to the scale of the overall project size). The potential impact is therefore considered to be low. These influences are largely outside of the control of the project, at the same time, they further emphasize the need to increase the resilience of the mangrove ecosystem.		
Mitigation Measure(s)		
The project already addresses these issues: - Activities 1.2.1, 2.1.1 and 2.1.2 aim to introduce climate-smart approaches to existing economic activities or new bio-enterprises based on climate-smart approaches, which would both lead to less contamination from harmful substances currently in use, e.g., in shrimp cultivation. - Activity 3.2.1 includes work with local governments, municipalities and prefectures that are responsible for territorial management and planning in the context of improved management of contamination.		
Selected Risk Factor 4 Project staff could be at risk due to tensions between certain actors within the project area that exist or may worsen as a result of project implementation.		
Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>
Description		
In the project area, there are tensions between shrimp farmers and fishermen, because some do not accept the existing rules. The project aims to work especially with local associations, which may lead to a further divide between members and non-members. Tensions between stakeholders within the project could disrupt activities if some stakeholders decide they do not want to participate. Furthermore, there is the risk of criminal gangs operating in remote mangrove areas. Ecuador's remote coastal areas are used to ship cocaine produced in neighboring countries further up the coast to Central America to the United States and to Europe. The risk exists throughout the project area but is more acute in the northern estuaries and some areas of the Guayas estuary. Violence between rival criminal gangs and with law enforcement is common, making it important that project staff and partners operating in the mangroves are not mistaken as being from rival gangs or being associated with law enforcement agencies. Despite recent gang violence in Guayaquil, the risk to project staff as a result of existing tensions is considered to be low and in case of actual security issues, CI's Security and Protection Plan applies to CI staff as well as project partners. CI already has a presence in the region, and there have been no gaps in operations, even with additional security precautions that were put into place (i.e. no travel after dark). CI Ecuador benefits from strong and experienced leadership, with a close relationship with the government and also with local partners, including communities where this project will work. CI Ecuador has regional and office level Security Focal Points, which helps to identify and assess risks quickly, efficiently, with the safety of CI, partners and beneficiaries the first priority. CI and partners also have strong relationships with the communities, which not only makes it easier to operate during uncertain times and provides CI with early warnings when conditions begin to deteriorate. In addition, the project is likely to reduce existing tensions through its planned activities. The established project-specific Grievance Redress Mechanism, if used by stakeholders as foreseen, can be used to track where tensions are aggravated, and responses will be identified on a case-by-case basis to address these tensions.		
Mitigation Measure(s)		
For the potential risk of tensions between stakeholders: CI will implement the project-specific Grievance Redress Mechanism to track grievances related to tensions between stakeholders and between stakeholders and the project team. The team will initiate spaces and procedures for stakeholder dialogue where tensions between stakeholders and the project team appear to have intensified as a result of the project and provide mediation to ensure improvement over time. The project ESMP also includes the requirement to screen all the community grant support that will be provided		

and consideration of tensions that the grant could create are to be assessed in the screening and mitigation measures developed if the risks are considered medium or high level.

For the work with AUSCEM organizations, CI and PUCSE will also be encouraging AUSCEM organizations to include more members to maximize participation of fishers within the associations and reduce the potential for conflict between members and non-members.

To address safety risks posed by criminal activity, CI's director for Safety and Security will conduct a thorough security assessment and mitigation plan during project start-up. Example mitigation measures include pre-activity community engagement to ensure local communities understand the project objectives and that there is no link to government efforts to combat criminal gangs; Hostile Environment Awareness training and guidance on avoiding and reporting criminal activity; and ensuring that it is easy to identify CI staff and partners as contributing to project objectives (rather than criminal or law-enforcement activity).

An additional mitigation measure will be an emphasis on selection of consultants and subgrantees who have a demonstrated history of operating in the project areas. PUCSE, the main project partner that will lead activities in the two northern sites, is well known in the northern areas and familiar with working in those regions and with local communities.

A Security Plan has been developed for this project and is included as Appendix 11 of the ESMP (**Annex 6**).

Selected Risk Factor 5 The project area is susceptible to earthquakes, subsidence, landslides, erosion, flooding or extreme climatic conditions.

Category	Probability	Impact
<u>Technical and operational</u>	<u>High</u>	<u>Low</u>
Description		
The project area is sensitive to erosion, flooding or extreme weather conditions, for example in El Niño or La Niña years. These natural extreme events have the potential to disrupt project operations; however, the impacts on project operations are expected to be low since the issues are likely to be relatively short-term by comparison to the length of the project.		
Mitigation Measure(s)		
The project includes several activities that are mitigation measures themselves, i.e., they can jointly reduce the overall impact of such extreme weather events or help improve the management of their impacts: - Local people's livelihoods will be improved through the development of new bio-enterprises (sub-activity 1.2.1); and, - The resilience and extent of the mangrove ecosystem will be increased, thus reducing the risk of impacts from extreme weather events (sub-activity 1.1.1), specifically coastal flooding. - Improve local planning, including identification of areas at risk of flooding and identification of priority natural infrastructure to protect and restore. Linked to the restoration activities (and related to risk factor 2 above), CI and PUCSE will consult with local stakeholders to ensure their knowledge about the influence of extreme weather events is taken into consideration in the establishment of bio-enterprises, if relevant (e.g. regarding the selection of sites to establish mangrove nurseries), as well as in the final confirmation of sites for mangrove reforestation.		
Selected Risk Factor 6 Political instability and changes in government agencies		
Category	Probability	Impact
<u>Governance</u>	<u>Medium</u>	<u>Low</u>
Description		
The political stability index has shown a steady improvement over the last decade and was at - 0.02 in 2014. The index measures perceptions of the likelihood of Ecuador's government becoming destabilized.		

Political risk in Ecuador is characterized by possible changes in government personnel in key positions or changing mandates of different ministries because of elections. One example is that the administration of mangrove concessions could be transferred from the MAATE to another Ministry, which could result in changes to concession regulations. Since some of the Outcome 3 activities relate to working at the subnational governments, changes at those levels could also impact the project operations. The main risk is that changes in personnel and/or government structures could cause delays in project operations. Another example is that a new administration may seek to alter FIAS's governing structure or rules, which may impact its independence of operations. FIAS has taken several steps to enhance good governance practices, including the expansion of its Board to provide for greater independence, consistent with best practice standards for conservation trust funds. A risk remains that some of these practices could be changed under a new administration.

A snap general election for both the executive and legislative branches took place in August 2023. All the leading political parties and candidates embraced environmental issues as key pillars of their platform so it seems unlikely that there would be backtracking on climate change commitments or support for mangrove conservation and restoration, the key focus areas of this proposal. The regulatory frameworks and programs around which the proposed project is built (Sustainable Use and Custody Agreements/AUSCEM and the Socio Manglar program) are enshrined in the constitution and regulatory frameworks and are unlikely to be impacted. Although ministers may change following the elections, ministerial level turn-over is not new for Ecuador and CI Ecuador has maintained strong relations at these levels throughout numerous previous administrations and through many different ministers. Local and provincial officials were not impacted by the 2023 elections, meaning that CI Ecuador's strong relations with government at those levels remained in place. Local and provincial elections will happen during the project implementation period but CI Ecuador's strong relations with those governments are expected to provide continuity for the project. A new administration has been in place in Ecuador since December 2023 and the engagement of the government, local associations and private sector remains.

Mitigation Measure(s)

CI has been able to maintain its activities during the last 20 years in the political context of Ecuador and has been able to collaborate consistently with successive administrations and with subnational governments. Elections were held in August 2023 and resulted in a presidential change in December 2023, however, not a change in the majority party.

All relevant ministries have been involved in the development of this project, including participation during Project Preparation Funding activities of Funding Proposal development. The project has been designed to address government priorities related to climate change, nature conservation and economic development and enjoys support from the ministries involved. MAATE is even providing co-financing for the project. In the case of any changes in mandate or personnel, CI will continue its extensive engagement with the government to ensure a smooth transition and to address possible changes as needed.

The risk that a new administration would alter FIAS's governing structure or rules is considered low risk and in the due diligence there are some measures highlighted to design and implement before the granting process. There are internal processes and policies that will be included in the grant to ensure compliance of use of GCF funds. The project budget also includes support for capacity building and enforcement support to FIAS.

The risk presented by changing priorities and approaches from new administrations warrants monitoring, although it has been reduced since FIAS's inception due to concerted outreach by FIAS and engagement with the donor community, resulting in reforms to FIAS's governance. It is unlikely that there would be dramatic additional reforms to FIAS so soon after the Government of Ecuador helped restructure it in 2017.

Of note, the vast majority of Fund decisions are made by an Administrative Board ("Junta Administrativa" or "JA") for each of FIAS's subsidiary funds (such as the Socio Bosque Fund). These JAs were created during financing agreements with donors, are largely determined according to the requirements in such agreements, and largely operate independent of the FIAS Board of Directors. The governance structure of each of these JAs is defined in their governing rules / operations manuals which may be subject to modification based on FIAS and donor agreements. The JAs are primarily responsible for supporting the execution of projects, making decisions on fund allocations and disbursements, consistent with the terms of any donor agreement. The JAs are responsible for the evaluation and oversight of donor

supported projects / funds. Accordingly, CI will ensure in its grant agreement with FIAS that the JA managing the Socio Manglar subaccount meets best practices, including any GCF requirements.

Selected Risk Factor 7 Prohibited Practices, including ML/FT

Category	Probability	Impact
<u>Prohibited practices</u>	<u>Low</u>	<u>Medium</u>
Description		
The Project involves a wide array of actors, including government, sub-grantees, vendors and service providers, and beneficiaries. Based on a calculation using multiple international risk indicators, CI assesses <i>Ecuador</i> to be a low-risk country for GCF prohibited practices, including terrorist financing and money laundering. CI Ecuador follows and is subject to CI's Anti-Money Laundering and Counter-Terrorism-Financing (AML/CFT) Policy and related procedures, which have been assessed as effective by the GCF. In addition, CI-Ecuador is subject to national laws related to AML/CFT, for which it has a country-specific Manual for Prevention of ML and FT. In compliance with the Organic Regulation to the Law of Prevention and Detection of Laundering Money and Financing Terrorism, CI-Ecuador's AML/CFT practices are supervised by a public entity, the Financial Analysis Unit (<i>Unidad de Análisis Financiero</i> [UAFFE]), which conducts a thorough review of financial procedures related to AML/CFT and to which CI-Ecuador must submit monthly reports. Finally, the project's major non-EE recipients of GCF funds, sub-grantees PUCESE and FIAS, were screened and cleared against international sanctions lists. Concerning impact, serious breaches in prohibited practices, including ML/FT, would impact the reputation of CI and the project.		

Mitigation Measure(s)

CI has several procedures and controls in place to prevent, mitigate, and identify risk. In order to comply with CI AML/CFT, U.S. law, and GCF fiduciary standards, CI applies rigorous CFT/AML screening to all CI funding recipients. CI also conducts an assessment of Prohibited Practices (PP) risks of grant recipients and includes any required mitigation measures in its legally binding grant agreements. All EE staff and subgrantees must also undergo training on PP (including red flags and how to report violations), and all grant and services agreements (used by both CI and sub-grantees) must include language on compliance with the GCF Policy on Prohibited Practices, including AML/CFT. Sub-grantees PUCESE and FIAS will receive capacity-building related to AML/CFT and prohibited practices from CI Ecuador, and compliance with PP policies and mitigation measures is part of project monitoring, including site visits and CI audits, throughout implementation.

To report and remedy risks, any CI employee; employee of an executing entity, grantee, partner, or vendor; or beneficiary may report a grievance or violation of these policies anonymously online or by telephone through CI's Ethics Hotline. All reported grievances will be investigated and addressed per CI's Anti-Fraud Policy & Guidelines for Investigations. Full details can be found on CI's website at <https://www.conservation.org/about/our-policies/reporting-illegal-or-unethical-conduct-statement>.

Selected Risk Factor 8 Unrealized or delayed co-financing

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>
Description		
Co-financing plays an important part in achieving the full impact of the proposed project. There is a medium risk of delayed or unrealized co-financing for the project, because MAATE's grant co-financing depends on the approval processes for government budgets. However, the committed co-financing is proposed for activity A.1.1.1 and therefore, the negative impact to the project would be limited to this activity. In addition, a portion of the co-financing committed by MAATE is 'in-kind' and related to staff salaries and therefore poses a low risk of not materializing.		
Mitigation Measure(s)		
The project team will periodically assess the degree of risk (both likelihood and impact) that co-financing will be funded and its associated impact on project implementation. CI will also conduct regular engagement with co-financing sources to understand and manage this risk. In the event that co-financing is not realized, CI will try to identify alternative sources. If that is not possible, CI would work with GCF within the framework of the project's FAA to propose adaptive		

management measures that may include reducing the scope of affected activities with minimal negative impact to project objectives.

Selected Risk Factor 9 Risk of inflation

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Low</u>
Description		
Although currently decreasing, Ecuador's inflation rate for 2022 was 3.7% according to the International Monetary Fund's Data Mapper (April 2023). Ecuador's average inflation rate over the period 2013 to 2022 was 1.8% with a maximum of 4.2% in 2017⁶⁴. This period includes several turbulent years for Ecuador's economy due to fluctuations in the global price of oil, on which Ecuador's economy is highly reliant. The IMF's forecast for inflation in Ecuador is 1.34% in 2025, rising to 1.54% in 2026 through to 2028⁶⁵. However, given the increases in inflation globally, the risk of rising inflation is considered a 'medium' risk to this project as it could impact the budget, co-financing and activity implementation. Higher than expected inflation would increase costs across the whole project, making it harder to achieve all of the project activities. High inflation rates would be accompanied by high interest rates at financial institutions, which could impact the likelihood of private sector shrimp farm companies taking loans to invest in climate smart shrimp practices (Activity 2.1.2).		
Mitigation Measure(s)		
To mitigate for the impacts of inflation, CI has included inflation considerations into its budget figures. The budget has been prepared assuming an inflation rate of 3%. The 3% figure is above the IMF's forecast but given the challenges of inflation forecasting and recent global increases in inflation, CI has used the 3% figure out of precaution. In addition, CI will regularly monitor inflation pressures on the project budget and continually seek to identify cost savings where possible.		

⁶⁴ <https://datos.bancomundial.org/indicador/FP.CPI.TOTL.ZG?locations=EC>; accessed 2 June 2023

⁶⁵ https://www.theglobaleconomy.com/Ecuador/inflation_outlook_imf/; accessed 2 June 2023

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

248. The Environmental and Social risk screening for the project indicated that a limited and adequate Environmental and Social Impact Assessment (ESIA) be conducted to identify potential project risks. This assessment resulted in the following:
249. The project is risk category B (moderate risk).
250. An Environmental and Social Impact Assessment and Environmental and Social Management Plan (ESMP), included as **Annex 6**, includes a summary of the project and the applicable legal and institutional framework, the risks identified, the action plan with mitigation measures and implementation modalities. Also included are a Grievance Redress Mechanism as well as:
- Natural Habitat Protection Plan;
 - Process Framework for Restriction of Access to Natural Resources;
 - Plan for Afro-descendant Communities and Cholos Pescadores;
 - Stakeholder Engagement Plan (**Annex 7**); and
 - Gender Action Plan (**Annex 8**).
251. The project does not intend to cause negative environmental or social impacts; on the contrary, it aims to increase the resilience of the mangrove ecosystem and benefit local communities. It is based on rights-based and inclusive approaches, including measures to ensure the participation of vulnerable groups, such as women, youth and others.
252. Some risks are, by their nature, beyond the control of the project. For example, climate change may impact the project area and there are localized impacts on the mangrove ecosystem from pollution in some places. The project includes a variety of activities to address these issues.
253. Some details of activities will be defined during the project and therefore risk screening and appropriate mitigation measures for these will be identified as part of the project activities. Screened activities that result in a category B risk rating will require further limited ESIA's and ESMPs to be developed as per the guidance provided in **Annex 6**. This applies to:
- Activities 1.2.1 and 1.2.2: It is necessary to identify which community livelihoods activities and small-scale enterprises to be supported in various locations near the time of the work. Once the options and locations for their development have been prioritized, their potential impacts must be assessed.
 - Activities 1.1.1: Project preparation included the identification of areas with potential for reforestation. The resulting map (see **Annex 2**) defines a potential of 10,000 ha while the project includes 4,850 ha as a target, so the detailed selection of sites for reforestation is pending. To address this, detailed reforestation plans will be developed as part of the reforestation activities including assessment of risks and measures to avoid environmental damage and negative social impacts.
254. Apart from those mentioned above, the most important risks and their mitigation measures are:

Environmental or Social Risk Factor	Mitigation Measures
Community Management groups (AUSCEMs) could impose restrictions on access to or use of natural resources and result in economic displacement (see	• Identify all stakeholders affected by these restrictions, with special attention to members of Afro-descendant communities, Cholos Pescadores and women-led households. • Consult with stakeholders on the need/measures for conservation and identify those measures that can lead to restrictions/economic displacement. • Consult with any affected stakeholders, applying a culturally appropriate and gender-sensitive approach, to develop a socioeconomic assessment and analysis showing quantifiable impact of restrictions on their livelihoods that can be used to inform, negotiate and identify solution options for inclusion in the refined Process Framework and the Plan for Afro-Descendent Communities and Cholos Pescadores. • Explore opportunities to maintain engagement of affected stakeholders in project activity 1.2.1.

ESMP, risks R3.1, R4.1 and RHR2).	• In cases where economic displacement is unavoidable, ensure that stakeholders remain involved throughout the process of defining and implementing the Process Framework for Restricting Access to Natural Resources (see Annex 6), including in the process for obtaining FPIC. • Seek FPIC from affected stakeholders, following an FPIC procedure that has been previously agreed in a participatory manner and in line with CI and GCF's ESMF guidelines for Applying FPIC. • Implement a Process Framework that has been agreed upon by local communities to which it applies, including the Afro-descendant communities and Cholos Pescadores (see Annex 6). • Implement the project-specific Grievance Redress Mechanism and trace complaints in the context of economic displacement to ensure satisfactory follow up and conclusion of complaints.
Women and other vulnerable groups, including those not affiliated with AUSCEMs, could be negatively impacted by not being able to participate in the same way as other stakeholders (see ESMP, risks RHR1 and RHR3).	• Investigate in more detail the potential adverse effects of the project on non-partners, with special attention to vulnerable groups such as women, youth, elderly fishermen, non-affiliates of AUSCEMs, members of Afro-descendant communities and Cholos Pescadores. • Conduct informational and awareness-raising sessions to clarify the potential benefits of joining an AUSCEM group, especially in the context of this project. • Identify ways to ensure that the project does not cause harm to local people unwilling to join an AUSCEM group and clearly communicate the efforts made to ensure this. More mitigation measures specifically to ensure gender equity are included in the gender action plan. The ESMP also includes mitigation measures to address the risk of Sexual Exploitation, Sexual Abuse and Sexual Harassment (SEAH).

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

255. During Project preparation, a gender assessment (see **Annex 8**) was undertaken to understand the differential vulnerability to, and impact of, climate change on women and men in the project area. From the findings of this assessment and an analysis of the proposed project design, a Gender Action Plan (GAP) was developed. The process to produce these components included a desktop review and analysis of available literature and relevant legislation, policy and programs in the country, as well as key informant interviews, household questionnaires, focus group discussions and consultation workshops with project stakeholders and beneficiaries.
256. The GAP (see **Annex 8**) identifies opportunities to close gender gaps, reduce gender-based violence (GBV) and the included actions are largely concentrated in the support of mangrove associations where direct collaboration with the women and men who depend on, and manage, the mangroves will occur. These include mangrove use agreements that integrate gender considerations, tailored and specific support to women in mangrove associations (including the two women's associations), setting targets (50/50) for women's and men's benefit from the adoption of diversified and climate resilient livelihoods options linked to mangroves, and ensuring that project staff and delivery partners are trained on gender equity principles and the importance of mitigating gender inequalities in the project, including gender-based violence.
257. To support implementation of the GAP, CI will employ a full-time Gender Manager to ensure robust gender analysis and lead the mainstreaming of gender into the project. The Gender Manager will develop and deliver appropriate capacity building for colleagues and delivery partners that is well-designed, clear, tailored to the context, and practical. The Gender Manager will accompany project teams to the field to ground truth progress on gender outcomes: meet with women/women's groups, collect qualitative information, help project teams to problem solve, propose adaptive changes and participate in report preparation. The manager will support the four estuaries and, in particular, the social coordinators in the implementation of the GAP. The manager will also collaborate with CI's global gender program to ensure consistency in gender approach across the institution and will be supported by CI's global Gender Advisor. The Gender Manager position will be based in Guayaquil city.
258. A complex set of historical, social, cultural, and economic factors influence the project sites' current gender norms and inequalities. There is a lack of political representation by women in decision-making, time deficit due to mostly reproductive labor impinging on their ability to engage in community, organizational, and project activities, and finally high rates of gender-based violence that further highlights inequalities and is an important barrier that

keeps women from exercising their voice and agency in mangrove conservation. This situation is compounded by a number of structural problems associated with education, health, the provision of utilities and limited employment opportunities.

259. In these mangrove-dependent communities, women are more vulnerable and experience higher rates of poverty than men. Men tend to control the main economic activity in the communities: the harvesting and sale of seafood. Often it is men who hold formal positions of leadership within communities and fishing associations.
260. There are distinct gender differences in mangrove use and management in the northern sites versus those in the south. In the north, women walk (often collectively) to the mangroves and are therefore more actively engaged; Mangrove associations in the north also have ~60% women members. The south is more male dominated; women's access and use of the mangrove areas is constrained by distance as the mangroves are far from the community and must be accessed by boat (often individually). Because they are not familiar with the mangroves, they are largely absent from the associations; mangrove associations in the south have ~15% women members.
261. Women living in mangroves work long days, which include family caregiving responsibilities, harvesting or fishing activities, cooking for the family and looking after the children. The average working day for women is around 14 hours and little societal value is placed on their caregiving or reproductive tasks. By contrast, men have much shorter days, fishing or harvesting for between four and five hours. This means that once their day's fishing or harvesting is over, they are relieved from work.
262. Rates of GBV are very high (67% in project sites) and could be perpetuated/increased by climate change impacts that result in economic disruption.
263. As a result of gender norms, men and women are impacted differently by climate related events. Rising sea levels and flooding – the two major effects cited in justifying climate change adaptation action in these project sites – can also have consequences like an increase in water salinity and water-borne diseases, which are likely to affect primarily women, who tend to be responsible for collecting water for the household and caring for sick family members. Furthermore, existing gender inequalities may be exacerbated by the dynamics of climate change in the project area.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

264. The project will be executed by CI-Ecuador, and therefore, financial management and procurement for this project will follow relevant CI policies and procedures, as well as relevant provisions in the Accreditation Master Agreement (AMA) signed between CI and the GCF.

Financial management:

265. CI uses Unit4 Business World software suite as its enterprise resource planning (ERP) system for financial accounting and monitoring in all its global operations. This system supports financial monitoring, reporting, human resources, procurement, and record-keeping in line with the requirements of CI's AMA. It supports a system of ledger accounts to separate and track income and expenditure by donor, project, and subaward. It also provides a coding system for monitoring and reporting of projects. In addition, all EEs and sub-grantees will be required to submit quarterly financial reports that will be systematically reviewed by the AE and EE, respectively.
266. An annual audit of the project will be conducted by an independent external audit firm in accordance with audit standards under U.S. GAAP and furnished to the GCF in accordance with CI's AMA.
267. Disbursements made to CI will be recorded to the ledger account for this GCF project, which can be tracked using the Business World system. Disbursements from the EE to sub-grantee will be made upon CI's receipt of acceptable progress and financial reports and any other requirements as included in the applicable grant agreement.
268. CI has procedures and controls in place to prevent risk of prohibited practices, including ML/FT (collectively, 'PP'). CI applies rigorous CTF/AML screening to all CI funding recipients. CI also assesses PP risks of grant recipients and includes mitigation measures in its grant agreements. All CI staff and subgrantees must also undergo training on PP (including red flags and how to report violations), and all grant and services agreements (used by both CI and sub-grantees) must include language on compliance with the GCF Policy on Prohibited Practices. Compliance with PP policies and mitigation measures is part of project monitoring, including site visits and CI

internal audits, throughout project implementation. Violations of PPs, fraud, or financial misuse may be reported anonymously online or by telephone through CI's Ethics Hotline. All reported grievances will be investigated and addressed per CI's Anti-Fraud Policy & Guidelines for Investigations.

269. Violations of Prohibited Practices, fraud, or financial misuse may be reported anonymously online or by telephone through CI's Ethics Hotline. All reported grievances will be investigated and addressed per CI's Anti-Fraud Policy & Guidelines for Investigations.

Procurement:

270. All procurement for this project must follow the Procurement Process Standards as defined in the CI Procurement Policy, which requires that all procurements of goods or services ensure the best value and comply with all donor funding terms and conditions, and that all goods be used only for their intended purposes and consistent with CI's Code of Ethics and suite of anti-fraud policies. CI's procurement standards are in line with GCF fiduciary standards.
271. All procurement done by the EE will follow CI's internal procurement controls and risk-mitigation measures, including procurement planning, multi-level purchasing approvals (including AE approval over a set threshold), quarterly reporting on procurement & equipment, periodic inventories, and post-award reviews. CI's Grant and Contracts Unit must approve contracts of US\$ 20,000 or more awarded by CI's Ecuador office, including for this project.
272. CI will assess the procurement policies of any subgrantee organization conducting procurement for this project. Consistent with CI's Procurement Policy, Project sub-grantees that meet CI's Procurement Process Standards as defined in the CI Procurement Policy will use with their own procurement policy and procedures for the project; otherwise, they will be required to use CI's Procurement Policy for the project in whole or in part. Detailed procurement requirements consistent with CI and GCF standards will be included in the grant agreement signed between CI and the grantee organization. The Project's sub-grantee PUCSE will conduct procurement to support the implementation of its Project activities, in line with these requirements and as described in **Annexes 10 and 19**.
273. CI will apply mitigation measures and monitor all procurement for this Project to ensure that any procured goods are used only for their intended purposes. It will provide training on procurement and prohibited practices for CI staff and subgrantees and will approve annual procurement plans and all procurements over a set threshold. It will also conduct monitoring through quarterly reporting on procurement & equipment and through periodic site visits.
274. An indicative plan for maintenance, operation, and disposition of GCF-funded equipment procured by CI Ecuador is found in **Annex 21**.

G.4. Disclosure of funding proposal

No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study
- ☒ Annex 3 Economic analysis in spreadsheet and narrative format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☒ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letters [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule

H.2. Other annexes as applicable

- ☒ Annex 14b Term sheet for the Socio Manglar subaccount and appendices
- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence/evaluation report for proposals that up-scale/replicate a pilot project
- ☒ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☒ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance)
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁶⁶
- ☒ Annex 23 Implementation arrangements
- ☒ Annex 25 Adaptation Beneficiaries
- ☒ Annex 26 Example of an AUSCEM Agreement

⁶⁶ Annex 22 is mandatory for mitigation and cross-cutting projects.



Oficio Nro. MAATE-SCC-2023-0535-0

Quito, D.M., 15 de noviembre de 2023

Asunto: Re: Funding proposal for the GCF by Conservation International Foundation regarding Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador

Señora
Mafalda duarte
Executive Director
GREEN CLIMATE FUND
En su Despacho

Dear Madam,

We refer to the project titled **Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador** in Ecuador as included in the funding proposal submitted by Conservation International Foundation to us on October 6th, 2023.

The undersigned is the duly authorized representative of Ministry of Environment, Water and Ecological Transition, the National Designated Authority of Ecuador.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

1. The government of Ecuador has no-objection to the project as included in the funding proposal;
2. The project as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ecuador;
3. In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We acknowledge that this letter will be made publicly available on the GCF website.

Atentamente,

Oficio Nro. MAATE-SCC-2023-0535-0

Quito, D.M., 15 de noviembre de 2023

Documento firmado electrónicamente

Mgs. Karina Maribel Barrera Moncayo
SUBSECRETARIA DE CAMBIO CLIMÁTICO

Copia:

Señor Magíster
Diego Gonzalo Quishpe Landeta
Director de Adaptación al Cambio Climático

Señorita Ingeniera
Gabriela Jeanneth Vargas Luna
Especialista de Políticas de Cambio Climático 3

Señora Secretaria
Teresa Del Pilar Pérez Toscano
Secretaria de la Subsecretaria

gv/db



Firmado electrónicamente por:
**KARINA MARIBEL
BARRERA MONCAYO**

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Conservation International Foundation (CI)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Republic of Ecuador. The project activities focus on the four main estuaries with mangroves: Cayapas Mataje, Muisine Cojimies, Guayas and Jambeli. The project area includes eight municipalities: El Guabo, Machala, Santa Rosa, Eloy Alfaro, Muisine, San Lorenzo, Guayaquil and Naranjal.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Thursday, June 13, 2024
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Ecuador.
Link to disclosure	English: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---disclosure95b281f63c7b42deb3eb7e37496adf0a.pdf?sfvrsn=d7c9c322_0 Spanish: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---es---disclosure1a80a7f4b9754b23a43f17cb7a678828.pdf?sfvrsn=b4f5ecff_0
Other link(s)	CI Website: English: https://www.conservation.org/gcf/projects/mangroves-for-climate Spanish: https://www.conservation.org/ecuador/nuestro-trabajo/programas/marino-y-costero

Remarks	An ESIA consistent with the requirements for a Category B project is contained in “Annex 6: Environmental and Social Impact Assessment (ESIA) and Environmental and Social Management Plan (ESMP)”.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity’s website	Thursday, June 13, 2024
Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Ecuador.
Link to disclosure	English: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---disclosure95b281f63c7b42deb3eb7e37496adf0a.pdf?sfvrsn=d7c9c322_0 Spanish: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---es---disclosure1a80a7f4b9754b23a43f17cb7a678828.pdf?sfvrsn=b4f5ecff_0
Other link(s)	CI Website: English: https://www.conservation.org/gcf/projects/mangroves-for-climate Spanish: https://www.conservation.org/ecuador/nuestro-trabajo/programas/marino-y-costero
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the “Annex 6: Environmental and Social Impact Assessment (ESIA) and Environmental and Social Management Plan (ESMP)”.
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity’s website	Indigenous People Plan (IPP) / Thursday, June 13, 2024

Language(s) of disclosure	English and Spanish
Explanation on language	Spanish is the official language of Ecuador, and the Government uses Spanish to communicate government publications (reports, policies, laws, minutes etc.) with the communities in the coastal areas.
Link to disclosure	English: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---disclosure95b281f63c7b42deb3eb7e37496adf0a.pdf?sfvrsn=d7c9c322_0 Spanish: https://www.conservation.org/docs/default-source/gcf/annex-06---ci-ecuador---esmp---es---disclosure1a80a7f4b9754b23a43f17cb7a678828.pdf?sfvrsn=b4f5ecff_0
Other link(s)	CI Website: English: https://www.conservation.org/gcf/projects/mangroves-for-climate Spanish: https://www.conservation.org/ecuador/nuestro-trabajo/programas/marino-y-costero
Remarks	The Plan for Afro-descendent Communities and Cholos Pescadores is included as Appendix 5 of “Annex 6: Environmental and Social Impact Assessment (ESIA) and Environmental and Social Management Plan (ESMP)”, which is the document published on the AE and CI-Ecuador’s websites.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Thursday, June 13, 2024
Place	The ESS documents were published in Spanish on the CI Ecuador website on June 13, 2024. CI-Ecuador website link: https://www.conservation.org/ecuador/nuestro-trabajo/programas/marino-y-costero All stakeholders who participated in the stakeholder consultation (see Annex 7, Summary of Consultations) have been informed by email that the document is available for consultation. An email address has been provided where comments can be sent to the project development team. To accommodate for those who may not have access to the internet, a computer, or a reliable power source, or would prefer to consult a physical copy of the document, CI and the Ministry of Environment, Water and Transition of Ecuador (MAATE) will be sharing the ESS document (Annex 6 of the Funding Proposal) by June 14, 2024 in the following locations: Marine, Coastal and Oceanic Direction office of MAATE. Av. Francisco de Orellana y Justino Cornejo, Guayaquil. 2nd Zonal Direction of MAATE. Av. Jaime Roldos Aguilera y Jr. Coronel centro de atención Ciudadana. Esmeraldas.

	7th Zonal Direction of MAATE. Calle Vela 1323 entre 25 de Junio y Sucre Edificio Ex Predesur - 3er piso. Machala. Provincial Government office of Esmeraldas. Avenida 10 de Agosto entre Bolívar y Pedro Vicente Maldonado. Esmeraldas. Provincial Government office of Guayas. Av. Juan Illingworth 108 y Av. Simón Bolívar. Guayaquil. Provincial Government office of El Oro. Avenida Junin
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, July 15, 2024

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP235

Proposal name:	Mangroves for Climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador
Accredited entity:	Conservation International Foundation (CI)
Country:	Ecuador
Project/programme size:	Small

I. Overall assessment of the Secretariat

- The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
Mangroves model: The project involves government, local communities and the private sector in supporting community-based enterprises and community mangrove management teams undertaking income generation while restoring and sustainably managing mangroves based on a results-based payment scheme. In addition to mitigation benefits, the mangroves provide ecosystem services that protect coastal communities and upstream communities from salinization caused by floods and sea level rise.	
Community grants: The project is working with a trust fund and government finance that aims to empower local communities to sustainably manage large areas of mangroves via community management agreements that grant reward payments for their management activities.	
Transformation of the private sector: The project will accelerate the transformation of the shrimp industry from being a driver of mangrove deforestation to a driver of preservation, while shrimp farmers will be able to access a premium price for certified shrimp cultivation.	

Financial sustainability model: The project will help to capitalize the trust fund subaccount on results-based payments for community mangrove management, with additional funding to be contributed annually from private and government sources. The trust fund is expected to run in perpetuity, hence generating impact beyond project implementation and even beyond the 20-year project lifespan.	
Scalability and replicability: The project may allow Ecuador to become a demonstration site for climate-smart shrimp farming that can be scaled and replicated across different regions, especially due to the project's link with the Global Mangrove Alliance.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity, and, if considered appropriate, subject to the conditions set out in Annex II of document GCF/B.39/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. The project background, as set out in the funding proposal, underscores the pressing need to address the impacts of climate change on Ecuador's coastal regions. Climate change projections indicate increasing risks, such as sea level rise, extreme weather events and coastal flooding, that threaten local communities and economic activities. Mangroves play a crucial role as natural barriers against erosion and flooding, yet ongoing deforestation exacerbates these risks. Despite recent conservation efforts, mangrove loss continues, with adverse consequences for carbon sequestration and biodiversity.

4. The proposed interventions aim to address these challenges by engaging local communities, the private sector and government entities in the protection and restoration of the mangroves. Through effective mangrove management and restoration, the project seeks to reduce greenhouse gas (GHG) emissions, to enhance flood protection for vulnerable coastal populations and to improve livelihoods. It aligns with Ecuador's national climate change strategies and contributes to global initiatives such as the Global Mangrove Alliance.

5. The project's components focus on increasing the area of mangroves under community-based management, engaging the private sector in sustainable practices and strengthening governance frameworks. By integrating these efforts, the project aims to achieve multiple climate mitigation and adaptation goals, including reducing GHG emissions, protecting ecosystems and enhancing infrastructure resilience.

6. Moreover, the project's alignment with the objectives of the Global Mangrove Alliance underscores its potential for knowledge-sharing and global impact. By leveraging international partnerships and best practices, the project aims to contribute significantly to global efforts to

conserve and restore mangroves, ultimately benefiting both Ecuador's coastal communities and the broader global environment.

7. The project will directly benefit 89,600 individuals through enhanced access to climate finance, adoption of climate-smart aquaculture practices, and mangrove restoration to mitigate landslide and flooding risks. Moreover, the long-term conservation and sustainable management of mangroves will indirectly benefit 3.4 million people, spanning 44 per cent of coastal provinces and 20.1 per cent of the country's population, as a result of the climate-resilient practices.

8. The total mitigation impact from mangrove conservation, restoration and preservation amounts to an estimated GHG emission reduction of 732,111 tonnes of carbon dioxide equivalent (t CO₂ eq) by the end of the implementation period and 4,609,470 t CO₂ eq over the project's 20-year lifespan, equivalent to USD 3.2 per t CO₂ eq for GCF finance. This project's estimated contribution to the nationally determined contribution (NDC) stands at 732,111 t CO₂ eq or 1.2 per cent over the project implementation period, rising to 4,609,470 t CO₂ eq or 7.4 per cent over the 20-year project duration.

9. The proposal includes a comprehensive budget totalling USD 45.9 million, with a significant portion (USD 36.4 million) sought from the GCF as a grant. The remaining USD 9.5 million is anticipated to be sourced from co-financing by the Ministry of Environment, Water and Transition of Ecuador and the Conservation International Foundation (CI), in the form of both in-kind contributions and grants.

2.2 Component-by-component analysis

Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs), and with protected areas implementing climate adaptation plans (total cost: USD 26 million; GCF cost: USD 19.5 million)

10. Component 1 of the project aims to increase mangrove areas under effective and climate-adapted management (AUSCEMs), including through community-based management and protected areas implementing climate adaptation plans. Despite the effectiveness of community-based stewardship in protecting mangroves, there are gaps in coverage that can be addressed through outreach and providing assistance in organizing, planning and strengthening relationships with government programmes that recognize AUSCEMs. The component is crucial to ensure that existing AUSCEM associations have the organizational capacity to sustain their stewardship efforts.

11. Community-based mechanisms for mangrove conservation are often economically precarious; therefore, generating stronger local economic benefits based on mangrove conservation is essential for ensuring sustainability. Restoration efforts under community stewardship will contribute to enhancing mangrove coverage and associated benefits.

12. To achieve these goals, output 1.1 focuses on reducing flood risk for vulnerable populations and reducing GHG emissions, through strengthening community-based mangrove management. Activities include providing training, technical assistance and equipment to AUSCEM groups, supporting the expansion of AUSCEMs, and assisting with the application process for incentives under the country's Socio Manglar (SM) programme.

13. Output 1.2 aims to improve livelihood activities and economically productive community businesses to incentivize participation in mangrove conservation and restoration. This involves providing technical and business development support to mangrove community associations, particularly focusing on women, youth and other vulnerable groups. In addition, financial mechanisms will be established and consolidated to support micro- and small-scale

enterprises, including cash grants and assistance in accessing credit and investment from existing financial institutions.

14. Overall, these efforts seek to strengthen community-based mangrove conservation and management, reduce deforestation, increase economic benefits for local communities and enhance resilience to climate change.

Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations (total cost: USD 10.2 million; GCF cost: USD 8.9 million)

15. The project aims to engage the private sector in promoting improved shrimp production methods to reduce mangrove deforestation and advance mangrove conservation and restoration in Ecuador. CI is the accredited entity (AE), and it will collaborate with Ecuador's National Conservation Agency to provide training on sustainable shrimp production (SSP) and promote climate-smart shrimp (CSS) farming methods, including mangrove restoration on farms. CI will also engage consultants to help small- and medium-sized shrimp farms access private finance.

16. Seed funding will be provided by the Aquaculture Stewardship Council and CI to establish a subaccount within Ecuador's Socio Bosque Fund to support the SM programme for the incentive payment. The project aims to catalyse investment in CSS farming methods, leading to reduced GHG emissions, increased profitability and mangrove restoration.

17. Technical assistance will be provided to shrimp farms for adopting SSP and CSS standards, with a focus on sustainable intensification and mangrove restoration. The project will facilitate partnerships and access to finance for shrimp farms transitioning to sustainable practices, including through the Eco Business Fund. It will also establish agreements with businesses to contribute to mangrove conservation and to finance the SM programme.

18. In addition, the project will disseminate knowledge through various platforms and workshops, develop a marketing strategy for CSS farming, and seek to mobilize private investment into the SM subaccount, supported by funds from the GCF. These efforts aim to strengthen private sector engagement in mangrove conservation and restoration while promoting SSP practices in Ecuador. With successful management, the SM subaccount is expected to create a long-term financial mechanism that attracts private sector investment and provides incentive payments for sustainable mangrove conservation and restoration.

Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies and legal enforcement (total cost: USD 6.2 million; GCF cost: USD 5.5 million)

19. Actions by local communities and the private sector to protect mangroves and maintain flood protection benefits as a nature-based climate change adaptation and mitigation strategy are contingent upon the Government of Ecuador establishing the necessary governance, regulatory and enabling conditions. These conditions are essential for deterring and controlling illegal activities and ensuring that climate considerations are integrated into planning and investment decisions.

20. By providing better climate change and mangrove forest information to decision-makers and stakeholders, particularly focusing on public entities, the project will directly contribute to increasing climate knowledge for decision-making, laying the groundwork for achieving project outcomes related to mangrove management and climate resilience.

21. Improved legal and regulatory capacity, focusing on planning, preparedness and enforcement, will support sustaining mangroves and enhancing awareness and resilience

among beneficiaries, primarily in the eight priority municipalities of the project area. This will directly contribute to strengthening institutional and regulatory systems for climate-responsive planning and development.

22. To achieve these objectives, the project includes several activities under output 3.1, such as monitoring mangrove conditions and socioeconomic impacts, integrating remote sensing and community participation for emissions reduction monitoring, and conducting socioeconomic impact evaluations. Output 3.2 entails harmonizing legal and regulatory frameworks at the local and sectoral level, incorporating climate resilience and mitigation strategies, and enforcing regulations with a focus on human rights.

23. Activities under output 3.2 include supporting local governments to improve planning instruments that integrate climate change adaptation and mangrove management, strengthening regulatory frameworks and law enforcement by relevant agencies, and generating proposals for regulatory changes in various sectors. This will involve analysing legal frameworks, facilitating discussions on legal reforms and providing training to judiciary staff. In addition, the project will develop an online and mobile communications application (app) for reporting deforestation and other impacts on mangroves, thereby enhancing transparency and enforcement efforts.

III. Assessment against investment criteria

3.1 Impact potential

Scale: N/A

24. The project aims to directly benefit 89,600 individuals, with an additional 3.4 million indirect beneficiaries. Furthermore, it targets an estimated GHG emission reduction of 4,609,470 t CO₂ eq over its 20-year duration.

25. The AUSCEM project seeks to address both adaptation and mitigation challenges posed by climate change, particularly in vulnerable coastal regions. By leveraging the protective capacities of mangrove ecosystems, the project endeavours to yield significant benefits for both local communities and the global environment. This project represents a multifaceted approach to addressing climate change impacts at both the local and the global scale, highlighting the crucial role of mangrove ecosystems in fostering resilience and mitigating environmental degradation. The project offers both adaptation and mitigation benefits, focusing on reducing flooding risks and enhancing livelihoods through improved mangrove management.

26. **Adaptation impacts:** Direct adaptation benefits include reduced flooding risk and increased resilience for vulnerable populations, while there are a number of indirect adaptation benefits for selected municipalities with high mangrove cover and populations at risk of flooding, including:

- (a) Potential benefits for 89,600 people in mitigating flood risks, according to flood risk modelling estimates;
- (b) A USD 280 million per year reduction in economic asset losses caused by extreme climate-related disasters;
- (c) Approximately 3.4 million indirect beneficiaries from improved planning processes in mangrove-rich municipalities;
- (d) 41,500 individuals (50 per cent female, 50 per cent male) expected to adopt climate-resilient livelihoods or experience reduced flood risk;
- (e) 156,633 hectares of mangroves protected and strengthened, covering all of Ecuador's mangroves; and

- (f) Restoration efforts covering protected areas, sustainable land-use programmes and shrimp farms.

27. **Mitigation impacts:** Mitigation benefits arise from the decreased GHG emissions resulting from a reduction in mangrove deforestation and an increase in carbon sequestration from restored mangroves, amounting to an estimated GHG emission reduction of 732,111 t CO₂ eq during the implementation period and 4,609,470 t CO₂ eq over the 20-year project lifespan.

3.2 Paradigm shift potential

Scale: N/A

28. This project aims to transform Ecuador's approach to valuing and utilizing critical natural infrastructure, particularly mangrove ecosystems, to protect coastal communities from flooding. By integrating the adaptation and mitigation benefits of mangroves into governance and economic frameworks, the project offers an innovative solution to reducing GHG emissions and enhancing the livelihoods of vulnerable populations. It addresses the current undervaluation of climate risks and ecosystem services provided by mangroves, which is the result of inadequate information, institutional coordination and private sector investment.

29. The project employs a multifaceted strategy, including strengthening public sector frameworks, fostering public-private partnerships and empowering local communities, with a focus on leveraging the "green infrastructure" of mangroves. This approach is cost-effective and scalable, and includes lessons, guidance and tools that can be used for other coastal communities threatened by climate change. Comprehensive monitoring and reporting mechanisms will facilitate learning and knowledge-sharing throughout the project, with mid-term and final evaluations to identify lessons learned and disseminate findings.

30. The project aims to create an enabling environment and policy framework by strengthening regulatory frameworks and fostering dialogue and collaboration across sectors at the national and subnational level. Capacity-building initiatives will empower local communities to participate in innovative business models and to engage in land-use planning processes with local government. The project emphasizes consensus-building among stakeholders through participatory processes, facilitating agreements for the sustainable management of mangroves. Private sector participation will be enhanced through support for land-use planning and the adoption of climate-smart practices, particularly mangrove conservation and restoration.

31. The project focuses on scaling up successful mangrove conservation and restoration activities in Ecuador, addressing the lack of basic land-use planning instruments in the project area. Participatory planning processes will be facilitated, documented and shared with the national government, with recommendations for legal and regulatory adjustments as needed.

3.3 Sustainable development potential

Scale: N/A

32. **Environmental co-benefits:** The project aims to reduce deforestation, enhance restoration and improve management practices across Ecuador's entire mangrove area, which spans 156,633 hectares. In addition to mitigating climate change and reducing flood risks, mangroves offer further ecosystem services; for example, they support biodiversity conservation, acting as habitats for numerous plant and bird species. Mangroves also serve as vital nursing grounds for fisheries, fostering the spawning, breeding and early development of various fish species. Thus mangroves play a significant role in fish production and provide substantial economic value through fisheries-related ecosystem services. The total ecosystem service value of mangroves, including biodiversity and other services, is estimated to be substantial, providing essential benefits to local economies.

33. **Social co-benefits:** In addition to mitigating flood risks, the project will yield various social co-benefits, particularly for mangrove-dependent communities and associations, as follows:

- (a) Improved social cohesion and organizational capacity among 60 associations with AUSCEMs, which will enhance the ability of communities to engage with government stakeholders, resolve internal conflicts and develop additional community activities;
- (b) Enhanced mangrove management and planning are likely to integrate social and environmental considerations, including water management and waste disposal, benefiting human health and well-being; and
- (c) Collaborative efforts to control illegal activities in mangrove areas between community-based associations and government authorities will enhance local security, generating collateral benefits.

34. **Economic co-benefits:** Mangrove conservation and restoration offer a cost-effective solution for climate change adaptation, generating economic benefits for the national and local government, the private sector and communities by avoiding the costs of extreme climate events. The shrimp production sector stands to benefit by accessing new markets through sustainable and CSS production, leading to increased export revenue and employment opportunities.

35. Communities will experience reduced flooding, along with minimized impacts on infrastructure, housing, crops and livelihoods. Diversification of livelihoods and adoption of adaptive fishing techniques will enhance the resilience, economic stability and food security of marginalized coastal fishing communities. Mangrove ecosystems form the foundation for diversified local economies, with artisanal fishing playing a significant role in sustaining livelihoods and contributing to marine fisheries.

36. **Gender-sensitive development impact:** The project will directly contribute to empowering women and other vulnerable groups in economic activities and advancing their participation and leadership in organizations and enterprises, ensuring a gender-sensitive approach to development initiatives.

3.4 Needs of the recipient

Scale: N/A

37. Ecuador has experienced notable economic growth from 2007 and 2018, resulting in a reduction in the levels of poverty and inequality and the ongoing expansion of the middle class. However, the economy's heavy reliance on the oil sector presents a significant vulnerability, particularly evident during periods of low oil prices and the appreciation of the US dollar.

38. With over 3,630 km of coastline, Ecuador's five coastal provinces are home to approximately 54 per cent of the population. These areas face significant risks from sea level rise, saltwater intrusion, storm surges, flooding and storms. Coastal regions of Ecuador are disproportionately affected by poverty, with 45 per cent of the population in the project area living below the extreme poverty line. Unsafe housing conditions persist due to rapid urbanization and persistent poverty.

39. The challenges of coastal development are compounded by the threats of climate change, yet Ecuador lacks an integrated plan to protect mangroves and wetlands from the combined impacts of sea level rise and intensified storms. The project addresses these pressing needs by aligning with Ecuador's national climate change strategy and plan, with the aim of supporting the national and local government, coastal communities and crucial economic sectors such as shrimp farming and fisheries.

3.5 Country ownership

Scale: N/A

40. The project is in alignment with the NDC in terms of contributing to both mitigation and adaptation goals through the aim of protecting mangrove ecosystems. The project is also aligned to other national policies, including the national climate change strategy, the conservation of the national forest estate, and REDD+ efforts. Moreover, the project is linked to the national measurement, reporting and verification and monitoring and evaluation (M&E) systems for NDC implementation; thus the results will be aggregated into the national biennial transparency reports that the national government will submit under the United Nations Framework Convention on Climate Change.

41. In addition, country ownership is firmly established through active government participation in project management structures. Both the Ministry of Environment, Water and Ecological Transition and the Vice-ministry of Aquaculture and Fisheries will be part of the project management advisory committee and the technical committees guiding each project component. This involvement enhances ownership and promotes awareness of the technological solutions proposed by the project, setting the foundation for replication and scalability in the future.

3.6 Efficiency and effectiveness

Scale: N/A

42. The project provides the most value of the scenarios considered over both the project implementation period (6.5 years) and the project lifespan (20 years), and it gives a high benefit-to-cost ratio (8:1). The project is expected to generate USD 299 million of economic value over and above the “without project” scenario over the 20-year project lifespan. As such, the project as designed provides excellent value for money because it will generate significant benefits to society over both the implementation period (6.5 years) and the expected project lifespan (20 years).

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

43. The project aims to achieve GHG emission reductions while providing flood protection and enhancing the livelihoods of the communities through coordinated efforts to protect and restore the mangrove ecosystem in Ecuador. The project interventions are intended to have positive impacts on people and the environment. Any unintended adverse impacts are expected to be minor and site-specific, and can be managed through careful design and standard mitigation measures. The environmental co-benefits due to reduced deforestation, restoration and management of the mangrove ecosystem include enhanced habitats, particularly for bird and fish species, and enhanced biodiversity and ecosystem services. The social co-benefits include reduced flood risks, improved capacity for mangrove management and better human health and well-being.

44. **Environmental and social risk category and safeguards instrument.** The project has been classified as environmental and social (E&S) risk category B. The Secretariat confirms the categorization and that it is within the AE’s E&S risk accreditation level. Since the interventions are several and located in multiple sites involving various communities and development partners, and are not yet identified, the AE has prepared an environmental and social impact assessment and an environmental and social management plan (ESMP) which together identify the potential risks and impacts and possible mitigation measures and set out the procedures for assessing and managing E&S risks and impacts of the individual interventions (subprojects).

45. Compliance with GCF's Environmental and Social Safeguards (ESS) Standards
46. The following paragraphs describe the project's compliance with the GCF interim ESS standards:
47. **ESS1: Assessment and management of environmental and social risks and impacts.** An environmental and social impact assessment and ESMP (included as annex 6 to the funding proposal) provide a summary of the project activities and outcomes and the applicable legal and institutional framework, the biophysical and social environment, the identified potential risks and impacts, the mitigation measures and action plan, and the implementation and monitoring mechanisms by which the project will be implemented. The main risk pertains to potential restriction or loss of access to mangrove resources for some groups or individuals resulting in economic displacement. However, economic displacement, if any, is expected to be minor because the planning and management of mangrove areas will be undertaken by the communities themselves, who are the traditional users and beneficiaries of the resource.
48. **ESS2: Labour and working conditions.** The project will mostly involve the community members themselves implementing the activities under the project. However, those who will be working on the subprojects may be exposed to drowning hazards, poisonous animals and disease vectors. Mitigation measures will include assessing the health and safety risks to workers and developing prevention and emergency response planning for these activities. The project's screening procedure also includes questions regarding whether the subproject has a risk of potentially involving or lead to working conditions that do not meet national or international labour laws and regulations. In this regard, the project will ensure the health and safety of subproject-related workers and protect workers' rights. Further, the project also excludes activities that might include the use of forced labour or child labour.
49. **ESS3: Resource efficiency and pollution prevention.** The project is not expected to contribute to pollution but rather will support farms to reduce the use of materials that could generate and discharge pollutants to the ecosystem. Hence, the project's activities are expected to lessen the pollution caused by the shrimp aquaculture farms and improve their resource efficiency approaches. The project's activities include technical assistance for the development and promotion of CSS aquaculture practices.
50. **ESS4: Community health, safety and security.** The mangrove restoration/reforestation activity of the project may expose community members to physical hazards, poisonous animals and disease vectors, given that mangrove forests are home to some potentially dangerous animals and vectors that may result in diseases such as malaria, dengue, leptospirosis and vibriosis. Community health, safety and security risk assessments will be conducted, and community health, safety and security plans will be implemented on each of the subprojects. The measures that will be implemented include first aid training for all project staff and guidance on how to address communities' safety related concerns.
51. **ESS5: Land acquisition and involuntary resettlement.** The AE has determined that the project would not involve acquisition of land but could potentially result to restriction or loss of access for some groups or individuals to mangrove resources. A process framework for restriction of access to natural resources is included as procedural guidance should community associations put in place restrictions on access to the mangrove resources for both members and non-members of these associations. The framework includes eligibility criteria, measures to assist affected people, and a conflict resolution and complaint mechanism.
52. **ESS6: Biodiversity conservation and sustainable management of living natural resources.** The AE has determined the applicability of ESS6 given that the project will support various forest use and conservation activities that may take place in and/or around protected areas and buffer zones. The project is not expected to result in the introduction of invasive species of flora and fauna, since the project aims to use only native species and enhance the ecosystem services that mangrove forests provide. The project includes mangrove restoration

activities using native and locally sourced species within the areas managed by the AUSCEMs, some of which are within protected areas. Additional restoration activities will also take place on private shrimp farms. The project will sign conservation agreements with the local communities that are to be supported by the project, particularly those that will oversee establishing forestry units and conduct reforestation activities. Assessment of the environmental, social and economic viability of the restoration/reforestation activities will be conducted to ensure that the activities are not only technical feasible but are environmentally appropriate, socially beneficial and economically viable.

53. The application of the GCF Indigenous Peoples Policy takes into consideration Afro-Ecuadorians and cholo artisanal fisher communities who are located within the project area. In addition to mainstreaming their engagement throughout the main funding proposal documents, the Plan for Afro-descendent Communities and Cholos Pescadores provides a specific instrument for ensuring that the rights of these groups are considered and strengthened throughout and beyond the project life cycle. In the preparation of the funding proposal, consultations were undertaken with both Indigenous Peoples and local communities, with outcomes included in the plan. In line with their roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and executing entities (EEs).

54. **ESS8: Cultural heritage.** The project is not expected to implement activities that might affect cultural heritage. As part of its exclusion list, it will also not support activities that will “involve the removal, alteration, or disturbance of any non-replicable or critical cultural heritage, or the use of intangible cultural heritage without the free, prior and informed consent of the communities to whom it belongs.”

55. **Sexual exploitation, abuse and harassment (SEAH).** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from SEAH in GCF-financed activities. The AE provided information on SEAH safeguards during the proposal development and outlined the SEAH risks in the ESMP, as well as providing some SEAH mitigation measures in the gender action plan. The AE is committed to ensuring a safe, trusted and respectful environment within its organization and global offices, as well as in its programmes and projects. This is in alignment with the CI SEAH policy which is applicable to its projects. The project recognizes potential SEAH risks, especially community engagements that will involve working directly with marginalized and vulnerable groups/communities. The AE will ensure that, in cases where SEAH incidents occur (or have a high probability of occurring), the rights and dignity of survivors are upheld in a confidential and gender-responsive manner. The AE has therefore identified potential SEAH risks and mitigation measures in the ESMP.

56. **Implementation arrangements.** The project will have a project steering committee that will provide guidance and support management regarding project implementation. A project management advisory committee will also be established to provide advice on technical matters. CI Ecuador will be the EE for all activities of this project and will be responsible for project execution, management and reporting of project’s activities. A project management unit will also be established and will be responsible for overall coordination, management and implementation of the project on the ground. The project will have dedicated full-time staff, including a safeguards manager and a gender manager among other full-time staff who will be responsible for safeguards and gender-related matters. An estuary field unit will be established in each of the project’s four target estuaries to be directly involved in the daily activities of the local associations.

57. **Stakeholder engagement and information disclosure.** The AE has engaged stakeholders through workshops and targeted meetings at the national and local level. The consultation process involved key project stakeholders such as the men and women of local mangrove associations and other disadvantaged or vulnerable stakeholders. Community organizations managing AUSCEMs were also involved. The issues raised in the consultation process were considered in the development of the project. A stakeholder engagement plan has

been prepared to identify stakeholders, define participation strategies, establish procedures and methodologies, and identify issues and concerns that could affect project implementation.

58. **Grievance redress mechanism (GRM).** The project has described a project-level GRM that will be promoted during the awareness-raising and capacity-building activities of the project. It describes the types of complaints that can be addressed, who can raise a complaint, the contact points at the community level, and the responsible authorities in handling complaints. The GRM also describes the ability for stakeholders to access CI's Ethics Hotline that employees, grantees and other partners and beneficiaries can access to report incidents anonymously. The GRM also allows for stakeholders to access the GCF's Independent Redress Mechanism. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples Focal Point will be available for assistance at any stage, including before a claim has been made.

4.2 Gender policy

59. The AE provided a gender assessment and action plan with the funding proposal, and therefore complies with the requirements of the GCF Gender Policy. The assessment report presents information on the differential impact of and vulnerability to climate change for women and men in the targeted project area. The gender assessment is based on secondary data, as well as consultation workshops that were held between 2016 and 2019. However, the majority of the secondary data is from outdated sources (i.e. 2010–2012). The Secretariat therefore recommends that, during early project implementation, the AE may need to make some revisions to the documents in annex 8 of the funding proposal if there are any major changes in the demographics and/or socioeconomic statuses that may have a bearing on this specific project.

60. The gender action plan for the project outlines various interventions in order to ensure that anticipated gender outcomes are realized. The emphasis is on interventions that will aim at increasing women's participation in ensuring diversified and climate-resilient livelihood options in mangroves, as well as increasing women's leadership, skills building within associations and gender quotas, among other things. The Secretariat recommends and suggests that the AE include the following provisions: (i) engage local women and fisherwomen in participatory project planning and implementation, including utilizing their knowledge of marine and coastal resources in assessments, and their traditional practices in natural resource management; (ii) ensure that women-led businesses have improved financial and market access and profitability, and address barriers within families and households by involving men; (iii) build critical awareness among women and disadvantaged groups on their rights as women and fisherfolk; and (iv) define common goals and develop strategic alliances between women's groups and other organizations/associations for more effective advocacy for policy change. Gender balance should be considered when forming stakeholder groups; this may involve including female minority leaders and managers to encourage participation and capacity-building. Also, the AE needs to ensure that financial leakage is eliminated and that financial resources actually reach the needy, poor, vulnerable and disadvantaged communities including women and girls.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

61. The project aims to coordinate efforts across the public, private and community sectors to protect and restore mangroves in Ecuador. This initiative plans to reduce mangrove loss, resulting in a reduction in GHG emissions amounting to 4.6 Mt CO₂ eq over the project lifespan. It will also accelerate mangrove recovery to enhance flood protection for 89,600 people and

improve the livelihoods of 41,500 people. The project activities are structured around three main components: (i) reducing flood risks and increasing resilience while also reducing GHG emissions and increasing sequestration; (ii) engaging the private sector in emission reductions and providing financial support for mangrove conservation; and (iii) creating conditions to sustain reductions in mangrove deforestation and increase restoration efforts by strengthening governance and coastal management policies. The project, with a total cost of USD 45.93 million, will be funded by USD 36.4 million from the GCF as grant, with additional co-financing totalling USD 9.53 million (grants and in-kind) for a co-financing ratio of 1:0.26.

4.3.2. **Accredited entity/executing entity capability to execute the current programme (low risk)**

62. CI, through its CI-GCF agency, will serve as the AE with CI Ecuador overseeing project execution. With a track record in conservation spanning three decades and covering Galapagos, marine/coastal regions, Andes Choco and the Amazon, CI has a history of supporting the creation and management of protected areas and collaborating with artisanal fisher associations for sustainable resource management. GCF has collaborated with CI through previous projects (i.e. FP158 and FP026) in addressing mitigation, adaptation and sustainable livelihoods in Africa.

4.3.3. **Programme-specific execution risks (medium risk)**

63. **Security risk:** The Secretariat notes that there are reports of threats from criminal gangs engaging in violent crimes and illicit activities across Ecuador. The coastal areas, particularly Guayaquil, are reportedly entangled in the cocaine trade, with drug traffickers exploiting weak law enforcement and port vulnerabilities to smuggle drugs, exacerbating security concerns in the country, potentially disrupting project implementation. The AE also acknowledges these challenges posed by criminal gangs operating in remote mangrove areas and the exploitation of Ecuador's coastal areas for cocaine trafficking, underscoring potential impacts on project activities. In its mitigation plan, the AE has developed a security plan (annex 6, ESMP appendix 11), and CI's Safety and Security Director will conduct a security assessment and develop a mitigation plan during project startup. The mitigation measures will include pre-activity community engagement, hostile environment awareness training and ensuring easy identification of CI's staff.

64. **Political instability:** The political landscape in Ecuador has been characterized by instability in the past and remains unstable following the October 2023 election. Newly elected President Noboa was voted into office without a majority in the National Assembly. To secure political stability, the President formed a coalition with two other political parties in the radical left and conservatives. Nonetheless, CI's 20 years of experience operating in Ecuador's political context provide a robust foundation for implementing mitigation measures. CI has strong relationships with local and provincial government officials, and it is expected that this will help to deliver proactive engagement with various stakeholders, including local communities, project partners and relevant authorities, to help address political challenges.

65. **Project viability (and concessionality):** The use of grants (i.e. 100 per cent concessionality) is justified given the country context where public and private funding for mangrove conservation and restoration efforts is limited and insufficient. Vulnerable populations in Ecuador are directly exposed to heightened flood risks exacerbated by climate change, and they lack adaptive capacity due to poverty. Grant funding is therefore warranted to create the enabling conditions for sustaining capacity and incentivizing private sector investments in mangroves, ensuring the integration of climate resilience measures and fostering long-term sustainability.

4.3.4. Compliance risk (medium risk)

66. Ecuador is not currently subject to United Nations Security Council resolutions, nor is it currently identified on the Financial Action Task Force lists of high-risk jurisdictions or jurisdictions under increased monitoring relating to anti-money laundering (AML) and countering the financing of terrorism (CFT).

67. The funding proposal noted a risk of criminal gangs operating in project areas and rated the probability as “medium”. The mitigating measures described in the security plan include staff training, mapping of high-risk areas, and a country threat and vulnerability assessment, as well as cooperation with local partners and authorities. The funding proposal also noted that project areas are located along a drug shipping route. The AE provided further information on its AML/CFT controls in annex 20 to the funding proposal and confirmed that its AML/CFT policies include sanctions screening of subgrantees and EE (CI Ecuador) employees.

68. Furthermore, as the political and security situation in Ecuador has evolved since the proposal was first submitted, the AE confirmed that the security plan has been updated accordingly, with additional security costs including in the budget.

69. Given the aforementioned risks, as well as the ongoing uncertainty within Ecuador, this project is deemed to be medium risk from a compliance perspective.

4.3.5. GCF portfolio concentration risk (low risk)

70. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is immaterial.

4.3.6. Recommendation

71. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	The funding proposal has an overall risk assessment of Medium based on the issues highlighted. It is recommended that the Board consider the above factors in its decision.
Accredited entity (AE)/executing entity (EE) capability	Low	
Project-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

72. CI, through its CI-GCF agency, will serve as the AE for the project. CI-GCF will also be responsible for providing support and guidance to the EE; monitoring project results and outputs; reporting to the GCF; and project closure and evaluation.

73. CI will self-execute this project, acting through its country office in Ecuador (referred to in this proposal as “CI” or “CI Ecuador”), which will be the EE for all project activities.

74. Therefore, financial management and procurement for this project will follow relevant CI policies and procedures, as well as relevant provisions in the accreditation master agreement (AMA) between CI and the GCF.

75. An annual audit of the project will be conducted by an independent external audit firm in accordance with audit standards under United States Generally Accepted Accounting Principles and provided to the GCF in accordance with CI’s AMA.

76. Disbursements made to CI will be recorded to the ledger account for this GCF project, which can be tracked using the Business World system. Disbursements from the EE to subgrantees will be made upon CI's receipt of acceptable progress and financial requirements as included in the applicable grant agreement.

77. All procurement for this project must follow the procurement process standards defined in CI's procurement policy, which requires that all procurement of goods or services ensures the best value and complies with all donor funding terms and conditions, and that all goods will be used only for their intended purposes and consistently with CI's code of ethics and suite of anti-fraud policies. CI's procurement standards are in line with the GCF fiduciary standards.

4.5 Results monitoring and reporting

78. As a cross-cutting project, the funding proposal aims to achieve a combination of mitigation and adaptation outcomes spanning three GCF results areas: forestry and land use for mitigation; most vulnerable people and communities, and ecosystems and ecosystems services for adaptation. The expected mitigation outcome is an estimated GHG emission reduction of 4,609,470 t CO₂ eq over the project's 20-year lifespan and 732,111 t CO₂ eq by the end of the implementation period. The expected adaptation beneficiaries amount to 89,600 and 3.4 million direct and indirect beneficiaries respectively. The AE submitted the necessary annex to demonstrate the calculation methods used for these estimates.

79. The theory of change has been improved and refined to clearly articulate the results chain that connects the proposed activities, outputs, outcomes and impact. It covers pathways that will lead to the expected climate outcomes and co-benefits.

80. The logical framework is consistent with the requirements of the Integrated Results Management Framework. The project is expected to realize its paradigm shift potential through a combination of building community knowledge and resources for mangrove management and livelihoods, strengthening the enabling environment for mangrove management, and strengthening collaboration between the private sector and government on mangrove protection and restoration. Core indicator 2 (adaptation beneficiaries), supplementary indicator 2.1 (adoption of climate-resilient livelihood options), supplementary indicator 3.1 (change in expected losses), core indicator 4 (natural resources brought under improved management practices), supplementary indicator 4.1 (natural resources restored) and core indicator 1 (GHG emissions reduced) have all been incorporated into the logical framework and relevant means of verification, targets and assumptions have been specified.

81. The M&E plan, which includes indicator-specific approaches for data collection and data sources, and an evaluation plan, are well provided for in the project budget and AE fees respectively. The budget allocation for the M&E activities is consistent with the requirements of the GCF Evaluation Policy.

4.6 Legal assessment

82. The AMA was signed with the AE on 13 July 2017, and was amended and restated on 15 August 2022, which became effective on 17 August 2022, for the second accreditation term.

83. The AE has provided a legal certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the project pursuant to clause 4.13 of the AMA.

84. The proposed project will be implemented in Ecuador, a country in which GCF is not provided with privileges and immunities. This means that, among other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further

assessed. The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Ecuador on 23 February 2023.

85. It is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP235

Proposal name:	Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador
Accredited entity:	Conservation International Foundation
Country:	Ecuador
Project/programme size:	Small

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

- The Mangroves for climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador project is a cross-cutting project aimed at strengthening the resilience of Ecuador's coastal areas to the negative impacts of climate change, while also significantly reducing greenhouse gas (GHG) emissions. Over an implementation period of six-and-a-half years, this small-sized project will bring together the public, private, and community sectors to focus on the conservation and restoration of mangrove ecosystems, which are crucial for coastal protection, biodiversity, and local livelihoods. This project meets the urgent need for climate adaptation and mitigation in regions prone to extreme weather and rising sea levels, providing direct benefits to Ecuador's most vulnerable coastal communities.
- The project interventions are centered around three main components:
 - Community empowerment:** Enhance the management and restoration of mangroves through community-led initiatives, thereby improving local livelihoods and ensuring sustainable ecosystem services;
 - Private-sector engagement:** Mobilize the shrimp farming sector to adopt and finance climate-smart practices that align with mangrove conservation efforts; and
 - Create and enabling environment:** Develop and enforce policies that integrate mangrove conservation into broader climate strategies, supported by improved governance and legal frameworks.
- The execution of the proposed project is expected to i) protect and manage 156,633 ha of mangrove ecosystems; ii) reduce GHG emissions by 4.6 million tonnes of carbon dioxide equivalent (Mt CO₂ eq) over its 20-year lifespan; and iii) enhance the climate resilience of 89,600 people by reducing flood risks through ecosystem-based adaptations.
- The total budget for the proposed project is USD 45.9 million. A substantial portion of this budget, USD 36.4 million, is requested as a grant from GCF. Co-financing of USD 9.5 million will be provided by Ecuador's Ministry of the Environment, Water and Ecological Transition (Ministerio del Ambiente, Agua y Transición del Ecuador (MAATE)), and Conservation International (CI) in the form of grants and in-kind contributions.
- This project is vital not only for its environmental, economic, and social benefits but also as a model for integrated, multisectoral climate action. It aligns with the national climate strategies of Ecuador and supports global efforts under the United Nations Framework Convention on Climate Change and its Paris Agreement.

6. With CI as the accredited entity (AE) and executing entity (EE), the proposed project will be implemented through a structured management approach involving a steering committee, a management unit, and localized implementation teams, ensuring efficient delivery and adherence to the planned outcomes.

1.2 Impact potential

Scale: High

7. The project aims to significantly reduce vulnerability for the 89,600 direct and 3.4 million indirect project beneficiaries, which represents about 1.8 per cent and 20.1 per cent of Ecuador's coastal and total populations, respectively. This wide reach focuses on the most vulnerable groups, particularly in impoverished rural coastal areas. However, detailing strategies to ensure that the project benefits reach the most marginalized people within these groups, including tailored measures for varying vulnerability levels, could have enhanced the project's adaptation impact.

8. From a climate-change mitigation perspective, the proposed project can reduce 4.6 Mt CO₂ eq over its 20-year lifespan, primarily through the restoration and conservation of mangrove forests, which are significant carbon sinks.

9. By prioritizing natural mangrove barriers, the proposed project avoids the long-term liabilities of climate-change-vulnerable constructions. Mangroves' inherent resilience and flood mitigation capabilities boost the adaptive capacity of the exposed communities. The continuation of activities initiated by the proposed project could integrate hybrid approaches, combining green and grey infrastructure where natural solutions might fall short.

10. The proposed project directly causes GHG emission reductions from mangrove deforestation and carbon enhancement through restoration. Both activities combined aim to result in a mitigation impact of 732,11 t CO₂ eq during the project's six-and-a-half year implementation and 4.6 Mt CO₂ eq during its 20-year lifespan. The proposal includes a clear project method and details the calculations behind the proposed project, clarifies measurement, reporting and verification (MRV) at the project scale as well as how it is linked to the national MRV system.

11. Furthermore, the proposed project indirectly supports low-emission practices by promoting sustainable shrimp-farming techniques within the private sector.

12. The proposed project will bolster institutional frameworks and regulatory systems by embedding climate resilience in local and national governance. This involves improving the community management of mangroves and supporting legal frameworks that encourage private-sector conservation efforts.

13. The proposed project promotes the generation and use of climate information in decision-making by partnering with scientific entities like the International Research Centre on El Niño Southern Oscillation. However, establishing robust feedback mechanisms to adapt and update practices based on the latest climate data could have further solidified this aspect.

14. By emphasizing community-based management and partnerships, the project raises awareness and adaptive capacity among local populations. Educating communities on climate threats and sustainable practices is key to the project's strategy, empowering stakeholders to actively engage in conservation and adaptation efforts.

15. Overall, the independent Technical Advisory Panel (iTAP) considers the impact potential for both adaptation and mitigation of the proposed project to be high.

1.3 Paradigm shift potential

Scale: Medium to High

16. The proposed project advocates a collaborative approach, engaging public, private, and community partnerships in mangrove management and restoration. By integrating the adaptation and mitigation benefits of mangrove ecosystems into economic and governance frameworks, the proposed project will create new market opportunities, especially in the shrimp aquaculture sector, to promote sustainable practices.
17. The project's theory of change is coherent and well-structured, focusing on increasing mangrove coverage for carbon sequestration and enhanced coastal resilience, directly supporting global low-carbon development pathways.
18. Furthermore, the iTAP welcomes the inclusion in activity 3.1.1. of an explicit system for MRV of mangrove cover and carbon stocks, including remote sensing, community participation and making subgrants to competitively selected universities or non-governmental organizations to collaboratively generate geospatial data and capacity that will be integrated both into the project monitoring and evaluation (M&E) plan and the national REDD+ MRV systems. This enhanced capacity should form an important building block for any future private and/or public sector efforts to mobilize resources based on mangrove carbon sequestration.
19. The M&E plan targets the ecological, social, and economic impacts of the proposed project. However, the documentation (the funding proposal and its annex 11) lacks extensive details on this plan. Improving this aspect, notably establishing a clear and fit-for-purpose feedback mechanism, could better ensure that lessons learned are captured and effectively disseminated, promoting adaptive management and replication in other contexts.
20. The proposed project involves creating a USD 4 million endowment in the Socio Manglar subaccount (a new financing window under the Socio Bosque Fund), securing long-term financial support for mangrove conservation. This strategic financial planning ensures the continuation of project benefits beyond the funding period. However, ongoing contributions and financial sustainability will require continuous reinforcement.
21. The proposed project promotes new business activities and markets, particularly in climate-smart shrimp farming. The potential for market creation could have been further expanded by exploring additional ecosystem services provided by mangroves that could be monetized (if the legal and regulatory framework in Ecuador allows it). By incorporating mangrove conservation into shrimp-farming practices, the project alters incentives for market participants by decreasing the environmental impact of shrimp farming, a major contributor to mangrove loss. Increasing awareness and highlighting the benefits of these changes among the broader industry could further shift market incentives. In addition, greater diversification of economic activities could also offer communities greater flexibility, which would not only help reduce their dependence on climate-sensitive sectors, but would also increase their resilience to adverse climatic conditions, including extreme events such as intense rainfall.
22. The proposed project will address barriers such as the lack of coordinated management and inadequate enforcement of environmental regulations. However, barriers such as those relating to communities' legal rights seem to be beyond the project's reach. Furthermore, long-term financing depends to some extent on private-sector participation, and the project's strategy for achieving this is lacking in detail.
23. Overall, the iTAP rates the paradigm shift potential of the proposed project as medium to high.

1.4 Sustainable development potential

Scale: High

24. The proposed project demonstrates significant potential for improving environmental quality and conservation. It emphasizes the restoration and preservation of mangrove ecosystems, which are vital for carbon sequestration, biodiversity conservation, and acting as

natural defences against coastal erosion and flooding. This aligns with Sustainable Development Goal (SDG) 13 (Climate action) and 15 (Life on land). The project aims to reduce 4.6 Mt CO₂ eq over its duration, thereby enhancing air quality and soil stability through increased mangrove coverage. Given that the impacts of the project's planned activities on air quality are less well documented, the M&E plan could incorporate this aspect.

25. The proposed project is designed to directly benefit around 41,500 individuals by supporting climate-resilient livelihoods through improved fisheries management and other ecosystem services provided by mangroves. It places special emphasis on gender-specific impacts, aiming to empower women by improving their access to resources and decision-making processes, aligning with SDG 5 (Gender equality). However, ongoing efforts in gender-focused capacity-building and ensuring equitable access to benefits are crucial.

26. This project also promotes economic externalities by creating jobs, enhancing local industries (especially fisheries and ecotourism), and encouraging sustainable practices such as climate-smart shrimp farming. It is expected to bolster local and national economic resilience, aligning with SDG 8 (Decent work and economic growth). By integrating local industries with conservation efforts, the project aims to establish a sustainable development model that can attract private investment, thereby enhancing the financial sustainability of conservation initiatives. Achieving long-term economic benefits will require strong partnerships and ongoing engagement with the private sector. Furthermore, to foster innovation and skills development within the local job market, the proposed project could have also considered strengthening frameworks for collaboration between industry and academia.

27. The proposed project caters to the needs of both women and men by promoting gender equality through inclusive participation in activities and decision-making processes. It places special emphasis on empowering women in the community, acknowledging their role in the economy, and in managing natural resources. Interventions are specifically designed to reduce women's vulnerability to climate-change impacts by improving their access to resources and information. However, the proposed project could also have developed more tailored interventions that address the specific vulnerabilities of women in coastal communities, potentially by expanding gender-focused educational and economic opportunities.

28. The proposed project effectively addresses multiple facets of sustainable development, including environmental protection, social inclusion, and economic growth, with a particular focus on gender equality. To fully realize these potentials, continuous monitoring and adaptive management, robust local participation, and sustained funding and technical support will be essential.

29. Overall, the iTAP rates the sustainable development potential of the proposed project as high.

1.5 Needs of the recipient

Scale: High

30. Ecuador's coastal region, especially its extensive mangrove ecosystems, faces significant climate hazards such as sea-level rise, extreme El Niño events, and increased precipitation variability. These factors amplify coastal flooding risks, endangering local communities and economic activities that rely heavily on fisheries and aquaculture. This vulnerability is intensified by the ongoing deforestation of mangroves, which are crucial for flood control and erosion prevention. The project's focus on restoring and conserving mangrove areas directly addresses these vulnerabilities and risks.

31. The proposed project focuses on areas that encompass a significant portion of Ecuador's coastal population, which is directly threatened by climate-induced flooding. Approximately 3.4 million people in coastal provinces are indirectly affected by these climate risks, with substantial assets at stake, including the shrimp aquaculture industry, a major economic driver

of the targeted coastal communities. Thus, the project's interventions are essential in protecting these populations and economic resources.

32. The project identifies and aims to support groups particularly vulnerable to climate impacts, integrating strategies that address their needs into the broader national climate and development frameworks. However, the effectiveness of these strategies should be closely evaluated to ensure that they result in tangible benefits.

33. The targeted regions experience high poverty rates and limited economic diversification, making them more vulnerable to climate impacts. By enhancing mangrove ecosystems that support fisheries and local economies, the proposed project promotes sustainable livelihoods and economic development, thereby reducing the socioeconomic vulnerability of these communities.

34. The project is well-aligned with the needs of Ecuador's climate-vulnerable coastal populations, addressing both immediate and long-term climate change related risks. Its comprehensive approach to enhancing both human and ecological resilience is promising.

35. Therefore, the iTAP evaluates the need of the recipient for the proposed project as high.

1.6 Country ownership

Scale: High

36. The proposed project directly supports Ecuador's objectives for low-emission and climate-resilient development as outlined in its national adaptation plan and nationally determined contributions. It aims to help achieve Ecuador's nationally determined contribution targets by reducing GHG emissions and promoting community-managed mangrove conservation through the Agreements for Sustainable Use and Custody of the Mangrove Ecosystem.

37. The Organic Environmental Code of Ecuador and local regulations offer a solid policy foundation for the proposed project. However, these policies require strengthening to ensure effective mangrove conservation and restoration. The project aims to foster a supportive environment for reducing mangrove deforestation and enhancing restoration through improved governance and climate adaptation strategies.

38. CI has extensive experience in managing climate-change initiatives. Alongside local partners, they have participated in previous mangrove conservation efforts, further enhancing their credibility and capability. The proposed project offsets a new opportunity to enhance local capacities further to ensure long-term sustainability and scalability of mangrove conservation and restoration initiatives in the country.

39. The project development process included consultations with civil society, local communities, the private sector, and other stakeholders, with a focus on gender equality and sustainable livelihoods. It intends to maintain engagement with these groups throughout the project, adhering to environmental and social safeguards. However, creating more explicit mechanisms for continuous stakeholder input and feedback during implementation could improve adaptiveness and inclusiveness.

40. The project engages in in-country institutions such as MAATE, along with local governments and community organizations, fostering local ownership and accountability. Strengthening these local institutions to effectively manage increased responsibilities will be crucial for successful project outcomes.

41. The proposed project is designed to complement and synergize with other national and international initiatives. This ensures that the project's activities are integrated and contribute to broader conservation and climate-resilience efforts.

42. The proposed project showcases strong country ownership by aligning with national priorities, receiving institutional support, engaging stakeholders, and integrating with national systems. The track record of the AE/executing entity further reinforces this assessment.

43. Overall, the iTAP considers the country ownership of the proposed project to be high.

1.7 Efficiency and effectiveness

Scale: Medium to High

44. The proposed financial structure, with a USD 36.4 million GCF grant and total funding of USD 45.9 million (co-financing ratio is 1:0.21), aims to tackle major obstacles like insufficient financing for long-term mangrove conservation and community engagement. However, the heavy dependence on grant funding indicates a need for a more diversified financial approach to ensure sustainability after the project's completion.

45. The project's reliance on grants reflects high concessionality, which is essential due to its focus on vulnerable communities and ecosystem services that offer public benefits. However, the proposed project could have more clearly outlined the steps taken to minimize the level of concessionality needed to ensure viability.

46. The proposed project is designed to leverage additional private and public investment rather than replace it. Specifically, it seeks to attract private-sector investment in sustainable shrimp farming and mangrove conservation. However, it would benefit from clearer metrics on the expected investment amounts and sources.

47. The project expects to reduce 4.6 Mt CO₂ eq over its lifetime, with an investment cost of roughly USD 10 per t CO₂ eq based on total funding. To fully evaluate its relative efficiency, this cost-effectiveness could have been more explicitly compared to similar projects.

48. The proposed project shows a commitment to long-term financial sustainability and technological integration, particularly through the use of the best available technologies and practices in mangrove restoration and shrimp farming. Furthermore, the M&E methodology on which MAATE, universities and community monitors will be trained goes beyond measuring the area replanted with propagules, and the survival rate of propagules, to include the complex hydrological dynamics of mangrove ecosystems as a contributor to mangrove losses and gains. This reflects global best practice in mangrove monitoring, and will hopefully be shared by CI with other AEs overseeing GCF projects involving mangrove restoration. However, providing more detail on financial planning for the post-GCF phases would offer stronger assurance of long-term viability.

49. Overall, the iTAP assesses the efficiency and effectiveness of the proposed project as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

50. The proposed project is well-structured with a solid theory of change to ensure long-term results and impact. In order to improve it, the iTAP suggests that the AE takes into account the following aspects:

- (a) **Diversification of financial strategies for long-term sustainability:** While the project's dependence on grant funding is understandable given its focus, it is crucial to develop a more diversified financial approach. This could involve establishing partnerships with private-sector investors, and exploring other sustainable financing mechanisms. Providing clearer metrics on expected investment amounts and sources will strengthen financial planning and ensure the project's viability beyond its initial funding period;

-
- (b) **Promotion of economic diversification to enhance long-term resilience and mitigation impact:** To further enhance the project's impact and financial viability, it is essential to promote economic diversification within the targeted regions. This can be accomplished by developing and supporting alternative livelihood opportunities beyond fisheries and shrimp farming. By fostering a more diversified local economy, the proposed project can reduce communities' dependence on climate-sensitive sectors and increase their resilience to adverse climatic conditions. Additionally, providing training and resources for these alternative economic activities can help create more stable and sustainable income sources for local populations; and
- (c) **Integration of comprehensive monitoring, evaluation and learning plans:** Establishing a clear and fit-for-purpose feedback mechanism within the M&E framework will help capture lessons learned and disseminate them effectively. This will promote adaptive management, enabling the project to adjust strategies as needed and enhance replicability in other regions and sectors.
51. Based on the analyses presented in the above paragraphs, the iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP235)

Proposal name:	Mangroves for Climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador
Accredited entity:	Conservation International Foundation (CI)
Country:	Ecuador
Project/programme size:	Small

Impact potential

iTAP evaluation: High

CI notes iTAP's positive assessment with thanks.

CI acknowledges that further work can be done during implementation to ensure that the project benefits for coastal communities planned under Component 1 reach the most vulnerable people within the target communities. The project design allows for this flexibility especially through the activities planned under Project Output 1.2 "Improved livelihood activities and more economically productive community businesses".

Paradigm shift potential

iTAP evaluation: Medium to High

CI agrees with the importance placed on the long-term financial sustainability of the project activities to ensure long-term benefits. This particularly applies to continuing to support community efforts to manage mangroves. Activity 2.2.1 is designed to address this need by engaging with the private sector to increase the support companies operating in Ecuador provide for mangrove conservation and restoration.

Sustainable development potential

iTAP evaluation: High

CI agrees with the need for gender-focused capacity-building and ensuring equitable access to project benefits. These issues are highlighted in Annex 8, the Gender Assessment and Action Plan. The Plan highlights how activities are designed to ensure that women and men can equitably benefit from community-based management and local economic development initiatives.

CI also notes the suggestion to strengthen collaboration between industry and academia. Activity 2.1.1, on the promotion of climate-smart aquaculture, includes a sub-activity to disseminate knowledge gathered by the project with partners, including in academia. CI will examine ways of strengthening this sub-activity during implementation.

Needs of the recipient
iTAP evaluation: High CI notes iTAP's positive assessment with thanks. CI notes the recommendation to evaluate the effectiveness of strategies to address the needs of the most vulnerable to climate change. Activity 3.1.1. includes a sub-activity to evaluate the socio-economic impact of project activities.
Country ownership
iTAP evaluation: High CI notes iTAP's positive assessment with thanks. CI acknowledges the recommendation to create explicit mechanisms for continuous stakeholder input and feedback during implementation to improve adaptiveness and inclusiveness. Annex 7 includes the Stakeholder Engagement Plan, which explains how stakeholders will be engaged in each activity during project implementation. CI will examine ways of further strengthening mechanisms for stakeholder input and feedback during project implementation.
Efficiency and effectiveness
iTAP evaluation: Medium to High CI acknowledges the challenge of ensuring long-term financial sustainability to bring together the public, private and community sectors to secure environmental services provided by mangroves. Public funding for protection of mangroves is a strategic investment that provides substantial environmental, social and economic returns. CI agrees that diversified financing of mangrove protection is needed and has built this into the project design. The activities of Component 2 seek to 1) integrate mangrove conservation and restoration as standard practice in shrimp aquaculture operations, and 2) attract private-sector investment for community management of mangroves.
Overall remarks from the independent Technical Advisory Panel: CI thanks the iTAP for the thorough review of the Funding Proposal package, the recommendations and the overall positive assessment. The assessment document provides an excellent summary of the project as it relates to the GCF's six assessment criteria. We welcome the recommendation to approve this Funding Proposal. CI especially notes the different aspects identified for strengthening the project, including the three main points highlighted in the "overall remarks" section. We are confident that the



project design allows for all the points that have been raised to be addressed during project implementation.

Mangroves for Climate: Public, Private and Community Partnerships for Mitigation and Adaptation in Ecuador

ANNEX 8: Gender Assessment Report and Gender Action Plan

*Prepared by Conservation International and Saraswati Rodriguez, Asesoramiento
Ambiental Estratégico (AAE) through a PPF grant from the Green Climate Fund*

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Acronyms and abbreviations

AFD: French Development Agency

FFLA: Fundación Futuro Latinoamericano (Latin American Future Foundation)

GCF: Green Climate Fund

GGCA: Global Gender and Climate Alliance

GIZ: German Corporation for International Cooperation

INEC: National Institute of Statistics and Censuses

MAATE: Ministry of Environment, Water and Ecological Transition of Ecuador

NDC: Nationally determined contribution

NGO: Non-governmental organization

UNDP: United Nations Development Programme

UTPL: Universidad Técnica Particular de Loja

USFQ: Universidad San Francisco de Quito

UIDE: Universidad Internacional del Ecuador

UASB: Universidad Andina Simón Bolívar

IUCN: International Union for Conservation of Nature

WEDO: Women's Environment and Development Organization

Executive summary

The mangroves along Ecuador's coast provide important economic services to local communities including food, income sources, and protection from flooding and storms. Across all project sites, women and men depend on the natural resources they obtain from the mangroves.

A complex set of historical, social, cultural, and economic factors influence the project sites' current gender norms and inequalities: Firstly, there is a clear lack of political representation by women in decision-making fora. Secondly, women are overburdened with work, especially reproductive labor, and this limits their ability to engage in community, organizational, and project activities. And thirdly, gender-based violence is high and further highlights the inequalities at the household level and an important additional barrier that keep women from using their voice and agency. This situation is compounded by a number of structural problems associated with education, health, the provision of utilities and limited employment opportunities.

In these mangrove-dependent communities, women are more vulnerable and experience higher rates of poverty than men. Men tend to control the main economic activity in the communities: the harvesting and sale of seafood. Often it is men who hold formal positions of leadership within communities and fishing associations. As Trevino & Murillo-Sandoval (2021) highlight in a study of mangroves in Esmeraldas province, the gendered use of mangrove resources, and societal gender norms, influence the strategies that men and women use to respond to mangrove deforestation in the face of growing shrimp aquaculture, with women being disproportionately affected by the loss of mangrove forests.

There are distinct gender differences with respect to mangrove use and management in the northern project sites versus those in the south. In the north, women are able to walk (often collectively) to the mangroves together and are therefore more actively engaged in mangrove use and management. Mangrove associations in the north have, on average, 60% women members. Men have a dual role, they work in crab and black cockle extraction as well as in fisheries outside the mangrove area and in the estuary. Engagement in decision-making within the fishing associations in the southern project sites is more male dominated with women engaging much less. In the south, women's difficulty in participating in decision-making derives from the fact they are not familiar with the mangrove or legally recognized as part of the communities or productive organizations. This can be attributed to their access and use of the mangrove areas which are far from the community and must be accessed by boat. Usually, women's mangrove activities in the south are more individual (each person arranges their own access to the area). Mangrove associations in the south have, on average, 15% women members.

Women living in mangroves work long days, which include family caregiving responsibilities, harvesting or fishing activities, cooking for the family and looking after the children. The average working day for women is around 14 hours and little societal value is placed on their caregiving or reproductive tasks. By contrast, men have much shorter days, fishing or harvesting for between four and five hours. This means that once their day's fishing or harvesting is over, they are relieved from work.

Rates of gender-based violence are high (67% in project sites) and could be perpetuated/increased by climate change impacts that result in economic disruption.

As a result of gender norms, men and women are impacted differently by climate related events. Rising sea levels and flooding – the two major effects cited in justifying climate action in these project sites – can also have consequences like an increase in water salinity and water-borne diseases, which are likely to affect primarily women, who tend to be responsible for collecting water for the household and caring for sick family members. Likewise, men (in their primary role as fishers and farmers) experience climate impacts such as more unpredictable weather and changing fish populations, which require farther (more dangerous) travel). Furthermore, existing gender inequalities may be exacerbated by the dynamics of climate change in the project area.

Our analysis included extensive review of relevant secondary information related to gendered uses of mangroves, the policy environment that supports gender equality in Ecuador, and statistical information for the project sites. This research was complemented with primary information from project sites gathered through surveys (involving 80 men/19 women), community workshops (involving 223 men/40 women) and semi-structured key informant interviews (involving 5 men/2 women).

The analysis points to several key areas of intervention for the project, as laid out in the project's gender action plan. These include mangrove use agreements that integrate gender considerations, tailored and specific support to women in mangrove associations (including the two women's associations), setting targets (50/50) for women's and men's benefit from the adoption of diversified and climate resilient livelihoods options linked to mangroves, and ensuring that project staff and delivery partners are trained on gender equity principles and the importance of mitigating gender inequalities in the project, including gender-based violence.

1. Introduction

1.1 Project background

Climate change will result in significant impacts on local environmental conditions along the coast of Ecuador, including a rise in sea levels, El Niño events, and changes in the intensity and variability of rainfall, flooding, and atmospheric temperatures. Based on models generated by NASA, climate change could cause sea levels to rise by up to 27 cm by 2100 (NASA-GISS, n.d.). Considering that the 1997-1998 El Niño event led to a 42 cm increase in sea levels, periods of heavier rainfall are expected to worsen the rise in sea levels.

The risk of flooding is a major concern for the coast of Ecuador. The Dartmouth Flood Observatory has recorded 22 major floods since 1987, displacing over 506,000 people (Dartmouth Flood Observatory, n.d.). Based on current assessments, 39% of areas within 5 km of the Ecuadorian coast are already at high risk of annual flooding (Secretaría Nacional de Gestión de Riesgos, 2018).

Ninety-nine percent (99%) of mangroves are concentrated along the shores of the country's 4 main estuaries: Cayapas-Mataje, Muisne, Guayas, and Jambelí. These areas are home to 2.9 million people and have very high poverty rates. These people rely heavily on fishing and harvesting crabs and black clams.

There are 4,556 registered members of mangrove management associations devoted almost exclusively to these activities (Ministry of Environment of Ecuador, 2020). These management associations cover approximately 43% of Ecuador's mangrove forests, meaning that the true number of people who fish and gather seafood from the mangroves may be at least double the number of registered members of associations (Zambrano, N., personal communication). With 4.04 family members dependent on each fisher (INEC, 2010), the total population directly dependent on fishing in mangroves is likely to be around 43,000.

Despite their importance, almost 25% of Ecuador's mangroves have been deforested since 1969 (CLIRSEN, 1999, 2007; Ministry of Environment of Ecuador, 2019). In 2018, there were 156,633 ha of mangrove remaining in mainland Ecuador. While the introduction of new regulations requiring shrimp farms to restore mangrove areas in order to operate and export legally has contributed to an increase in net reforestation, there continues to be significant gross deforestation.

The ongoing mangrove deforestation in these areas is the result of a combination of (i) an increase in illegal mangrove logging, (ii) poor enforcement of environmental regulations that penalize causing damage to mangroves, (iii) pollution from liquid and solid waste, and (iv) poor spatial planning by local governments for mangrove conservation (Carvajal & Santillán, 2019).

Given their importance, and in the face of future impacts from climate change, the government of Ecuador has made it a priority to reduce the risks faced by coastal communities and promote the ecosystem services provided by mangroves in its National Climate Change Strategy 2012 – 2025, its National Climate Change Plan 2015-2018, and other national frameworks.

In this context, this project seeks to build resilience and a coordinated adaptive capacity along the entire coast and in the sectors most vulnerable to climate change, in particular coastal communities and the fishing/shrimp sectors. The project has 3 components:

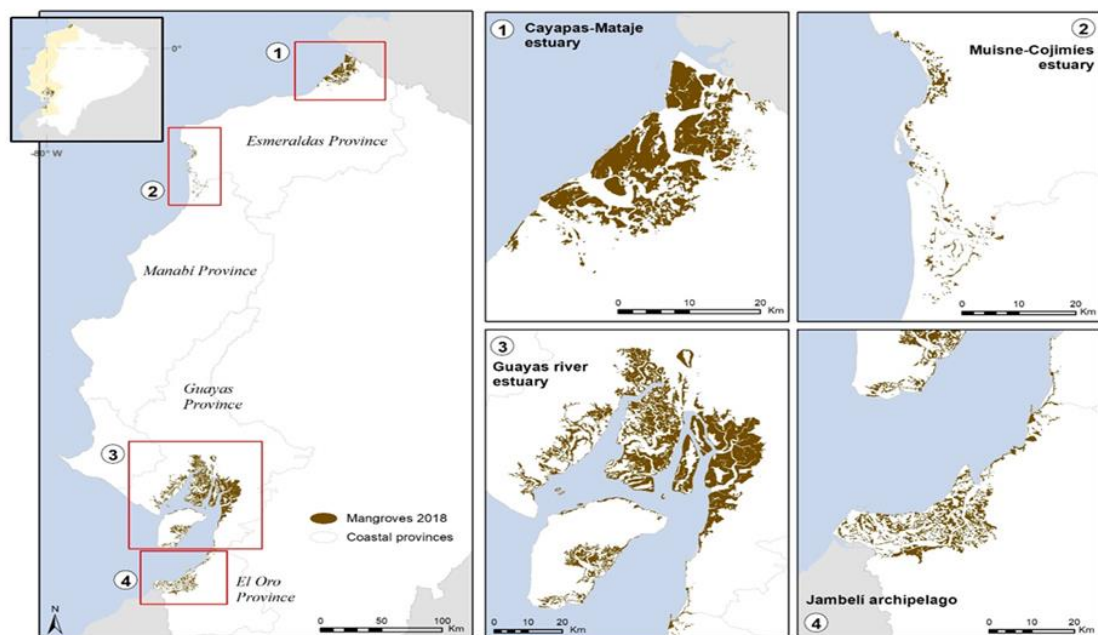
Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and protected areas implementing climate adaptation plans.

Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.

Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.

Ecuador's mangroves provide important economic benefits to local communities who harvest white fish, black clams, crabs, fish, shrimp, oysters, mussels, blue crabs and sea snails for self-consumption and sale. The forest is also used for beekeeping and logging/charcoal production CIFEN/AAE (2019).

Figure 1. Map showing the principal mangrove areas in Ecuador and the four estuaries/areas that make up the project area.



Mangrove dwellers are a heterogeneous group in terms of race. In the northern estuaries, the mangrove-dependent population is chiefly Afro-Ecuadorian, while in the southern provinces, mangrove populations are made up of *mestizos* (mixed-race population) and *cholo* fishermen (*cholos pescadores*; Latorre, 2012).

Occupational identities as crab or seafood harvesters, combined with situations of social vulnerability and poverty, play a more important role in self-identification than ethnicity. In general, mangrove dwellers experience low wages, informal employment and exploitation by intermediaries, malnutrition, low levels of education and inadequate infrastructure and public services (Latorre, 2012).

Despite the mangrove dwellers' age-old occupation of the area and shared management practices, this ecosystem, like the entire intertidal zone, is considered a "National Asset for Public Use" (*bien nacional de uso público*). This means that the mangrove swamp is owned by the state and renders the intertidal zone inalienable, immovable, and imprescriptible.

Table 1. Location and name of project sites

Province	Name of estuary
Northern project areas	
Esmeraldas	(1) Muisne (2) Cayapas-Mataje
Southern project areas	
Guayas	(3) Guayas River
El Oro	(4) Jambelí

From a gender perspective, the two northern estuaries are generally similar to each other, with both men and women actively engaged in mangrove use and management, while the southern two estuaries are more male dominated with women engaging much less than in the north. With this in mind, we have organized this gender assessment and action plan to focus on these two specific areas (north and south).

1.2 Purpose and methodology

The purpose of this gender assessment and action plan is to identify key information on the roles, activities, needs, opportunities, interests and rights of men, women, boys and girls that live in the project area in order to understand how the project can support their resilience to climate change. In doing so, the gender assessment exposes the gaps and inequities that exist between males and females, particularly in relation to the environment and climate change. Finally, we also took an intersectional lens, recognizing that many other social characteristics (e.g., wealth, education, marriage status, age, etc.) also contribute to women's and men's climate vulnerability and resilience. Within this framework, the gender assessment took a phased approach to data collection and analysis:

Collection of secondary information. This process involved reviewing existing documentation from government, academics, and organizations that work in this area. This includes statistical information, background data, information on the area, and documents and bibliographic references to support the gender analysis. A list of the key documents can be found at the end of this document. The documents include information of previous consultancies carried out in mangrove areas, reports of the Fishing Authority, and press bulletins related to production and marketing activities. Statistical information from official sources, including the national census (2010) and living condition surveys (2014 & 2019), were used to determine the characteristics of the social and economic conditions of the people living in the estuaries.

Collection of primary information. Primary information was collected through various instruments determined based on the stakeholder group and as described in this image:



1.2.1 Workshops

In August and September 2019, we conducted 4 workshops¹ in 3 of the estuaries (Muisne, Guayas and Jambelí) with men and women from local associations that focus on crab and cockle harvest. The workshops included Afroecuadorian communities and Cholos Pescadores in Guayas and Jambelí. Following the observation above that women are more engaged in mangrove use in the north, it is unsurprising to see (in Table 2) that women were much more present in the workshop in the north compared to those in the south.

The main objectives were:

- Obtain participants' general information related to time of residence in the area, economic activity, organization and activities of women and men in mangrove areas.
- Obtain information of potential reforestation sites, types of mangroves to plant and their benefits.
- Identify the perceptions that men and women have of the impacts of current climate conditions (wind, rain, presence of water holes, temperature, etc.) and marine conditions, on their communities and mangrove area concessions.
- Identify what kind of mangrove products are currently used, who uses them, and how they could be used in the future.

Table 2. Participation in community stakeholder workshops

Workshop/Location	Stakeholders	Men	Women	TOTAL
Northern region				
Bunche – Muisne – Esmeralda	Members and leaders from the communities of Bunche – Muisne, Chamanga, Pedro Carbo, Salinas and Bellavista. Representatives of the German Corporation for International Cooperation GmbH (GIZ), Ministry of	13	25	38

¹ Workshop reports can be found in the Stakeholder Engagement section

	Environment (MAE) and the Tourism Association.			
Southern region				
Guayas estuary (Puerto Roma)	Members and leaders from the communities of Puerto Roma, Cerritos de los Morreños, Santa Rosa, Bellavista, Puerto Salinas, Puerto Libertad and Guayaquil.	134	1	135
Guayas estuary (Balao)	Leaders from the Asociación de Cangrejeros Balao, Nuevo Porvenir, Puerto El Morro Marines, Puerto El Morro Apamupen, Puerto El Morro Porteño and Puerto El Morro Forjadores de Futuro.	59	6	65
Jambeli (Puerto Bolívar)	Members and leaders from the communities of Costa Rica, Mar de Galilea, Isla Pongalilla, Amor y Esperanza, Puerto Bolívar Union, 17 de Enero, 24 de Octubre.	17	8	25

Some gender-related information was also generated during the project's stakeholder engagement workshops, facilitated by the MAE and CI. The main objective of the workshops was to develop an analysis of the conservation policies and use of the mangrove ecosystem, and their contribution in the generation of ecosystem services and the wellbeing of the population. While gender as a theme was included as one topic, it was not extensive. Two workshops were held in the Guayas estuary (southern) in 2016:

Table 3. Participation in policy stakeholder workshops

Location	Stakeholders	Men	Women	TOTAL
Playas del Villamil, Provincia del Guayas	Representatives from the Secretary of Aquaculture, Association of Shellfish and Artisanal Fisherman Las Huacas, Ministry of Environment (MAE), Ministry of Tourism, National Institute of Statistics and Census, German Corporation for International Cooperation GmbH (GIZ), Secretariat of Water, Provincial Directorate of Guayas Environment, CONDESAN, National Institute of Fishery, Catholic University of Guayaquil, Association of Users of the Manglar Cerrito de Los Morreños, SENPLADES, Litoral Polytechnic University, Municipality of Guayaquil, Conservation International (CI-Ecuador), Association of Crabbers and Artisanal Fisherman 21 de Mayo – Puerto Roma, National Institute of Biodiversity, Association of Crabbers 6 de Julio and National Secretariat of Planning and Developing.	21	11	32

Ciudad de Guayaquil	Representatives from the Ministry of Environment (MAE), National Institute of Biodiversity, CIIFEN, SENAGUA, Ministry of Tourism, National Institute of Fishery, ESPOL, Conservation International (CI-Ecuador), TNC and CONDESAN.	8	9	17
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1.2.2 Semi structured interviews

To supplement the workshops and desk review, targeted interviews were conducted with 7 representatives from Muisne, Guayas and Jambeli estuaries. The aim of the interviews was to increase stakeholder input into the identification of social and environmental risks of the planned project. Interview questions therefore included the following risk-related topics:

- Discrimination against women on the basis of their gender, especially with regard to participation in the design and implementation and access to opportunities and benefits.
- Limitations in women's ability to use, develop and protect natural resources, taking into account the different roles and positions of men and women in accessing environmental goods and services.
- Threats to women in the context of occupational health and safety.
- Other gender equality concerns related to the project.
- Indirect impacts on biodiversity or the ecosystem services on which the livelihoods of affected communities depend.
- Adverse effects in terms of inequality or discrimination for affected populations.
- Possible restrictions in the availability, quality and access to basic resources or services.
- Adverse impacts on “indigenous peoples” (i.e., here Afro-Ecuadorians and Cholos Pescadores).
- Negative effects on the development priorities of “indigenous peoples” as defined by themselves.
- Adverse impacts on the enjoyment of human rights by the affected population and particularly by marginalized groups.
- Aggravation of conflict or generation of violence between affected communities and individuals.

The interviews confirmed the risks that had been identified during the workshops and provided further detail on the situation in different places and the unique perspectives of women and men. The following table provides an overview of the interviewees².

² Interviewees were asked whether the information they provided could be recorded and did not request to remain anonymous.

Table 4. Individual interviews (2 men, 5 women)

Male president of a Fishing Cooperative with 150 members (30 women) in the south. Members are crab collectors, fishers and women linked to activities such as gastronomy and crab pulp.
Male, former president of an Association in the south that has 105 members who are fishers and collect shells and crabs. At the moment, all members are men, but the plan is to involve women at some point in crab pulping.
Afro-Ecuadorian female member of a Women's Forum in the north.
Female president of a fishers association in the south (40 men + 20 women)
Afro-Ecuadorian female president of a Women's Forum in the north with 35 members.
Female former President of an Association in the south. 67 men + 200 women.
Female president of an association of fishers (1 men + 19 women) in the north.

1.2.3 Surveys

Another important source of primary information was surveys conducted with 99 local leaders and association members (80 males/19 females) during the same time as the workshops. The surveys gathered information about: (i) mangrove benefits like landscape, recreation, air purification, capture of commercial species and for self-consumption, bird habitat, obtaining honey and wood for house construction. (ii) the perception of changes in the weather (iii) drivers of impacts on mangrove ecosystem (iv) mangrove restoration, and (v) ecosystem services.

Table 5. Surveys conducted with community associations

Area	Association	Local leaders	Members
Northern	Balao		27
	Puerto El Morro		7
	Naranjal		2
	Miraflores		1
	Puerto Baquerizo		1
	Recinto Nuevo Porvenir		1
	Puerto Roma	6	9
	Machala		9
	Puerto Bolivar	7	4
	Isla Pongalillo		2
	Isla Costa Rica		1
	Sauces II		1
Southern	Naranjal	1	1
	Puerto El Morro	1	1
	Bunche	1	2
	Bunche	1	1
	Pedro Carbo	1	1
	Salina-Muisne	1	1
	Chamanga	1	1

	Pedro Carbo	1	1
	Muise	1	3
Total		22	77

The gender assessment and subsequent gender action plan follow the Guidelines outlined in the Green Climate Fund (GCF) Mainstreaming Gender in Green Climate Fund Projects and the Updated Gender Policy and Action Plan 2018–2020, as well as Conservation International-GCF's Gender Policy in their latest (November 2020) Environmental and Social Management Framework. The purpose of the assessment is to ensure that the proposed project is gender responsive, meaning it will substantially help to overcome historical gender biases. Measures have been put in place by this project to ensure that both women and men benefit and that the burden on women in particular, as a result of their gender roles, is not increased but reduced by the project activities.

2. Situation of men and women in coastal Ecuador

2.1 Legal status of women across Ecuador

This section describes the national legal framework in terms of gender and the international legal framework for gender and the environment that the country has been following and with which it has been bringing itself into line.

2.1.1 National legal framework

Ecuador has made significant progress in terms of national regulations to support women's rights. For over 10 years, in the Constitution of the Republic of Ecuador, the principle of non-discrimination contained in Paragraph 4 of Article 66 has been further underpinned by Paragraph 2 of Article 11, which prohibits discrimination based on certain suspect classifications including sex.

Article 11 of the Constitution includes the concepts of direct and indirect discrimination. It makes reference to direct discrimination with the phrase "that has the purpose of impairing or nullifying the recognition, enjoyment or exercise of rights" and indirect discrimination with the phrase "that has the effect of impairing or nullifying the recognition, enjoyment or exercise of rights."

Also included are a series of obligations incumbent upon the state to achieve effective gender equality. These are a series of affirmative actions promoting true equality for rights holders in a situation of inequality. They include developing and implementing policies to achieve equality between men and women; including a gender perspective in plans and programs; promoting equal representation of men and women in leadership and decision-making positions; bringing about conditions that guarantee the rights and principles recognized in the Constitution, in particular equality in diversity and non-discrimination, with an obligation to prioritize actions for groups needing special attention due to ongoing inequalities, exclusion, discrimination or violence; and adopting the necessary measures to guarantee equality for women in access to employment, professional and career training and advancement, fair pay, and self-employment initiatives.

Also enshrined in the Constitution of the Republic are the principles of parity, equity and rotation of power as cornerstones of the electoral system. Therefore, consideration must be given to alternating men and women in electoral lists. Similarly, the Constitution establishes the principle of parity between men and women in appointing judicial officers.

2.1.2 Agenda for Women

As of 2018, the National Council for Gender Equality, a national authority with powers aimed at mainstreaming gender equality in the country, has a key instrument to implement policy: the National Agenda for the Equality of Women and LGBTI People 2018-2021, which includes lines of action, policies and actions to be undertaken by institutions and civil society based on the principle of fair treatment.

The guide is divided into several lines of action that include autonomy and a culture of peace, and addresses issues relating to a life free of violence, actions in education and public policies in health, sexual and reproductive rights, or in fields such as sport. Another line of action is centered on sustainability of life and combines issues such as reproduction and sustainability, while also establishing an environmental policy that sets forth the need to generate statistics and promote conservation and reforestation programs and initiatives with the goal of improving ecosystem services and water sources.

The agenda also includes a section on leadership and the transformation of sociocultural patterns and discusses the monitoring and assessment of actions undertaken to promote equality.

2.1.3 Gender policy of the Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE)

The Ministry of Environment, Water and Ecological Transition of Ecuador (MAATE) is responsible for complying with international instruments signed by Ecuador in matters relating to environmental management and sustainable development. However, the country has made some important advances in gender.

The MAATE must therefore actively define policies, mechanisms, and strategies conducive to introducing interculturality and gender perspectives in programs, projects, and actions that come under its area of authority.

In this context, the General Coordinating Office for Environmental Planning defined the “Strategy for the Institutionalization of Interculturality and Gender Perspectives” as a methodological approach and a mechanism for advocacy within the National Environmental Policy (PAN). This initiative aims to comply with Constituent Assembly legislative decrees and urgently address issues that have been put off or rendered invisible. The goal is to recognize spaces to exercise citizen rights and consolidate democratic processes, strengthening the institutional framework of the National Councils for Equality (Peoples and Nationalities, Women; Afro-Ecuadorians, the Montubio people, among others), proposals linked to the democratizing role of the state with a decentralized and territorial approach, and the application of the cross-cutting themes of gender equity, interculturality and the environment.

The MAATE has proposed this strategy as a policy proposal to include social perspectives in environmental management. It is also consistent with major strategic lines proposed by central government and reflected in various public management instruments.

Furthermore, the constitution clearly shows the state’s obligation to ensure fair treatment of men and women and of different peoples and ethnic groups, and respect universal human rights inherent to every human being, including the right to a safe environment.

On January 30th, 2019, the environment minister, Tarsicio Granizo, and the deputy resident representative of the United Nations Development Programme (UNDP), Nuno Queiros, signed the Project Appraisal Committee, an enabling document with which it will be possible to begin a broad process to include a gender approach in developing Ecuador’s nationally-determined contributions (NDCs). This project was undertaken within the framework of the NDC-SP (NDC Support Programme).

Also in 2019, the MAATE and the National Council for Gender Equality signed an interinstitutional cooperation framework agreement, through which the two institutions committed to leading and ensuring the inclusion of gender equality policies to create and strengthen effective, efficient interventions in climate change-related projects. This agreement is the starting point for the joint development of the Gender and Climate Change Action Plan, which will serve to define national priorities and goals that respond to the specific needs of, and differential impacts faced by, women, men, and diverse peoples in the face of climate change, taking into consideration age, ethnicity, disability and/or human mobility.

2.1.4 An NDC with a gender perspective

Ecuador is defining its NDC through a participatory process led by the MAE, with support from a number of aid agencies: the United Nations Development Programme (UNDP), the Food and Agriculture Organization of the United Nations (FAO), the French Development Agency (AFD), the NDC-Partnership, the German Corporation for International Cooperation (GIZ), Euroclima Plus, the Consortium for the Sustainable Development of the Andean Ecoregion (CONDESAN), and Cooperación Española (FIIAPP).

The NDC Support Program is being implemented through, among other groups, the Technical Roundtable on Gender and Climate Change, which is led by the MAE with support from the UNDP. This group also includes representatives from Fundación Futuro Latinoamericano (FFLA), which provides facilitation support, the National Council for Gender Equality, the German Corporation for International Cooperation (GIZ), Grupo Faro, CARE, Universidad Técnica Particular de Loja (UTPL), Universidad San Francisco de Quito (USFQ), the Rosa Luxemburg Foundation, Heifer International, the International Union for Conservation of Nature (IUCN), and Fundación Altrópico. The Technical Roundtable on Gender and Climate Change was founded in response to a need to establish links between academia, non-governmental organizations and international aid, and create a space for interaction between different agents that contribute to the study, incorporation and strengthening of a gender approach in climate action in Ecuador.

In addition, with the support of Corporación Grupo Randi Randi (CGRR), a participatory methodology was developed to include a gender approach.

a) National Climate Change Adaptation Plan (PLANACC) and gender mainstreaming.

The PLANACC establishes a need to identify and create programs and activities based on areas of work or sectors that evidence has shown, in theory and practice, to be most vulnerable to climate change and to cause the most economic, social and environmental harm to the country as a result of the impacts of climate change. In this respect, one of the implementation criteria mentioned by the plan is the relationship with the social, economic or environmental system. Equally, the sectors prioritized for climate change adaptation in Ecuador can be grouped according to their relationship with the social, economic and environmental systems (MAE, 2012, 87).

In this sense, and taking into account the scope of adaptation initiatives, there are more opportunities to include women and sectors considered at risk or vulnerable.

2.2 Political participation at national, provincial, and local levels

Female voting in Ecuador increased from 76% to 84% between 2009 and 2018. However, since the National Assembly was established in 2009, it has not included an equal number of men and women: in 2009, only 40 (32.3%) of the 124 elected assembly members were women. (National Electoral Council, 2019)

In the project areas, there is low female representation at the parish and mayoral level. It is interesting to note the differences of gender parity in parish candidates verses election: it is clear that women candidates are not as successful in being elected. It is also interesting to note there is not much difference in northern vs. southern project sites. Women are highly underrepresented at the mayoral level in both candidates and elected positions, with particular starkness in the northern sites where there are no female mayors.

Table 6. Parish board members in the project areas

Province	Estuary	Candidates		Elected	
		Female	Male	Female	Male
Northern sites					
Esmeraldas	Muisne Cayapas	705 (44%)	900	81 (28%)	204
Southern sites					
El Oro	Jambelí	485 (44%)	610	69 (28%)	176
Guayas	Guayas	446 (45%)	539	50 (34%)	95

Source: National Electoral Council, 2019

Table 7. Mayors in the project areas

Province	Estuary	Candidates		Elected	
		Female	Male	Female	Male
Northern sites					
Esmeraldas	Muisne Cayapas	7 (15%)	40	0 (0%)	7
Southern sites					
El Oro	Jambelí	11 (16%)	58	1 (7%)	13
Guayas	Guayas	35 (17%)	166	5 (20%)	20

Source: National Electoral Council, 2019

Looking at the gender gap in politics can be a useful indication of women's perceived standing in society, as well as highlight some of the historical inequalities that continue to persist. In this instance, it is clear that the project areas have larger gender gaps than the rest of the country, especially when it comes to actually being elected to a position of power and influence. The reasons are likely multifaceted, including long-held societal/gender norms that privilege men in elected positions, fewer women who have achieved the level of education needed to

successfully compete, and unequal burden of household chores that often keep women from pursuing leadership outside of the home.

2.3 Socio-economic information

At a national level, the statistics and gender diagnosis presented in the Agenda for Women 2018-2021 (National Council for Gender Equality, CNIG) have shown that relations between men and women have been asymmetrical and have gone unchallenged throughout history. This has been a root cause of inequalities. At a macro level, and especially in rural settings, women have found themselves relegated to the private sphere, with little chance to undertake productive labor, while men have greater opportunities to engage in the public sphere, in making decisions that affect the community and their families, and in business dealings with external stakeholders.

Overburdening caused by the roles played by women in the home is another issue identified by community members. “In general women are responsible for the home, children, husbands, and family members, both in terms of providing general care and financially. Women have the most important role in society because they play a part in all the different spheres of activity: the home, productive labor, and the community,” (data collection workshop, comment by one female participant). Likewise, men (in their roles as fishers) also face increasing socio-economic challenges such as reduced fish populations, drug trafficking, and illegal fishing. This is forcing fishers to travel even farther out to sea where open waters are dangerous. Men’s fishery-based livelihoods are changing as Beitz & Gaibor (2018) explain, the lack of employment in recent years has driven more men to enter the cockle fishery.

The national unemployment levels have shown a clear decline over the last five years and stood at 4.4% in March 2017. The unemployment rate was higher for women than for men – in 2019 5.7% of women in the economically active population were unemployed, while for men this figure was 3.8%. Overall, female labor force participation is limited, at 49% compared to 79% of men. In the artisanal fishing sector, the conditions of poverty and extreme poverty are high in all the cases analyzed.

Table 8. Demographics in project site areas

Municipality	Estuary	Pop-ulation ¹	% men	% women	Ethnic Groups (% of total)					% Poverty ¹
					Mestizos	Afro descendents	Montubio	Other	White	
El Guabo	Jambelí	59.536	50,7	49,3	91,2	3,9	1,7	0,7	2,3	74%
Machala	Jambelí	306.309	49,2	50,8	86,8	6,8	1,7	0,9	3,8	56%
Santa Rosa	Jambelí	80.299	49,2	50,8	94	2,6	0,8	0,7	1,9	56%
Eloy Alfaro	Cayapas Mataje	46.305	49,2	50,8	11,3	67,9	2,2	18,1	0,6	94%
Muisne	Muisne Cojimies	36.426	50,6	49,4	53,4	38,8	3,2	2,7	1,8	98%
San Lorenzo	Cayapas Mataje	48.391	47,7	52,3	9,9	83,9	0,6	5,1	0,4	84%
Guayaquil	Guayas	2.746.403	48,9	51,1	85,2	6,9	2,3	1,5	4,1	47%

Naranjal	Guayas	83.691	50,1	49,9	86	4,5	5,8	0,8	2,9	74%
	Total	3.407.360								

Source: 1 INEC 2022; 2 INEC 2010

Communities in the northern Cayapas-Mataje and Muisne-Cojimies estuaries (and the northern area more broadly) are generally small: 74% have between 4 and 28 households, only 13% have more than 130 households, and the largest (Pampanal) has only 225 households. The average number of household members is 4.46. Houses are usually small and 88.7% are built from wood, 55% of which are made from mangrove wood. Others are made from a combination of wood and other materials, like brick and concrete. Almost all roofs are made of corrugated iron (Ocampo-Thomason, 2006)

Communities in the southern Guayas and Jambeli estuaries, are bigger than those in the north. The average number of members (and families) in each association is 128. The largest association based on the number of members is Hualtaco at Jambeli estuary with 369 households. The average number of people per household is 4.05. Another characteristic is that in the south there is a larger number of families that are female as head of the family (33% in the north vs. 36% in the south) as Table 9 shows. This situation is important to highlight as only 19% of the associations has more than 30% women as members.

Widespread machismo is one hurdle to achieving equity in the productive, reproductive and community spheres in these areas. “The fact that men are considered masters of the home amounts to a lack of education because it’s machismo; we are equal with different ways of thinking, but the woman’s main role is in the home. But when women are left on their own, they take on both roles,” (data collection workshop, comment by one female participant) (MAE, 2018).

Table 9. Household size and demographics in project sites

	Homes		Economically active population and occupation	
Estuaries	Average Size	Index of average female head in the municipality	EAP (% of total)	Occupation (% of EAP)
Northern				
Cayapas-Mataje	4,3	39,0	38,03	91,78
Muisne-Cojimies	4,31	26,85	33,75	90,23
Southern				
Guayas	3,9	34,6	43,11	91,81
Jambelí	3,7	37,5	43,26	94,56
Total	4,05	34,50	39,54	92,10

Source: INEC 2010

Across all project sites, women and men depend on the natural resources they obtain from the mangroves, namely crab, black clam, and resources from fishing in general. In exceptional circumstances, such as during the closed season, men find they need to earn an income outside the home. In all project areas, men and women depend on the outside market as they do not have basic commodities. In most cases, family money is used to cover these expenses. Income specifically obtained from small businesses and selling essentials can be used by women for expenses considered luxuries. To support the family, basic foodstuffs, clothing, and other consumables need purchasing. The specific gendered uses and management of mangroves is detailed in a section below.

2.4 Education

Since 1990, the education gap in Ecuador between females and males has narrowed. On average, males and women attend school for 7 years with difference of 2 years between the estuaries at north and south. Illiteracy rates nationally are 6,7 males, 9,6 females with a difference in gender that we didn't find in the estuaries.

In the project sites:

Table 10. Education in project sites

Estuaries	Illiteracy			Years of schooling		
	% total	% men	% women	Total	Men	Women
Northern						
Cayapas-Mataje	13,29	13,55	13,02	6,92	6,78	7,07
Muisne-Cojimies	14,19	14,50	13,85	6,67	6,63	6,51
Southern						
Guayas	3,90	3,80	3,99	8,36	8,36	8,36
Jambelí	4,38	4,31	4,45	8,24	8,17	8,23
Average	8,94	9,04	8,83	7,55	7,49	7,54

Source: INEC 2010

Inequality in education and literacy can be a product of gender inequality in society and can directly influence women's ability to participate, engage, and influence climate activities. In this situation, however, it is encouraging to see that literacy and schooling is relatively equal between males and females in the project sites.

Education is limited to the primary level. Secondary education can only be pursued in the larger towns of the cantons, which are several kilometers away. This is a hindrance for women because it limits their access to education – "leaving" their communities requires money for transport. In addition, as victims of violence and sexual harassment, they feel vulnerable at having to travel long distances on paths with little infrastructure before reaching a means of river or land

transport. Lastly, women's responsibility toward caregiving makes this their priority within their communities, leading them to give up the chance to pursue education.

The rural population does not have access and full coverage to attend primary, secondary and higher education. Formal education presents limitations, especially at high school, whose percentages of attendance are lower in comparison to elementary school, in most rural parishes (Rengel 2020). In the case of islands in the Guayas estuary this situation is even worse, where parents prefer not to send their kids to the school on another island because of security issues with piracy (personal communication).

2.5 Time use/division of labor

In all project areas, caregiving dynamics are similar and, for the most part, are the responsibility of women. Described below is an example of a daily routine encompassing the different experiences in the project areas during field data collection.

Women living in mangrove swamps work long days, which begin with family caregiving responsibilities and then harvesting or fishing activities. They return to cook for the family and look after the children, and then cook dinner and finally rest. The average working day for women is around 14 hours, and little societal value is placed on caregiving or reproductive tasks, rarely seeing them as a social investment.

By contrast, men have much shorter days, fishing or harvesting for between four and five hours. This means that once their day's fishing or harvesting is over, they are relieved from work. This situation changes when they work externally and find themselves compelled to work as day laborers in shrimp farms or on a country estate or ranch (*finca*), in which case they leave home at around 7 a.m. and return around 4 or 5 p.m. They take a packed lunch and return home at different times depending on how far away their place of work is.

In this context, spare time is virtually unheard of for women. They rarely leave home, while men get together on the playing areas in the afternoon or have time to rest after lunch. Problems with alcohol were mentioned, but not ubiquitous. Some interviews made mention of the fact that when decisions on how use financial resources are left entirely to men, a good share is spent on alcohol.

These observations and surveys are validated by a gendered time use survey conducted earlier (INEC 2010) in the northern province of Esmeraldas. The study highlights a similar unequal division of labor, with women performing – on average – 2 hours of labor more than men. The difference in male and female time use is especially notable in unpaid labor, particularly housework, which is primarily undertaken by women, for an average of four hours a day compared to just one hour a day for men.

Together, this data reveals the excessive burden faced by women in work both inside and outside the home. In sum, they spend their time tending to household members, obtaining resources to support their families, and engaging in community activities. The situation becomes yet more complex when women are the head of a household where resources need to be obtained on a daily basis in order to subsist.

2.6 Health and gender-based violence

Life expectancy in Ecuador is approximately 76.4 years (73.7 for men and 79.1 for women). The maternal mortality rate across Ecuador is 59/100,000 live births; infant mortality is approximately 17 deaths/1,000 births. Healthcare is available in the main towns of each canton, but accessing it requires having the financial means to leave the area. Pregnant women, for instance, leave the communities one or two months before they are due and stay with relatives so they can receive care in nearby hospitals in the cities of Machala, Guayaquil or Esmeraldas.

Table 11. Incidence of gender-based violence

Location	Percentage
National	64.9
Urban	65.7
Rural	62.8
Esmeraldas (north)	68.2
Guayas (south)	63.1
El Oro (south)	70.2
Afroecuatorians	71.8
Mestizas	65.1
Indigenous	64.0
Montuvias	58.4
Global*	35
Perú	70

Source: INEC, 2020, ENCUESTA NACIONAL SOBRE RELACIONES FAMILIARES Y VIOLENCIA DE GÉNERO CONTRA LAS MUJERES – ENVIGMU 2019 Retrieved in January 2020

*WHO.2013. Estimaciones globales y regionales de la violencia contra la mujer.

Rates of gender-based violence are very high and are linked to the socioeconomic dynamics of exclusion and poverty. The table above presents the magnitude of the issue and manifestations of violence most common. Further, it reveals that incidents are relatively similar in both urban and rural areas. Rates of violence in the project sites are similar to those at the national rate, which is nearly twice as high as the global average of 35%.

The importance of gender-based violence was underscored during our field data gathering: one female leader reported that almost all women had endured this kind of violence at some point in their life. Many women attending the workshop nodded in agreement, their body language confirming that they had experienced or witnessed this type of violence.

The increase in gender-based violence in disaster situations and places experiencing economic tension has been well documented. Both of these factors are closely linked to climate change on the coast of Ecuador. When communities and families are unable or unprepared to deal with the effects of climate change and climate-related disasters, this gives rise to heightened social and financial tensions that may cause gender violence to escalate. Disasters increase existing gender inequality and given that Ecuador already has a high rate of gender violence at over 60%, the phenomenon is likely to worsen as the effects of climate change become more severe. At

the same time, as Ecuador remains stuck in the grip of the COVID-19 pandemic, the effects of which will be felt for many years to come, it is also important to stress that gender-based violence has increased drastically throughout the country as a result of the shutdowns and economic tension. A 2017 report by the Red Cross on gender violence on the Ecuadorian coast (the Manabí and Esmeraldas provinces) following the 2016 earthquake found a lack of gender sensitivity in the disaster response mechanisms and called for better prevention and an improved response to gender-based violence in disasters.

If gender-based violence is rooted in, and triggered by, economic stress, then relieving that stress can help to mitigate and reduce the incidents of GBV. Proactively working with men and women to adapt and buffer their livelihoods to climate change impacts, which can then hopefully reduce economic stress, can be an important mitigation tactic.

2.7 Assets and infrastructure

Local dynamics in regard to economic and financial matters and other assets are similar across the project areas. Women have scarce opportunities to access resources because they are unable to access credit, with the exception of savings plans and small loans for household consumption. Women have no capital to post as collateral to access credit (information from interviews, August 2019). “There are many resources here, in tourism, in fishing, there are belief systems, there are many other options possible. Sadly, there is no way to put them into effect because we cannot access credit,” (data collection workshop, comment by one female participant) (MAE, 2018).

Men, on the other hand, have higher earning potential and greater opportunities to access formal loans. Additionally, benefits were recorded from the rural social security program for small-scale agricultural workers, *Seguro Social Campesino*, in one area in the north (Muisne).

As far as infrastructure is concerned, the field visits showed there was electricity in all sites, while Muisne and Jambeli had running, non-potable water, and in Puerto Roma in Guayas Estuary water has to be managed on the islands, generally by collecting rainwater. Some women sell fresh water they have collected or purified water that is brought in from the mainland.

In the areas close to cities, they do have access to cellular phone and Internet services, which has helped connectivity, especially in Muisne and Guayas Estuaries. The service is weak and at times non-existent, but it does at least enable communication with neighboring areas.

The roads to the main towns of the cantons are usually paved, with gravel roads serving to connect communities. Transport services are limited both in number and frequency. This creates a sense of vulnerability for women, as they have to walk long days, often without public lighting.

Speedboats are available to travel by river between Jambelí and Machala, and between Guayaquil and Puerto Roma and other islands. In Muisne, transport is available from the back road to the town center. River transport is safer because boarding takes place in the town center and the place of arrival has better infrastructure.

3. Access, use, and management of mangroves.

3.1 Importance of mangroves to local populations

The mangroves provide a livelihood for fishing communities, mestizos, settlers, and Afro-Ecuadorians, who all live in the project areas. The information on ecosystem services that was generated in participatory workshops with the community showed that it was the provisioning services that the population most valued, for the consumption and sale of products derived from mangroves. The community reported that the effects of climate change are impacting the provision of these ecosystem services. The regulating service was identified in coastal protection in both the northern and southern project sites, and in the provision of clean air, with reforestation efforts undertaken by both men and women, in the case of southern Guayas. The cultural services, meanwhile, are geared toward the development of ecotourism and the use of ancestral knowledge to cure diseases with the bark and roots of mangroves. These activities are performed both by men (extraction) and women (preparation), while in the case of ecotourism, it is women who feature most prominently, through the cuisine they offer tourists.

While outdated, an excellent account of gendered use of natural resources comes from the northern Esmeraldas province (Veach 1996). In this example, ocean fishing is done exclusively by men with highly specialized knowledge and from motorized canoes, often being their main source of income. Women generally do the bulk of the fish processing, with men usually marketing the fish. Other livelihood activities performed primarily by men include carpentry, agriculture, logging, charcoal production, and hunting. In contrast, resource use done primarily by women includes collecting concha (mollusks) which, while the income potential is less than fishing, is considered to be more reliable.

More recent research with mangrove communities in Esmeraldas highlighted the impacts that expanding shrimp aquaculture is having on gendered use of mangrove resources, with collapsing fishing livelihoods leading more men into mangrove exploitation and displacing women's traditional mangrove-dependent resources and livelihoods (Trevino & Murillo-Sandoval 2021). This overuse of the mangrove resources is leading to depletion of mangrove cockle populations and putting the community in a vulnerable position as everyone is increasingly dependent on this fishery to subsist (ibid).

3.2 Participation and decision-making in mangrove management

Women are largely inactive in discussions and decision-making in their communities, while men are usually more vocal in meetings and better represented in the boards. It is at these meetings that decisions about things like monthly payments of the members, defining the rules for crab extraction (amounts extracted by each member, defining the extraction areas, or number of hours in the areas). Some organizations, particularly those associated with their activities (e.g., mollusk collection), include female members, and this can be an important place to build their capacities. Certain roles within mixed organizations are more feminized, such as secretary and treasurer which tend to have less clout in decision-making. Within community processes for mangrove management, women's specific interests are rendered invisible, which can deepen divides in terms of access, use, and control. Women's limited participation can be attributed to several reasons:

1. One recurring problem in women's participation is their lack of leadership. In the words of one community leader, "women know everything, the only thing they lack is leadership, they already know the rest." The fact that women do not feel they have that capacity makes them uneasy. Certain restrictions faced by women have to do with the issue of leadership, as many women still fear coming into confrontation in assemblies and standing up for their rights as women.
2. Women are held back by their families/spouse and the household responsibilities they largely shoulder. "you can't go because you have to cook for me."
3. Transportation and accessibility to meeting spaces is also limited. Interviews revealed that only men hold knowledge about fishing and how to operate motorboats, and this knowledge allows them to move around the area with ease and facilitates their participation in various decision-making fora. Women do not own the canoes or outboard motors, only their labor power.

It is important to mention that women engage to a greater extent when it comes to attending meetings involving securing one or more basic services for coastal communities. In other words, women are overrepresented in activities that support community services and works but are lacking in community decision-making regarding natural resource management in the mangrove ecosystem.

The mangrove management organizations in project sites were established with the aim of promoting the use of mangroves and accessing communal benefits. The mangrove use agreements are drawn up by the associations, which have a specific objective. While there are no formal limitation in terms of access rights, as mentioned above, due to customary practices, women have limited decision-making power even though they may be association members. In one community, access to discussions and decision-making regarding the mangrove is particularly challenging as women's responsibilities are geared toward caregiving; there is no interest or receptiveness toward identifying spaces to provide or promote activities associated with women's use or exploitation of mangrove resources.

One other organizational aspect that should be mentioned is that there are community boards (organizations directly associated with the mangroves) with boards of directors in associations generally made up of 5 people, who were selected by all members of the Association. Leadership positions are generally held by men. The women who participate do so as secretaries or board members with no additional specific decision-making tasks.

There are also organizations of fishermen, crabbers and black clam harvesters. They also have boards of directors that handle and address sporadic needs to achieve and improve on their objectives. Depending on the area, women play a minimal role on the board, but this depends on the particular nature of each organization.

Below is a table that describes the number and gender make-up of the mangrove associations in the four project estuaries. There is a remarkable difference in gender make-up between the northern and southern estuaries. In the south, women's difficulty in participating in decision-making derives from the fact they are not familiar with the mangrove or legally recognized as part of the communities or productive organizations. This can be attributed to their access and use of the mangrove areas. In the north, women can walk to the mangroves in groups, but in the south the mangrove areas are far from the community, so they have to take a boat. Usually, women's mangrove activities in the south are more individual (each person arranges their own

access to the area). This is why women are less involved in mangrove-related activities in the south.

Table 12. Association membership in project sites

Project site	Women's representation in mangrove associations	
	Total # of associations	Average % of women in associations
Northern		
Muisne	6	60%
Cayapas-Mataje	14	62% 1 association entirely female
Southern		
Guayas River	23	11%
Jambelí	24	20% 1 association entirely female
Source: MAAE 2020		

3.2.1 Northern estuaries (Cayapas-Mataje + Muisne)

Cayapas-Mataje estuary

The population in the northern Cayapas-Mataje estuary is chiefly Afro-Esmeraldian and in sociocultural and local development terms is dependent on exploiting natural resources from the mangroves, particularly black clams and blue crabs, and fishing in general.

In Cayapas-Mataje, men and women work harvesting black clams and other resources. In terms of natural resource management, women are especially remarkable as they have great knowledge of how to carry out this arduous task. Lastly, it is women who control the sale of black clams and use the earnings to purchase certain products to feed their families.

Typically, each community in the northern Cayapas-Mataje estuary has specific areas for black clam harvesting, which are usually respected by other communities. Traditionally, black clam harvesters use small wooden canoes with oars, which limits the distances they travel to harvest. In addition, women prefer to harvest in areas close to their communities. Fishermen tend to alternate areas, leaving some areas undisturbed for a few weeks so that the black clams are able to recover. Traditionally, harvesters return the “mother” (with young) so she may reproduce, and they also return small clams (under 4.5 cm long) to allow them to grow.

Black clam harvesting has traditionally been performed by women. One reason for this is the fact that black clams are gathered individually by hand, so no investment of capital is required. The harvesting areas are close to communities, which allows women to take their children with them and makes it easier for them to combine black clam harvesting with housekeeping chores. Black clam harvesting is the most popular form of work undertaken by women in the area. In the 99 households that reported black clam harvesting as their main livelihood, 82.4% of harvesters are women (Ocampo-Thomason, 2006). In addition to the women who harvest, a further 2.4% of women purchase black clams to sell them on to external intermediaries. One of

the most significant findings is that 20% of female-headed households depend almost exclusively on black clam harvesting, and in 10% of mixed households, both spouses harvest black clams as their main, and in some cases only, livelihood.

Black clam harvesting is the livelihood of 16% households. Other shellfish are also gathered at the same time as black clams, but they have little market value and are used to provide protein for household members (Ocampo-Thomason, 2006).

Interestingly, today 17.6% of black clam harvesters are men. The interviews revealed that men have taken up black clam harvesting as they have no other source of income, having been driven out of farming after selling their land to the shrimp industry (Ocampo-Thomason, 2006).

In Cayapas-Mataje, day-to-day livelihoods revolve around productive labor and livelihoods. They derive their income primarily from black clam harvesting (known as “concheo”), crab harvesting, and fishing. To a lesser extent, they perform agricultural work like sowing and picking coconuts or other plants, or gathering smaller mangrove shellfish, but these activities are more for self-consumption, subsistence, or barter for other foods.

The work carried out by Cayapas-Mataje associations is traditional in nature. For example, no processing technologies are used after the resources are harvested. Some expenditure is incurred by men for the purposes of aquaculture, but women face gender discrimination due to the scant value placed on their work, which is evident in the restrictions they face in accessing credit and other financial services. Women encounter structural challenges in accessing credit and the financial system itself. For example, only 9% of the loans for female heads of household did not require any type of guarantee, compared to 29% of male heads of household. Women have, on average, 10% less access than men to an account at a financial institution. In addition, the average labor income of a man with a job is 1.27 times higher than the income earned by a woman with a job in Ecuador (BCE 2018), 31% of female heads of household access credit from informal sources, while this happens to 25% of men. This situation occurs at different levels and in urban and rural areas.

Muisne

Muisne Cojimíes is notable for the high level of participation by women as members and leaders of these associations.

Table 13. Gender roles reported in Muisne-Cojimíes

Activity	Women	Men
Caregiving tasks	Feeding, childcare, help with homework, among others.	Help at times
Black clam harvesting (<i>concheo</i>)	Cooking, fishing, and selling	Fishing and selling

Crabs, including blue crabs and mangrove ghost crabs	Fishing, selling, and cooking	Fishing, sorting, and selling
Catching, selling and sorting	Cooking, fishing, and selling	Fishing, sorting, and selling
Tourism	Culinary activities, running tourism businesses, trail guides	Operating small boats guiding tours, providing tourist presentations, culinary activities
Seedbeds for species	Group cleaning work, management workshops	Group cleaning work, management workshops
On the mangroves	Reforestation activities	
Use of mangroves (medicine)	Strip the bark off to make infusions	
Coastal protection	Refuse collection	

Source: Survey on Perception of Environmental Standards in Mangroves, CIFEN / AAE, August 2019

Most land in the mangrove area is communally owned. However, women are only minimally included in decision-making because they do not feature explicitly in the regulatory framework. Furthermore, cultural practices maintain the sexual division of labor and continue to exclude women from mangrove conservation activities.

Often, it is men who hold formal positions of leadership within communities and fishermen's associations. Women, on the other hand, are responsible for homemaking, caring for family members, and cooking; very few are able to generate their own income through sales in small businesses like retail stores.

In Muisne, women engage in housework and caregiving activities but are also involved in collecting black clams. Crab and black clam harvesting is the main economic activity, performed on a daily basis in the morning. Men work harvesting crabs in the mangrove and also engage in small-scale fishing. Women as young as 13 get involved in harvesting the black clams. In this area it is also common to find women who are the heads of their households and who, for various reasons, take charge of productive labor.

Those being the two main economic activities associated with the mangrove swamp, men and women have developed adequate know-how to improve harvesting in tough conditions: mud, brackish water, mosquitoes and poor mobility.

It was reported that a hundred clams fetch \$2.50 USD and a string of 20 crabs \$3.00 USD. The women do not get involved in selling what they harvest – that is the responsibility of the men. However, situations arise in which intermediaries come into the communities to purchase the seafood gathered.

In these communities, men have a greater chance of engaging in productive labor in the surrounding areas. Both black clams and crabs are sold on the local market to intermediaries who transport the products to large cities in the north of the country.

There is a month-long closed season between February and March, with a permanent ban on harvesting black clams below a certain size. The aim of these bans is to ensure the reproduction, growth and conservation of the species. The local population is aware of the dates and regulations of the closed season as the Ministry of Environment (the controlling agency) works to disseminate the dates and specifics of the closed seasons.

“The women are well aware that they must only collect black clams greater than 4.5 cm, as provided by law, and the rest must be returned to the mangrove. Similarly, they know the dates of the closed seasons for crabs by heart (even though they may not know why it is forbidden to harvest them). As for fishing, they know they can engage in artisanal fishing without using chemicals, and they use traditional methods of fishing, with a casting net or “wide-meshed” fishing net. Other than that, they know little or nothing about environmental conservation, resource management, recycling, proper waste disposal, and being kind to animal and forest species,” (interview with female community leader).

During the closed seasons, men and women from the local populations struggle to access economic resources as there are limited job opportunities, especially for women, who have less training or formal education and have caregiving responsibilities. It was reported in the interviews that very few women are able to leave the home to engage in paid work, as women bear full responsibility for reproductive activities. Furthermore, the existing work opportunities can be summarized as follows:

- Shrimp farms, where shrimp are bred and harvested. This work is carried out in the water, by day and in full-day shifts; this is also due to the use of pollution-causing chemicals. The nature of the work means that it is mostly considered a man’s job.
- Farm work, which includes sowing, reaping, cleaning, and harvesting crops. Working hours are from 7 a.m. to 4 p.m., which a short lunch break. These tasks are considered a man’s job and very few women access this type of work on private land.
- State institutions generally look to employ people with at least a high school diploma. Given the requirements and difficulties women experience in accessing formal education, a very low percentage of women are able to access this kind of work.

Meanwhile, men work as day laborers on the crops of nearby farms and in stocking and tending shrimp. Men are able to travel to nearby towns to offer their labor as construction workers or any other type of service. However, this is more difficult for women as it is not culturally accepted.

This analysis clearly shows gender inequalities in the use of, access to and control of mangrove resources in the project’s three regions. With the goal of guaranteeing gender equity and sustainability in mangrove conservation, substantial efforts will be deployed to ensure that both men and women participate and reap the full benefits of this project. This analysis includes a detailed gender plan and will contain specific recommendations, activities, staffing details, and the budget.

3.3 Southern estuaries (Guayas + Jambeli)

Guayas

The mangrove forest in Guayas is notable for being very near the city of Guayaquil. The mangroves can be reached by land and water, and this has shaped the socio-productive landscape among the population. The Guayas River estuary contains 80% of the country's shrimp farms.

In the Guayas River archipelago, the local population possesses ancestral community property under land legislation that ensures ownership of community land and the recognition, allocation and titling of the lands and territories of indigenous communes, communities, peoples and nationalities, the Afro-Ecuadorian people and Montubio people, under the Constitution and international agreements and other instruments regarding collective rights (Organic Law on Rural Lands and Ancestral Territories, 2016). However, the land is located in protected areas over which the Ministry of Environment has influence, and therefore they are required to draw up an Agreement on the Sustainable Use and Stewardship of the Mangrove Ecosystem (AUSCEM).

Due to socio-cultural dynamics, women and men have different experiences in terms of access to and use and control of natural resources, as summarized in the following table.

Women	Men
They do not know the mangrove forest and it does not appeal to them as a place to visit or get to know. They operate in populated areas. Women do not take part in decision-making processes and take primary responsibility for care giving activities.	Men know the mangrove forest, use it and decide which areas will be harvested for mangrove ghost crabs. They sign the mangrove use agreements.

The area is subject to a ban on crab and black clam harvesting for one month between February and March to enable the species to reproduce and grow. During this period, the population is able to support itself financially through other activities, particularly tourism. A significant number of men work in the community as service providers for the construction, agro-exports and shrimp farms. In comparison, this is more common in the south because in the north these economic options do not exist.

On the other hand, due to the geographical position of the Puerto Roma and Balao area (an estuary with remnants of mangroves), the population devotes itself fully to harvesting crabs and black clams. This work is the responsibility of men, who go out to the surrounding areas on a daily basis to harvest. This work takes up about 4 hours of their day. They also fish for their own consumption.

Women, on the other hand, are responsible for caregiving duties. Some attempt to earn an income from removing and selling crab pulp. They also work selling clothes or cosmetics or invest in small stores that serve the basic needs of the local population. Single women generally live

with their parents or family and rejoin the family nucleus. The following table further clarifies the different roles of women and men in the Guayas estuary.

Table 14. Gender roles reported in the Guayas estuary area

Activity	Women	Men
Caregiving tasks	Feeding, childcare, help with homework, among others.	Help at times.
Black clam harvesting (<i>concheo</i>)	Sale of clams.	Harvesting.
Crabs	Removal and sale of crab pulp.	Harvesting and sale.
Oysters	Shucking.	Harvesting and sale.
Tourism		Tour guides, boatmen, provide demonstrations, take tourists fishing.
Beekeeping	No involvement.	Cleaning and inspection, performed by the over-60s.
Use of mangrove bark	Strip the bark off the mangrove and make infusions to control sugar levels.	
White fishing	Prepare products to sell to tourists, door-to-door sales, the scales are used to make handicrafts.	Fishing and sale.
Exploratory activities	Guides.	Work as guides.
Seedbeds	Collect seeds for reforestation.	Reforestation, refuse collection.
Trading	Clothing, cosmetics, stores.	

Source: Survey on Perception of Environmental Standards in Mangroves, CIFEN / AAE, August 2019.

In terms of vulnerabilities or impacts, a lack of water for human consumption and difficulties securing healthy food that meets the population's nutritional requirements are plain to see. Lastly, it is worth pointing out that the population is highly dependent on larger towns to obtain food, reducing opportunities to fulfill dietary needs. The inhabitants see themselves as cut off from the economic system as they live in an island environment that can lead to difficulties accessing services, workspaces, and life options (interview with a female leader from Guayas).

One other important subsistence activity is gathering honey for sale. For women, this remains difficult and they do not get involved in the sales process.

The problems cited in terms of vulnerability to climate change include flooding and a loss of livelihoods caused by environmental pollution.

The mangrove forests of the province of El Oro are located in the Jambelí estuary, a special management zone on the outskirts of Puerto Bolívar, an area accessed by boat and inhabited by a number of communities. Socioeconomically it is challenging for those living in the area, as services are limited and they are unable to produce food, and so they are plainly dependent on the mainland. This situation directly affects gender dynamics.

The characteristics of the management zone complicate land ownership as there is no land titling. Populations or communities living in the areas are not formal owners but have houses and buildings that have been maintained for over 50 years, since the first settlers arrived. In addition to the local population, men and women from the city of Machala come to the mangrove to fish for a living.

In this particular context, women have limited opportunities to engage in fishing; many work shucking crabs. Those living in the areas surrounding Jambelí may fish and sell what they catch.

In the Puerto Bolívar area, black clam and crab harvesting takes place in the Guayas River estuary, where there are remnants of mangrove forests. Over the last 20 years, the local population has come under intense pressure from shrimp farming companies that have sought to cut down mangrove forests for their own benefit.

In this area, the productive labor carried out by men and women can be characterized as follows.

Table 15. Gender roles reported in the Jambeli project area

Activity	Women	Men
Caregiving	Mostly the responsibility of women.	Help out, without taking full responsibility.
Tourism	Food preparation.	Boat rides, transportation within and outside the islands, guiding and leisure activities around the islands.
Mangrove ghost crab and lobster harvesting	Pulp removal, some sell to small restaurants locally.	Use traps to catch crabs. Dive in rocky coastal areas to catch lobster.
Black clam harvesting	Collect clams from the mangrove swamp and sell them.	Harvesting.
Fishing	Some women fish with their husbands. This is not their explicit responsibility, but rather they “help out”.	Fishing in the estuary. Net or rod fishing by day; they only work a few hours in the morning.
Oyster	Collection and sale.	
Group cleaning		Collect waste, sort it and sell whatever can be recycled.
Source: Survey on Perception of Environmental Standards in Mangroves, CIFEN / AAE, August 2019.		

Women engage in artisanal fishing more frequently here than in other areas and, in fact, some are members of artisanal fishing organizations. As mentioned in the table, the type of fishing equipment used in the area allows women to access and take part in fishing activities. In this sense, women experience some difficulties in accessing, using, and controlling natural resources. Women do not own the canoes or outboard motors, only their labor. This means that in community mangrove management processes, their own specific interests become invisible, which may exacerbate divides in terms of access, use, and control.

Lastly, the Jambelí River estuary has one association made up exclusively of women and some other associations have female leaders.

4. Vulnerability and adaptation to climate change

It is clear that, as a result of gender norms, men and women are impacted differently by climate related events. Global evidence has shown that in places with high gender inequality, women and children are much more likely to die because of natural disasters. Rising sea levels and flooding – the two major effects cited in justifying climate action in these project sites – can also have consequences like an increase in water salinity and water-borne diseases. Both are more likely to directly affect women, who tend to be responsible for collecting water for the household and who have the responsibility of caring for sick family members. Likewise, men, who are primarily responsible for sea fishing and agriculture are negatively impacted by increasingly unpredictable weather.

4.1 Impacts and response to past climate-related events.

The project's areas of influence are places of high climate risk and vulnerability. As islands, they run the risk of flooding and pests that would affect the human population, together with pollution from waste brought over by currents from the mainland.

In order to understand how previous climate-related events have impacted women and men in the project sites, we asked workshop respondents to describe their past experiences. The data collection process included identifying perceptions of climate change vulnerability. This exercise was devised based on the last 20 years and the characteristics identified.

Table 16. Historical climate impacts on women and men

Project site	Year	Event	Differentiated impacts	
			Women	Men
Northern (Muisne)	1970	El Niño phenomenon	The fish left and they were unable to work. They emigrated elsewhere and worked the land (cacao, rubber, banana); economic impact, loss of life and disease.	Less black clam harvesting, they moved to other areas to collect black clams (an hour and a half away). They emigrated to work in Atacames (in restaurants and cafeterias) and “ditched their children.”
	2000 - 2001	La Niña phenomenon	Many emigrated to other countries (Spain, USA, and Chile) and were left jobless. They were no longer able to sell larvae to shrimp farmers.	Economic impacts; they had no work or income; migration, many men left the country; loss of life. They were no longer able to sell larvae to shrimp farmers.
			Good quality and quantity of clams, but prices on the local market fell.	When fish were abundant, prices on the local market fell.
Southern (Guayas)	1980	El Niño phenomenon	Epidemics and diseases. Children were underweight and malnourished.	Loss of livelihoods. Areas of mangrove flooded. Lack of job opportunities. Shrimp farms had no production so were not hiring.
	2005	La Niña phenomenon	Impact on families' financial situation; they were unable to sell seeds to the shrimp farms,	Fish left due to the red tide. There were no crabs because the ground “went hard”.

			the economic impact led to difficulties providing for families.	Difficulty fishing due to the death of species, higher water temperature. Very heavy rainfall preventing fishing.
	2009	Ocean current	Often they had to help push the boats.	Rivers filled with silt and boats were unable to move, so they had to push the boats or wait for high tide.
Southern (Jambeli)	1998	El Niño phenomenon	They were left with no sources of work due to the loss of crabs; overburdened with household cleaning.	They sought work outside the mangrove, such as on banana plantations and in the construction industry.
	2004	Swell	Impact on families' financial situation.	Loss of crabs and black clams in the area, loss of source of work.
	2017 and 2019	Red tide	Women who fished for these species were left without work.	Direct impact on people working with shellfish.

Source: Field data collection through interviews and participatory workshops, August 2019.

4.2 Climate projections in project sites and implications for women and men

Climate change will result in significant impacts on local environmental conditions along the coast of Ecuador, including a rise in sea levels, El Niño events, and changes in the intensity and variability of rainfall, flooding, and atmospheric temperatures.

Based on models generated by NASA, climate change could cause sea levels to rise by up to 27 cm by 2100 (NASA-GISS, n.d.). Considering that the 1997-1998 El Niño event led to a 42 cm increase in sea levels, periods of heavier rainfall are expected to worsen the rise in sea levels.

The risk of flooding is a major concern for the coast of Ecuador. The Dartmouth Flood Observatory has recorded 22 major floods since 1987, displacing over 506,000 people (Dartmouth Flood Observatory, n.d.). Flooding was particularly severe during the El Niño events of 1972/73, 1982/83, 1997/98, and in 2008 and 2012, affecting between 184,000 and 222,000 ha (Ocles Padilla, 2018). Flooding of coastal areas will increase as a result of the sea level rise (SLR), particularly due to extreme sea level (ESL) events in which periodic storm surges and wave setup (an increase in the mean water level due to the presence of breaking waves) are worsened by projected increases in the relative sea level (Kirezci et al., 2020). Higher sea levels will lead directly to an increase in coastal flooding in Ecuador (Reguero et al., 2015).

Given their regular contribution to sea level rise, the El Niño events are of particular concern. The mean projected increase in sea levels due to climate change is only comparable to sea level changes experienced during El Niño events. The increase in sea levels, together with exceptionally high rainfall associated with El Niño on the coast of Ecuador, limits the discharge of river systems into the ocean, causing flooding inland. This will lead to impacts on drainage along the entire coastal region, which are not fully captured in existing models of coastal and river flooding. Based on current assessments, 39% of areas within 5 km of the Ecuadorian coast are already at high risk of annual flooding (Secretaría Nacional de Gestión de Riesgos, 2018).

Existing gender inequalities may be exacerbated by the dynamics of climate change in the project area. That said, the impacts and challenges are expected to be greater in the following areas:

- Having access to water fit for human consumption. At present they only have rainwater or tap water, pointing to substantial difficulty in adequately securing this resource, especially if droughts become longer or unpredictable. The water problem affects the general population because they lack adequate water, both in terms of quantity and quality, however it is women who are responsible for collecting water and therefore directly impacts on their time burden.
- Depending on the market for food of adequate quantity and quality. This may be a challenge and could widen gaps in access for men and women because all the basics need to be purchased with money, and the money is usually controlled by men.
- Women and men in four estuaries live in uncertain socioeconomic contexts and situations of extreme poverty, and lack access to general resources: land, tools, and formal education for gainful employment. This situation hinders their access to options to improve their living conditions and the dynamics of violence. While this situation is true of both, as explained in previous sections, women do tend to be at higher risk and with fewer assets.
- Men during non-take season are usually forced to search for work in other municipalities and provinces they live, leaving their families behind. Main sources of work are banana plantation, construction and commerce.
- The most vulnerable groups are elder men, divorced, widowed or single women, who do not have the financial support of members of a community organization or of a family with male income earners. They find themselves compelled to adopt extremely insecure livelihoods. The problem of single, widowed and divorced women is complex and in this respect the interviews did not provide sufficient information. The elderly related to artisanal activities usually can't generate incomes during their last years and don't have social networks to relay on.
- The gap in access to employment may be worsened by a lack of access to formal education and natural resources. For example, the women in the communities visited have no formal education, making it difficult for them to access key information to make appropriate, timely decisions and manage risks, and they are excluded from formal education settings. This occurs as a result of the lack of education available and because their relationship with the market is different.
- Artisanal fisheries in general and mangrove fisheries in particular, perpetuate conceptions of masculinity that, prevent fishermen from taking initiatives to break the cycle of poverty and environmental degradation. The pressure that derives from the social obligation to comply with the role of suppliers, pushes fishermen to irresponsible fishing behavior, justifying the non-observance of environmental regulations. Attitudes of negligence and conformism emanate from the need to prove to be a man and not fall into attitudes that show weakness or any trait considered feminine, repressing feelings to reaffirm their masculine position, defining themselves as someone without fear or pain, with a heroic attitude, giving little importance to the consequences.

Furthermore, it should be borne in mind that women will continue to spend a substantial amount of time on caregiving, lessening their chances of being able to truly participate in decision-making and in managing the use and exploitation of mangrove resources.

Conclusion and implications for project

This gender assessment highlights the complex set of historical, social, cultural, and economic factors that influence the project site's current gender norms and inequalities. Firstly, men and women are vulnerable and susceptible to the impacts of climate change in differentiated ways; they also have different coping mechanisms. Secondly, there is a clear lack of political participation by women in community-based mangrove/fishery decision-making fora. Thirdly, the data presented suggest that women are overburdened with work, especially reproductive labor, and this limits their ability to engage in community, organizational, and project activities. And finally, gender-based violence is quite high and further highlights the inequalities at the household level and an important additional barrier that keep women from using their voice and agency. This situation is compounded by a number of structural problems associated with education, health, the provision of utilities and limited employment opportunities. All these factors come together to shed light on a violation of rights, with gender inequities affecting women of all ages.

In these mangrove-dependent communities, while both men and women are vulnerable to climate change impacts, women often have higher vulnerability due to higher rates of poverty, fewer livelihood alternatives, and less access to decision-making/information spaces compared to men. Men tend to control the main economic activity in the communities: the harvesting and sale of seafood. Often it is men who hold formal positions of leadership within communities and fishing associations. Women, on the other hand, take care of housekeeping, caregiving for family members and cooking; very few are able to generate their own income.

Taking a holistic and proactive approach, the project's gender action plan will ensure that the projects' goals and activities take into account the complex realities that women and men face and help to improve the resilience of the mangrove-dependent communities.

Gender evaluation for the project					
Root causes	Sociocultural norms regarding land access do not recognize women as organization members or landowners.	Limited work opportunities for women and men in mangrove areas, becoming more difficult with climate change.	Lack of formal education for both sexes.	Structural and sociocultural dynamics are maintained through control over income.	Culturally, caregiving is seen as a woman's job.
Implications /causes	Difficulties participating in decision-making.	Women & men find it more difficult to generate income; turn to changes in livelihoods (e.g., migration, men enter traditionally female fisheries – black clam, cockle)	Difficulty making informed, knowledgeable decisions on conservation and work-related issues.	Gender-based violence.	Women devote long hours to caregiving duties, limiting their ability to access other environments and activities.
Immediate effects	Women do not participate in decisions on the mangrove.	Families do not have sufficient food to meet their nutritional requirements. Additional pressure on resources.	Women are left out of discussions on the environment and biodiversity conservation.	Women do not participate in community life.	Women are confined to the home and are highly dependent on a partner or male family members.

Effects	Women of all ages fail to access conservation benefits (including increased CC resilience)	Total dependence on the market for subsistence. Reduced efficiency of women's harvest.	Women of all ages fail to access conservation benefits (including increased CC resilience)	Policies and decision-making does not necessarily take into account women's needs.	Women do not gain access to socio-productive activities.
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Gender Action Plan

Given the gender assessment above, there are several areas where this project can integrate gender-responsive actions to help close gender gaps in mangrove management in order to support more efficient, effective, and equitable mangrove conservation. These opportunities are largely concentrated in the support to mangrove associations where direct collaboration with the women and men who depend on, and manage, the mangroves will occur. Activities within the gender action plan have been included in the project's budget, as noted. The project will have one CI safeguards specialist and one CI gender specialist in the project management unit, two CI field social specialists in the Guayas and Jambeli estuaries, and one PUCESE staff in the northern estuaries (4 CI staff costs total USD 1,575,580, 1 PUCESE staff cost USD 85,546). Together, these individuals will drive the implementation and accountability of the GAP, in collaboration with the entire project team and partners. Travel for these positions has also been included in the project budget (4 CI Staff Travel USD 95,202). CI's global Gender Advisor will also support the project's implementation and monitoring of the GAP (USD 66,531 in staff cost, USD 6,704 in travel).

Project Component 1: Mangrove areas under effective and climate-adapted management increased, including through community-based management (AUSCEMs) and protected areas implementing climate adaptation plans.							
Gender outcome: Women and men are able to equitably benefit from community-based management and local economic development initiatives, thus contributing to increased resilience to climate impacts (as monitored by the project monitoring system livelihoods assessment & mangrove cover).							
Output	Activity	Indicators	Baseline	Goal	Timeline	Entity responsible	Budget
1.Community-based mangrove conservation and management contributes to closing gender gaps and reducing exposure to flood risk of 3,465 vulnerable people (est. 50%M/50%F).	1.1 Revise existing mangrove use and stewardship agreements (and support new agreements) that supports women's and men's equal engagement in management and grants women and men equal rights to concessions and use of NTFPs in protected areas. This may include indicating a minimum number of seats on the association body for women.	Number of revised/new agreements with language that allows for equal decision-making, rights to concessions, and use of NTFPs.	No prior information. 59 agreements	100% of the 59 existing agreements and 100% of new agreements include gender considerations.	Third year of project implementation.	Mangrove technician + gender/social specialists.	Included in AUSCEM training workshops (Activity 1.1.1). Total USD 297,801
	1.2 Work closely with the project's mangrove association partners to support and encourage women's equitable and meaningful participation, recognizing and responding to potential elevation of GBV.	Percentage of women association members who have received tailored leadership training (e.g., technical skills, public speaking, etc.) and productive activities.	0	50% (1078 women association members - total)	End of project	Gender and safeguard specialist, social specialists.	Included in AUSCEM training workshops (Activity 1.1.1; Total USD 297,801) and local exchange visits for women leaders in AUSCEMs (Activity 1.2.1; Total USD 51,477)

		Number of mangrove associations and individuals that have received training on gender, joint decision making, reducing GBV, etc.	0 associations	60 associations	By end of project	Gender and safeguard specialist, social specialists.	Included in AUSCEM training workshops (Activity 1.1.1; Total USD 297,801) and specialists' duties
			0 men/0 women	800 association members			
		Number of women in associations	1078 (24% of all association members)	50% of all association members are women	By end of project		
	1.3 Document and share lessons learned, examples of women leaders, success stories, etc. to inform and influence other mangrove conservation activities to be gender-responsive	Number of communication pieces that share information on gender approach in this project	0	At least 2/year	Annual	Gender & SG specialist	Costed as part of the specialists' duties.
2. Socio-manglar investment plans include gender considerations	2.1 Contribute to the development of Socio-Manglar investment plans to ensure gender considerations are incorporated in community mangrove conservation/management initiatives.	Number of Socio Manglar investment plans that include gender-responsive activities and the equitable input of women and men.	0	4 (one for each estuary)	Third year of project implementation.	Gender and safeguard specialist. Social specialist.	Costed as part of the specialists' duties.
3. Males and females in mangrove areas benefit from adoption of diversified, climate resilient options and business practices linked to mangroves	3.1 Provide tailored technical support for current economic activities, or new activities that are priorities for women and men.	Number of males and females in mangrove areas benefiting from the adoption of diversified, climate resilient livelihood options and	0	41,500 (50% female/50% male)	Second year	Gender and safeguard specialist. Social specialist.	Costed as part of the specialists' duties.

		business practices linked to mangroves					
	3.2 Include gender consideration into the competitive fund for associations.	% of accepted proposals that have gender considerations, as relevant.	0	100% of relevant projects			
	3.3 Provide tailored support to the 2 women-only associations to develop strong proposals.	# of proposals developed by the women associations.		2 proposals			
4. Project team has the technical capacity to advance gender-responsive project activities.	4.1 Provide ongoing gender training/dialogue to CI and MAAE personnel	Number of regular training/refreshers/discussion sessions given to CI and MAAE personnel on gender and mitigating/reducing GBV in the project.	0	8	2x/year in years 2, 3, 4,5	Gender and safeguard specialist.	Costed as part of the specialists' duties; 8 workshops on awareness raising and capacity building on gender included (total USD 17,014)
	4.2 Ensure project's Grievance Redress Mechanism is gender sensitive, prepared to respond to GBV-related incidents.	Number/% of grievances that are related to gender		N/A – no target, simple reporting			
Project Component 2: The private sector becomes a transformational agent for change by reducing GHG emissions and providing financial support to conserve and restore mangroves that increase climate resilience for other coastal populations.							
Gender outcome: Shrimp aquaculture sector is more inclusive of communities, including women, minorities, and vulnerable men.							
Output	Activity	Indicator	Baseline	Goal	Timeline	Responsible	Budget

5. Aquaculture farms owners are more aware of the need for inclusion.	5.1 Sensitize the members of the Chamber of Aquaculture on the Aquaculture Stewardship Council's social impact standards on inclusion, human rights, and equity.	Percentage of aquaculture farm owners that attend the training.	0	30% of membership	Year 2, 3, 4	Gender and safeguard specialist & Private Sector	Costed as part of the specialists' duties. Workshop costs on inclusivity, human rights and equity USD 6,269 plus trainings on climate smart shrimp (total USD 273,963)
Project Component 3: Create the enabling conditions for sustaining reductions in mangrove deforestation and increased mangrove restoration by strengthening governance, climate change adaptation strategies, coastal management policies, and legal enforcement.							
Gender outcome: Local governments have included gender considerations & targets in their PDOT (Coastal Development + Zoning Plans)							
6. Support local governments to improve and/or implement PDOTs that are gender sensitive.	6.1 Provide guidance and support to local governments on how to apply a gender approach to PDOT implementation.	# of subnational PDOTs with gender approach	0	9	Year 2,3	Gender and safeguard specialist & Adaptation specialist	Costed as part of the specialists' duties. Workshop costs covered under Activity 3.2.1 (total USD 39,884)
7. Project's socioeconomic surveys are gender disaggregated and analyzed to identify gender-related changes in mangrove-dependent communities,	7.1 Survey instrument is gender informed	Gender & safeguard specialist is involved in survey design (Y/N)	N	Y	Each year	Gender and safeguard specialist	Costed as part of the specialists' duties. Socio economic surveys budgeted separately

which is then shared with relevant governments to inform policy.							
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Appendix 1: Interview and data collection guide

1. Objective:

- Identify the dynamics of participation, decision making.
- Show the impacts differentiated by gender

2. Agenda

Time	Activity	Methodology
30 min	Presentation of the activity Awareness through the dynamic "I never e ..."	The activity will be presented, additionally, the ice will be broken. Participants will be asked to make a circle. A ball will be handed out at random and will request that when they receive the ball, they can say their names and which organization they come from. In addition, they are asked to mention something that they were not allowed to do as children because they were women and men". The facilitator will make the first presentation as an example.
40 min	Community map	How has the landscape in which they live changed and what are the effects of climate change? Show differences in women and men both in the illustration and in the evidence of impacts.
30 min	Identification of benefits and risks with a gender perspective	Work will be done on identifying the main negative impacts of climate change for women and men, as well as identifying the main benefits and risks. They will be recorded on paper. To close the activity, the session will be closed based on the following guiding questions: - What are the coincidences and differences between the responses of the groups regarding heat, rain and wind? - What are the coincidences and differences between the responses on impacts on crops and animal husbandry? - Compare the responses of the different groups and analyze who has greater responsibility for maintaining crops and raising animals that are affected by changes in the climate, men or women? - How does this affect the daily work of women related to taking care of the home and their diet? - Do these impacts represent a threat to the food security of the family and the community? Why?
40 min	Prioritization of gender issues for women and men.	Once the problems and possible benefits and risks have been identified, the impact areas (community, productive, reproductive and community) will be identified. We will use the tree of life and post it for prioritization. The prioritization will be based on:

		- Pink: priority 1, the most important to address and improve the impacts of climate change on the lives of women. - Orange, priority 2, moderately important to address and improve the impacts of climate change on the lives of women. - Green: priority 3, nothing important to address and improve the impacts of climate change on women's lives.
	Conclusions and closing	

Matrix: Identification of impacts of climate change			
Differentiated impacts on nature		Differentiated impacts for women and men	
Landscape / natural resources	Animals	Women	Men

Adaptation / incentive measure	Activities	Who participates? (% men,% women)	How do women and men participate?	How are the benefits shared?	Obstacles to the participation of women	Strategies for the participatory inclusion of women
Improve irrigation (systems, technification).	Activity 1					
	Activity 2					
	Activity 3					
Vegetable nursery	Activity 1					
	Activity 2					
	Activity 3					
	Activity 4					

Appendix 2: Sample Job Description for Gender Specialist

Summary:

Conservation International is looking for a Gender Specialist to be based in Guayaquil City. This position is responsible for oversight, coordination, and technical support for the implementation of the project's Gender Action Plan (GAP).

Key Responsibilities:

- Coordinate/oversee the implementation and execution of the GAP within the framework of the GCF Gender Mainstreaming Policy.
- Develop and deliver appropriate capacity building for colleagues and delivery partners that is well-designed, clear, tailored to the context, and practical.
- Monitor gender-related outcomes through established project-specific indicators and through qualitative information gathering.
- Accompany project teams to the field to ground truth progress on gender outcomes: meet with women/women's groups, collect qualitative information, help project teams to problem solve, propose adaptive changes, participate in reports and other activities.
- Identify opportunities/entry points to support and encourage a diversity of women's leadership through project activities.
- Develop relationships with gender and women-focused NGOs/CSOs/government entities in order to learn and share lessons.
- Communicate internally and externally to share materials, project results, and lessons learned among project/program stakeholders and external audiences.
- Be the point person for any gender, Sexual Exploitation and Abuse and Harassment (SEAH) or Gender-based Violence (GBV) related grievance that may come through the project's grievance mechanism.
- Collaborate with CI's global gender program to ensure consistency in gender approach across the institution.

Required Qualifications.

- Experience in leading the implementation of gender/social analyses and gender action plans in conservation/climate/NRM-related projects involving Indigenous and rural populations.
- Work experience in implementing projects with a gender approach.
- Experience/understanding of taking a survivor-centered approach to respond to GBV issues.
- Strong facilitation and training skills.
- Bachelor's degree with a focus in social science, international development, or related discipline, ideally with postgraduate studies (specializations, diplomas, master's degree, etc.).

- Spanish and English language proficiency. (TOEFL, IELTS, TOEIC certificate etc.)
- Excellent skills in written and verbal communication.
- Executive and analytical capacity to implement projects, manage multiple tasks and adapt to a dynamic environment.
- Ability to work well in an interdisciplinary, multicultural, diverse, and dynamic team environment.
- Strategic, creative, flexible, and able to work under general supervision and take initiative to solve problems.
- Competencies with Microsoft Office 365.

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